Wavence 25

Alarm Clearing User Manual

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Issue 01

Alarm Clearing User Manual

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Table of Contents

Summary of changes 18

1 About this document 19

2 FCC part 15 subpart B 22

3 Maintenance philosophy 24
- 3.1 Troubleshooting 24
- 3.2 Before going to site checklist 25
- 3.3 Troubleshooting basics 26
- 3.4 Path problems on a commissioned link 28
- 3.5 Path problems on a new link 29
- 3.6 Troubleshooting TDM configuration problems 31
- 3.7 Troubleshooting Ethernet problems 32

4 Safety awareness 36
- 4.1 Safety EMC-EMF-ESD norms and equipment labeling 36
- 4.2 Safety rules 36
- 4.3 Labels indicating danger, forbidding, command 37

5 Electrostatic-sensitive devices 49
- 5.1 Electrostatic-sensitivity 49
- 5.2 Handling circuit packs 50

6 Product support information for the ANSI market 51
- 6.1 Customer service telephone and email support 51
- 6.2 Product documentation 51
- 6.3 Training 52
- 6.4 Technical support center 52
- 6.5 Repair and return services 53
- 6.6 Spare parts and replacement cards 53
- 6.7 Return for credit or warranty exchange procedure 54
- 6.8 Service center 54
- 6.9 Installation and maintenance services 55

7 Alarm and abnormal conditions overview 56
- 7.1 Overview 56
- 7.2 Friendly names 56
- 7.3 Facilities and equipment 59

8 Alarm and abnormal condition clearing procedures 61
- 8.1 Alarms and abnormal conditions 61

9 Supporting information 122
- 9.1 Additional support tools for maintenance and trouble clearing 122

10 1588 TC Fail alarms 125
- 10.1 General 125

10.2 Clear 1588 TC Fail alarms 125

11 AIM Read Fail alarm 127
11.1 General 127

12 AIS alarms 128
12.1 General 128
12.2 Clear AIS alarms 131
12.2.1 Clearing AIS alarms 131
12.2.2 AU4-AIS on Channelized E1 facility 132
12.2.3 TU12-AIS on Channelized E1 facility 132
12.2.4 AIS on E1 in on Channelized E1 facility 133
12.2.5 AIS on E1 out on Channelized E1 facility 133
12.2.6 AIS on E1 or DS1/3 in on E1, DS1 or DS3 facility 133
12.2.7 AIS on E1 or DS1/3 out on E1, DS1 or DS3 facility 134
12.2.8 AIS on EleSdh/OptSdh-in on STM-1/OC-3 facility 134
12.2.9 AIS on EleSdh/EleSdh-out on OC-3 facility 135

13 Announce Packet Loss 136
13.1 General 136
13.2 Clear Announce Packet Loss alarms 136

14 ATPC high power timeout alarms 138
14.1 General 138
14.2 Clear ATPC high power timeout alarms 139

15 Bandwidth degrade for protection switch alarms 141
15.1 General 141
15.2 Clear bandwidth degrade for protection switch alarms 141

16 Bandwidth Over Flow alarms 143
16.1 General 143
16.2 Clear bandwidth over flow alarms 143

17 Battery Failure alarms 145
17.1 General 145
17.2 Clear battery failure alarms 146

18 CLA alarms 148
18.1 Purpose 148
18.2 General 148
18.3 Procedure 148

19 Clock Class Degraded 151
19.1 Purpose 151
19.2 General 151
19.3 Procedure 151

20 Clock failure alarms 154
20.1 Purpose 154
20.2 General 154

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20.3 Procedure 155

21 Combiner loss of alignment alarms 162
21.1 Purpose 162
21.2 General 162
21.3 Procedure 162

22 Critical Link Event alarms 168
22.1 Purpose 168
22.2 General 168
22.3 Procedure 168

23 DC & PfoE Cables present 171
23.1 General 171
23.2 Procedure 171

24 Degrade alarms on sync source 172
24.1 Purpose 172
24.2 General 172
24.3 Procedure 174

25 Degrade alarms on LAG 177
25.1 Purpose 177
25.2 General 177
25.3 Procedure 178

26 Delay Response Loss 179
26.1 Purpose 179
26.2 General 179
26.3 Procedure 179

27 DemXPIC LOS 182
27.1 Purpose 182
27.2 General 182
27.3 Procedure 183

28 Demodulator fail alarms 185
28.1 Purpose 185
28.2 General 185
28.3 Procedure 186

29 Dialog failure alarms 188
29.1 Purpose 188
29.2 General 188
29.3 Procedure 189

30 Duplicate Address Detected alarms 195
30.1 General 195
30.2 Procedure 196
30.2.1 Duplicate address detected - TMN Ethernet MGMT Port 197
30.2.2 Duplicate address detected - Port #4 TMN Ethernet 199

30.2.3 Duplicate address detected - TMN In-Band Port #1/2 201
30.2.4 Duplicate address detected - PPP IP/RF Port 202

31 Early warning indication alarms 206
31.1 Purpose 206
31.2 General 206
31.3 Procedure 207

32 ETH OAM loss of continuity alarms 208
32.1 Purpose 208
32.2 General 208
32.3 Procedure 208

33 Eth OAM Mismerge Condition alarms 210
33.1 Purpose 210
33.2 General 210
33.3 Procedure 210

34 Eth OAM Remote Defect Indication alarms 212
34.1 Purpose 212
34.2 General 212
34.3 Procedure 212

35 Eth OAM Unexpected Level alarms 214
35.1 Purpose 214
35.2 General 214
35.3 Procedure 214

36 Eth OAM UnexpectedMEPID alarms 216
36.1 Purpose 216
36.2 General 216
36.3 Procedure 217

37 Eth OAM Unexpected Period alarms 218
37.1 Purpose 218
37.2 General 218
37.3 Procedure 218

38 Excessive Environmental Temperature alarms 220
38.1 Purpose 220
38.2 General 220
38.3 Procedure 221

39 Fan Communication Problem (MPT-HLS) 223
39.1 Purpose 223
39.2 General 223
39.3 Procedure 223

40 Fan Fail (MPT-HLC) 226
40.1 Purpose 226
40.2 General 226

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40.3 Procedure 226

41 High BER alarms on radios 230
41.1 Purpose 230
41.2 General 230
41.3 Procedure 231

42 High BER alarms on SDH cards 233
42.1 Purpose 233
42.2 General 233
42.3 Procedure 234
42.3.1 High BER on EleSdh/OptSdh-in on STM-1/OC-3 facility 234
42.3.2 High BER on EleSdh/OptSdh-out on STM-1/OC-3 facility 235
42.3.3 High BER on Channelized E1-in on STM-1 facility 235

43 Housekeeping alarms 237
43.1 Purpose 237
43.2 General 237
43.3 Procedure 238

44 Incompatible frequency alarms 240
44.1 Purpose 240
44.2 General 240
44.3 Procedure 241

45 Incompatible modulation alarms 243
45.1 Purpose 243
45.2 General 243
45.3 Procedure 244

46 Incompatible Shifter alarms 246
46.1 Purpose 246
46.2 General 246
46.3 Procedure 247

47 Incompatible Power alarms 249
47.1 Purpose 249
47.2 General 249
47.3 Procedure 250

48 Internal Communication Problem alarms 252
48.1 Purpose 252
48.2 General 252
48.3 Procedure (MPT) 253
48.4 Procedure (UBT) 258

49 LAG port down alarms 259
49.1 Purpose 259
49.2 General 259
49.3 Procedure 259

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50 LAG Rx Problem 261
50.1 Purpose 261
50.2 General 261
50.3 Procedure 261

51 LAG Traffic Down 263
51.1 Purpose 263
51.2 General 263
51.3 Procedure 263

52 LAG Traffic Loss 265
52.1 Purpose 265
52.2 General 265
52.3 Procedure 265

53 LAG TX Problem 267
53.1 Purpose 267
53.2 General 267
53.3 Procedure 267

54 Link Identifier Mismatch alarms 269
54.1 Purpose 269
54.2 General 269
54.3 Procedure 270

55 Log Upload failed 271
55.1 Purpose 271
55.2 General 271
55.3 Procedure 271

56 Loop Problem alarms 272
56.1 Purpose 272
56.2 General 272
56.3 Procedure 272

57 Loss of alignment alarms 278
57.1 Purpose 278
57.2 General 278
57.3 Procedure 278

58 Loss Ethernet Synchronization Message Channel alarms 282
58.1 Purpose 282
58.2 General 282
58.3 Procedure 283

59 Loss Of CESoETH Frame alarms 285
59.1 Purpose 285
59.2 General 285
59.3 Procedure 286

60 Loss Of Frame alarms on facilities 294

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60.1 General 294
60.2 Procedure 295
60.2.1 Loss of frame on E1/DS1/DS3/EleSdh/OptSdh-in facility 296
60.2.2 Loss of frame on DS1/DS3/E1/Ele/OptSdh-out facility 300

61 Loss of frame alarms on radios 308
61.1 General 308
61.2 Procedure 308

62 Loss of multi-frame alarms 316
62.1 Purpose 316
62.2 General 316
62.3 Procedure 316

63 Loss of pointer alarms 318
63.1 Purpose 318
63.2 General 318
63.3 Procedure 318

64 Loss of protection alarms 320
64.1 Purpose 320
64.2 General 320
64.3 Procedure 320

65 Loss of signal alarms on facilities 322
65.1 Purpose 322
65.2 General 322
65.3 Procedure 323

66 Loss of signal alarms on Ethernet 327
66.1 Purpose 327
66.2 General 327
66.3 Procedure 328

67 Loss of signal alarms on radios 330
67.1 Purpose 330
67.2 General 330
67.3 Procedure 331

68 Loss of signal synchronization alarms 333
68.1 Purpose 333
68.2 General 333
68.3 Procedure 334

69 Loss of signal alarms on LAG 336
69.1 Purpose 336
69.2 General 336
69.3 Procedure 337

70 Low input voltage alarms 339
70.1 Purpose 339

70.2 General 339
70.3 Procedure 339

71 MAC Status defect 351
71.1 Purpose 351
71.2 General 351
71.3 Procedure 351

72 Missing Diversity Alignment Configuration 352
72.1 Purpose 352
72.2 General 352
72.3 Procedure 352

73 Memory Card Forbidden to use 353
73.1 Purpose 353
73.2 General 353

74 OAM discovery not completed alarms 354
74.1 Purpose 354
74.2 General 354
74.3 Procedure 354

75 Power supply failure alarms 357
75.1 Purpose 357
75.2 General 357
75.3 Procedure 358

76 PPP fail alarms 366
76.1 Purpose 366
76.2 General 366
76.3 Procedure 366

77 Provisioning mismatch alarms 369
77.1 Purpose 369
77.2 General 369
77.3 Procedure 369

78 Radio encryption mismatch alarms 371
78.1 Purpose 371
78.2 General 371
78.3 Procedure 371

79 Rate mismatch alarms 374
79.1 Purpose 374
79.2 General 374
79.3 Procedure 375

80 Remote Defect Indication alarms on radios 377
80.1 Purpose 377
80.2 General 377
80.3 Procedure 378

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81 Remote defect indication alarms on facilities 379
81.1 Purpose 379
81.2 General 379
81.3 Procedure 380

82 Replaceable unit missing alarms 381
82.1 Purpose 381
82.2 General 381
82.3 Procedure 382

83 Replaceable unit problem alarms 384
83.1 General 384
83.2 Procedure 386
83.2.1 Replaceable unit problem on CorEvo 387
83.2.2 Replaceable unit problem on MSS-E/HE/XE 389
83.2.3 Replaceable unit problem on P2E3DS3, P32E1DS1, or SDHACC 390
83.2.4 Replaceable unit problem on EASv2 392
83.2.5 Replaceable unit problem on FAN, A-FAN, E-FAN, or MPT-HLS fan 395
83.2.6 Replaceable unit problem on SFP 396
83.2.7 Replaceable unit problem on MPT-HLC, or MPT-HLS 398
83.2.8 Replaceable unit problem on UBT ODU 403
83.2.9 Replaceable unit problem on RPS SFP on UBT ODU or MPT-HLS 405
83.2.10 Replaceable unit problem on RPS or XPIC module on UBT ODU 406
83.2.11 Replaceable unit problem on Diversity receiver module on MPT-HLC or MPT-HLS 407

84 Replaceable unit type mismatch alarms 412
84.1 Purpose 412
84.2 General 412
84.3 Procedure 414

85 Retiming buffer overflow alarms 415
85.1 Purpose 415
85.2 General 415
85.3 Procedure 416

86 RF Switch Fail alarms 417
86.1 Purpose 417
86.2 General 417
86.3 Procedure 417

87 RPS path fail 421
87.1 Purpose 421
87.2 General 421
87.3 Procedure 421

88 Rx diversity missing alarms 426
88.1 Purpose 426
88.2 General 426
88.3 Procedure 427

89 Rx Overload alarms 431
89.1 Procedure 431

90 Rx problem alarms 432
90.1 Purpose 432
90.2 General 432
90.3 Procedure 432

91 Secondary battery failure alarms 434
91.1 Purpose 434
91.2 General 434
91.3 Procedure 435

92 Service selective drop alarms 437
92.1 Purpose 437
92.2 General 437
92.3 Procedure 437

93 Signal Label Mismatch alarms 439
93.1 Purpose 439
93.2 General 439
93.3 Procedure 440

94 Software download failed 441
94.1 Purpose 441
94.2 General 441
94.3 Procedure 441

95 Standby version mismatch alarms 442
95.1 Purpose 442
95.2 General 442
95.3 Procedure 443

96 Sync Packet Loss 444
96.1 Purpose 444
96.2 General 444
96.3 Procedure 444

97 Synchronization signal fail alarms 447
97.1 Purpose 447
97.2 General 447
97.3 Procedure 448

98 Threshold crossing alarms 453
98.1 Purpose 453
98.2 General 453
98.3 Procedure 456
98.3.1 TCA on radio facility 457
98.3.2 TCA on DS1/E1/OptSdh/EleSdh-in 457
98.3.3 TCA on DS1/E1/OptSdh/EleSdh-out 458

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99 Trace identifier mismatch alarms 459
99.1 Purpose 459
99.2 General 459
99.3 Procedure 459

100 Traffic down alarms 461
100.1 Purpose 461
100.2 General 461
100.3 Procedure 461

101 Traffic loss alarms 463
101.1 Purpose 463
101.2 General 463
101.3 Procedure 463

102 Unavailable time alarms 466
102.1 Purpose 466
102.2 General 466
102.3 Procedure 468

103 Tx problem alarms 470
103.1 Purpose 470
103.2 General 470
103.3 Procedure 470

104 Unconfigured equipment present alarms 472
104.1 Purpose 472
104.2 General 472
104.3 Procedure 473

105 Underlying resource unavailable alarms 475
105.1 Purpose 475
105.2 General 475
105.3 Procedure 476

106 Underlying resource unavailable service alarms 478
106.1 Purpose 478
106.2 General 478
106.3 Procedure 479

107 Version mismatch alarms 480
107.1 General 480
107.2 Procedure 480
107.2.1 Version mismatch (License) 481
107.2.2 Version mismatch (Software) 481
107.3 XPIC Cable Loss 482
107.4 Purpose 482
107.5 General 482
107.6 Procedure 483

List of Figures

Figure 1 Trap destination IP 33
Figure 2 Debug Info, ping selection 34
Figure 3 IP input 34
Figure 4 3 packets sent and recieved 35
Figure 5 Compliance boundaries for Wavence transceivers with 1ft (30 cm) antenna: 42
Figure 6 Anti static protection device kit 44
Figure 7 Electrostatic-sensitive sign 49

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List of Tables

Table 1 Naming conventions 20
Table 2 Friendly Name/Entity Syntax 57
Table 3 Friendly name syntax definitions 58
Table 4 Craft Terminal's Active Alarms alarm types 62
Table 5 Abnormal Conditions 103
Table 6 CorEvo Card LED indications 109
Table 7 Transport Card LED indications 111
Table 8 Ethernet LED indications 112
Table 9 MPT-HLC transceiver LED alarm indications 113
Table 10 MPT-HLS transceiver front panel indicator details 117
Table 11 TMN network troubleshooting 120
Table 12 Supporting information 123
Table 13 1588 TC Fail alarms 125
Table 14 AIS alarms 130
Table 15 Announce Packet Loss alarms 136
Table 16 ATPC High Power Timeout alarms 139
Table 17 Bandwidth Degrade for Protection Switch alarms 141
Table 18 Bandwidth Over Flow alarms 143
Table 19 A battery fail alarms 145
Table 20 CLA alarms 148
Table 21 Clock Class Degraded alarms 151
Table 22 Clock failure alarm 154
Table 23 Combiner loss of alignment alarm 162
Table 24 Critical link event alarm 168
Table 25 Degrade alarms 173
Table 26 Degrade alarms 177
Table 27 Delay Response Loss alarms 179
Table 28 DemXPIC LOS alarms 182
Table 29 Demodulator fail alarms 185
Table 30 Dialog failure alarms 189
Table 31 Duplicate address detected alarms 196
Table 32 Early Warning Indication alarms 206
Table 33 Eth OAM Loss of Continuity alarms 208
Table 34 Eth OAM MisMerge Condition alarms 210
Table 35 Eth OAM Remote Defect Indication alarms 212
Table 36 Eth OAM Unexpected Level alarms 214
Table 37 Eth OAM UnexpectedMEPID alarms 216
Table 38 Eth OAM Unexpected Period alarms 218
Table 39 Equipment alarms 220
Table 40 Fan fail alarm 223

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Table 41 Fan fail alarm 226
Table 42 High BER alarms on radio equipment 230
Table 43 High BER alarms on SDH facilities 234
Table 44 Housekeeping alarms 237
Table 45 Incompatible Frequency alarms 240
Table 46 Incompatible modulation alarms 243
Table 47 Incompatible Shifter alarms 246
Table 48 Incompatible Power alarms 250
Table 49 Internal communication problem alarms 253
Table 50 LAG Port Down alarms 259
Table 51 Link Identifier Mismatch alarms 269
Table 52 Loop Problem alarms 272
Table 53 Loss of alignment alarms 278
Table 54 Loss ESMC alarms 283
Table 55 Loss Of CESoETH Frame alarms 285
Table 56 Loss Of Frame alarms 295
Table 57 Loss of frame radio alarms 308
Table 58 Loss of multi-frame alarms 316
Table 59 Loss of Pointer alarms 318
Table 60 Loss Of Protection alarms 320
Table 61 Loss Of Signal alarms 323
Table 62 Loss of Signal alarms on Ethernet 328
Table 63 Loss of Signal alarms on radios 331
Table 64 Loss Of Signal Sync alarms 333
Table 65 Loss of Signal alarms on LAG 337
Table 66 Low Input Voltage alarms 339
Table 67 OAM discovery not completed alarms 354
Table 68 Power Supply Failure alarms 358
Table 69 PPP Fail alarms 366
Table 70 Provisioning Mismatch alarms 369
Table 71 Radio encryption mismatch alarms 371
Table 72 Rate mismatch alarms 374
Table 73 Remote Defect Indication alarms 378
Table 74 Remote Defect Indication alarms 379
Table 75 Replaceable Unit Missing alarms 382
Table 76 Replaceable Unit Problem alarms 385
Table 77 Replaceable Unit Type Mismatch alarms 413
Table 78 Retiming Buffer Overflow alarms 415
Table 79 RF Switch Fail alarms 417
Table 80 RPS path fail alarms 421
Table 81 Rx diversity missing alarm 426
Table 82 Rx Problem alarms 432

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Table 83 Secondary Battery Failure alarms 434
Table 84 Service Selective Drop alarms 437
Table 85 Signal Label Mismatch alarms 440
Table 86 Standby Version Mismatch alarms 442
Table 87 Sync Packet Loss alarms 444
Table 88 Synchronization Signal Fail alarms 448
Table 89 Threshold crossing alarms 455
Table 90 Trace Identifier Mismatch alarms 459
Table 91 Traffic Down alarms 461
Table 92 Traffic Loss alarms 463
Table 93 Unavailable Time alarms 467
Table 94 Tx Problem alarms 470
Table 95 Unconfigured equipment present alarms 473
Table 96 Underlying Resource Unavailable alarms 476
Table 97 Underlying Resource Unavailable Service alarms 478
Table 98 Version Mismatch alarms 480

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Summary of changes

The following has changed in this release. Changes between document issues are cumulative. Therefore, the latest document issue contains all changes made to previous issues.

Changes between release Wavence 24, Issue 01 and Wavence 25, Issue 01

Removed phased out hardware throughout the document.

Added/updated Wavence 25 alarms.

General: AIM Read Failure (EEPROM OK) alarm is automatically cleared after one hour.

Changes between release Wavence 23A, Issue 01 and Wavence 24, Issue 01

Alarms and abnormal conditions, chapter updated with new alarms for MSS-XE, MSS-E, MSS-HE.

Changes between release Wavence 23, Issue 01 and Wavence 23A, Issue 01

Alarms and abnormal conditions, chapter updated with new alarms for UBT, MSS-XE, MSS-E, MSS -HE and CAHD.

Changes between release Wavence 22A, Issue 01 and Wavence 23, Issue 01

Alarms and abnormal conditions, chapter updated with new alarms for MSS-XE, MSS-E, MSS-HE and CAHD.

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1. About this document

Purpose

This guide provides information on how to interpret and resolve alarms on the Wavence.

Intended audience

This document is intended for individuals who are responsible for maintaining and troubleshooting the Wavence.

Safety information

For your safety, this document contains safety statements. Safety statements are given at points where risks of damage to personnel, equipment, and operation may exist. Failure to follow the directions in a safety statement may result in serious consequences.

For additional safety precautions, see the *Wavence Product Information Manual.*

Technical support

If you purchased a service agreement for your Wavence and related products from a distributor or authorized reseller, contact the technical support staff for that distributor or reseller for assistance. If you purchased an Nokia service agreement, check this link for instructions to contact NOKIA Support personnel.

Online survey

You can give direct feedback regarding Customer Documentation through this short form :

↗ https://forms.office.com/Pages/ResponsePage.aspx?id=URdHXXWWjUKRe3D0T5YwsCMrxVsgZAtPphmbs9GKxM9UREkzM1AwS0s4N0w4UUtRNFcyRUVLS0o1Uy4u

Documentation feedback

If you have questions or comments about this documentation, please contact:

documentation.1.feedback@nokia.com

Naming conventions in the documentation

The generic terms described in Table: Naming conventions are used when the topic applies to all of the related items listed. If the topic is specific to one or a subset of the items listed, the specific names will be stated in the guide.

Table 1: Naming conventions

Term	Refers to
CAHD	Refers to Carrier Aggregation High Density card.
Core	Refers to CorEvo card
CorEvo	Refers to both: • CorEvo-1G card • CorEvo-10G card
EAC	Refers to EAC and EAC-10G modules.
EAS	Refers to EASv2 card
MPT-HLS	Refers to MPT-HLS standard
MSS	Refers to the following: • MSS-4 • MSS-8
PDH	Refers to all of the following: • E1 • DS1 • DS3
SDH	Refers to both: • STM-1 • OC-3
UBT	Refers to UBT-T/T XP, UBT-S and UBT-m

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Audience

This guide is intended for network installers and system administrators who are responsible for installing, configuring, or maintaining networks. This guide assumes you are familiar with electronic and networking technologies.

Information symbols

Observe the warnings and notes to avoid injury or equipment damage during installation and maintenance. Follow the safety procedures and guidelines when working with and near electrical equipment. Warning statements and notes are provided in each chapter.

The following symbols are contained in the Wavence documentation.

⚠ DANGER!

DAMAGE
indicates that the described activity or situation may result in serious personal injury or death; for example, high voltage or electric shock hazards.

⚠ CAUTION!

DAMAGE
indicates that the described activity or situation may, or will, cause equipment damage or serious performance problems.

⚠ CAUTION!

DAMAGE
indicates that the described activity or situation may, or will, cause service interruption.

ⓘ Note:

provides information that is, or may be, of special interest.

2. FCC part 15 subpart B

Wavence unlicensed radio

The JF6-9558H/6933B-9500MPT (MPT-HL) unlicensed radio provides fast deployment of service with microwave radio. No license and small antennas (no FCC and Industry Canada requirements) allow immediate turn-up. After the license is received, the unlicensed MPT-HL radio can be easily converted to the lower 6 GHz licensed band.

The JF6-9558H/6933B-9500MPT unlicensed radio operates in the 5725-5850 Information, Scientific, and Medical (ISM) band in accordance with FCC Part 15.247 and IC RSS-210. This unlicensed radio, although operating in the same band as a spread spectrum radio, operates using narrower bandwidths than spread spectrum.

FCC Class B compliance statement

The JF6-9558H/6933B-9500MPT unlicensed radio has been tested and found to comply with the limits for a Class B digital device, pursuant to Part 15 of the FCC Rules and IC RSS-210. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.

FCC Class B requirements

⚠ WARNING!

Regulatory compliance warning: Physical changes or modifications to the JF6-9558H/6933B-9500MPT (unlicensed) radio are strictly prohibited.

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⚠ CAUTION!

Changes or modifications not expressly approved by Nokia could void the authority to operate the JF6-9558H/6933B-9500MPT (unlicensed) radio.

⚠ CAUTION!

Installation, Turn-Up, Maintenance, and Operation Instruction supplied with the JF6-9558H/6933B-9500MPT (unlicensed) radio require strict adherence for continued part 15 of the FCC Rules and IC RSS-210 compliance.

This device complies with part 15 of the FCC Rules and IC RSS-210. Operation is subject to the following three conditions: (1) this device may not cause harmful interference. (2) This device must accept any interference received, including interference that may cause undesired operation. (3) This device must be professionally installed.

Cet appareil radio est conforme Á IC RSS-210. Son fonctionnement respecte les trois conditions suivantes: 1) cette radio ne cause pas d"interférences néfastes, 2) cette radio peut recevoir des interférences, ainsi que des interférences qui peuvent causer des opérations non désirées, et 3) cette radio doit être installée par des Professionnels.

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3. Maintenance philosophy

Maintenance of the system consists of repair and preventive maintenance. Repair covers isolation of faults, troubleshooting to the circuit card level, and subsequent replacement of the faulty circuit card. Circuit card repair is not covered. Preventive maintenance covers periodic routine tasks that are performed to ensure continued proper operation of the system.

The system incorporates both software- and firmware-based diagnostics to continuously monitor system integrity. Also available are diagnostics that can be manually invoked. Problems detected by diagnostics are reported as alarm/event messages displayed at a user terminal and/or lighted alarm indicators on the equipment. These messages and lights are the primary trouble analysis tools for fault isolation.

The instruction "contact next level of technical support for assistance" appears throughout this manual. The intent of this statement is to escalate the problems that are beyond the scope of this Task Oriented Practice (TOP) to a higher level of expertise. Users who have local interim levels and/or procedures should implement those in response to this instruction. See the Introduction for customer access (HELP) contacts.

3.1 Troubleshooting

Overview

Troubleshooting procedures generally assume the following conditions:

- An alarm indication was received through the Craft Terminal's Active Alarms, Craft Terminal or a lighted alarm indicator.
- Maintenance personnel are familiar with system operation and use of the Craft Terminal's Active Alarms and Craft Terminal.
- Maintenance personnel have system administration privileges or a system administrator is available.
- Replacement components (spares) are available and fully functional.

The following guidelines and information should be observed during troubleshooting:

- Trouble clearing begins with an analysis of alarm indications. The correct chapter is referenced

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in the index.

- Clear upstream problems first. This may also clear downstream problems. Upstream/downstream imply directionality. If there is a Radio TX RF issue: downstream would be the far end RX and upstream would be the EAS to Radio interface.
- As a rule, no main circuit card should be physically removed unless traffic was manually switched to protection by entering the appropriate Craft Terminal command. Exceptions to this are addressed in specific chapters.
- As a rule, no protection circuit card should be physically removed unless a manual lockout switch issued by entering the appropriate Craft Terminal command. Exceptions to this are addressed in specific chapters.
- As a rule, in unprotected radio configurations physically removing a circuit card will cause a loss of traffic. Exceptions to this are addressed in specific chapters.
- To avoid unnecessary service interruption, exercise care when removing circuit cards, performing commands, and doing other maintenance tasks.
- The procedures in this manual are used to aid in clearing alarms for which protection is available. If alarms exist on both a card and its mate card, or if alarms exist on a card and its mate card is not available for protection, always contact the next level of technical support for assistance with clearing the alarms.
- Always follow electrostatic-sensitive device procedures when handling circuit cards.
- After circuit card replacement, allow time for alarms to clear. As a general rule, allow a minimum of 120 seconds to clear. Some alarms may take longer to clear. Allow time for the alarm to clear before assuming the replacement did not clear the trouble.
- When replacing a circuit card does not clear the trouble, remove the replacement circuit card and return the original circuit card to service.
- When troubleshooting procedures fail to clear the fault, an obscure or multiple fault is assumed to exist.

3.2 Before going to site checklist

Overview

Where possible, before going to site obtain the following information:

- Does the fault require immediate attention?
- Determine who is the best-placed person to attend the fault.
- Confirm the nature and severity of the reported fault, its location, Wavence type, frequency

band, high/low end ODU or MPT-HL Transceiver, capacity modulation and configuration (non protected, protected, diversity). Ask:

  - Is just one Wavence link affected, or a number of links in the same geographical area?
  - Is the path down completely or is traffic passing but with a BER alarm?
  - Is only one or a number of tributaries affected?
  - Could the fault be in the equipment connected to Wavence, rather than in Wavence? Are there alarms on other, connected equipment?
  - Is it a hard or intermittent fault?
  - Do alarms *confirm* which end of an alarmed link is faulty?
- Could the weather (rain, ice, high wind, temperature) be a factor in the reported fault?
  Note: If the fault suggests a rain fade or other weather related fade condition and it matches the prevailing weather conditions, do not take any action until the weather abates.
- Does link history suggest any fault trends?
  - Does the fault history for the link indicate a likely cause?
  - Is the Wavence link newly installed?
  - Has there been any recent work done on the link?
- Ensure that you have with you:
  - Appropriate spares. Where an equipment failure is suspected, these should include replacement Cards/plug-ins, MPT-HL Transceiver and ODU. If an ODU is suspected then local/national climbing safety requirements must be adhered to.
  - A laptop PC and an Ethernet cable. Wavence Node/Terminal IP address and the addresses for any remote sites to be accessed are needed.
  - If login security has been enabled, you need the ‘engineer" password for the local and also any remote sites to be accessed.
    If you use TACACS+ and the TACACS+ server connection is down, the locally saved user/pwd will be used only.
  - Any special test equipment that may be needed, such as a BER tester.
  - Toolkit.
  - Key(s) for access to the site.

3.3 Troubleshooting basics

Overview

This section provides general guidance on Wavence troubleshooting:

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Check front-panel LED indications. These provide summary alarm indications, which can help narrow down the location and type of failure. See Alarm and abnormal condition clearing procedures for more information.

> **Note:**
> Where a Status LED on a plug-in is off (unlit), but power to the MSS is confirmed by LEDs on other plug-ins, check the seating of the affected plug-in.

Check Main Screen. When logging into Wavence with Craft Terminal, the opening screen is the Main Screen. Use the information provided to check for severity and problem type. See Table: Craft Terminal's Active Alarms alarm types for probable cause and recommended action.

Check the basics first.

- For example, if multiple alarms are present, and these include power supply voltage or hardware alarms, always check their cause before looking at resultant down-stream path failure or path warning (signal) alarms.
- Similarly, if a path-related failure is indicated (no hardware or software alarms), investigate the path. Go to the WebCT Radio Dashboard page to check supporting data, such as low RSL and incidence of intermittent pre-failure BER alarms, which if present are evidence of a path -related failure. See Troubleshooting path problems in this chapter for more information.

Check if symptoms match the alarm. Alarms reflect the alarm state, but in exceptional circumstances an alarm may be raised because of a failure to communicate correctly with the alarm source, or a failure in alarm management processing. Always check to see if symptoms match the alarm, using LED indications and the Craft Terminal.

Check if recent work may be a cause. Recent work at the site may be a cause or contributing factor. Check for a configuration change, software upgrade, power recycling (reboot), or other site work:

- Many hardware alarms are only initiated as a loss-of-communications alarm during a reboot, software upgrade, or reconfiguration. By not being able to communicate with the Core, their settings cannot be loaded. The fault may be at the hardware device (most likely), communications to it, or the Core.
- Hardware/software compatibility alarms will be raised when a new plug-in is installed that needs a later version of Wavence software.
- Hardware incompatible alarms will be raised when a plug-in is installed in a slot that has been configured for a different plug-in.

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MSS before an ODU. If there is doubt about whether a fault is in the MSS or ODU, always troubleshoot the MSS first.

MSS cards are hot-pluggable. There is no need to power-down before replacing, but traffic will be lost unless the plug-in is protected.

Plug-in restoration and SW upgrade time. Ensure adequate time is allowed for services to resume when a plug-in is replaced.

Troubleshooting path problems

A path-related problem, with the exceptions of interference, reflection, or multipath, is characterized by traffic being similarly affected in both directions. Generally, if you are experiencing only a one-way problem, it is not a path problem.

> **Note:**
> A path extends from ODU or MPT-HL antenna port to ODU or MPT-HL antenna port.

Normally a path problem is signaled by a reduced RSL, and depending on its severity, a high BER.

Only in worst case situations, such as an antenna knocked out of alignment, will a path fail completely, and stay that way.

For weather-related problems, such as rain or ducting, the path problem will disappear as the weather returns to normal.

3.4 Path problems on a commissioned link

Overview

A path problem on an existing link, one that has been operating satisfactorily may be caused by:

- Weather-related path degradation
- Changed antenna alignment or antenna feed problem
- New path obstruction
- Interference from other signal sources

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Weather-related path degradation

If BER alarms are fleeting/not permanent and RSL returns to its normal, commissioned level after the alarm is cleared, rain, diffraction, or multipath fading is indicated. Rain fade is the likely cause of fade for links 13 GHz and higher. Diffraction and multipath/ducting for links 11 GHz and lower. If these alarms are persistent, there could be a problem with the link design or original installation.

Changed antenna alignment or antenna feed problem

If RSLs do not return to commissioned levels after a period of exceptionally strong winds, suspect antenna alignment. Also, check the antenna for physical damage, such as may occur with ice-fall. For a remote-mounted ODU or MPT-HL, check its antenna feeder.

New path obstruction

Where all other parameters check as normal, and the path has potential for it to be obstructed by construction works, view/survey the path for possible new obstructions.

Interference from other signal sources

Interference usually affects traffic in just one direction. Unlike other path problems, RSL is not affected. If suspected, check for new link installations at, or in the same geographical area, as the affected site. Ultimately, a spectrum analyzer may have to be used to confirm interference, which is not an easy task given the need to connect directly to the antenna port, after removing the ODU or MPT-HL.

The following tool may be used:

- Wavence Spectrum Analyzer tool for UBT/HLC

For more information, see the Release Notes delivered with each tool.

3.5 Path problems on a new link

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Overview

For a new link, potential problems can extend to also include:

- Incorrect antenna alignment
- Mismatching antenna polarizations
- Incorrect path calculations
- Reflections
- IDU-ODU cable BER

Incorrect antenna alignment

One or both antennas incorrectly aligned. See the Wavence Turn-up Manual, Antenna Polarization Verification Antenna Feedhorn Verification - XPD Procedure.

Mismatching antenna polarizations

The link can't be aligned with the calculated receive level. Difference is around 30dB.

The polarization has to be checked manually at the mounted UBT.

Incorrect path calculations

If the RSLs are too low or too high, antenna alignment is correct, and Tx power settings are correct, check the path calculations used to determine the link performance. A good calculation match is +/- 2dB. Disagreements in excess of 3 dB should be investigated.

Reflections

Reflection (path cancellation) problems may not have been picked up at the path planning stage, particularly if the survey was a simple line-of-sight. If suspected, resurvey the path.

IDU-ODU cable BER

If just BER is present, this is not seen in PM counters. The radio PM can be perfect but the data is not passing properly. If the data is not passing properly, this can only be seen if:

- the incoming data from the ETH ports are not forwarded to the radio links
- WebCT is very slow when managing the remote Wavence node - ECFM is indicating issues

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(CCM, SOAM PM)

3.6 Troubleshooting TDM configuration problems

Overview

Configuration problems should only occur during the setup of a new link, or reconfiguration of an existing link. The more common problems may be broadly categorized as:

- Compatibility Problems
- Incorrect circuit connections
- Incorrect ID naming and commissioning
- Incorrect/incompatible trib settings

Compatibility problems

The two alarms that may activate are Configuration Not Supported and SW/HW Incompatible:

Configuration Not Supported: The plug-in installed is not enabled or is incorrect for the configuration.

Incorrect circuit connections

If the circuits are incorrectly connected, then no alarms are activated. An incorrect circuit assignment means the expected end-to-end connectivity will not happen. For every node carrying the lost circuit(s), check the circuit assignments again.

Take extra care when configuring ring circuits.

Where the problem is not obvious, run the tributary loopback BER test using the BER tester from your equipment, to track a single circuit through a Wavence network, beginning at the node closest to the node applying the BER test.

Incorrect ID naming and commissioning

All traffic-carrying circuits must have a unique flow ID for the cross-connect capability to operate.

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Incorrect/incompatible trib settings

Trib line interface settings incorrect, or line levels incompatible. While no alarm activates for an incorrect setting, its effect may result in line levels being too low (LOS alarm), or too high, resulting in a high BER.

Ensure the tributary type, unbalanced/balanced and configured impedance are correct.

3.7 Troubleshooting Ethernet problems

Overview

This section gives general guidance on troubleshooting problems related to the Ethernet ports on the CorEvo, EAC, EASv2 cards for the UBT.

The most common Ethernet problems are network and connectivity related and therefore always check the following first:

- Verify link partner capability, provisioning, and connection
- Verify radio provisioning matches link partner
- Verify cabling between radio and link partner

The LEDs on the card's front panel for each Ethernet connector are a good indicator of correct connectivity and activity on the Ethernet port. See Table: Ethernet LED indications for LEDs that are locally at the alarmed site.

In order for the green Link LED to light:

1. Cable must be connected to Ethernet port
2. Ethernet port must be enabled (provisioned Enabled on Core Settings Screen)
3. Speed and mode (on Core Settings Screen) must be provisioned the same as the link partner.

The yellow LED opposite the green on the connector indicates activity only. The flashing yellow LED is not an indicator of signal type or quality.

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Troubleshooting TMN problems

The most common TMN problems are network related and first alert is normally observed by improper operation at the SNMP master. Always check the following first:

- Verify master is properly registered in NE to receive traps.
- Verify SNMP version matches system requirements.
- Verify correct community string and privileges.
- Verify proper network routing.
- Check the TMN routing table at the Wavence
- Perform Ping tests from Wavence to server and from server to Wavence (ping is typically done first):
  - Copy the NFM-P server IP address

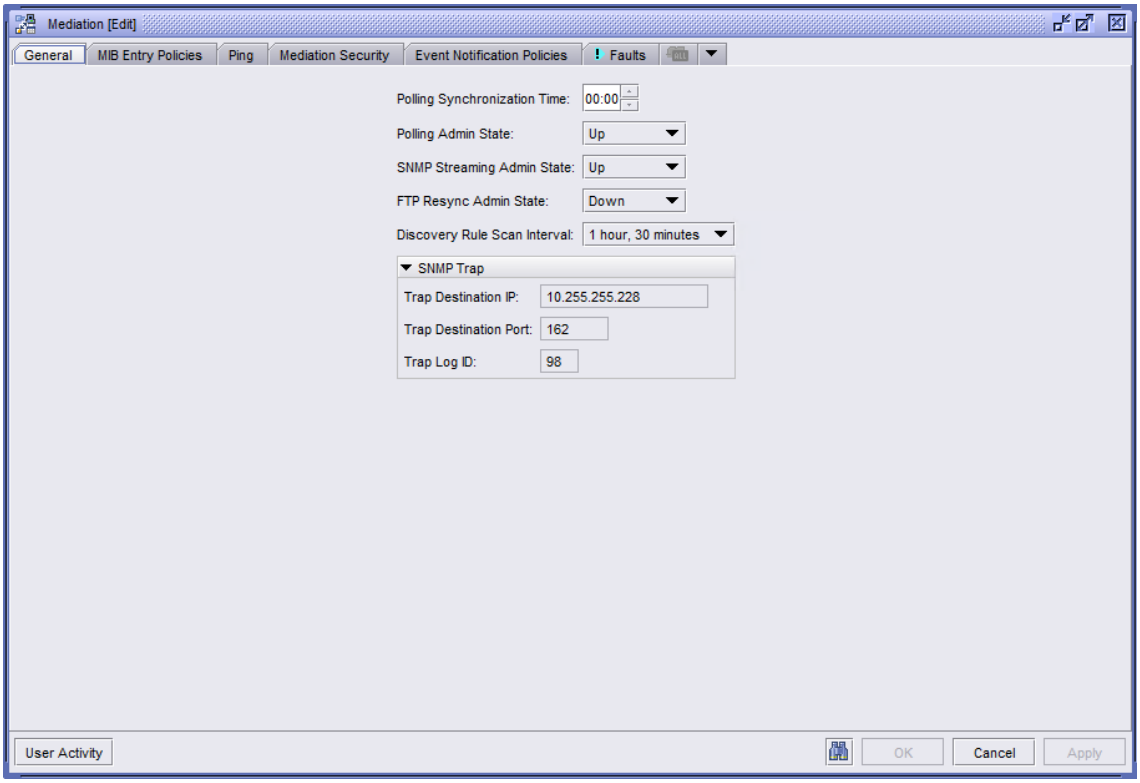
    Ping from NFM-P to the NE.

    Enhanced check from NFM-P Linux:
    - ping, traceroute to NE
    - ping with MTU 2000 to the NE
    - SNMP Get to the NE

    At the traceroute output, log in to the last seen IP/node and check the routing tables.

    Compare with the routing table of the Wavence.

    Figure 1: Trap destination IP

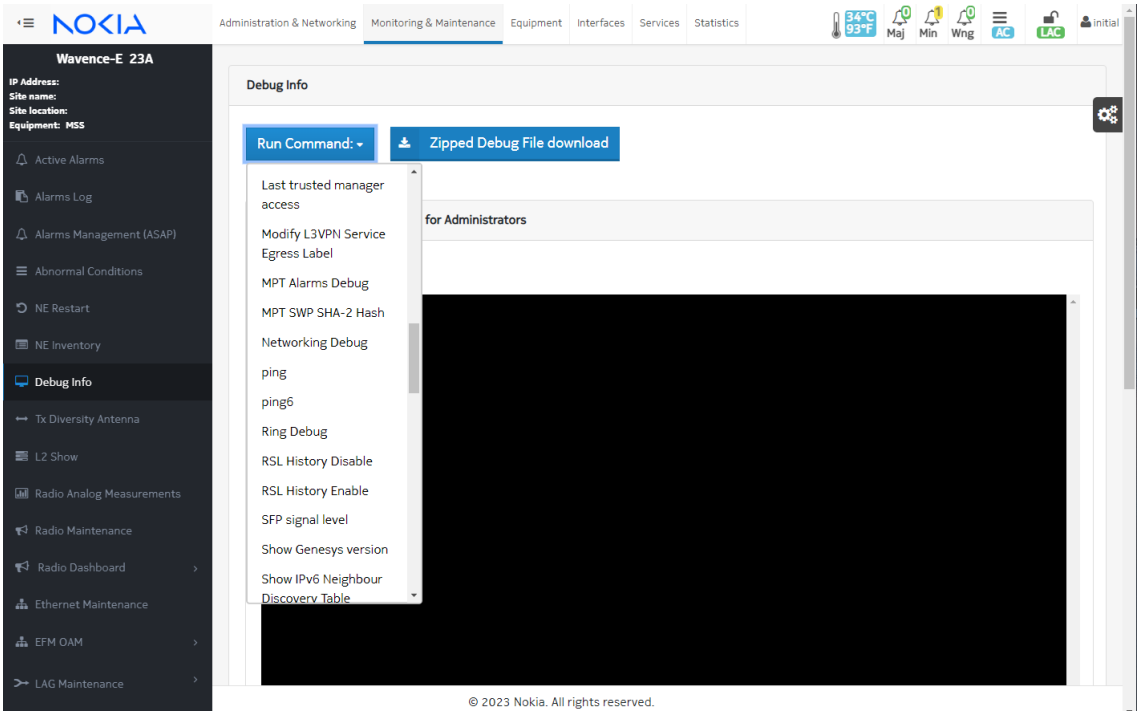


    The IP address can be found at `General > SNMP Trap > Trap Destination IP`

inside the NFM-P.

- Go to `Monitoring & Maintenance > Debug Info` in WebCT

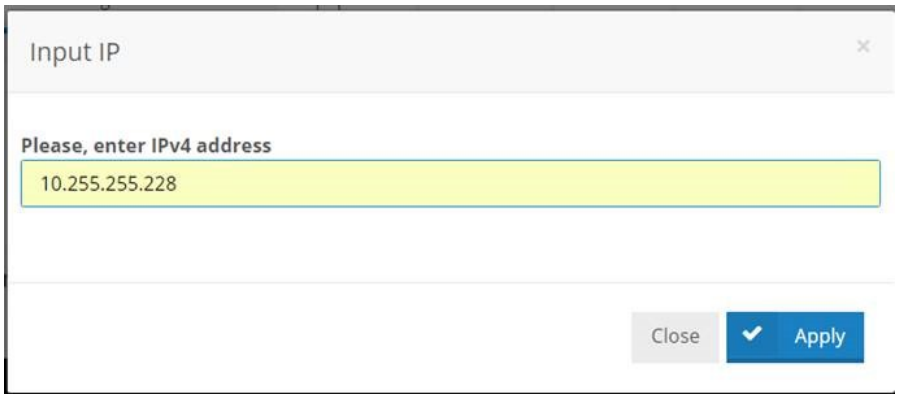
Figure 2: Debug Info, ping selection



Select the command "Ping"

- Enter the IP address in the given field

Figure 3: IP input



- Check that 3 packets are received

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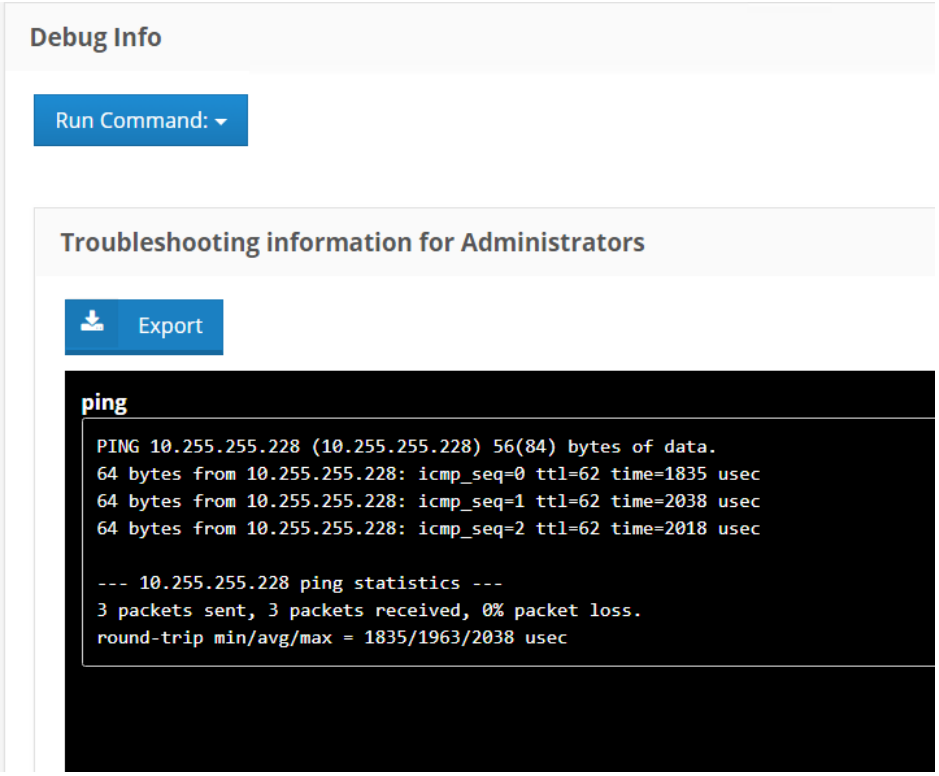
Figure 4: 3 packets sent and recieved



See Table: TMN network troubleshooting for information about TMN network troubleshooting.

4. Safety awareness

4.1 Safety EMC-EMF-ESD norms and equipment labeling

Overview

This section describes the equipment labeling and the norms mandatory or suggested that must be considered to avoid injuries on persons and/or damage to the equipment.

- Safety rules
- Electromagnetic compatibility (EMC norms)
- Equipment protection against electrostatic discharges
- Equipment protection against electrostatic discharges

4.2 Safety rules

General rules

Before performing any installation, turn-on, tests or operation and maintenance operations, read carefully the related sections of this manual, specifically:

- Hardware Installation
- Commissioning
- Maintenance and Upgrade

Observe safety rules

When equipment is operating, do not access the equipment parts which are protected with Cover Plate Shields removable with tools.

If you must access the equipment parts when it is operating, this is allowed exclusively to service personnel, where for Service Personnel or Technical assistance is meant:

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- "Personnel which has adequate Technical Knowledge and experience necessary to be aware of the danger that he might find in carrying out an operation and of the necessary measurements to reduce danger to minimum for him and for others".
- The Service Personnel can only replace the faulty units with spare parts.
- The Service Personnel is not allowed to repair: hence the access to the parts not specified is not permitted.
- The keys and/or the tools used to open doors, hinged covers to remove parts which give access to compartments in which are present high dangerous voltages must belong exclusively to the service personnel.

For the eventual cleaning of the external parts of the equipment, absolutely do not use any inflammable substance or substances which in some way may alter the markings and inscriptions.

It is recommended to use a slightly wet cleaning cloth.

The Safety Rules stated in the handbook describe the operations and/or precautions to observe to safeguard service personnel during the working phases and to guarantee equipment safety, that is, not exposing persons, animals, things to the risk of being injured/damaged.

Whenever the safety protection features have been impaired, *REMOVE POWER*.

To cut off power proceed to switch off the power supply units as well as cut off power station upstream (rack or station distribution frame). The safety rules described in this handbook are distinguished by the following symbol and statement:

4.3 Labels indicating danger, forbidding, command

Overview

Follow the instructions printed on the labels affixed to the units and assemblies.

- Dangerous electrical voltages
- Risks of explosions
- Moving mechanical parts
- Equipment connection to earth
- Heat-radiating mechanical parts
- Microwave radiations (EMF norms)
- Laser precautions

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Pay attention to the information stated in the following, and proceed as instructed.

> **Note:**
>
> The symbols presented in following paragraphs are all the possible symbols that could be present on Nokia equipment, but are not all necessarily present on the equipment this manual refers to.

Dangerous electrical voltages

Labeling:

The following warning label is affixed next to dangerous voltages (>42.4 Vp; >60 VDC).

If it is a Class 1 equipment connected to mains, then the label associated with it will state that the equipment will have to be grounded before connecting it to the power supply voltage, for example:

Safety instructions:

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DANGER!

DAMAGE

Carefully observe the specific procedures for installation / turn-up and commissioning / maintenance of equipment parts where D.C. power is present, described in the relevant installation / turn-up and commissioning / maintenance documents and the following general rules:

- Personal injury can be caused by -48VDC. Avoid touching powered terminals with any exposed part of your body.
- Short circuiting, low-voltage, low-impedance, DC circuits can cause severe arcing that can result in burns and/or eye damage. Remove rings, watches, and other metal jewelry before working with primary circuits. Exercise caution to avoid shorting power input terminals.

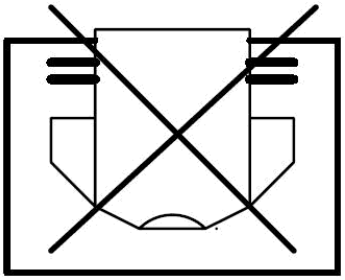
Risks of explosions

Labeling:

This risk is present when batteries are used, and it is signaled by the following label:

Safety instructions:

Therefore, slits or apertures are made to let air circulate freely and allow dangerous gases to down flow (battery-emitted hydrogen). A 417-IEC-5641 Norm. compliant label is affixed next to it indicating that the openings must not be covered up.



Moving mechanical parts

Labeling:

The following warning label is affixed next to fans or other moving mechanical parts:

Safety instructions:

Before carrying out any maintenance operation see that all the moving mechanical parts have been stopped.

Equipment connection to earth

Labeling:

Terminals for equipment connection to earth, to be done according to international safety standards, are pointed out by the suitable symbol:

Safety instructions:

The position of earth connection terminals is specified in the Hardware Installation section.

Heat-radiating mechanical parts

Labeling:

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DANGER!

DAMAGE

Safety instructions: Carefully observe the specific procedures for installation / turn-up and commissioning / maintenance of equipment parts where heat-radiating mechanical parts are present, described in the relevant installation / turn-up and commissioning / maintenance documents and the following general rule: Personal injury can be caused by heat. Avoid touching powered terminals with any exposed part of your body.

The presence of heat-radiating mechanical parts is indicated by the following warning label in compliance with IEC 417 Norm, Fig.5041:

Microwave radiations (EMF norms)

Equipment emitting RF power (Reminder from site preparation procedure):

The site must be compliant with ICNIRP guidelines or local regulation if more restrictive.

Safety instructions:

The following rules should be strictly applied by Customer:

- Non authorized persons should not enter the compliance boundaries, if any, for the general public.
- Compliance RF boundaries, if any, related to Electro Magnetic Field exposure must be marked.
- Workers should be allowed to switch-off the power if they have to operate inside compliance boundaries.
- Assure good cable connection.
- Install the antenna as high as possible from floor or area with public access (if possible the cylinder delimiting the compliance boundaries, if any, or the cylinder corresponding to the transmission area directly in front of antenna with the same diameter as the antenna, more than 2 meters high).
- Install the antenna as far as possible from other existing equipment emitting RF power.

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Always remember that someone standing in front of the Wavence antenna may cause traffic shutdown.

Place the relevant stickers:

EMF emission warning sign

On the site when applicable (when people can cross the compliance boundaries and/or the transmission area of the antenna, that is, roof top installation)

- Warning label "Do not stand on the antenna axis"

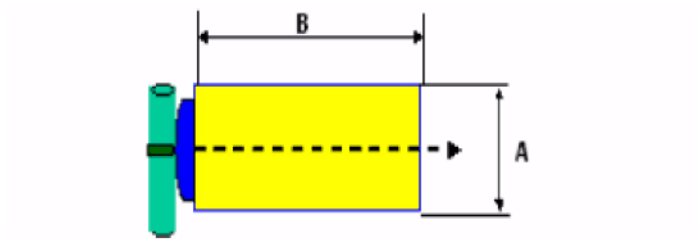
On the mast (front side)

- EMF emission warning sign (Yellow and black) to be placed at bottom of antenna, visible by someone moving in front of the antenna (roof top installation)

On the antenna (rear side)

- EMF emission warning sign, placed on the antenna.

Figure 5: Compliance boundaries for Wavence transceivers with 1ft (30 cm) antenna:



System	Modulation	Worst Configuration	A (m)	B (m)

Electromagnetic compatibility (EMC norms)

The equipment's EMC norms depend on the type of installation being carried out such as cable termination or grounding, and on the operating conditions.

Before carrying out any installation, turn-on, tests & operation and maintenance operations, read carefully the related sections of this manual, in particular:

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- Hardware Installation
- Maintenance and Upgrade

The norms set down to guarantee EMC compatibility, are distinguished inside this manual by the symbol and term:

ATTENTION **EMC Norms**

1. EMC General Norms - Installation
   - All connections (towards the external source of the equipment) made with shielded cables use only cables and connectors suggested in this manual or in the relevant Plant Documentation, or those specified in the Customer's "Installation Norms" (or similar documents)
   - Shielded cables must be suitably terminated
   - Install filters outside the equipment as required
   - Ground connect the equipment utilizing a conductor with proper diameter and impedance
   - Mount shields (if utilized), previously positioned during the installation phase, but not before having cleaned and degrease it.
   - Before inserting the shielded unit proceed to clean and degrease all peripheral surfaces.
   - Screw fasten the units to the subrack.
   - To correctly install EMC compatible equipment follow the instructions given.
2. EMC General Norms - Turn-on, Tests & Operation
   - Preset the electrical units as required to guarantee EMC compatibility
   - Check that the equipment is operating with all the shields properly positioned.
   - To properly use EMC compatible equipment observe the information given
3. EMC General Norms - Maintenance
   - Before inserting the shielded unit, which will replace the faulty or modified unit, proceed to clean and degrease all peripheral surfaces.
   - Clean the dummy covers of the spare units as well.
   - Screw fasten the units to the subrack.

Equipment protection against electrostatic discharges

Before removing the ESD protections from the equipment, observe the precautionary measures stated. Make sure that the ESD protections have been replaced and after having terminated the maintenance and monitoring operations.

Most electronic devices are sensitive to electrostatic discharges, to this concern the following

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warning labels have been affixed:

Observe the precautionary measures stated when having to touch the electronic parts during the installation/maintenance phases.

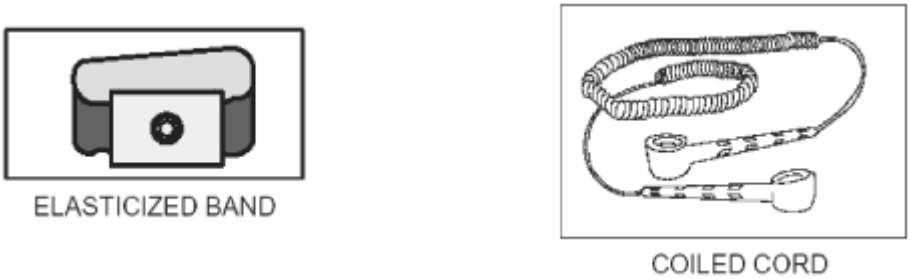
Workers are supplied with anti static protection devices consisting of an elasticized band worn around the wrist and a coiled cord connected to the elasticized band and to the stud on the subrack. See Figure: Anti static protection device kit.

Cautions to avoid equipment damage

Anti static protection device kit

Whenever is necessary to handle spare parts and cards out of their own box, this kit, as shown in Figure: Anti static protection device kit , must be always warn and its termination must be connected to a grounded structure, to avoid the possible damage of the electronic devices for electrostatic discharges.

Figure 6: Anti static protection device kit



Screw fixing

In normal operation conditions, all screws must be always tightened to avoid item detachment and to ensure the equipment EMI-EMC performance.

- The screw tightening torque must be:
  - 2.8 kg x cm (0.28 Newton x m) ±10%
  - 2.4317 in lb. (0.2026 ft lb.) ±10%
- Exceeding this value may result in screw breaking.

MSS-ODU cable disconnection / connection

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To avoid equipment damage, always unplug the card first before disconnecting the MSS-ODU cable (at MSS or ODU side).

Fan, equipment rack, and equipment cabinet precautions

DANGER!

DAMAGE

- Keep your fingers away from the rotating fan blades. Pull the fan-tray card by the thumbscrews only and wait for the fan blades to stop spinning before you attempt to remove the fan-tray card completely from the chassis.
- The Wavence requires at least two people to support, align, and attach the MSS-8 shelf or MPT-HL shelf to an equipment rack. To prevent equipment damage or personal injury, make sure you have enough help.
- The Wavence requires at least three people to lift and carry an equipment rack populated with Wavence equipment. To prevent equipment damage or personal injury, make sure you have enough help or the appropriate lifting and transporting equipment.
- The Wavence requires at least two people to align and secure an equipment rack populated with Wavence equipment. To prevent equipment damage or personal injury, make sure you have enough help.
- To prevent personal injury and equipment damage due to unbalanced loading of the equipment rack or cabinet, make sure the equipment rack or cabinet is properly secured to the floor, ceiling, or other rigid structure before mounting the Wavence in it. For approved methods of securing the equipment rack, read the equipment-rack installation instructions or contact the equipment-rack manufacturer.

When installing the equipment observe the following:

Electrical precautions

DANGER!

DAMAGE

- Install the Wavence in a restricted-access area only. Entrance to a restricted-access area is intended for qualified or trained personnel and access to it is controlled by a locked barrier.
- The chassis does not contain main overcurrent protection devices. The user must provide circuit breakers or fuses and disconnects between the power source and the Wavence. Each power feed from a source (-48 V dc and Return) requires a 25-amp dc -rated fast-trip circuit breaker or fuse and disconnect. Circuit breakers or fuses must meet applicable local and national electrical safety codes and be approved for the intended application.
- Make sure you connect the node to a -48 V dc source that is electrically isolated from the ac source and is reliably connected to earth ground.
- For personal safety, make sure you connect and secure the installation site"s frame -ground (earth ground) wire to the frame-ground terminal on the Wavence before you connect any other wires to the node.
- A dc-power source provides high energy, which can cause serious injury or equipment damage. Only Nokia qualified personnel should connect the dc power to the Wavence. To prevent serious injury or equipment damage, make sure the power source cables are de-energized before you handle or connect them to the node.

Take appropriate safety precautions when performing procedures on electrical equipment. Hazardous electrical potentials are present when system power is on.

Some procedures in this manual require working with small conductive objects, such as screwdrivers, fuses, washers, screws, and nuts. When working on a chassis at the top of an equipment rack, a dropped object that falls into a lower chassis can cause physical damage and electrical short circuits. To prevent this, place a piece of paper or other cover over the lower chassis to catch fallen objects. Remove the paper or other cover when work is complete.

Laser precautions

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DANGER!

DAMAGE

- The use of controls and/or adjustments, or the performance of procedures other than those specified herein may result in hazardous infrared radiation exposure.
- Laser infrared radiation is not in the visible spectrum; therefore, it is not visible to the naked eye or with laser safety glasses. Although it cannot be seen, laser radiation may be present.
- Never look directly into an unterminated fiber-optic connector unless it is absolutely known that no optical power is being emitted by the connector.
- Never look into a broken optical fiber cable unless it is absolutely known that no laser radiation is present.
- Never look at an optical fiber splice, cable, or connector unless it is absolutely known that no laser radiation is present in the fiber. Laser radiation can come from a fiber -optic transmitter, an Optical Time Domain Reflectometer (OTDR), or other optical test equipment.
- Never look directly into an unterminated optical connector or cable with a magnifier/microscope unless it is absolutely known that no laser radiation is being emitted from the connector or cable. A magnifier or microscope greatly increases the laser radiation hazard to the eyes.
- This system normally operates as a Class I Laser Product (no hazard). However, during servicing operations, when optical connectors are being connected, disconnected, or handled without dust covers, it is possible to be exposed to Class IIIb laser radiation, which can cause eye damage.
- Everyone within a 10-foot radius of an unterminated optical fiber or connector that is connected to a powered transmitter must wear laser safety goggles or eye shields.

Verify that laser labels on equipment state that the system conforms to all applicable standards of 21 CFR 1040.10. If there are no danger labels, contact Nokia

The invisible infrared radiation emitted by the fiber-optic transmitter can cause eye damage. Observe local office procedures and the following dangers:

Laser safety goggles or eye shields are not required if the following work rules are strictly followed:

1. Always remove electrical power from fiber-optic transmitters before disconnecting fiber-optic connectors in the path between the transmitter and the receiver.

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2. Never connect an unterminated optical cable to a fiber-optic transmitter. Always connect fiber -optic cables to fiber-optic receivers, test sets, or some other termination first.

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5. Electrostatic-sensitive devices

5.1 Electrostatic-sensitivity

Overview

⚠ WARNING!

Common plastic, white foam, cellophane, and masking adhesive tapes must not come in contact with ESDs or their packaging.

An Electrostatic-Sensitive Device (ESD) can withstand voltage spikes of only 10 to 100 volts and can be damaged or effectively destroyed by a discharge that might go unnoticed by a technician. Some devices have built-in protection. However, because this protection is effective only against the lower levels of electrostatic charges, a false sense of security often prevails.

Common plastics (synthetic insulating materials), clothing, and paper or cardboard are the most common sources of static charges.

Observe special precautions when the ESD sign is displayed.

Figure 7: Electrostatic-sensitive sign

The following items are examples of ESDs:

- MOS (Metal Oxide Semiconductor) capacitors, transistors, Integrated Circuits (ICs)
- CMOS (Complementary Metal Oxide Semiconductor) transistors, ICs
- JFET (Junction Field Effect Transistors)

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- IGFET (Insulated Gate Field Effect Transistors)

5.2 Handling circuit packs

Overview

⚠ WARNING!

Heel straps are effective only while standing on conductive or electrostatic-dissipative surfaces.

Although the risk of damage to an ESD is reduced considerably after it is assembled into a circuit designed to protect sensitive components, take the following precautions to reduce static charges to harmless levels:

- Handle all circuit packs as ESDs unless they are known not to contain electrostatic-sensitive parts.
- Wear ground straps, wrist (PN 1AD012470001) before and while touching or handling circuit packs containing ESDs.
- Cover surfaces with resistance to ground in excess of 100 megohms, such as ordinary tile, with properly grounded static dissipative runners.
- Workbenches must be earth-grounded, and work surfaces must be covered with an antistatic or static dissipative material bonded to the bench (bolt). A field service kit (PN 1AD068980001) or equivalent can be used if an adequate workbench is not available.
- ESDs are delivered with protective packing (containers or conductive foam). The devices should remain in their original containers until needed. Store (even temporarily), pack, and ship circuit packs in antistatic bags or containers.
- Containers with ESDs must contact the antistatic work surface, and the wrist strap must be connected before parts are removed from packaging. Devices must be handled by their bodies.
- Do not handle printed circuit board or components unnecessarily. Use plastic handle.
- Do not use synthetic bristled brushes or acid brushes to clean circuit packs.
- Handle failed circuit packs with same precautions as good circuit packs.

ESDs are protected when properly packaged in conductive or antistatic packaging. Acceptable packaging is marked as either conductive or antistatic.

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6. Product support information for the ANSI market

6.1 Customer service telephone and email support

Visit the customer support section on Nokia's website to find telephone and email contact information for your country:

↗ https://customer.nokia.com/support/s/

After-hours emergency telephone support is available. An emergency is defined as an out-of -service, traffic-affecting problem or a nonoperating alarm system on traffic-bearing systems.

6.2 Product documentation

Product documentation is available through DOC Center at:

↗ https://documentation.nokia.com/aces/cgi-bin/au_doc_list.pl

Product documentation updates appear on DOC Center before they are available in any other format.

At Nokia Support portal web site, ↗ https://customer.nokia.com/support/s/au_doc_list.pl, follow the on-screen instructions to register for access and obtain a log in ID. In addition to accessing product documentation, the Nokia Support Portal allows the user to view the following:

- Application notes
- Configuration notes
- Data collections
- Frequently Asked Questions (FAQs)
- General information books
- General Release Documents (GRDs)
- Installation documents
- Methods of Procedure (MOPs)
- Product Change Notifications (PCNs)
- Product Information Bulletins (PIBs)
- Product manual updates

- Software patch and software load documents
- Software Update Documents (SUDs)
- Technical bulletins
- Training documents
- Urgent Product Warnings (UPWs)

6.3 Training

Equipment training is available to all customers. Courses are available at the training facilities in Plano, Texas. If a customer cannot attend a standard course, the Training department can arrange a course for a specific requirement and conduct it at the customer's facility. For further information, call customer service telephone support and ask for a training coordinator.

6.4 Technical support center

The Technical Support Center (TSC) staff provides high-quality technical assistance. Customers can expect effective telephone assistance when their crafts and maintenance personnel have been trained by Nokia's Training department and are equipped with adequate test equipment, spares, and documentation at the site.

After-hours emergency telephone support

Emergency support is available after-hours for emergencies

An emergency is defined as an out-of-service, traffic-affecting problem or a nonoperating alarm system on traffic-bearing systems.

Nonemergency is defined as installation turn-ups, application questions, traffic cutover, routine maintenance, or other non-service-affecting maintenance. All non-service-affecting, after-hours telephone services are billable to the customer.

Please provide the operator with the following information:

- Company name
- Caller name
- A telephone number where caller can be reached
- A brief description of the problem, including the product involved

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After-hours nonemergency telephone support

After-hours telephone support to address new installations, system expansions, system operations, system application, or other non-service-affecting issues is available by contacting Nokia.

On-site technical support

On-site technical support is available when an issue cannot be resolved remotely. This determination is usually made by Nokia TSC during the problem investigation process. These services may or may not be billable to a customer. This depends on several factors such as what type of Service Level Agreement a customer has with Nokia and the age of the product.

6.5 Repair and return services

As part of a comprehensive technical support program, NOKIA provides factory repair services for equipment. This service is available both during and after the warranty period through NOKIAapos;s Return and Repair department.

6.6 Spare parts and replacement cards

For spare parts, spare cards, card exchange, and in-warranty replacement on a routine or emergency basis, call customer service telephone support.

Provide the following information:

- Customer Contact Name
- Customer Company Name
- Customer telephone number
- Customer E-mail Address
- A brief description of the problem, including product line, part number, and quantity of parts needed

For emergency assistance after normal business hours, call customer service telephone support, ask the operator for Emergency Parts Assistance, and provide the operator with the required information. The operator will contact an appropriate individual to respond.

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6.7 Return for credit or warranty exchange procedure

Returned equipment must have a Parts Request (PR) number. Obtain an PR number by calling Nokia.

No equipment should be returned without an PR number. The following information is required:

- Description and quantity of equipment to be returned
- Reason for return
- Order number the equipment was purchased against and approximate date of purchase

6.8 Service center

The Service Center tests, repairs, and modifies all cards (both in and out of warranty). Cards received for repair or modification are returned promptly.

Return for repair procedure

Refer to Return for credit or warranty exchange procedure for information on obtaining an PR number. Notification to the Service Center and issuance of an PR number by Nokia personnel must be made prior to shipment of parts. The following information must be furnished with the request for return authorization:

- Purchase order number or requisition number
- Description and quantity of equipment to be returned
- Reason for return:
  - Modification required
  - Defective equipment to be repaired
- Warranty status (in or out of warranty) and warranty date stamped on unit
- Specific nature of problem
- Name and telephone number of person who identified problem
- Special instruction/information

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Shipping instructions for repair, credit, or warranty exchange

Return equipment or parts prepaid to the address provided below. The PR number must be prominently marked on the shipping label, the packing list, and any correspondence regarding the order.

- Include company name, address, and name of person to contact in case of a question.
- Include specific reason for return. (This aids prompt processing.)
- Include the same requisition number or purchase order number that was furnished with request for return authorization.
- Include type number and part number of unit.
- State whether equipment is in or out of warranty.
- Furnish shipping address for return of unit, if applicable, or other pertinent details.
- Mail purchase order, if applicable, to address shown under Return for Repair Procedure, Attention: Service Center.

For technical support, contact your local customer support team. See the Support web site (http://support.alcatel-lucent.com/support/) for contact information.

6.9 Installation and maintenance services

Engineering and installation service

Nokia has experience in central office, outside plant, and customer premises applications, and specializes in flexible scheduling and high-quality service. Qualified staff are in place nationwide, so an installation can be started and completed promptly.

Contract maintenance service

Nokia field service is well-suited for private networks of any size.

Factory-trained service technicians are qualified on similar systems before they are allowed to maintain customer equipment. They have direct access to additional technical support around the clock and to all necessary tools and test equipment.

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7. Alarm and abnormal conditions overview

7.1 Overview

Retrieve alarms and abnormal conditions

To retrieve alarms using WebCT, execute the following sequence from the main menu bar:

Monitoring & Maintenance Domain>Active Alarms

7.2 Friendly names

Naming overview

Equipment friendly names are constructed using the following structure:

r<rack number>s<subrack number>b<board number>d<daughter>

The syntax for equipment Friendly Names / Entities is described in Table: Friendly Name/Entity Syntax.

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Table 2: Friendly Name/Entity Syntax

Location	Syntax	Values	Description
Rack	r	r01	Rack number
Subrack	s	s1	Subrack number for MSS-4, and MSS-8 shelves
		s[101,102]	Subrack number for MPT-HLS units
		s[3-8][5-8]	Subrack number for MPT-HLC transceiver connected to EASv2 card (EASv2 slot number and Ethernet port number associated with MPT-HLC Transceiver)
		s1[7-8]	Subrack number for MPT-HLC, MPT-HLS Transceiver connected to CorEvo card
Board	b	board#[1-10]	Board/slot number of equipment in alarm
		b#[1-9]	
		Slot#[1-9]	
Daughter	d	daughter#[07-08]	Daughter SFP port number of equipment in alarm associated with CorEvo card
		daughter#[05-08]	Daughter SFP port number of equipment in alarm associated with EASv2 card
		daughter#[01-02]	Daughter SFP port number of equipment in alarm associated with SDHACC card

Radio facility friendly names are constructed using the following structure:

f<facility name>rd<radio direction>b<board number>p<port number>c<channel number>

DS1, DS3, and Ethernet facility friendly names are constructed using the following structure:

f<facility name>b<board number>p<port number>

The syntax for facility Friendly Names / Entities is described in Table: Friendly name syntax definitions.

Table 3: Friendly name syntax definitions

Location	Syntax	Values	Description
Facility Name	f	DS-1-in, out	DS1 facility and direction
		DS-3-in, out	DS3 facility and direction
		E1-in, out	E1 facility and direction
		EleSdh-in	STM-1/OC-3 facility and direction (supported by electrical SFP)
		OptSdh-in	STM-1/OC-3 facility and direction (supported by optical SFP)
		MAU	Ethernet signal
		Radio	Radio facility
Radio Direction	rd	Dir#[3-8].[5-8]	Radio direction of MPT-HLC, and MPT-HLS radio channel in alarm connected to EAS card
Board	b	board#[1-9]	board/slot number of MSS-8 (1-9), MSS-4 (1-5) equipment in alarm
		b#[1-9]	
		Slot#[1-9]	
Port Number	p	Port#[1-32]	E1/DS1 port number of facility in alarm on P32E1DS1
		Port#[1-2]	DS3 port number of facility in alarm
		Port#[1-2]	STM-1/OC-3 port number of facility in alarm
		Port#[1-8]	Ethernet port number of facility in alarm associated with EASv2 and CorEvo
Channel Number	c	Ch#1	Main radio channel in alarm
		Ch#0	Spare radio channel in alarm
ERPS	ERP	ERP[1-2]#[1-2]	Ethernet Ring Protection Switch

The following are examples to illustrate how to interpret the probable cause and friendly name combinations:

Example: Probable Cause: Replaceable Unit Missing

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- Friendly Name: r01s1/board#8
  - Translation: Missing card at r01 (rack 01) s1 (subrack 1 (MSS-8 Shelf)) /board#8 [missing card provisioned for slot #8]

Example: Probable Cause: Replaceable Unit Type Mismatch

Example: Probable Cause: Loss Of Signal

- Friendly Name: DS1-in/slot#5/Port#22
  - Translation: DS1-input, slot #5, Port #22 [Loss Of incoming DS1 Signal, on P32E1DS1 card in slot #5, port #22]

Example: Probable Cause: Loss Of Frame

- Friendly Name: DS3-in/slot#6/Port#2
  - Translation: DS3-input, slot #6, Port #2 [Loss Of DS3 Frame on P2E3DS3 card in slot #6, port #2]

7.3 Facilities and equipment

Overview

The Facilities/equipment column in the troubleshooting chapters list the entities to which an alarm type can apply. Depending on the entities affected, some alarm types have more than one corresponding procedure.

In the Facilities/equipment column, entities are listed using the following definitions:

- EAC: EAC and EAC-10G cards
- CorEvo:
  - Core Switching Module card, second generation
  - Ethernet facility supported by the CorEvo card
- EASv2:
  - 8-Port Ethernet Access Switch, second generation
  - Ethernet facility supported by the EASv2 card
- FAN: MSS-4/8 Fan card
- MSS: MSS-4/8 shelf
- MPT-HLC:

    - MPT-HL Transceiver card, second generation cubic
    - MPT-HLC: RF facility supported by MPT-HLC Transceiver card
- MPT-HLS:
    - MPT-HLS Transceiver card
    - MPT-HLS: RF facility supported by MPT-HLS Transceiver card
- SFP: Small Form Factor Pluggable equipped on listed equipment type
- P2E3DS3: 2-Port DS3 card
- P32E1DS1: 32-Port DS1 card
- DS3: DS3 facility supported by P2E3DS3 card
- DS1: DS1 facility supported by P32E1DS1 card
- SDHACC: 2-Port SDH card
- STM-1/OC-3: STM-1/OC-3 facility supported by SDHACC card
- UBT: Ultra Broadband Transceiver

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8. Alarm and abnormal condition clearing procedures

8.1 Alarms and abnormal conditions

Alarms reported by the Alarm Manager

Table: Craft Terminal's Active Alarms alarm types describes the alarms that are reported by the Alarm Manager and the corresponding procedure to clear the alarm.

Service Affecting (SA) and Non Service Affecting (NSA) alarms

An alarm, and the related Severity, is SA when the condition which brings to the raise of the alarm is affecting the capacity of the system to deliver traffic (if the alarm is related to a transport entity) or to deliver synchronization (if the alarm is related to an entity involved in synchronization management).

The alarm is NSA if the condition which brings to the raise of the alarm is not affecting the delivery of traffic or synchronization (i.e. a Dem Fail alarm raised only on Spare radio is NSA if the radio is RPS protected, Dem Fail alarms raised on Main radio and Spare radio are both SA since in that condition the radio is no more able to receive any traffic from remote end)

Alarm severity can be: Major (MJ), Minor (MN) or Warning (WN).

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Table 4: Craft Terminal's Active Alarms alarm types

Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
1588 TC Fail	UBT-S UBT-m UBT-T - channel A UBT-T - channel B UBT-I MPT-HLC MPT-HLS	Reported when, due to radio impairments, TC time stamping can not be guaranteed due to alignment of the local and remote counters. Verify that no other radio path alarms, such as Demodulator Fail. If radio path alarms exist, troubleshoot accordingly.	MJ	MN	1588 TC Fail alarms
AIM Read Fail	UBT-S UBT-T	It reports that AIM data have not been read.	MN	WN	Alarms available only in WebCT.AIM Read Fail alarm
AIS	Channelized E1 E1 DS1 DS3 STM-1 OC-3 MSS-E MSS-HE/XE	Reported due to one of the following: • Channelized E1 High Order Path • Channelized E1 Low Order Path • AIS incoming to the PDH card from the demarcation panel. • AIS incoming to the SDH card from the demarcation panel. • AIS detected from the CorEvo switch • AIS detected by the receive circuits, indicating an upstream failure Troubleshoot the input or upstream source of the signal and check the input cabling.	MJ	none	AIS alarms
AIS	CorEvo MSS-E MSS-HE MSS-XE	A MEP receives an AIS frame.	MN	none	AIS alarms

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Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
Announce Packet Loss	CorEvo	Indicates that an OC Slave has stopped receiving PTP Announce messages from the Boundary Clock port working as Master or OC Master. Check for radio path alarms and clear. For the associated VLAN, check for and clear any alarms that exist along the path to either the Boundary Clock or OC Master. Check the configured announce message rate.	MJ	MN	Announce Packet Loss
Announce Packet Loss	MSS-XE MSS-E MSS-HE/XE	Reported by an OC -Slave when there is a lack of reception of Announce messages from a Master (loss of the channel carrying the traceability information)	MJ	MN	Announce Packet Loss

Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
ATPC High Power Timeout	UBT-S UBT-m UBT-T channel A UBT-T channel B UBT-I MPT-HLC	Reported for a ATPC High Power Time -out. When ATPC is enabled, the local transmitter was operating continuously at full power for 5 minutes, and the transmitter was reduced to its minimum power setting and this condition notification was generated. This condition will remain active until a message is received that the active receiver at the other end of the link is no longer requesting additional transmitter power. If the local transmit power is at its minimum power setting, troubleshoot the RF path from the near end transmitter to the far end receiver.	MJ	none	ATPC high power timeout alarms
Bandwidth Degrade for Protection Switch	Core MSS-E MSS-HE MSS-XE	An ERPS Instance has entered a Protection state. Retrieve the alarms for the radio or ETH link direction on each node and troubleshoot accordingly.	none	WN	Bandwidth degrade for protection switch alarms

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Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
Bandwidth Over Flow	MPT-HLC MPT-HLS	While the radio was not communicating with the switch, TDM/SDH cross -connections were provisioned with a total bandwidth which exceeds the net bandwidth capacity of the radio profile. Verify the cross -connections for the radio with Bandwidth Over Flow alarm and delete TDM2TDM and/or TDM2ETH cross-connections until the total bandwidth is equal to or less than the net bandwidth capacity of the radio profile.	MJ	none	Bandwidth Over Flow alarms
Battery Failure[1]	MSS-8 MPT-HLS MSS-E MSS-HE/XE	An A battery feed failed or is missing. Verify the following: • That there is no blown fuse providing power to the A -power feed • The shelf power cable connections • the A-power connections between the BDFB and power distribution panel. • The A-power BDFB breakers	none	MN	Battery Failure alarms

Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
CLA	MPT-HLS/HLC Radio Access Card UBT-I	A detected CLA allows the transmitters in a 1+1 HSB radio configuration (both in ATPC and in RTPC) to be switched based on HBER, Demodulator Fail, Receive Fail alarm information from the remote receivers. Perform the Clear CLA command.	MJ	Ac	CLA alarms
Clock Class Degraded	CorEvo	Indicates that a boundary Clock has entered into holdover. Check for and clear any related Sync Packet Loss, Synchronization Signal Fail, and Announce Packet Loss alarms. Cross-check the parent/GM clock parameters. If 1588 frequency is based on the node check for the Sync-E path if equal to the PtP frequency/path.	WN	WN	Clock Class Degraded
Clock Class Degraded	MSS-XE MSS-E MSS-HE	Reported by a Boundary Clock when the time recovery algorithm of PTP Clock enters in holdover	WN	WN	Clock Class Degraded

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Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
Clock Failure	UBT-S UBT-m UBT-T channel A UBT-T channel B UBT-I MPT-HLS MPT-HLC	The UBT ODU/ or MPT-HLC/MPT-HLS is not able to lock the air Tx or Rx symbol rate to the NE clock. Power cycle the radio. Unable to lock the user clock.	MJ	MN	Clock failure alarms
Combiner Loss of Alignment	MPT-HLS/HLC radio with diversity combiner	Non-alignment status detected between main and diversity flows in configurations with combiner. Verify that no other radio path alarms (such as loss of signal) exist, and troubleshoot accordingly.	none	MN	Combiner loss of alignment alarms
Critical Link Event	Ethernet port	The Ethernet port"s link partner has declared a link fault condition. Determine all the alarms declared against the Ethernet port and troubleshoot accordingly.	MJ	MN	Critical Link Event alarms
	EAC, EAC1G2G5 and EAC10G User Ethernet Ports	Reported when EFM OAM is enabled on a User Ethernet interface and Critical Link Event condition is received by Remote DTE.			
DC & PfoE Cables present	UBT-S UBT-m UBT-T UBT-T XP	Voltage detected on both DC and PfoE power cables	None	MN	For more information, please contact Nokia TAC.

Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
Degrade	CorEvo sync SFP P32E1DS1 OC-3 STM-1 User Ethernet Port EAC, EAC1G2G5 and EAC10G User Ethernet Ports; Electrical (RJ45) and Optical (SFP) E1/DS1 Access Card MPT-HLS and Cubic UBT-S UBT-m UBT-T UBT-T XP RLAG on CAHD RLAG UBT-I MSS-E MSS-HE MSS-XE MPT-HLS MPT-HLC	Frequency of the source signal is degraded. Managed only if the interface has been configured as a primary or secondary synchronization source. • If Sync-in, verify that the Sync reference source (GPS or other sync source) is functioning correctly. • If Radio port, verify that no other radio path alarms exist and troubleshoot accordingly. • If PDH or SDH port, verify that no LOS alarm exists on the sync reference port and troubleshoot accordingly.	MN	WN	Degrade alarms on sync source
Degrade alarms on LAG	Core RLAG: • UBT-S • UBT-m • UBT-T • UBT-I MPT-HLS MPT-HLC	A member(s) of an Ethernet or Radio LAG group has no activity. Degrade is declared when the number of active ports in the LAG are greater than zero and less than the LAG size. Identify the alarms that are declared against members of the LAG group and troubleshoot accordingly.	MN	none	Degrade alarms on LAG
	LAG (CA mode 1b on: MSS-E, MSS-HE, MSS-XE UBT-S UBT-m UBT-TUBT-T XP	Signal Degrade on the LAG Interface (Radio)	MN	none	

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Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
Delay Response Loss	CorEvo	Indicates that either a Boundary Clock port working as Slave or an OC Slave has stopped receiving PTP Delay Response messages from either the Boundary Clock port working as Master or the OC Master. Check for radio path alarms and clear. For the associated VLAN, check for and clear any alarms that exist along the path to either the Boundary Clock or OC Master. Check if the VLAN tagging and untagging is configured at the ETH port working with the VLAN of this dedicated PTP port.	WN	WN	Delay Response Loss
Delay Response Loss	MSS-E MSS-HE MSS-XE	Reported when there is a lack of reception of PTP. Delay Response messages from a Master.	WN	WN	Delay Response Loss
Dem XPIC LOS	UBT-S UBT-m UBT-I MPT-HLS MPT-HLC	Bad or missing cable/cable connection at XPIC connector on UBT ODU, MPT-HLS/MPT -HLC Verify that the XPIC cable is properly terminated, securely connected, and the cable is not damaged.	MN	none	DemXPIC LOS

Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
Demodulator Fail	MPT-HLS MPT-HLC	Loss of receive signal, bad receive signal, or internal receive circuit failure. Verify that the RF path is clear.	MJ	MN	Demodulator fail alarms
	UBT-S UBT-m UBT-T channel A UBT-T channel B	Internal receive circuit failure. The demodulation function failure represents *the synchronization missing at received side*.	MJ	MN	
Dialog Failure	UBT-S UBT-m UBT-T UBT-I MPT-HLC MPT-HLS	Temporary Loss of communication with UBT-ODU/. MPT-HLC/MPT-HLS Wait for the appropriate time -out period to expire before proceeding. This alarm can be raised and cleared multiple times if the ICU-ODU cable is not working properly.	none	WN	Dialog failure alarms

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Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
Duplicate Address Detected	TMN MGMT Port • CorEvo TMN Port#4 • CorEvo TMN In-Band Port [1-2] Radio	Duplication of IPv6 Global Unicast address detected on TMN management port, Port#4, In-Band port 1-2 Verify the desired IPv6 Configuration value for the NE according to site engineering documentation. Reprovision if not correct.	none	MN	Duplicate Address Detected alarms
	VPRN Interface	Duplication of IPv6 Global Unicast address or duplication of IPv6 Link Local address detected.	none	MN	
	TWAMP Light Reflector	Duplication of IPv6 Global Unicast address or duplication of IPv6 Link Local address detected.	none	MJ	
	MSS-E MSS-HE MSS-XE	Duplication of IPv6 Global Unicast address detected	none	MN	
Early Warning Indication	MPT-HLS MPT-HLC	10E-9 BER detected No action is required at this time. Monitor the receive signal for increased degrading.	none	WN	Early warning indication alarms
	UBT-S UBT-m UBT-T channel A UBT-T channel B UBT-I	10E-9 BER detected The Early Warning alarm shall be computed/reported in 1+1 configurations to support hitless switching. Early Warning shall also be computed/reported in an unprotected configuration.	none	WN	

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Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
Eth OAM Loss of Continuity	CorEvo MSS-E MSS-HE MSS-XE	MEP received no CCM frames from a peer MEP during an interval equal to 3.5 times the CCM transmission interval configured at the MEP. If MIPs were created on the MEP, use the Link Trace command using WT CLI Tool. Reported when R -APS MEP receives no valid CCM frames from a remote R -APS MEP during an interval equal to 3.5 times the CCM transmissionperiod configured at the R -APS MEP. Also raised if the remote ETH port has a LoS.	MN	none	ETH OAM loss of continuity alarms
Eth OAM MAC Status Defect	MSS-E MSS-HE MSS-XE	Reported if a MEP receives a CCM frame with correct MD/MEG Level, correct MAID/MEG, correct MEP ID and correct CCM Interval, but reporting a failure in its Port Status TLV or Interface Status TLV.	MN	none	
Eth OAM Mismerge Condition	CorEvo	MEP received a CCM frame with correct MD Level but incorrect MA name. Retrieve MA name for both ends of the MEP using WT CLI Tool and reprovision the values.	MN	none	Eth OAM Mismerge Condition alarms

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Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
Eth OAM Remote Defect Indicator	CorEvo MSS-HE MSS-E MSS-XE	MEP received a CCM frame with the RDI field set. Check the far end of the MEP for one or more of the RDI set conditions and troubleshoot accordingly.	MJ	none	Eth OAM Remote Defect Indication alarms
Eth OAM Unexpected Levels	CorEvo	MEP received a CCM frame with the incorrect MD Level/MEG Level. MEP receives a CCM frame with correct MD/MEG Level but with incorrect MAID/MEG Retrieve the MD Levels and reprovision.	MN	none	Eth OAM Unexpected Level alarms
Eth OAM Unexpected CCM Connectivity	MSS-E MSS-HE MSS-XE	Reported if either: A MEP receives a CCM frame with incorrect MD/MEG Level; A MEP receives a CCM frame with correct MD/MEG Level but with incorrect MAID/MEG.	MN	none	

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Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
Eth OAM Unexpected MEPID	CorEvo MSS-E MSS-HE MSS-XE	Received MEPID value is different than the expected MEPID. MEP received a CCM frame with correct MD/MEG Level, correct MAID/MEG but with unexpected MEP ID MEP received a CCM frame with correct MD/MEG Level, correct MAID/MEG, correct MEP ID, but with a CCM Interval value different than itsown CCM Interval. Determine which MEPID value(s) are incorrect and reprovision. Check if ECFM is configured to detect an ETH loop. If a loop-detection is configured then the dedicated VLAN is looped in the network and blocking the loop is required.	MN	none	Eth OAM UnexpectedMEPID alarms
Eth OAM Unexpected Period	CorEvo	Received MEP Time -Out period value is different than the expected MEP Time -Out period. Reprovision the MEP Transmission Interval values if they are incorrect.	MN	none	Eth OAM Unexpected Period alarms

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Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
Excessive Environmental Temperature	MPT-HLC	Excessive Temperature on module. Verify that the MPT -HL fan card is equipped and operating properly.	none	WN	Excessive Environmental Temperature alarms
	MPT-HLS	Excessive Temperature on module.	MJ	WN	
	MSS-E MSS-HE MSS-XE	Excessive Temperature on module.	none	WN	
	MSS-4/8	Excessive Temperature on module.	none	WN	
Fan Communication Problem	MPT-HLS	Problem communicating with the MPT-HLS fan has been detected. Verify that all the MPT-HLS fans are equipped and operating properly.	none	MN	Fan Communication Problem (MPT-HLS)
Fan Fail	MPT-HLC UBT-I	Failure of at least one MPT-HLC Transceiver Fan Unit detected Verify that ALL the MPT-HLC fans are equipped and operating properly.	none	MN	Fan Fail (MPT-HLC)

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Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
Firmware Download On Going	CorEvo MSS-E MSS-HE/XE EASv2 P2E3DS3 P32E1DS1 SDHACC SDHCHAN MPT-HLC MPT-HLS EAC EAC1G2G5 EAC10G UBT-S UBT-m UBT-T UBT-I	Download Status. A download of an FPGA or other software unit on the addressed card. **Note:** When firmware download is completed on all HW units, the standby unit operational status changes to the `enabled` status. In case the firmware download is not completed for some IDU units after 30 minutes, the standby unit's operational status changes to `enabled` and the FirmwareDownloadOnGoing alarm is raised. In such case contact the local TAC.	none	WN	No action required.
Flash Card Failure	CorEvo MSS-E MSS-HE/XE	An indication that the SNMPv3 agent had a failure writing the set-up file to the flash card. Indicates a failed flash card on the CorEvo, MSS-E or MSS-HE/XE .	none	MN	For more information, please contact Nokia TAC
Flash Card Realignment in Progress	CorEvo	Flash Card realignment of Standby Core in progress	none	WN	No action required

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Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
High BER	MPT-HLS MPT-HLC	Bit Error Rate threshold (10E-4) exceeded on receiver input circuits on MPT-HL, MPT-HLC transceiver. Verify that there are no weather-related conditions.	MN	WN	High BER alarms on radios
	UBT-S UBT-m UBT-T channel A UBT-T channel B UBT-I	Bit Error Rate threshold (10E-4) exceeded on receiver input circuits on Radio Access card/UBT. The high BER alarm becomes significant and must be reported if there are no defect in the previous step of the Rx chain.	MN	WN	
High BER	SDHACC	Bit Error Rate threshold (10E-5) exceeded on receiver input circuits on SDHACC card from demarcation panel for the STM-1/OC-3 facility. Troubleshoot the upstream source of the incoming STM-1/OC-3.	MN	WN	High BER alarms on SDH cards
Housekeeping Alarm	Fan MSS-HE MSS-XE	Housekeeping alarm input is active. Verify the equipment that is associated with housekeeping alarm input which is in the alarm state and troubleshoot appropriately. Housekeeping Alarm. Indicates the presence of an input housekeeping alarm.	none	MN	Housekeeping alarms

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Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
Incompatible Frequency	UBT-S UBT-m UBT-T channel A UBT-T channel B UBT-I MPT-HLC MPT-HLS	Configured frequency is outside of the supported frequency range. Reprovision the frequency of the radio to the required frequency of the microwave link.	MJ	MN	Incompatible frequency alarms
Incompatible Modulation	MPT-HLC MPT-HLS	Configured modulation is not supported by MPT -HLC/HLS Reprovision the modulation of the radio to the required modulation of the microwave link.	MJ	MN	Incompatible modulation alarms
	UBT-S UBT-m UBT-T channel A UBT-T channel B UBT-I	Configured Modulation not supported by UBT.	MJ	MN	
Incompatible Shifter	MPT-HLC MPT-HLS	Configured shifter value not supported by MPT-HLS, or MPT-HLC. Reprovision the shifter value of the radio to the required shifter value of the microwave link.	MJ	MN	Incompatible Shifter alarms
	UBT-S UBT-m UBT-T channel A UBT-T channel B UBT-I	Configured shifter value not supported by UBT.	MJ	MN	

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Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
Incompatible Tx Power	MPT-HLC MPT-HLS	Configured RTPC/ATPC Tx power value or ATPC Rx power threshold value not supported by MPT-HLS, MPT -HLC. Reprovision the RTPC/ATPC Tx power levels and/or ATPC Rx power threshold level of the radio to the required transmit power level of the microwave link.	MJ	MN	Incompatible Power alarms
	UBT-S UBT-m UBT-T channel A UBT-T channel B UBT-I	Configured TX power value not supported by UBT.	MJ	MN	
Internal Communication Problem	MPT-HLC MPT-HLS	Loss of communications with MPT-HLS, or MPT-HLC. Correct radio port provisioning and MPT fiber cable connections per site documentation.	MJ	MN	Internal Communication Problem alarms
	UBT-S UBT-m UBT-T UBT-I	Loss of communication with UBT. Check the IDU-ODU cable. If the ODU is powered, replace the IDU-ODU cable, port at the IDU or ODU.	MJ	MN	
Inter Radio Unit Communication Failure	UBT-S UBT-m UBT-T UBT-I	Loss of Communication between mated UBTs.	none	MN	Alarms available only in WebCT.Procedure (UBT)

Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
LAG Port Down	UBT-S UBT-m UBT-T UBT-I MPT-HLS MPT-HLC	Reported when a member of an L1 LAG is removed from the LAG group Identify the alarms that are declared against members of the LAG group and troubleshoot accordingly.	MN	WN	LAG port down alarms
	RLAG CA mode 1b on MSS-E/HE/XE UBT-S UBT-m UBT-T UBT-T XP	Loss of Signal on the LAG Interface (Radio)	MJ	none	
	RLAG (CA mode 1b on MSS-E, MSS-HE, MSS-XE): UBT-S UBT-m UBT-T UBT-T XP	Reports a link out of the L1 LAG.	MN	WN	
LAG Rx Problem	RLAG (CA mode 1b on MSS-E, MSS-HE, MSS-XE): UBT-S UBT-m UBT-T UBT-T XP	Reports traffic loss due to issues at RX blocks or to malformed packets passing through the LAG direction.By default, this alarm is disabled. When disabled, this alarm is not considered to determine the L1 LAG operative state.	MJ	MN	For more information, please contact Nokia TAC
LAG Traffic Down	RLAG (CA mode 1b on MSS-E, MSS-HE, MSS-XE): UBT-S UBT-m UBT-T UBT-T XP	Reports a stable traffic loss due to a complete unavailability of the L1 LAG data path.By default, this alarm is disabled. When disabled, this alarm is not considered to determine the L1 LAG operative state.	MJ	MN	For more information, please contact Nokia TAC

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Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
LAG Traffic Loss	RLAG (CA mode 1b on MSS-E, MSS-HE, MSS-XE): UBT-S UBT-m UBT-T UBT-T XP	Reports packet loss due to an extreme latency difference over L1 LAG members or due to errors on the data path.By default, this alarm is disabled. When disabled, this alarm is not considered to determine the L1 LAG operative state.	MN	WN	For more information, please contact Nokia TAC
LAG TX Problem	RLAG (CA mode 1b on MSS-E, MSS-HE, MSS-XE): UBT-S UBT-m UBT-T UBT-T XP	Reports traffic loss due to issues at TX blocks or to malformed packets.By default, this alarm is disabled. When disabled, this alarm is not considered to determine the L1 LAG operative state.	MJ	MN	For more information, please contact Nokia TAC
Link Identifier Mismatch	MPT HLS/MPT-HLC connected to CorEvo	Link identifier number provisioned on MPT-HLS/ MPT -HLC card settings screen is different from link identifier number provisioned at the far end of hop. Reprovision the Link Identifier values according to the provisioned Link Identifier values for the near end and far end of the hop for the alarmed radio.	MJ	none	Link Identifier Mismatch alarms
	UBT-S UBT-m UBT-T channel A UBT-T channel B UBT-I	Link Identifier Mismatch detected on the radio aggregate signal (Rx side).	MJ	none	

Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
Loop Problem	MPT-HLS MPT-HLC	Loss of ATPC command path between far end transmitter and local receiver. Perform a protection switch.	none	MN	Loop Problem alarms
Loss of Alignment	MPT-HLS MPT-HLC	Delay/loss of alignment between main and protect RPS RF paths detected. Remove and reapply power to the radio.	none	MN	Loss of alignment alarms
Loss Ethernet Synchronization Message Channel	CorEvo EASv2 EAC, EAC1G2G5 and EAC10G User Ethernet Ports; Both Electrical (RJ45) and Optical (SFP) UBT-S UBT-m UBT-T UBT-I MSS-E MSS-HE MSS-XE MPT-HLS MPT-HLC	Reported if there is a loss of the Ethernet Synchronization Message Channel. The alarm is applicable to Ethernet ports and Radio interfaces configured with SSM support enabled and is assigned as a synchronous clock source input. Verify no radio or Ethernet alarms are declared against the radio direction declaring Loss ESMC.	MN	none	Loss Ethernet Synchronization Message Channel alarms
Loss of CESoETH Frame	DS1 E1 DS3 OC-3 STM-1 Channelized E1 MSS-E MSS-HE/XE	Packets are not being received by the DS1, DS3, E1, OC-3, or STM-1 emulation circuits on the receive circuits from the switch matrix. Verify that there are no current path or circuit alarms associated with the facility in alarm for the near end, the far end, and all intermediate NEs.	MJ	none	Loss Of CESoETH Frame alarms

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Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
Loss of Frame	DS1 E1 DS3 OC-3 STM-1 Channelized E1 MSS-E MSS-HE/XE	Loss of DS1, DS3, E1, OC-3, or STM-1 frame detected incoming to the PDH/SDH card from the demarcation panel for the DS1, DS3, E1, OC-3, or STM-1 facility. Verify that there are no active circuit alarms associated with the facility in alarm at the near end and client end NE of the facility. Loss of frame detected for RX dierection Loss of frame detected for TX dierection	MJ	none	Loss Of Frame alarms on facilities
Loss of Frame	MPT-HLS MPT-HLC	Far end TX problems, RF path problems, or local circuit failures have caused BER to increase to the point that frames are being lost. If weather-related problems exist, wait for the weather to clear and recheck alarms.	MJ	MN	Loss of frame alarms on radios
Loss of Multiframe	Channelized E1	Loss of multi-frame detected for the VC4 (Tx Direction). Troubleshoot the source of input STM-1 signal that is associated with the channelized E1 to the SDHCHAN card.	MJ	none	Loss of multi-frame alarms

Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
Loss of Pointer	Channelized E1	Loss of pointer detected for the AU4 or TU12(Tx Direction). Troubleshoot the source of input STM-1 signal that is associated with the channelized E1 input signal to the SDHCHAN card.	MJ	none	Loss of pointer alarms
Loss of Protection	MPT-HLS	The radio direction is affected by a failure that radio protection (RPS) can"t recover. Only reported if RPS is provisioned. Determine all of the radio alarms that are declared against the radio direction declaring the Loss Of Protection and troubleshoot accordingly.	MJ	none	Loss of protection alarms
Loss of Signal	DS1 DS3 E1 OC-3 STM-1	No DS1, DS3, E1, OC-3 or STM-1 input signal detected. Troubleshoot source of input to PDH, or SDH card.	MJ	none	Loss of signal alarms on facilities

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Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
Loss of Signal	TMN MGMT Port • CorEvo • MSS-E • MSS-HE • MSS-XE TMN Port#4 • CorEvo CorEvo EASv2	Loss of Signal is declared when either: • no Ethernet input signal detected on the Ethernet port • a communication problem with the remote peer (that is, a link down condition) Verify that the link partner is operating normally. Ethernet link is not established. **Note:** Note: When you connect a PC to TMN Local interface on NIM with DHCP disabled and then disconnect it, LOS alarm is raised. After enabling DHCP, LOS alarm does not clear.	MJ	none	Loss of signal alarms on Ethernet
	MSS-E MSS-HE MSS-XE	Generated when no signal is present on the incoming Sync-In port	none	MN	

Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
Loss of Signal	EAC, EAC1G2G5 and EAC10G User Ethernet Ports; EAC, EAC1G2G5 and EAC10G Ethernet Ports configured as Radio; Electrical (RJ45) and Optical (SFP) MSS-E MSS-HE/XE Core Ethernet Ports configured as Radio Both Electrical (RJ45) and Optical (SFP). EASv2 Ethernet Ports configured as Radio Both Ele CAHD Ethernet Ports configured as Radio Both Ele	Loss of Ethernet (LOS) signal (the signal missing) when no Ethernet input signal is detected on the port. Also Reported when there is a communication problem with the remote peer (i.e. link down condition).	MJ	none	Loss of signal alarms on radios
	Coupler Ethernet Interface: UBT-T UBT-S UBT-m UBT-I	Loss of Ethernet (LOS) signal (the signal missing) when no Ethernet input signal is detected on the Ethernet Coupler Port. Also for a communication problem with the remote peer (i.e.link down condition).	MJ	MN	
Loss of Signal	CorEvo MSS-E MSS-HE/XE	Generated when no signal is present on the incoming Sync-In port. Verify sync source provisioning at both the sync source and the alarming card.	none	MN	Loss of signal synchronization alarms

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Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
Loss of Signal	CorEvo RLAG: • UBT-S • UBT-m • UBT-T • UBT-I	ALL members of an Ethernet or Radio LAG group have no activity. Loss of Signal is declared when the number of active ports in the LAG equals zero. Identify the alarms that are declared against members of the LAG group and troubleshoot accordingly.	MJ	none	Loss of signal alarms on LAG
Loss of Signal	TMN ports TMNEthernet/MngtPort for MSS-E and MSS-HE/XE	Loss of signal on the Ethernet TMN Interface	none	MN	
Low Input Voltage	UBT-S UBT-m UBT-T UBT-T XP	Reported when the input voltage drops below the configured threshold.	MJ	none	Low input voltage alarms
Mac Status Defect	CorEvo	MEP received a CCM frame with correct MD/MEG Level, correct MAID/MEG, correct MEP ID and correct CCM Interval, but reporting a failure in its Port Status TLV or Interface Status TLV.	MN	none	For more information, please contact Nokia TAC
Missing Diversity Alignment Configuration	UBT-I	Reported when not having a "calibrated" state and Diversity is enabled	None	MN	Align Space Diversity as described in *WebCT User Manual*, chapter *Aligning Space Diversity*.

Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
OAM Discovery Not Completed	Ethernet port EAC, EAC1G2G5 and EAC10G User Ethernet Ports	The Ethernet port (EFM) OAM discovery process is not in operational status. If the Ethernet first mile feature is desired, verify/provision the feature on the Ethernet port link partner associated with the Ethernet port in alarm.	MJ	MN	OAM discovery not completed alarms
Power Supply Failure	EASv2 EAC EAC1G2G5 MSS-E MSS-HE/XE	MPT Power Supply Failure detected, either an Open -Circuit or Over -Current condition on an EASv2, EAC or EAC1G2G5 power supply port. Terminate, repair, or replace power supply cable as required. For MSS-XE: Power Supply Fail detection of an Open-Circuit or Over-Current condition on a power supply port. It must be reported only after an UBT has been provisioned on the MSS-XE.	MJ	MN	Power supply failure alarms
Power Supply Failure	UBT-S UBT-m UBT-T UBT-T XP	UBT Power Supply Fail detection of an Open-Circuit or Over-Current condition on a power supply port.	MJ	MN	Power supply failure alarms

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Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
PPP Fail	UBT-S UBT-m UBT-T UBT-I MPT-HLS MPT-HLC	TMN over PPP Fail Verify that no other radio path alarms (such as Demodulator Fail) exist, and troubleshoot accordingly.	none	WN	PPP fail alarms
Provisioning Mismatch	CorEvo MSS-E MSS-HE/XE	The wrong flash card is installed and the current database is in conflict with the license string on the card. Replace the flash card with the correct flash card for the license.	MN	none	Provisioning mismatch alarms
Radio Encryption Mismatch	MPT-HLS MPT-HLC UBT-S UBT-m 80 UBT-T channel A UBT-T channel B	Radio encryption is enabled on one end of the radio hop and not on the other end of the hop. Or the radio encryption key provisioned on one end is different than the key provisioned on the other end of the hop. If radio encryption is desired, provision radio encryption on the far end using the encryption key.	MJ	none	Radio encryption mismatch alarms

Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
Rate Mismatch	CorEvo EAC SFP User Ethernet Ports EAC1G2G5 SFP User Ethernet Ports EAC10G SFP User Ethernet Ports Core10G SFP Radio Ports EAC SFP Radio Ports EAC1G2G5 SFP Radio Ports EAC10G SFP Radio Port MSS-E SFP Radio ports MSS-HE/XE SFP Radio ports MSS-E SFP User Ethernet Ports MSS-HE/XE SFP User Ethernet Ports MSS-XE SFP Radio Ports SFP on CAHD	Reported when the configured speed of an optical 10 GbEth/1 GbEth User Ethernet interface or an optical 1 GbEth User Ethernet interface is not compatible with the provisioned SFP transceiver. If the equipped SFP supports the required Ethernet interface rate, verify/provision Ethernet interface port in alarm. Reported when the configured speed of an optical Radio interface is not compatible with the provisioned SFP transceiver	MJ	MN	Rate mismatch alarms
Remote Defect Indication	UBT-S UBT-m UBT-T UBT-I MPT-HLS MPT-HLC	Remote Defect Indication (RDI) in a received radio direction. Alarm is reported when an RDI is received on a given radio direction from the remote end of radio link.	none	MN	Remote Defect Indication alarms on radios

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Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
Remote Defect Indication	OC-3 STM-1	Remote Defect Indication (RDI) in a received SDH Direction. Alarm is reported when an RDI is received on a given SDH direction from the remote end of SDH/OC-3 link. Determine all of the facility alarms that are declared against the far end STM-1/OC-3 facility associated with the STM-1/OC-3 facility declaring the Remote Defect Indication (RDI) and troubleshoot accordingly.	MJ	none	Remote defect indication alarms on facilities
Replaceable Unit Missing	CorEvo FAN A-FAN E-FAN EASv2 P2E3DS3 P32E1DS1 SDHACC SFP RPS SFP	Card/SFP is missing from slot (Only declared, if the card/SFP has been provisioned. Card/SFP could be unreachable). Install the card/SFP type corresponding to slot provisioning.	none	MN	Replaceable unit missing alarms
	EAC EAC1G2G5 EAC10G CAHD	Card is missing from slot. Reported to be provisioned board in slot. Board could be unreachable.	MJ	MN	
	SFP EAC card SFP EAC1G2G5 card SFP EAC10G card SFP on CAHD	SFP is missing from Board. Reported to be provisioned board	MJ	MN	
	MSS-E MSS-HE/XE	SFP is missing from MSS	MJ	MN	

Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
Replaceable Unit Problem	CorEvo FAN A-FAN E-FAN EASv2 P2E3DS3 P32E1DS1 SDHACC SFP RPS SFP XPIC Module Rx Diversity Module	Card failed. Perform a protection switch if applicable.	none	MN	Replaceable unit problem alarms
	EAC EAC1G2G5 EAC10G CAHD	Card failed. Only set if the board has been provisioned.	MJ	MN	
	SFP EAC card SFP EAC1G2G5 card SFP EAC10G card SFP on CAHD	SFP Module Failed. Only set if the board has been provisioned.	MJ	MN	
	UBT-S UBT-m UBT-T UBT-I	Card failed. Only set if the board has been provisioned / enabled.	MJ	MN	
	UBT-S UBT-m UBT-T UBT-I	Card failed. TX_Fault indication is raised by the SFP (if supported). Hence, the transmission capability of this SFP is problematic	MJ	MN	
	UBT-I	Card failed detected for the Rx Diversity module for UBT-I. Only set if the module has been provisioned	None	MN	
UBT-T channel A UBT-T channel B	One of the carriers of the TWIN is in failure. Only if the carrier is enabled.	MJ	MN		
	MSS-E MSS-HE/XE	Card failed	MJ	MN	
	SFP MSS-E SFP MSS-HE/XE	SFP module failed	MJ	MN	

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Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
Replaceable Unit Type Mismatch	Core Card	Card/ODU in slot does not match card/ODU configured in CorEvo memory. Replace the installed card with card type provisioned in Core memory.	none	MN	Replaceable unit type mismatch alarms
	SFP	The plugged SFP does not match the the configured one (related to Sync-IO and ToD SFPs).	MJ	MN	
	E1/DS1 Access Card Radio Access Card DS3 Access Card				
	EASv2	Card/ODU in slot does not match card/ODU configured in CorEvo or CAHD memory. Replace the installed card with card type provisioned in Core memory.			
	EAC EAC1G2G5 EAC10G CAHD				
	SDH Module				
	FAN A-FAN E-FAN				
	MPT-HLS MPT-HL v2 Cubic				
	SubRack	Subrack mnemonic does not macth with EAC or CAHD configured in DB.			
	Traffic SFP on: UBT-S UBT-m UBT-I UBT-T MSS-E MSS-HE/XE	Radio physically attached to the slot/port does not match radio type provisioned.			
	Coupling port SFP on: UBT-S UBT-T UBT-m UBT-I	The plugged SFP does not match the the configured one (related to Sync-IO and ToD SFPs). The plugged SFP on the coupling port is not supported.			

Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
Retiming Buffer Overflow	E1 Channelized E1	Retiming buffer overflow for channelized E1/E1tributary. Troubleshoot any possible synchronization alarms.	MJ	none	Retiming buffer overflow alarms
RF Switch Fail	UBT-I MPT-HLS MPT-HLC	RF switch fail detected Correct all cable and cable connections as required.	none	MN	RF Switch Fail alarms
RPS Path Fail	MPT-HLS MPT-HLC	Determine all radio alarms that are declared against the radio direction declaring RPS Path Fail and troubleshoot accordingly.	MJ	MN	RPS path fail
Rx Diversity Missing	Radio with Rx Diversity combiner (MPT-HLS, MPT-HLC, UBT-I)	Combiner function enabled but Rx Diversity module Missing Using the Craft Terminal, from the radio main view, Settings tab, select the SD check-box (unchecked).	none	MN	Rx diversity missing alarms
Rx Overload	UBT-S UBT-m UBT-T channel A UBT-T channel B UBT-I	Input overload condition at the RF front end detected.	MN	WN	Alarms available only in WebCT.Rx Overload alarms

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Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
Rx Problem	RLAG: • UBT-S • UBT-m • UBT-T • UBT-I MPT-HLS MPT-HLC	Reported when a loss of traffic occurs due to issues at the RX blocks or to malformed packets on the L1 LAG data path. Identify the alarms that are declared against members of the LAG group and troubleshoot accordingly By default, this alarm is disabled. When disabled, this alarm is not considered to determine the L1 LAG operative state.	MJ	MN	Rx problem alarms
Secondary Battery Fail	MSS-8 MPT-HLS MSS-E MSS-HE MSS-XE	B battery feed failed or is missing. Verify the fuse and power cables providing power to the B-power feed	none	MN	Secondary battery failure alarms
Service Selective Drop	OC-3 STM-1	Indicates an OC-3 or STM-1 Flow is selectively dropped due to a reduction of Radio L1 LAG throughput. Identify the alarms that are declared against members of the LAG group associated with the affected OC-3 or STM-1 and troubleshoot accordingly.	MJ	MN	Service selective drop alarms

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Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
Signal Label Mismatch	Channelized E1	Signal Label Mismatch detected for the AU4 or TU12(Tx Direction). Troubleshoot the source of input STM-1 signal that is associated with the channelized E1 input signal to the SDHCHAN card.	MJ	none	Signal Label Mismatch alarms
Software Download Failed	MSS-E MSS-HE MSS-XE	Software Download Conditional self-test Fail	MJ	none	For more information, please contact Nokia TAC
Standby Version Mismatch	MPT-HLS MPT-HLC	Standby software version on MPT-HLC does not match software version on CorEvo. Verify that the Firmware Download On Going is declared against the radio. If it is ongoing, wait 30 minutes.	none	WN	Standby version mismatch alarms
	UBT-S UBT-m UBT-T UBT-I	Standby Software version on UBT does not match software version on Core.	none	WN	

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Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
Sync Packet Loss	CorEvo MSS-E MSS-HE MSS-XE	A Boundary Clock/OC Slave has stopped receiving PTP Sync messages from either a Boundary Clock or OC Master. Check for radio path alarms and clear. For the associated VLAN, check for and clear any alarms that exist along the path to either the Boundary Clock or OC Master. For MSS product family: Reported when there is a lack of reception of PTP Sync messages from a Master (loss of the packet timing signal)	WN	WN	Sync Packet Loss
Synchronization Signal Fail	CorEvo User Ethernet port DS1 E1 OC-3 STM-1 RLAG on CAHD EAC, EAC1G2G5 and EAC10G User Ethernet Ports Electrical (RJ45) and Optical (SFP) UBT-S UBT-m UBT-T UBT-I MSS-E MSS-HE MSS-XE MPT-HLS MPT-HLC	Synchronization Signal Fail is reported if the received source signal is not suitable as a Sync Source. Managed only if the interface has been configured as a primary or secondary synchronization source. Verify Degrade, Loss ESMC, or Loss of Signal are not also declared against Sync source declaring Synchronization Signal Fail. Troubleshoot any other synchronization alarms.	MN	WN	Synchronization signal fail alarms

Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
Threshold Crossing Alarm	DS1 E1 OC-3 STM-1 UBT-S UBT-m UBT-T channel A UBT-T hannel B UBT-I MSS-E MSS-HE/XE MPT-HLS MPT-HLC	Threshold Crossing Alarm (TCA) indicates alarm threshold has been exceeded. Applies to performance monitor parameters with an accumulation period of 24 hours and 15 minutes. Verify the radio channel power levels and determine the local weather conditions that could affect radio performance.	MJ	MN	Threshold crossing alarms
Threshold Crossed	UBT-S UBT-m UBT-T channel A UBT-T channel B UBT-I UBT-T XP channel A UBT-T XP channel B	Threshold Crossing Alarm (TCA) is emitted on counter basis. Applies to performance parameters with an accumulation period of 24 hours and 15 minutes.	none	MN	Threshold crossing alarms
Traffic Down	RLAG: • UBT-S • UBT-m • UBT-T • UBT-I MPT-HLS MPT-HLC	Reported when a loss of traffic occurs and the L1 LAG data path is unavailable. Identify the alarms that are declared against members of the LAG group and troubleshoot accordingly. By default, this alarm is disabled. When disabled, this alarm is not considered to determine the L1 LAG operative state.	MJ	MN	Traffic down alarms

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Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
Traffic Loss	RLAG: • UBT-S • UBT-m • UBT-T • UBT-I MPT-HLS MPT-HLC	Reported when packet loss occurs due to an extreme latency difference over L1 LAG members or when errors occurs on the L1 LAG data path. Determine the desired LAG membership and provision LAG membership accordingly. By default, this alarm is disabled. When disabled, this alarm is not considered to determine the L1 LAG operative state.	MN	WN	Traffic loss alarms
Tx Problem	RLAG: • UBT-S • UBT-m • UBT-T • UBT-I MPT-HLS MPT-HLC	Reported when a loss of traffic occurs due to issues at the TX blocks or to malformed packets on the L1 LAG data path. Identify the alarms that are declared against members of the LAG group and troubleshoot accordingly. By default, this alarm is disabled. When disabled, this alarm is not considered to determine the L1 LAG operative state.	MJ	MN	Tx problem alarms

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Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
Unavailable Time	DS1 E1 OC-3 STM-1 UBT-S UBT-m UBT-T channel A UBT-T channel B UBT-I MSS-E MSS-HE/XE MPT-HLS MPT-HLC	A period of unavailable time begins when 10 consecutive SES events are detected. Alarms are associated with the performance monitoring port. Determine local weather conditions that could affect radio performance.	MJ	MN	Unavailable time alarms
Unconfigured Equipment Present	CorEvo FAN E-FAN A-FAN EASV2 P2E3DS3 P32E1DS1 SDHACC SFP EAC EAC1G2G5 EAC10G SFP EAC card SFP EAC1G2G5 card SFP EAC10G card MSS-E MSS-HE MSS-XE CAHD	Card in MSS shelf slot 2 through 8 is not provisioned (enabled) but is present in slot. SFP is not provisioned (enabled) but is present in card.	none	WN	Unconfigured equipment present alarms
	CAHD	Card in slot is not provisioned (enabled) but is present in the slot	none	WN	

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Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
Underlying Resource Unavailable	OC-3 STM-1 Channelized E1 E1 User Ethernet port SFP in User Ethernet port MPT-HLS MPT-HLC	Underlying Resource Unavailable on Interface - transmission resources affected by equipment failure. - Alarm URU These alarms are reported by the MIB for the 1350 OMS and are not reported by the alarm manager. Evaluate and troubleshoot other alarms on the equipment reporting the Underlying Resource Unavailable alarm.	MJ	none	Underlying resource unavailable alarms
Underlying Resource Unavailable Service	Channelized E1 E1 EASv2 MPT-HLS MPT-HLC	Underlying Resource Unavailable Service on E1 Interface - transmission resources affected by equipment failure - Alarm URUS Evaluate and troubleshoot other alarms on the equipment reporting the Underlying Resource Unavailable service alarm.	MJ	none	Underlying resource unavailable service alarms
Version Mismatch	CorEvo MPT-HLS MPT-HLC	Software version on CorEvo, MPT-HLC, or MPT-HLS does not match software version on CorEvo.	none	MN	Version mismatch alarms
	UBT-S UBT-m UBT-T UBT-I	Software version on UBT does not match software version on Core.	none	MN	
Log Upload Failed	MSS	Reported by system when SFTP failed or due to internal SW error	MN	MN	For more information, please contact Nokia TAC

Probable cause	Facilities/ equipment	Description and possible solution	Service Affecting (SA)	Non Service Affecting /Abnormal condition (NSA/Abn)	See
XPIC Cable Loss	UBT-S UBT-m UBT-T UBT-T XP	XPIC Cable Connection defect detected	MN	none	For more information, please contact Nokia TAC

For Alarms available only in WebCT, please refer to *Wavence WebCT User Manual*.

Abnormal conditions

Table: Abnormal Conditions provides information about abnormal conditions.

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Table 5: Abnormal Conditions

Operation	Name and equipment	Description and solution
Core facing PDH loopback	Loopback on PDH: • DS1/E1 on P32E1DS1 - DS1 Port#[1-32] Internal - E1 Port#[1-32] Internal • DS3 on P2E3DS3 - DS3 Port#[1-2] Internal	The associated entity has a core facing PDH loopback activated at the near end. Using the Craft Terminal, release the loopback. If the condition does not clear, contact support.
Core facing SDH loopback	Loopback on SDH: • STM-1/OC-3 on SDHACC - STM-1 Port Internal - OC3 Port Internal	The associated entity has a core facing SDH (STM-1/OC3) loopback activated at the near end. Using the Craft Terminal, release the loopback. If the condition does not clear, contact support.
Digital core facing loopback	Loopback on Radio: • MPT-HLC, or MPT-HLS connected to CorEvo - Direction #1.[7-8] • MPT-HLC or MPT-HLS connected to EASv2 - Direction #[3-8].[5-8] UBT-S UBT-m UBT-T UBT-I	The associated entity has a loopback activated at the near end. Using the Craft Terminal, release the loopback. If the condition does not clear, contact support.

Operation	Name and equipment	Description and solution
Forced adaptive modulation	Radio: • MPT-HLC, or MPT-HLS connected to CorEvo - Direction #1.[7-8] - Channel #1 • MPT-HL connected to EASv2 - Direction #[3-8].[5-8] - Channel #1 • MPT-HLC or MPT-HLS connected to EASv2 - Direction #[3-8].[5-8] - Channel #1	Manual Adaptive Modulation command has been executed. Using the Craft Terminal, release the Forced Adaptive Modulation. If the condition does not clear, contact support.
	UBT-S UBT-m UBT-T channel A UBT-T channel B UBT-I	Reported when a Manual Forced Adaptive. Modulation command is executed. It shall be cleared when the request is released.
Forced Switch - EPS	Equipment Protection: • CorEvo - Main#1 Slot#1 • P2E3DS3, P32E1DS1, SDHACC - Main#1 Slot#[3,5,7] • MPT-HLC or MPT-HLS connected to CorEvo - Main#1 Slot #1 Port #7 • MPT-HLC or MPT-HLS connected to EASv2 - Main#1 Slot #[3,5,7] Port #[5-8] UBT-S UBT-m UBT-T	A Forced Equipment Protection Switch (EPS) has been executed for the protection group. Using the Craft Terminal, release the Forced Switch. If the condition does not clear, contact support.
Forced switch - RPS	Rx Radio Protection: • MPT-HLC or MPT-HLS connected to CorEvo - Main#1 Slot #1 Port #7 • MPT-HLS or MPT-HLC connected to EASv2 - Main#1 Slot #[3,5,7] Port #[5-8]	A Forced Rx Protection Switch (RPS) has been executed for the protection group. Using the Craft Terminal, release the Forced Switch. If the condition does not clear, contact support.

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Operation	Name and equipment	Description and solution
Forced switch - TPS	HSB Protection: • MPT-HLC or MPT-HLS connected to CorEvo - Main#1 Slot #1 Port #7 • MPT-HLC or MPT-HLS connected to EASv2 - Main#1 Slot #[3,5,7] Port #[5-8]	A Forced Tx Protection Switch (TPS) has been executed for the protection group. Using the Craft Terminal, release the Forced Switch. If the condition does not clear, contact support.
Line facing circuit loopback	Loopback: • MPT-HLC, or MPT-HLS connected to CorEvo - Direction #1.[7-8] • MPT-HLC or MPT-HLS connected to EASv2 - Direction #[3-8].[5-8]	The associated entity has a loopback activated at the near end. Using the Craft Terminal, release the loopback. If the condition does not clear, contact support.
Line facing PDH loopback	Loopback on PDH: • DS1/E1 on P32E1DS1 - DS1 Port#[1-32] External - E1 Port#[1-32] External • DS3 on P2E3DS3 - DS3 Port#[1-2] External	The associated entity has a line facing PDH (DS1, DS3, or E1) loopback activated at the near end. Using the Craft Terminal, release the loopback. If the condition does not clear, contact support.
Line facing SDH loopback	Loopback on SDH: • STM-1/OC-3 on SDHACC - STM-1 Port Internal - OC3 Port Internal	The associated entity has a core facing SDH (STM-1/OC3) loopback activated at the near end. Using the Craft Terminal, release the loopback. If the condition does not clear, contact support.

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Operation	Name and equipment	Description and solution
Lockout switch - EPS	Equipment Protection: • CorEvo - Spare#0 Slot#2 • P2E3DS3, P32E1DS1, SDHACC - Spare#0 Slot#[4,6,8] • MPT-HLC or MPT-HLS connected to CorEvo - Spare#0 Slot #1 Port #8 • MPT-HLC or MPT-HLS connected to EASv2 - Spare#0 Slot#[4,6,8] Port #[5-8] UBT-S UBT-m UBT-T	A Lockout Equipment Protection Switch (EPS) has been executed for the protection group. Using the Craft Terminal, release the lockout switch. If the condition does not clear, contact support.
Lockout switch - RPS	Rx Radio Protection: • MPT-HLC or MPT-HLS connected to CorEvo - Spare#0 Slot #1 Port #8 • MPT-HLC or MPT-HLS connected to EASv2 - Spare#0 Slot#[4,6,8] Port #[5-8]	A Lockout Rx Protection Switch (RPS) has been executed for the protection group. Using the Craft Terminal, release the Forced Switch. If the condition does not clear, contact support.
Lockout Switch - TPS	HSB Protection: • MPT-HLC or MPT-HLS connected to CorEvo - Spare#0 Slot #1 Port #8 • MPT-HLC or MPT-HLS connected to EASv2 - Spare#0 Slot#[4,6,8] Port #[5-8]	A Lockout Tx Protection Switch (TPS) has been executed for the protection group. Using the Craft Terminal, release the lockout switch. If the condition does not clear, contact support.
Rx main squelch	Radio: • MPT-HLC or MPT-HLS connected to CorEvo - Direction #1.[7-8] - Channel #[0-1] • MPT-HLC or MPT-HLS connected to EASv2 - Direction #[3-8].[5-8] - Channel #[0-1]	Main receiver has been squelched. Using the Craft Terminal, release the squelch. If the condition does not clear, contact support.

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Operation	Name and equipment	Description and solution
Rx Div squelch	Radio: • MPT-HLC or MPT-HLS connected to CorEvo - Direction #1.[7-8] - Channel #[0-1] • MPT-HLC or MPT-HLS connected to EASv2 - Direction #[3-8].[5-8] - Channel #[0-1] UBT-I	Diversity receiver has been squelched. Using the Craft Terminal, release the squelch. If the condition does not clear, contact support.
Tx Mute	Radio: • MPT-HLC or MPT-HLS connected to CorEvo - Direction #1.[7-8] - Channel #[0-1] • MPT-HLC or MPT-HLS connected to EASv2 - Direction #[3-8].[5-8] -Channel #[0-1] UBT-S UBT-m UBT-T channel A UBT-T channel B	Manual or automatic mute command has been executed. Using the Craft Terminal, release the TxMute. If the condition does not clear, contact support.
ACM Degrade	UBT-S UBT-m 80 UBT-T channel A UBT-T channel B UBT-I	Reported when a Radio Link has not operated at the maximum provisioned modulation for more than one hour. Non Service Affecting
Automatic Tx Mute	MPT UBT-S UBT-m 80 UBT-T channel A UBT-T channel B UBT-I	The local entity has an Automatic TX Mute enabled.

LED indicators and TMN troubleshooting

The following tables provide information about LED indicators. See:

- Table: CorEvo Card LED indications for CorEvo card LEDs
- Table: Transport Card LED indications for transport card LEDs
- Table: Ethernet LED indications for Ethernet card LEDs
- Table: MPT-HLC transceiver LED alarm indications for MPT-HLC transceiver LEDs
- Table: MPT-HLS transceiver front panel indicator details for MPT-HLS transceiver LEDs

See Table: TMN network troubleshooting for TMN Networking troubleshooting information.

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Table 6: CorEvo Card LED indications

LED	Indication	Probable cause	Corrective action
Status	OFF	Card Not Provisioned	Provision card
		Not Powered	Verify MSS-4/8 shelf fuse is not blown or missing. Verify Power cable(s) are properly t terminated at both ends. Verify voltage level at the power distribution panel.
		Not Equipped	Verify that the card is properly installed.
	Green Blinking	Configuration Downloading, SW Booting, or flash card realignment.	Wait for configuration download or software boot to complete.
	Green	Card Properly Equipped and Provisioned. currently In -Service/Active.	Normal Operating condition. No Action required.
	Yellow	Card Properly Equipped and Provisioned as EPS. Currently in Standby.	Normal operating condition for Standby card. No Action required.
	Red	Card Fail	Replace card
		Card Mismatch	Check alarms for replaceable unit type mismatch, see Replaceable unit type mismatch alarms.

LED	Indication	Probable cause	Corrective action
Alarm in -formation	Off	No alarms, warnings, or abnormal conditions.	Normal Operating condition. No Action required.
	Yellow Blinking	At least one abnormal condition present in the NE.	Retrieve abnormal condition using WebCT, see Table: Abnormal Conditions , and troubleshoot accordingly.
	Yellow	At least one alarm with Warning severity present in the NE. An abnormal condition may also be present.	Retrieve alarms using the Craft Terminal's Active Alarms, see Table: Craft Terminal's Active Alarms alarm types , and troubleshoot accordingly.
	Red	At least one alarm with Major or Minor severity present in the NE. An alarm with Warning severity or an abnormal condition may also be present.	Retrieve alarms using the Craft Terminal's Active Alarms, see Table: Craft Terminal's Active Alarms alarm types , and troubleshoot accordingly.
	Red and yellow transition	At least one alarm with Major or Minor severity and at least one alarm with Warning severity is present in the NE.	Retrieve alarms using the Craft Terminal's Active Alarms, see Table: Craft Terminal's Active Alarms alarm types , and troubleshoot accordingly.

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Table 7: Transport Card LED indications

LED	Indication	Probable cause	Corrective action
Status	OFF	Card Not Provisioned	Provision card
		Not Powered	Verify MSS-4/8 shelf fuse is not blown or missing. Verify Power cable(s) are properly terminated at both ends. Verify voltage level at the power distribution panel.
		Not Equipped	Verify that the card is properly installed.
	Green Blinking	Configuration Downloading, SW Booting, or flash card realignment.	Wait for configuration download or software boot to complete.
	Green	Card Properly Equipped and Provisioned. currently In -Service/Active.	Normal Operating condition. No Action required.
	Yellow	Card Properly Equipped and Provisioned as EPS. Currently in Standby.	Normal operating condition for Standby card. No Action required.
	Red	Card Fail	Replace card
	Red Blinking	Card Mismatch	Check alarms for replaceable unit type mismatch, see Replaceable unit type mismatch alarms.
PFoE	Yellow	Port is emitting power according to the configuration.	When the LED for the port is yellow, the port is emitting power. These LEDs are just indicators and should match what is configured for that port.
	OFF	Port is not emitting power according to the configuration.	

Table 8: Ethernet LED indications

LED	Indication	Description	Probable cause	Corrective action
ETH Link	Green LED Not Lit	Loss of Ethernet communication with link partner.	Cable between link partner and Ethernet port is disconnected/ broken.	Connect/repair cable.
			Speed/Mode provisioning mismatch between link partner and Ethernet port.	Check local Ethernet provisioning screen. Check link partner provisioning.

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Table 9: MPT-HLC transceiver LED alarm indications

LED	Indication	Probable cause	Corrective action
Status	OFF	Card Not Provisioned	Provision card
		Not Powered	Verify MPT-HLC Transceiver fuse is not blown or missing. Verify Power cable is properly terminated at both ends. Verify voltage level at the power distribution panel.
		Invalid Configuration	Verify configuration provisioning and correct as required.
	Green Blinking	Configuration Downloading, or SW Booting. (Different blink rates should be used to identify each step of the startup process.)	Wait for configuration download or software boot to complete.
	Green	Card Properly Equipped and Provisioned. EPS or RPS currently In -Service/Active.	Normal Operating condition. No Action required.
	Yellow	Card Properly Equipped and Provisioned. EPS and RPS currently in Standby.	Normal operating condition for Standby MPT-HLC Transceiver. No Action required.
	Green/Red Blinking	Card properly equipped and provisioned. RX signal fault, EPS or RPS currently In-Service/Active. Traffic is probable affected.	Troubleshooting the Green/Red indicator includes checking for consistent and/or intermittent alarms on the RF path and checking RSL history for possible fading. If configuration is a 1+0, traffic is being affected. If 1+1, RPS switching occurs (and is forced to the standby radio if necessary) and EPS may be needed if a HW issue since Green indicates the radio is Active EPS. The RX issue could be a result of path, receive Hw, and far end transmitter path.
		Equipment problem at either near end or far end:	Check alarms. Check abnormal conditions. Check protection schemes.
		Path related problem:	Check RSL levels Check equipment alarms. Verify no weather related issue.
		PA switch at the far end is set to OFF ‘0" position:	Verify PA switch position at the far end and set to ON ‘1" position.

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LED	Indication	Probable cause	Corrective action
Status (continued)	Yellow/Red Blinking	Card properly equipped and provisioned, RX signal Fault, EPS and RPS currently in Standby. Traffic maybe affected:	Troubleshooting the Yellow/Red indicator includes checking for consistent and/or intermittent alarms on the standby RF path and checking RSL history for possible fading on diversity path. Yellow indicates the Radio Equipment Standby. RX issue can be a result of diversity path, and/or standby receive Hw. If the issue is occuring on both main and standby radios (Green/Red on Main and Yellow/Red on Standby), investigate far end transmitter as well.
		Equipment problem at either near end or far end:	Check alarms. Check abnormal conditions. Check protection schemes.
		Path related problem:	Check RSL levels Check equipment alarms. Verify no weather related issue.
	Red	Card Fail	Replace MPT-HLC Transceiver
	Red Blinking	MAP Communication Time-out: Communication between the MPT -HLC Transceiver and the EASv2 card is lost.	Check alarms for Internal Communication Problem, if present, see Internal Communication Problem alarms.

LED	Indication	Probable cause	Corrective action
Power Emission	Green LED ON	MPT-HLC Transceiver is emitting power as expected according to the known configuration.	No Action required
	Yellow	MPT-HLC Transceiver is not emitting power due to a forced squelch condition including manual PA switch or an auto-mute condition. Most probable causes:	Probable causes include: • A manual mute performed by user. (check abnormal conditions) • PA is turned off (check front panel of MPT. PA may be switched to "0"). There should be a Replaceable Unit Problem Alarm against the Radio. • Auto Mute due to configuration error or incompatibilities (e.g., shifter, frequency, power) • RF Switch Fail • Excessive temperature (check alarms) • XPIC Auto Mute Procedure • Radio Internal Communication Problem • Radio is not provisioned • Tx failure conditions that affect the Tx chain including PA output circuitry and MPT Card Fail. If unable to determine cause, please contact Nokia TAC.
		TX mute active	Release TX mute
		PA switch is positioned Off	Position PA switch to On position
	Red	MPT-HLC Transceiver is abnormally emitting power in contrast with the known configuration (Tx Power alarm is raised during normal operational mode	—
	Green LED OFF	MPT-HLC Transceiver is not emitting power for reasons other than "forced squelch".	Transmitter will be off (besides manual or auto muting) when: • The MPT is configured for protection and the Transmit protection switch (TPS) is in the standby state. • The MPT is currently going through a software/firmware reboot or download
		Transmit Protection Switch (TPS) is currently in standby state.	Verify MPT-HL Transceiver should be in standby state.
		Software booting or FPGA download in progress	Wait for software booting or FPGA download to complete

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Table 10: MPT-HLS transceiver front panel indicator details

LED	Indication	Probable Cause	Corrective Action
Status (S)	Off	Card not powered	• Verify that the MPT-HLS fuse is not blown or missing • Verify that the power cable is properly terminated at both ends • Verify voltage level at the TRU
	Green Blinking	FPGA Downloading, or SW Booting. (Different blink rates identify each step of the startup process.)	Wait for configuration download or software boot to complete.
	Green	In Service, Normal Operation, and Properly Provisioned. EPS or RPS currently In-Service/Active.	Normal operating condition. No action required.
	Yellow	Card Properly Equipped and Provisioned in 1+1, currently in Standby.	Normal operating condition for standby radio. No action required.
	Green/Red Blinking	Card Properly Equipped and Provisioned, Rx signal fault, EPS and RPS in Standby.	Check alarms. Check abnormal conditions. Check protection schemes.
	Yellow/Red Blinking	Card Properly Equipped and Provisioned, RX signal Fault, EPS and RPS currently in Standby. Traffic may be affected.	Check alarms. Check abnormal conditions. Check protection schemes.

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LED	Indication	Probable Cause	Corrective Action
Status (S) (continued)	Red	• HW Card Fail • Power supply failure (Auto-test failure) • FPGA failures • LO alarms • Modem Tx sync alarm • PA switch active • Diversity board card fail (only in combiner mode) • RF switch fail (only for the spare MPT-HLS in HSB configuration) In a PA critical temperature condition, the LED remains green.	Replace MPT-HLS transceiver
	Red Blinking	MAP Communication Time-out: Communication between the MPT-HLS Transceiver and the EASv2 cards is lost. This occurs in split-mount configuration only.	Check alarms for Internal Communication Problem, if present, see Internal Communication Problem alarms.

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LED	Indication	Probable Cause	Corrective Action
Power Emission	Off	MPT-HLS Transceiver is not emitting power, according to the known configuration, for example, Hot Standby.	Verify MPT-HL Transceiver should be in standby state.
	Green	MPT-HLS Transceiver is emitting power as expected according to the known configuration.	No action required
	Yellow	MPT-HLS Transceiver is not emitting power due to an operator command, including manual PA PSU switch or Tx mute.	Release Tx Mute or position PA PSU switch to On position
	Red	MPT-HLS Transceiver is abnormally emitting power, in contrast with the known configuration (for example, when Tx power alarm is raised during normal operative mode)	Because the Transceiver is abnormally emitting power, the Transmitter LED will be RED. The Radio will also be in alarm and the alarm typically is a Replaceable Unit Alarm. To troubleshoot, the following can be done: • Check ATPC and RTPC power setting. Reconfiguring and reapplying may be necessary. Attempt to reapply at a lower transmit power setting if path design allows for this. • Check the power cycle radio and monitor. • Replace radio. Could be temperature of PA. For more information, please contact Nokia TAC.

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Table 11: TMN network troubleshooting

Problem	Possible Cause	Possible Solution
Unusually slow communication in radio network	Normal network management traffic is saturating the communications channel. Polling radios for PM data or missed alarms too rapidly Multiple remote software downloads in process IP traffic other than network management traffic being routed through radio network	There may be too many radios being managed within a single region. Split the radio network management into different regions and back-haul the traffic for each region through separate channels. Poll the radios more slowly. Download to fewer radios at a time. Configure external routers to allow only network management related traffic through the Management network of the radios. Dynamic route updates (OSPF) may attempt to reroute high speed traffic through the TMN network if a high speed ink fails.
Unable to operate controls using SNMP	To perform control operations, the Manager must be registered as a craft device.	Register the Manager as a craft device. Manager registration type can be changed as needed to type ‘ct" to allow control operation and then be changed back to ‘nml" for normal operation.
Can Read SNMP objects but cannot Write to SNMP objects	Incorrect community string If the TMN Interface is configured for SNMPv2, the write community string is probably wrong.	Use the correct community string. Use the correct write community string.

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Problem	Possible Cause	Possible Solution
No traps being received from NE	Manager not registered in NE to receive traps Communication failure in network Traps are blocked by firewall	Register Manager with NE. Check network connectivity. Check redundant network paths and routing. Traceroute (tracert) is useful for locating path or routing faults. Check firewall provisioning to allow trap UDP ports.
Unable to communicate with the NE through the radio network (unable to ‘ping" the NE).	Possible communication path failure or routing failure within the radio network. Incorrect IP address on the PC.	Use traceroute (tracert) to help locate for communication path or routing problems. Verify PC IP address, subnet, and default route.
Can ‘ping" the TMN Interface but cannot communicate with the NE using SNMP, or can only see a few SNMP objects in the NE.	If using SNMPv2, using the wrong community string, username, or passphrase. Manager not registered or properly registered with the NE. The objects visible to an unregistered manager are generally: the 'system' MIB, a few objects under 'tsdimSnmpNEMibObject', the 'opticsIMMgrPollingInfoTable', the 'snmpUsmMib', and 'snmpVacmMib' objects.	Verify community string, username, and passphrase. Verify Manager registration with the NE and register or correct registration.

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9. Supporting information

9.1 Additional support tools for maintenance and trouble clearing

Overview

Table: Supporting information lists the supporting information that is required to perform maintenance and trouble clearing on the system. The following references are used:

- For information about operation and administration using the WebCT, see the Wavence WebCT User Manual.
- See the Wavence Product Information Manual to determine card location and type.

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Table 12: Supporting information

Supporting information	See
Alarm condition types	Wavence WebCT User Manual
Card locations	Wavence Product Information Manual
Electrostatic-sensitive devices	Electrostatic-sensitive devices
Safety awareness Laser precautions	Safety awareness
Maintenance philosophy	Maintenance philosophy
Operations support system	Wavence WebCT User Manual
Security/user authorization	Wavence WebCT User Manual
Alarm and event reports, Retrieve	Wavence WebCT User Manual
Clean fiber-optic connectors	Wavence Hardware Installation and Replacement Manual
Clock reference and operation mode selections	Wavence WebCT User Manual
Log into system	Wavence WebCT User Manual
Loopback (Core Facing)	Wavence WebCT User Manual
Manage cross-connections	Wavence WebCT User Manual
MIB Database Management (Backup and Restore)	Wavence WebCT User Manual
Password change	Wavence WebCT User Manual
Performance monitoring	Wavence WebCT User Manual
Protection switching controls	Wavence WebCT User Manual
Provision equipment	Wavence WebCT User Manual
Provision or edit ports	Wavence WebCT User Manual
Preventive maintenance (MPT-HL/MPT-HLC transceiver air filter replacement)	Wavence Hardware Installation and Replacement Manual
Protection upgrade and downgrade	Wavence Hardware Installation and Replacement Manual

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Supporting information	See
Resolve TCO Suite startup issues	Wavence Software Installation and NE Upgrade Manual
Software package rollback	Wavence Software Installation and NE Upgrade Manual
System power down and power up	Wavence Hardware Installation and Replacement Manual
Upgrade and downgrade radio profile	Wavence Hardware Installation and Replacement Manual
Upgrade unlicensed MPT-HL radio to lower 6 GHz radio	Wavence Hardware Installation and Replacement Manual
Upgrade MPT-HL to MPT-HL L1 LAG port	Wavence Hardware Installation and Replacement Manual

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10. 1588 TC Fail alarms

10.1 General

Purpose

This procedure describes how to identify and clear 1588 TC Fail alarms.

Additional information

A 1588 TC Fail alarm indicates that, due to radio impairments, TC time stamping can not be guaranteed due to the alignment of the local and remote counters.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: 1588 TC Fail alarms describes the equipment that can generate 1588 TC Fail alarms.

Table 13: 1588 TC Fail alarms

Friendly name/entity	Facility/equipment
Radio/Dir#1.[7-8]/Slot#1/Port#[7-8]/Ch#1	MPT-HLC or MPT-HLS connected to CorEvo
Radio/Dir#[3-8].[5-8]/Slot#[3-8]/Port#[5-8]/Ch#1	MPT-HLC or MPT-HLS connected to EASv2
N/A	UBT-S, UBT-m, UBT-T channel A, UBT-T channel B, UBT-I

10.2 Clear 1588 TC Fail alarms

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Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: 1588 TC Fail alarms for information.

3 Verify that no other radio path alarms, such as Demodulator Fail. If radio path alarms exist, troubleshoot accordingly.

4 Did other radio path alarms exist and were they cleared?

   1. If yes, go to step 5.
   2. If no, contact the next level of technical support for assistance

5 Retrieve the alarms using the Craft Terminal's Active Alarms.

6 Did the 1588 TC Fail alarm clear?

   1. If yes, the procedure is complete.
   2. If no, contact the next level of technical support for assistance.

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11. AIM Read Fail alarm

11.1 General

AIM Read Fail alarm applies to UBT-S and UBT-T and it indicates that AIM data have not been read.

There are two cases of AIM Read Fail alarm in WebCT:

- AIM Read Failure (EEPROM KO).
  Major alarm. It indicates that the link is down and the radio cannot not be configured at all. In order to clear the alarm, send UBT/AIM back to factory for repair.
- AIM Read Failure (EEPROM OK).
  Warning alarm. It indicates that the Radio link starts-up but it is not functional. In order to clear the alarm, send UBT/AIM back to factory for repair but with no sake of urgency. AIM Read Failure (EEPROM OK) alarm is automatically cleared after one hour.

In any case, it is recommended to sent the UBT to factory with its own attached AIM to assure the full analysis and clearing.

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12. AIS alarms

12.1 General

Purpose

This procedure describes how to identify and clear AIS alarms.

Additional Information

AU4-AIS is a High Order path alarm indication signal on a channelized E1 unit. The AU4-AIS is specified as all "1"s in the entire High Order path Administrative Unit. The alarm becomes significant and must be reported if an LOS, LOF, TIM, or MS-AIS alarm for the input is not present. As consequent action, any E1 extracted from STM-1 will be replaced with AIS (and then circuit emulated according to the provisioned settings).

TU12-AIS is a Low Order path alarm indication signal on a channelized E1 unit. The TU12-AIS is specified as all "1"s in the entire Low Order path Tributary Unit. The alarm becomes significant and must be reported if an LOS, LOF, TIM, MS-AIS or AU4-AIS, AU4-LOP, VC4-SLM or VC4-LOM alarm for the input is not present. As a consequent action, the relevant E1 will be replaced with AIS (and then circuit emulated according to the provisioned settings).

AIS (on channelized E1-in) is an alarm indication signal on one or more E1input signals incoming to the NE on the SDHACC card. The problem is associated with the E1 input signal to the SDHACC card.

AIS (on channelized E1-out) is an alarm indication signal detected by the receive circuit from the radio link on one or more of the channelized E1 lines, indicating an upstream failure.

AIS (on E1/DS1/3-in) is an alarm indication signal on one or more E1, DS1, or DS3 input signals incoming to the NE on the P32E1DS1 or P2E3DS3 cards, or MSS-E / MSS-HE/XE shelf. The problem is associated with the E1, DS1 or DS3 input signal to the P32E1DS1 or P2E3DS3 cards,or MSS-E / MSS-HE/XE shelf.

AIS (on E1/DS1/3-out) is an alarm indication signal detected by the receive circuit from the radio link on one or more of the E1, DS1, or DS3 lines, indicating an upstream failure.

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AIS (on Ele/OptSdh-in) is an alarm indication signal on one or more STM-1/OC-3 input signals incoming to the NE on the SDHACC card. The problem is associated with the input signal to the SDHACC card.

An AIS (on Ele/OptSdh-out) is an alarm indication signal detected by the receive circuit from the radio link on one or more of the STM-1/OC-3 lines, indicating an upstream failure.

See the Wavence Product Information Manual to determine card location and address.

Table: AIS alarms describes AIS alarms.

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Table 14: AIS alarms

Friendly name/entity	Facility/ equipment	Description
AU4/Slot#[3-8]/Port#1	Channelized E1	High Order Path alarm indication signal on a channelized E1 unit.
TU12#[1-3]-[1-7]-[1-3]/Slot#[3-8]/Port#1	Channelized E1	Low Order Path alarm indication signal on a channelized E1 unit.
E1-in/#[1-3]-[1-7]-[1-3]/Slot#[3-8]/Port1	Channelized E1	AIS detected incoming to the SDH-Chan card from the demarcation panel for the E-1 facility.
E1-out/#[1-3]-[1-7]-[1-3]/Slot#[3-8]/Port1	Channelized E1	AIS detected from the CorEvo switch matrix for the E-1 facility, indicating an upstream failure.
DS1-in/Slot#[3-8]/Port#[1-32]	DS1 on P32E1DS1	AIS detected incoming to the PDH card from the demarcation panel for the DS1 or DS3 facility.
DS3-in/Slot#[3-8]/Port#[1-2]	DS3	
E1-in/Slot#[3-8]/Port#[1-32]	E1 on P32E1DS1	
DS1-out/Sl#[3-8]/Port#[1-32]	DS1 on P32E1DS1	AIS detected from the CorEvo switch matrixot for the DS1 or DS3 facility, indicating an upstream failure.
DS3-out/Slot#[3-8]/Port#[1-2]	DS3	
E1-out/Slot#[3-8]/Port#[1-32]	E1 on (P32E1DS1)	
EleSdh-in/slot#[3-8]/Port[1-2]	STM-1/OC-3	AIS detected incoming to the SDHACC card from the demarcation panel for the STM-1/OC-3 facility.
OptSdh-in/slot#[3-8]/Port[1-2]		
EleSdh-out/slot#[3-8]/Port[1-2]	STM-1/OC-3	AIS detected from the CorEvo switch matrix for the STM-1/OC-3 facility, indicating an upstream failure.
OptSdh-out/slot#[3-8]/Port[1-2]		
MS/slot#[3-8]/Port1	STM-1	

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Friendly name/entity	Facility/ equipment	Description
DS1-out/Slot#[1]/Port#[1-16]	DS1 on MSS-E, MSS-HE/XE	AIS detected by the receive circuits, indicating an upstream failure
E1-out/Slot#[1]/Port#[1-16]	E1 on MSS-E, MSS -HE/XE	
DS1-in/Slot#[1]/Port#[1-16]	DS1 on MSS-E, MSS-HE/XE	AIS detected on Transmit Side
E1-in/Slot#[1]/Port#[1-16]	E1 on MSS-E, MSS -HE/XE	
Not Applicable	CorEvo	A MEP receives an AIS frame

12.2 Clear AIS alarms

12.2.1 Clearing AIS alarms

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: AIS alarms for information.

3 Troubleshoot any of the following:

1. AU4-AIS on Channelized E1 facility
2. TU12-AIS on Channelized E1 facility
3. AIS on E1 in on Channelized E1 facility
4. AIS on E1 out on Channelized E1 facility
5. AIS on E1 or DS1/3 in on E1, DS1 or DS3 facility
6. AIS on E1 or DS1/3 out on E1, DS1 or DS3 facility
7. AIS on EleSdh/OptSdh-in on STM-1/OC-3 facility
8. AIS on EleSdh/EleSdh-out on OC-3 facility

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12.2.2 AU4-AIS on Channelized E1 facility

Procedure

1 Troubleshoot the source of input STM-1 signal that is associated with the channelized E1 to the SDHCHAN card.

2 Verify the cabling that is associated with the input STM-1 signal to the SDHCHAN card.

3 Retrieve the alarms using the Craft Terminal's Active Alarms.

4 Did the AU4-AIS alarm clear?

   1. If yes, the procedure is complete.
   2. If no, contact the next level of technical support for assistance.

12.2.3 TU12-AIS on Channelized E1 facility

Procedure

1 Troubleshoot the source of input STM-1 signal that is associated with the channelized E1 to the SDHCHAN card.

2 Verify the cabling that is associated with the input STM-1 signal to the SDHCHAN card.

3 Retrieve the alarms using the Craft Terminal's Active Alarms.

4 Did the TU12-AIS alarm clear?

   1. If yes, the procedure is complete.
   2. If no, contact the next level of technical support for assistance.

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12.2.4 AIS on E1 in on Channelized E1 facility

Procedure

1 Troubleshoot the source of input that is associated with the input signal to the SDHACC card.

2 Verify the cabling that is associated with the input signal to the SDHACC.

3 Retrieve the alarms using the Craft Terminal's Active Alarms.

4 Did the AIS (on channelized E1-in) alarm clear?

   1. If yes, the procedure is complete.
   2. If no, contact the next level of technical support for assistance.

12.2.5 AIS on E1 out on Channelized E1 facility

Procedure

1 Troubleshoot the upstream source of the incoming channelized E1.

2 Retrieve the alarms using the Craft Terminal's Active Alarms.

3 Did the AIS (on channelized E1-out) alarm clear?

   1. If yes, the procedure is complete.
   2. If no, contact the next level of technical support for assistance.

12.2.6 AIS on E1 or DS1/3 in on E1, DS1 or DS3 facility

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Procedure

1 Troubleshoot the source of input that is associated with the input signal to the P32E1DS1 or P2E3DS3 cards, or MSS-E/HE/XE.

2 Verify the cabling that is associated with the input signal to the P32E1DS1 or P2E3DS3 cards.

3 Retrieve the alarms using the Craft Terminal's Active Alarms.

4 Did the AIS (on E1/DS1/3-in) alarm clear?

   1. If yes, the procedure is complete.
   2. If no, contact the next level of technical support for assistance.

12.2.7 AIS on E1 or DS1/3 out on E1, DS1 or DS3 facility

Procedure

1 Troubleshoot the upstream source of the incoming E1, DS1 or DS3.

2 Retrieve the alarms using the Craft Terminal's Active Alarms.

3 Did the AIS (on E1/DS1/3-out) alarm clear?

   1. If yes, the procedure is complete.
   2. If no, contact the next level of technical support for assistance.

12.2.8 AIS on EleSdh/OptSdh-in on STM-1/OC-3 facility

Procedure

1 Troubleshoot the source of input that is associated with the input signal to the SDHACC card.

2 Verify the cabling that is associated with the input signal to the SDHACC card.

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3 Retrieve the alarms using the Craft Terminal's Active Alarms.

4 Did the AIS (on EleSdh/OptSdh-in) alarm clear?

   1. If yes, the procedure is complete.
   2. If no, contact the next level of technical support for assistance.

12.2.9 AIS on EleSdh/EleSdh-out on OC-3 facility

Procedure

1 Troubleshoot the upstream source of the incoming STM-1/OC-3.

2 Retrieve the alarms using the Craft Terminal's Active Alarms.

3 Did the AIS (on EleSdh/EleSdh-out) alarm clear?

   1. If yes, the procedure is complete.
   2. If no, contact the next level of technical support for assistance.

13. Announce Packet Loss

13.1 General

Purppose

This procedure describes how to identify and clear Announce Packet Loss alarm.

Additional information

An Announce Packet Loss alarm indicates that an OC Slave has stopped receiving PTP Announce messages from either the Boundary Clock port working as Master or the OC Master.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Announce Packet Loss alarms describes the equipment that can generate the Announce Packet Loss alarm.

Table 15: Announce Packet Loss alarms

Friendly name/entity	Facility/equipment
PTPClock/PPPPort#[1-16/Peer#[1-16]	CorEvo

13.2 Clear Announce Packet Loss alarms

Procedure

1 Determine the location of either the Boundary Clock or the OC Master and the path from the OC Slave in the alarm to either the Boundary Clock or the OC Master.

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2 Using the Craft Terminal's Active Alarms, retrieve all alarms associated with the path to either the Boundary Clock or the OC Master.

3 Verify that no radio path alarms exist along the path to either the Boundary Clock or the OC Master. If radio path alarms exist, troubleshoot accordingly.

4 Did radio path alarms exist and were they cleared?

1. If yes, go to step 5.
2. If no, go to step 7.

5 Retrieve the alarms using the Craft Terminal's Active Alarms.

6 Did the Announce Packet Loss alarm clear?

1. If yes, the procedure is complete.
2. If no, go to step 7.

7 Using the craft terminal, determine the VLAN associated with the OC Slave. Retrieve all alarms associated with the VLAN along the path from either the Boundary Clock or the OC Master to the OC SLAVE.

8 Verify that no alarms exist along the path to either the Boundary Clock or the OC Master. If alarms exist, troubleshoot accordingly.

9 Did alarms exist and were they cleared?

1. If yes, go to step 10.
2. If no, contact the next level of technical support for assistance

10 Retrieve the alarms using the Craft Terminal's Active Alarms.

11 Did the Announce Packet Loss alarm clear?

1. If yes, the procedure is complete.
2. If no, contact the next level of technical support for assistance.

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14. ATPC high power timeout alarms

14.1 General

Purpose

This procedure describes how to identify and clear ATPC high power timeout alarms on radios.

Additional information

An ATPC high power timeout alarm indicates that the local transmitter was operating continuously at full power for 5-minutes, and the transmitter power was reduced to its minimum power setting.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: ATPC High Power Timeout alarms describes ATPC High Power Timeout alarms.

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Table 16: ATPC High Power Timeout alarms

Friendly name/entity	Facility/ equipment
RadioTxPower/Dir#1.[7-8]/Slot#1/Port#[7-8]/Ch#[0-1]	MPT-HLC connected to CorEvo
RadioTxPower/Dir#[3-8].[5-8]/Slot#[3-8]/Port#[5-8]/Ch#[0-1]	MPT-HLC connected to EASv2
RadioTxPower/Dir#1.[7-8]/Slot#1/Port#[7-8]/Ch#[0-1]	MPT-HLS connected to CorEvo
RadioTxPower/Dir#[3-8].[5-8]/Slot#[3-8]/Port#[5-8]/Ch#[0-1]	MPT-HLS connected to EASv2
N/A	UBT-S UBT-m UBT-T channel A UBT-T channel B UBT-I

14.2 Clear ATPC high power timeout alarms

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarm.

2 Review the output. See Table: ATPC High Power Timeout alarms for information.

3 Using the Craft Terminal, determine whether the local transmit power is at minimum power setting.

4 Is the local transmit power at its minimum power setting?

   1. If yes, go to step 5.
   2. If no, contact the next level of technical support for assistance.

5 Troubleshoot the RF path from the near end transmitter to the far end receiver.

6 Retrieve the alarms using the Craft Terminal's Active Alarms.

7 Did the ATPC high power timeout alarm clear?

1. If yes, the procedure is complete.
2. If no, contact the next level of technical support for assistance.

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15. Bandwidth degrade for protection switch alarms

15.1 General

Purpose

This procedure describes how to identify and clear Bandwidth degrade for protection switch alarms.

Additional information

Bandwidth Degrade for Protection Switch alarms indicates that an ERPS Instance has entered a Protection state.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Bandwidth Degrade for Protection Switch alarms provides information about bandwith degrade for protection switch alarms.

Table 17: Bandwidth Degrade for Protection Switch alarms

Friendly name/entity	Facility/ equipment
ERP#[1-2]#[1-2]	CorEvo

15.2 Clear bandwidth degrade for protection switch alarms

Procedure

1. Using the Craft Terminal's Active Alarms, retrieve all alarms.

2. Review the output. See Table: Bandwidth Degrade for Protection Switch alarms for information.

3. Retrieve the alarms for the radio direction on one node of the Ethernet Ring reporting the Bandwidth degrade using the Craft Terminal's Active Alarms.

4. Determine all of the radio alarms that are declared against the radio direction and troubleshoot accordingly.

5. Retrieve the alarms using the Craft Terminal's Active Alarms.

6. Did the Bandwidth degrade for protection switch alarm clear?

   1. If yes, the procedure is complete.
   2. If no, go to step 7.

7. Have all nodes in the Ethernet Ring been verified?

   1. If yes, contact the next level of technical support for assistance.
   2. If no, repeat steps 3 to 6 for the other nodes in the Ethernet Ring.

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16. Bandwidth Over Flow alarms

16.1 General

Purpose

This procedure describes how to identify and clear Bandwidth Over Flow alarms.

Additional information

A Bandwidth Over Flow alarm indicates that while the MPT-HLC/MPT-HLS was not communicating with the CorEvo, EASv2, TDM2TDM and/or TDM2ETH cross-connections were provisioned with a total bandwidth which exceeds the net bandwidth capacity of the radio profile.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Bandwidth Over Flow alarms provides information about Bandwidth Over Flow alarms.

Table 18: Bandwidth Over Flow alarms

Friendly name/entity	Facility type
Radio/Dir#1.[7-8]/Slot#1/Port#[7-8]/Ch#[0-1]	MPT-HLC or MPT-HLS connected to CorEvo
Radio/Dir#[3-8].[5-8]/Slot#[3-8]/Port#[5-8]/Ch#[0-1]	MPT-HLC or MPT-HLS connected to EASv2

16.2 Clear bandwidth over flow alarms

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Bandwidth Over Flow alarms for information.

3 Verify the cross-connections for the radio with Bandwidth Over Flow alarm and delete TDM2TDM and/or TDM2ETH cross-connections until the total bandwidth is equal to or less than the net bandwidth capacity of the radio profile.

> **Note:**
>
> Total Bandwidth provisioned is validated at the add/drop nodes of the network only. Pass-Through TDM2TDM and/or TDM2ETH flows supported by the NE are not included in the total bandwidth validation and must be included in the network design to determine the net bandwidth capacity of the radio profile required.

4 Retrieve the alarms Using the Craft Terminal's Active Alarms.

5 Did the Bandwidth Over Flow alarm clear?

1. If yes, the procedure is complete.
2. If no, contact the next level of technical support for assistance.

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17. Battery Failure alarms

17.1 General

Purpose

This procedure describes how to identify and clear Battery Failure alarms.

Additional information

A Battery Failure alarm indicates that the A battery feed has failed or is missing, when battery protection is available.

See the Wavence Product Information Manual to determine card location and address, and for information about supported voltage.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: A battery fail alarms provides information about battery fail alarms.

Table 19: A battery fail alarms

Friendly name/entity	Equipment type
r01/subrack#1	MSS-8
r01/subrack#[101-102]	MPT-HLS
r01/s[101-102]/[1-10]	MPT-HLS
MSS-E (Battery A)	MSS-E
MSS-HE/XE (Battery A)	MSS-HE/XE

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17.2 Clear battery failure alarms

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: A battery fail alarms for information.

3 Verify that there is no blown fuse providing power to the A-power feed.

4 Verify the shelf power cable connections between the MSS-8 or MPT-HLS and the power distribution panel.

5 Verify the A-power connections between the BDFB and power distribution panel.

6 Verify that the A-power BDFB breakers are in the ON position.

7 Retrieve the alarms using the Craft Terminal's Active Alarms.

8 Did the Battery Failure clear?

   1. If yes, the procedure is complete.
   2. If no, go to step 9.

9 Measure the voltage of the A-power feed.

10 Is the measured voltage within the supported voltage range (MSS-8: -40.8 VDC to -57.6 VDC, MPT-HLS: +20.4 VDC to +57.6 VDC or -20.4 VDC to -57.6 VDC, MSS-E / HE/XE: VDC to VDC)?

   1. If yes, contact the next level of technical support for assistance.
   2. If no, go to step 11.

11 Troubleshoot the A-power feed voltage problem.

12 Retrieve the alarms using the Craft Terminal's Active Alarms.

13 Did Battery Failure condition clear?

   1. If yes, the procedure is complete.

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2. If no, contact the next level of technical support for assistance.

18. CLA alarms

18.1 Purpose

This procedure describes how to identify and clear CLA alarms.

18.2 General

A CLA alarm indicates that the standby transmitter has/had a silent failure which caused the far end to request a HSB protection switch.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: CLA alarms provides information about CLA alarms.

Table 20: CLA alarms

Friendly name/entity	Facility type
Radio/Dir#1.[7-8]/ Slot#1/Port#[7-8]/Ch#[0-1]	MPT-HLC connected to CorEvo
Radio/Dir#[3-8].[5-8]/Slot#[3-8]/Port#[5-8]/Ch#[0-1]	MPT-HLC connected to EASv2
Radio/Dir#[3-8].[5-8]/Slot#[3-8]/Port#[5-8]/Ch#[0-1]	UBT-I

18.3 Procedure

Before you start

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Note:

When Tx mute operator command is issued on MPT-HLC, CLA algorithm continues to run.

Procedure

1 Using the id3DB-19291-AEAA-TQZZA-ai8rm379lm, retrieve all alarms.

2 Review the output. See Table: CLA alarms for information.

3 Attempt to Clear CLA alarm using the manual switch command?

CAUTION!

Performing a Clear CLA command on MPT-HLS, MPT-HLC or UBT-I will cause a loss of traffic on ALL facilities associated with the MPT-HLS, MPT-HLC or UBT-I declaring CLA alarm IF the silent failure still exist. The Clear CLA command Forces traffic onto the radio channel in alarm and resets the CLA alarm. If the silent failure still exists, ALL traffic will be dropped until the far end requests a HSB protection switch for the link. The traffic outage duration will be from 30 to 60 seconds. Schedule an appropriate maintenance window per local practices and procedures.

1. If yes, go to step 4.
2. If no, go to step 12.

4 Using the Craft Terminal, on the *Protection Schemes* tab for the local MPT-HLS, MPT-HLC or UBT-I, select *HSB Protection* for the channel declaring CLA alarm.

5 On the Commands tab, from the *New Value* drop down, choose Clear CLA.

6 Click on the Apply button. The Switch confirmation window appears.

7 Click on the OK button.

8 Are both receivers at the far end in alarm and traffic affected?

1. If yes, go to step 12.
2. If no, go to step 9.

9 Wait two minutes for the CLA alarm to be declared against the MPT-HLC, MPT-HLS or UBT-I.

10 Retrieve the alarms using Craft Terminal's Active Alarms.

11 Is the CLA alarm declared against the MPT-HLC, MPT-HLS or UBT-I?

1. If yes, go to step 12.
2. If no, the procedure is complete.

12 Replace the MPT-HLC, MPT-HLS or UBT-I CLA declared against. See the Wavence Hardware Installation and Replacement Manual.

13 Perform the Clear CLA command.

⚠ CAUTION!

Performing a Clear CLA command on MPT-HLC, MPT-HLS or UBT-I will cause a loss of traffic on ALL facilities associated with the MPT-HLC, MPT-HLS or UBT-I declaring CLA alarm IF the silent failure still exist. The Clear CLA command Forces traffic onto the radio channel in alarm and resets the CLA alarm. If the silent failure still exists, ALL traffic is dropped until the far end requests a HSB protection switch for the link. The traffic outage duration is 30 to 60 seconds. Schedule an appropriate maintenance window per local practices and procedures.

14 Are both receivers at the far end in alarm and traffic affected?

1. If yes, contact the next level of technical support for assistance.
2. If no, go to step 15.

15 Wait two minutes for the CLA alarm to be declared against the MPT-HLC, MPT-HLS or UBT-I?

1. If yes, contact the next level of technical support for assistance.
2. If no, the procedure is complete.

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19. Clock Class Degraded

19.1 Purpose

This procedure describes how to identify and clear Clock Class Degraded alarm.

19.2 General

A Clock Class Degraded alarm indicates that a boundary Clock has entered into holdover.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Clock Class Degraded alarms describes the equipment that can generate the Clock Class Degraded alarm.

Table 21: Clock Class Degraded alarms

Friendly name/entity	Facility/equipment
PTPClock	CorEvo

19.3 Procedure

Procedure

1 Determine the location of either the Boundary Clock working as Master or the OC Master, and the path from the Boundary Clock in the alarm, to either the Boundary Clock working as Master or the OC Master.

The 1588 Announce Messages shall be correctly received by the BC node in order to run the

BMCA (Best Master Clock Algorithm) and then select which is the PTP Port that has to work as Slave according to the following precedence order of comparison between local and received parameters from other 1588 nodes:

- Clock Class
- Clock Accuracy
- PTP Variance
- Priority 2
- Local Priority
- Clock Identity
- Distance

2 Using the Craft Terminal's Active Alarms, retrieve all alarms associated with the path to the Boundary Clock working as Master or the OC Master.

3 Check for reception of PTP Sync messages. Verify that no Sync Packet Loss alarms exist along the path to the Boundary Clock working as Master or the OC Master. If Sync Packet Loss alarms exist, troubleshoot accordingly.

4 Did Sync Packet Loss alarms exist and were they cleared?

1. If yes, go to step 5.
2. If no, go to step 7.

5 Retrieve the alarms using the Craft Terminal's Active Alarms.

6 Did the Clock Class Degraded alarm clear?

1. If yes, the procedure is complete.
2. If no, go to step 7.

7 Using the craft terminal, verify the CRU is locked to a reference synchronization source and the synchronization source has the right Quality Level needed by the Time Recovery.

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Note:

This step is only applicable if the following are true:

- Frequency Source of PTP Clock set to NE Clock
- Frequency Source of PTP Clock set to 1588 PTP
- 1588 PTP is selected as either the Primary or Secondary synchronization clock source

If the Frequency Source is set to 1588 PTP but 1588 PTP is not used as synchronization clock source, this procedure does not apply.

8 Verify that no Synchronization Signal Fail alarms exist along the path to the boundary clock Master. If alarms exist, troubleshoot accordingly.

9 Did Synchronization Signal Fail alarms exist and were they cleared?

1. If yes, go to step 10.
2. If no, go to step 12.

10 Retrieve the alarms using the Craft Terminal's Active Alarms.

11 Did the Clock Class Degraded alarm clear?

1. If yes, the procedure is complete.
2. If no, go to step 12.

12 Using the craft terminal, check that at least one PTP Port of the Boundary Clock is working as a Slave.

13 Did the Clock Class Degraded alarm clear?

1. If yes, the procedure is complete.
2. If no, contact the next level of technical support for assistance.

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20. Clock failure alarms

20.1 Purpose

This procedure describes how to identify and clear Clock Failure alarms.

20.2 General

A Clock Failure alarm indicates that the MPT-HLC, or MPT-HLS is unable to lock the air Tx or Rx symbol rate to the NE clock.

For UBT, a Clock Failure alarm indicates that UBT-T, UBT-S or UBT-m is unable to lock the user clock (Tx or Rx).

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Clock failure alarm provides information about Clock Failure alarms.

Table 22: Clock failure alarm

Friendly name/entity	Equipment type
Radio/Dir#1.[7-8]/Slot#1/Port#[7-8]/Ch#[0-1]	MPT-HLC or MPT-HLS connected to CorEvo
Radio/Dir#[3-8].[5-8]/Slot#[3-8]/Port#[5-8]/Ch#[0-1]	MPT-HLC or MPT-HLS connected to EASv2
N/A	UBT-S UBT-m UBT-T channel A UBT-T channel B UBT-I

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20.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Clock failure alarm for information.

3 Is the system configured with protection?

1. If yes, go to step 25.
2. If no, go to step 4.

4 Unprotected radio configuration

> **⚠ CAUTION!**
>
> Power Cycling an in-service UBT ODU, MPT-HLC, or MPT-HLS in an unprotected link causes a loss of traffic on ALL facilities that are associated with the UBT ODU, MPT -HLC, or MPT-HLS in alarm. Schedule an appropriate maintenance window per local practices and procedures.

> **⚠ CAUTION!**
>
> Replacing the UBT ODU, MPT-HLC, or MPT-HLS in an unprotected link causes a loss of traffic on ALL facilities that are associated with the UBT ODU, or MPT-HLC, or MPT -HLS in alarm. Schedule an appropriate maintenance window per local practices and procedures.

5 Is the traffic affected?

1. If yes, go to step 8.
2. If no, go to step 6.

6 Verify that it is appropriate to proceed with clear Clock Failure procedure or appropriate maintenance window is scheduled per local practices and procedures.

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7 Is it appropriate to proceed with clear Clock Failure procedure?

1. If yes, go to step 8.
2. If no, contact the next level of technical support for assistance.

8 Which card is in alarm?

1. If UBT ODU, go to step 9.
2. If MPT-HLC, go to step 16 .

9 UBT ODU

⚠ CAUTION!

Power Cycling the UBT ODU in an unprotected facility causes a loss of traffic on ALL facilities that are associated with the UBT ODU in alarm. Schedule an appropriate maintenance window per local practices and procedures.

Power cycle the UBT ODU.

1. If UBT ODU is powered through EASv2, power cycle the UBT ODU. From the UBT Main View, Power Source tab, Settings tab, disable power source. Wait 10 seconds and then reapply power source.If UBT ODU is powered through a power injector card/power injector box, disconnect the DC+Data cable connection to the UBT ODU at the power injector card/box. Wait 10 seconds and reconnect.
2. If UBT ODU is powered with connection to UBT Power Unit/UBT Extended Power Unit, disconnect the power cable connection to the UBT ODU at the UBT Power Unit/UBT Extended Power Unit. Wait 10 seconds and reconnect.
3. If UBT ODU is powered with connection to direct office power, locate the fuse protecting the UBT ODU and remove. Wait 10 seconds and reinstall the fuse.

10 Dialog Failure is declared against the UBT ODU. Wait for Dialog Failure condition to clear before proceeding.

11 Retrieve the alarms using the Craft Terminal's Active Alarms.

12 Did the Clock Failure alarm clear?

1. If yes, the procedure is complete.
2. If no, go to step 13.

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13 Replace the UBT ODU in alarm. See the Wavence Hardware Installation and Replacement Manual to replace UBT ODU.

> **⚠ CAUTION!**
>
> Replacing the UBT ODU in an unprotected link causes a loss of traffic on ALL facilities that are associated with the UBT ODU in alarm. Schedule an appropriate maintenance window per local practices and procedures.

14 Retrieve the alarms using the Craft Terminal's Active Alarms.

15 Did the Clock Failure alarm clear?

1. If yes, the procedure is complete.
2. If no, contact the next level of technical support for assistance.

16 MPT-HLC

> **⚠ CAUTION!**
>
> Power Cycling the MPT-HLC in an unprotected link causes a loss of traffic on ALL facilities that are associated with the MPT-HLC in alarm. Schedule an appropriate maintenance window per local practices and procedures.

17 Position the MPT-HLC Transceiver PA switch to the OFF '0" position.

18 Remove the fuse in the power distribution panel that is suppling battery power to the MPT -HLC transceiver.

19 Install the fuse in power distribution panel that is suppling battery power to the MPT-HLC transceiver.

20 Position the MPT-HLC Transceiver PA switch to the ON '1" position.

21 Wait for the status LED to turn steady green or amber.

22 Did the Clock Failure alarm clear?

1. If yes, the procedure is complete.
2. If no, go to step 23.

23 Replace the local MPT-HLC. See the Wavence Hardware Installation and Replacement Manual to replace the MPT-HLC transceiver.

> **⚠ CAUTION!**
>
> Replacing the MPT-HLC in an unprotected link causes a loss of traffic on ALL facilities that are associated with the MPT-HLC in alarm. Schedule an appropriate maintenance window per local practices and procedures.

24 Did the Clock Failure alarm clear?

1. If yes, the procedure is complete.
2. If no, contact the next level of technical support for assistance.

25 Protected radio configuration

26 Determine the status of UBT ODU/MPT-HLC/MPT-HLS declaring Clock Failure alarm.

1. If Active, go to step 27.
2. If Standby, go to step 28.

27 Is the traffic affected?

1. If yes, go to step 28.
2. If no, contact the next level of technical support for assistance.

28 Is the UBT ODU/MPT-HLC/MPT-HLS declaring Clock Failure alarm the main or spare?

1. If main, go to step 29.
2. If spare, go to step 33.

29 Perform a Forced Rx Radio Switch using the Craft Terminal.

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⚠ CAUTION!

Forced EPS/HSB/RX Radio switching traffic causes a disruption to traffic.

30 Perform a Forced Equipment Switch using the Craft Terminal.

31 Perform a Forced HSB Switch using the Craft Terminal.

32 Go to step 36.

33 Perform a Lockout Rx Radio Switch using the Craft Terminal.

⚠ CAUTION!

Forced EPS/HSB/RX Radio switching traffic causes a disruption to traffic.

34 Perform a Lockout Equipment Switch using the Craft Terminal.

35 Perform a Lockout HSB Switch using the Craft Terminal.

36 Which card is in alarm?

1. If UBT ODU, go to step 37.
2. If MPT-HLC/MPT-HLS, go to step 44.

37 UBT ODU

Power cycle the UBT ODU.

1. If UBT ODU is powered through EASv2, power cycle the UBT ODU. From the UBT Main View, Power Source tab, Settings tab, disable power source. Wait 10 seconds and then reapply power source.
2. If UBT ODU is powered through a Power Injector Card/Power Injector Box/UBT Extended Power Unit, disconnect the DC+Data cable connection to the UBT ODU at the power injector card/box. Wait 10 seconds and reconnect.
3. If UBT ODU is powered with connection to UBT Power Unit/UBT Extended Power Unit, disconnect the power cable connection to the UBT ODU at the UBT Power Unit/UBT Extended Power Unit. Wait 10 seconds and reconnect.

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4. If UBT ODU is powered with connection to direct office power, locate the fuse protecting the UBT ODU and remove. Wait 10 seconds and reinstall the fuse.

38 Dialog Failure is declared against the UBT ODU. Wait for Dialog Failure condition to clear before proceeding.

39 Retrieve the alarms using the Craft Terminal's Active Alarms.

40 Did the Clock Failure alarm clear?

1. If yes, go to step 53.
2. If no, go to step 41.

41 Replace the UBT ODU in alarm. See the Wavence Hardware Installation and Replacement Manual to replace the UBT ODU.

42 Retrieve the alarms using the Craft Terminal's Active Alarms.

43 Did the Clock Failure alarm clear?

1. If yes, go to step 53.
2. If no, contact the next level of technical support for assistance.

44 MPT-HLC/MPT-HLS

Position the MPT-HLC/MPT-HLS Transceiver PA switch to the OFF '0" position.

45 Remove the fuse in the power distribution panel that is suppling battery power to the MPT -HLC/MPT-HLS transceiver.

46 Install the fuse in the power distribution panel that is suppling battery power to the MPT -HLC/MPT-HLS transceiver.

47 Position the MPT-HLC/MPT-HLS Transceiver PA switch to the ON '1" position.

48 Wait for the status LED to turn steady amber.

49 Did the Clock Failure alarm clear?

1. If yes, go to step 54.

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2. If no, go to step 50.

50 Replace the MPT-HLC/MPT-HLS. See the Wavence Hardware Installation and Replacement Manual to replace MPT-HLC/MPT-HLS Transceiver.

51 Wait for the status LED to turn steady amber.

52 Did the Clock Failure alarm clear?

1. If yes, go to step 53.
2. If no, contact the next level of technical support for assistance.

53 Release the Forced/Lockout Rx Radio Switch using the Craft Terminal.

⚠ CAUTION!

Releasing Forced/Lockout EPS/HSB/RX Radio switch causes a disruption to traffic.

54 Release the Forced/Lockout Equipment Switch using the Craft Terminal.

55 Release the Forced/Lockout HSB Switch using the Craft Terminal.

The procedure is complete.

21. Combiner loss of alignment alarms

21.1 Purpose

This procedure describes how to identify and clear Combiner Loss Of Alignment alarms.

21.2 General

A Combiner Loss Of Alignment alarm indicates that a delay/loss of alignment between the main and diversity signal is detected by the combiner in combiner receiver configurations.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Combiner loss of alignment alarm provides information about Combiner Loss of Alignment alarms.

Table 23: Combiner loss of alignment alarm

Friendly name/entity	Equipment type
r01/subrack#1[7-8]	MPT-HLC connected to CorEvo
r01/subrack#[3-8][5-8]	MPT-HLC connected to EASv2
r01s[101-102]/board#[1-10]	MPT-HLS

21.3 Procedure

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Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Combiner loss of alignment alarm for information.

3 Retrieve the alarms using the Craft Terminal's Active Alarms, verify that no other radio path alarms (such as loss of signal) exist, and troubleshoot accordingly.

4 Did other radio path alarms exist and were they cleared?

   1. If yes, go to step 5.
   2. If no, go to step 7.

5 Retrieve the alarms using the Craft Terminal's Active Alarms.

6 Did the Combiner Loss of Alignment alarm clear?

   1. If yes, the procedure is complete.
   2. If no, go to step 7.

7 Using the Craft Terminal, check the RSL values for the main and diversity receiver for the MPT-HLC/MPT-HLS declaring combiner loss of alignment alarm and troubleshoot accordingly.

   If one of the RX is extremely low (below -90dBm) while the other is good, the loss of alignment can occur. In this case there is very likely some installation issue.

8 Retrieve the alarms using the Craft Terminal's Active Alarms.

9 Did the Combiner Loss of Alignment alarm clear?

   1. If yes, the procedure is complete.
   2. If no, go to step 10.

10 Perform an MPT restart on the NE: Supervision/Restart MPT.

11 Wait for the status LED to turn steady amber or green.

12 Retrieve the alarms using the Craft Terminal's Active Alarms.

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13 Did the Combiner Loss of Alignment alarm clear?

1. If yes, the procedure is complete.
2. If no, go to step step 14. Perform the NE restart on the NE: Supervision/Restart NE. Communication with the NE is lost during the NE restart and you are logged out of the system.

14 Perform the NE restart on the NE: Supervision/Restart NE. Communication with the NE is lost during the NE restart and you are logged out of the system.

> **⚠ CAUTION!**
>
> Performing the Restart NE command may cause a disruption to traffic.

15 Log into the NE.

16 Retrieve the alarms using the Craft Terminal's Active Alarms.

17 Did the Combiner Loss of Alignment alarm clear?

1. If yes, the procedure is complete.
2. If no, go to step 18.

18 Is the radio direction in a protected configuration?

1. If yes, go to step 19.
2. If no, go to step 28.

19 Verify that the MPT-HLC or MPT-HLS in alarm status is *Standby*.

20 Is the MPT-HLC or MPT-HLS in alarm main or spare?

1. If main, go to step 21.
2. If spare, go to step 25.

21 Perform a Forced EPS Switch using the Craft Terminal.

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⚠ CAUTION!

Forced/Lockout switching traffic causes a disruption to traffic.

22 Perform a Forced HSB Switch using the Craft Terminal.

23 Perform a Forced Rx Radio Switch using the Craft Terminal.

24 Go to step 28.

25 Perform a Lockout EPS Switch using the Craft Terminal.

⚠ CAUTION!

Forced/Lockout switching traffic causes a disruption to traffic.

26 Perform a Lockout HSB Switch using the Craft Terminal.

27 Perform a Lockout Rx Radio Switch using the Craft Terminal.

28 Inspect the RF cabling to the main and diversity receiver.

Note:

This step is applicable only to MPT-HLC.

⚠ CAUTION!

The following steps on an unprotected radio direction will cause a loss of traffic associated with the radio direction in alarm. Schedule an appropriate maintenance window per local practices and procedures.

Correct ALL issues identified.

29 Retrieve the alarms using the Craft Terminal's Active Alarms.

30 Did the Combiner Loss of Alignment alarm clear?

1. If yes, go to step 46.
2. If no, go to step 31.

31 Disconnect the data cable connection at the MPT-HLC/MPT-HLS to MSS. Then reconnect the cable.

32 Wait for the status LED to turn steady amber or green.

33 Retrieve the alarms using the Craft Terminal's Active Alarms.

34 Did the Combiner Loss of Alignment alarm clear?

1. If yes, go to step 46.
2. If no, go to step 35.

35 Position the MPT-HLC/MPT-HLS Transceiver PA switch to the OFF '0" position.

36 Remove the fuse in the power distribution panel that supplies battery power to the MPT -HLC/MPT-HLS transceiver.

37 Install the fuse in the power distribution panel that supplies battery power to the MPT -HLC/MPT-HLS transceiver.

38 Position the MPT-HLC/MPT-HLS Transceiver PA switch to the ON '1" position.

39 Wait for the status LED to turn steady amber or green.

40 Retrieve the alarms using the Craft Terminal's Active Alarms.

41 Did the Combiner Loss of Alignment alarm clear?

1. If yes, go to step 46.
2. If no, go to step 42.

42 Replace the MPT-HLC or MPT-HLS in alarm. See the Wavence Hardware Installation and Replacement Manual to replace the unit.

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43 Wait for the status LED to turn steady amber or green.

44 Retrieve the alarms using the Craft Terminal's Active Alarms.

45 Did the Combiner Loss of Alignment alarm clear?

1. If yes, go to step 46.
2. If no, contact the next level of technical support for assistance.

46 Is the radio direction in a protected configuration?

1. If yes, go to step 47.
2. If no, the procedure is complete.

47 Release the Forced/Lockout EPS Switch using the Craft Terminal.

48 Release the Forced/Lockout HSB Switch using the Craft Terminal.

49 Release the Forced/Lockout Rx Radio Switch using the Craft Terminal.

The procedure is complete.

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22. Critical Link Event alarms

22.1 Purpose

This procedure describes how to identify and clear critical link event alarms.

22.2 General

A critical link event alarm indicates the Ethernet port"s link partner has declared a link fault condition.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Critical link event alarm provides information about critical link event alarms.

Table 24: Critical link event alarm

Friendly name/entity	Equipment type
ETHLinkOAM/Slot#1/Ports[1-8]	Ethernet port on CorEvo
ETHLinkOAM/Slot#[3-8]/Ports[1-8]	Ethernet port on EASv2

22.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

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2 Review the output. See Table: Critical link event alarm for information.

3 Retrieve the alarms using the Craft Terminal's Active Alarms at the NE declaring critical link event.

4 Are there additional alarms declared against the Ethernet port in alarm?

1. If yes, go to step 5.
2. If no, go to step 11.

5 Determine all the alarms declared against the Ethernet port and troubleshoot accordingly.

6 Retrieve the alarms using the Craft Terminal's Active Alarms.

7 Did the critical link event alarm clear?

1. If yes, the procedure is complete.
2. If no, go to step 8.

8 Verify the operational status of the link partner and troubleshoot accordingly.

9 Retrieve the alarms using the Craft Terminal's Active Alarms.

10 Did the critical link event alarm clear?

1. If yes, the procedure is complete.
2. If no, go to step 11.

11 Verify the cabling between the link partner and the CorEvo/EASv2 card.

12 Retrieve the alarms using the Craft Terminal's Active Alarms.

13 Did the critical link event alarm clear?

1. If yes, the procedure is complete.
2. If no, go to step 14.

14 Is the Ethernet port configured with an SFP?

1. If yes, go to step 15.

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2. If no, contact the next level of technical support for assistance.

15 Replace the SFP at the CorEvo/EASv2 card.

16 Retrieve the alarms using the Craft Terminal's Active Alarms.

17 Did the critical link event alarm clear?

1. If yes, the procedure is complete.
2. If no, go to step 18.

18 Replace the SFP at the link partner.

19 Retrieve the alarms using the Craft Terminal's Active Alarms.

20 Did the critical link event alarm clear?

1. If yes, the procedure is complete.
2. If no, contact the next level of technical support for assistance.

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23. DC & PfoE Cables present

23.1 General

This procedure describes how to identify and clear DC & PfoE Cables present alarm.

This alarm indicates the radio is being powered by both PfoE and DC power. The alarm will be reported as minor because the radio will default to the DC power.

23.2 Procedure

Procedure

1 Check PfoE and DC provisioning/configuration.

2 Disable and/or unplug one of them.

Disabling/unplugging the DC power will cause the radio to power cycle since the radio was defaulting to the DC power. This will affect the traffic.

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24. Degrade alarms on sync source

24.1 Purpose

This procedure describes how to identify and clear degrade alarms on synchronization sources. For information about degrade alarms on Radio or Ethernet LAG, see Degrade alarms on LAG.

24.2 General

A Degrade alarm indicates that the frequency of the sync source signal is degraded.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Degrade alarms provides information about degrade alarms.

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Table 25: Degrade alarms

Friendly name/entity	Equipment type
Sync-in/Port#[7-8]	CorEvo sync SFP Port
SyncSource/ETH/slot#1/Port#[1-8]	CorEvo User Ethernet Port
N/A	EAC, EAC1G2G5 and EAC10G User Ethernet Port
SyncSource/ETH/slot#[3-8]/Port#[1-8]	EASv2 User Ethernet Port
Radio/Dir#[3-8].[5-8]/Slot#[3-8]/Port#[5-8]/Ch#[0-1]	MPT-HLC or MPT-HLS connected to EASv2
Radio/Dir#1.[7-8]/Slot#1/Port#[7-8]/Ch#1	MPT-HLC or MPT-HLS connected to CorEvo
RadioLAG#[1-14]	MPT-HLC or MPT-HLS connected to EAS
DS1-in/Slot#[3-8]/Port#[1-32]	P32E1DS1
E1-in/Slot#[3-8]/Port#[1-32]	P32E1DS1
EleSdh-in/slot#[3-8]/Port[1-2]	STM-1/OC-3
OptSdh-in/slot#[3-8]/Port[1-2]	
MS/slot#[3-8]/Port1	STM-1
N/A	UBT-S UBT-m UBT-T UBT-I
Sync-in/Port#[1-2]	MSS-E, MSS-HE/XE
Ethernet/Slot#[1]/Port#[1-5]	MSS-E User Ethernet Ports
Ethernet/Slot#[1]/Port#[1-8]	MSS-HE/XE User Ethernet Ports
DS1-in/Slot#[1]/Port#[1-16]	DS1 Tributary on MSS-E, MSS-HE/XE
E1-in/Slot#[1]/Port#[1-16]	E1 Tributary on MSS-E, MSS-HE/XE

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24.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Degrade alarms for information.

3 Degrade What is the sync source?

1. If Degrade - Sync-In, go to step 4.
2. If Degrade - Radio port, go to step 9.
3. If Degrade - DS1, E1, OC-3, or STM-1 port,, go to step 14.

4 Degrade - Sync-In

Verify that the Sync reference source (GPS or other sync source) is functioning correctly.

5 Verify the Sync-In cabling and connections to the sync-in port on the CorEvo card or MSS-E / HE /XE unit (main and spare, if equipped).

6 Verify the Sync cabling and connections to the sync-out port on sync source.

7 Retrieve the alarms using the Craft Terminal's Active Alarms.

8 Did the Degrade Defect alarm clear?

1. If yes, the procedure is complete.
2. If no, go to step 23.

9 Degrade - Radio port

Retrieve the alarms using the Craft Terminal's Active Alarms.

10 Verify that no other radio path alarms, such as Demodulator Fail. If radio path alarms exist, troubleshoot accordingly.

11 Did the other radio path alarms exist and were they cleared?

1. If yes, go to step 12.

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2. If no, go to step 23.

12 Retrieve the alarms using the Craft Terminal's Active Alarms.

13 Did the Degrade Defect alarm clear?

1. If yes, the procedure is complete.
2. If no, go to step 23.

14 Degrade - DS1, E1, OC-3, or STM-1 port

Retrieve the alarms using the Craft Terminal's Active Alarms.

15 Verify that no LOS alarm exists on the sync reference DS1, E1, OC-3, or STM-1 port. If a LOS alarm exists, troubleshoot accordingly.

16 Did an LOS alarm exist on the sync reference DS1, E1, OC-3, or STM-1 port and was it cleared?

1. If yes, go to step 17.
2. If no, go to step 19.

17 Retrieve the alarms using the Craft Terminal's Active Alarms.

18 Did the Degrade Defect alarm clear?

1. If yes, the procedure is complete.
2. If no, go to step 19.

19 Verify that the sync reference DS1, E1, OC-3, or STM-1 port frequency is correct. If the sync reference frequency is not correct, troubleshoot accordingly.

20 Was a sync reference DS1, E1, OC-3, or STM-1 port frequency problem identified and was it corrected?

If yes, go to step 21.

1. If no, go to step 23.

21 Retrieve the alarms using the Craft Terminal's Active Alarms.

22 Did the Degrade Defect alarm clear?

1. If yes, the procedure is complete.
2. If no, go to step 23.

23 Verify that the sync source provisioning at both the sync source and the alarming card is correct.

24 Retrieve the alarms using the Craft Terminal's Active Alarms.

25 Did the Degrade Defect alarm clear?

1. If yes, the procedure is complete.
2. If no, go to step 26.

26 Replace the card/SFP the Degrade Defect alarm is declared against. See the Wavence Hardware Installation and Replacement Manual.

27 Retrieve the alarms using the Craft Terminal's Active Alarms.

28 Did the Degrade Defect alarm clear?

1. If yes, the procedure is complete.
2. If no, contact the next level of technical support for assistance.

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25. Degrade alarms on LAG

25.1 Purpose

This procedure describes how to identify and clear degrade alarms on Radio or Ethernet LAG. For information about degrade alarms on synchronization sources, see Degrade alarms on sync source.

25.2 General

A Degrade alarm indicates that a member of an Ethernet or Radio LAG group has no activity. A Degrade is declared when the number of active ports in the LAG are greater than zero and less than the LAG size.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Degrade alarms provides information about degrade alarms.

Table 26: Degrade alarms

Friendly name/entity	Equipment type
EthLAG#[1-14]	CorEvo
RadioLAG#[1-14]	MPT-HLC or MPT-HLS
N/A	UBT-T, UBT-S, UBT-m, UBT-I

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25.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve ALL alarms.

2 Review the output. See Table: Degrade alarms for information.

3 Identify the alarms that are declared against members of the LAG group. If alarms exists, troubleshoot accordingly.

4 Did the alarms exist on the LAG group members and were they cleared?

   1. If yes, go to step 5.
   2. If no, contact the next level of technical support for assistance.

5 Retrieve the alarms using the Craft Terminal's Active Alarms.

6 Did the Degrade alarm clear?

   1. If yes, the procedure is complete.
   2. If no, contact the next level of technical support for assistance.

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26. Delay Response Loss

26.1 Purpose

This procedure describes how to identify and clear Delay Response Loss alarm.

26.2 General

A Delay Response Loss alarm indicates that either a Boundary Clock port working as Slave or an OC Slave has stopped receiving PTP Delay Response messages from either the Boundary Clock port working as Master or the OC Master.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Delay Response Loss alarms describes the equipment that can generate the Delay Response Loss alarm.

Table 27: Delay Response Loss alarms

Friendly name/entity	Facility/equipment
PTPClock/PTPPort#[1-16]/Peer#[1-16]	CorEvo

26.3 Procedure

Procedure

1 Determine the location of either the Boundary Clock or OC Master, and the path from either the Boundary Clock or OC Slave in the alarm to either the Boundary Clock or OC Master.

2 Using the Craft Terminal's Active Alarms, retrieve all alarms associated with the path to either the Boundary Clock or OC Master.

3 Verify that no radio path alarms exist along the path to either the Boundary Clock or OC Master. If radio path alarms exist, troubleshoot accordingly.

4 Did radio path alarms exist and were they cleared?

1. If yes, go to step 5.
2. If no, go to step 7.

5 Retrieve the alarms using the Craft Terminal's Active Alarms.

6 Did the Delay Response Loss alarm clear?

1. If yes, the procedure is complete.
2. If no, go to step 7.

7 Using the craft terminal, determine the VLAN associated with either the Boundary Clock or OC Slave. Retrieve all alarms associated with the VLAN along the path from either the Boundary Clock or OC Master, to either the Boundary Clock or OC Slave.

8 Verify that no alarms exist along the path to either the Boundary Clock or OC Master. If alarms exist, troubleshoot accordingly.

9 Did alarms exist and were they cleared?

1. If yes, go to step 10.
2. If no, go to step 12.

10 Retrieve the alarms using the Craft Terminal's Active Alarms.

11 Did the Delay Response Loss alarm clear?

1. If yes, the procedure is complete.
2. If no, go to step 12.

12 Using the craft terminal, retrieve OC Master and Slave provisioning. Using site documentation determine the correct provisioning. If provisioning errors exist, correct accordingly.

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13 Did provisioning errors exist and were they corrected?

1. If yes, go to step 14.
2. If no, contact the next level of technical support for assistance.

14 Did the Delay Response Loss alarm clear?

1. If yes, the procedure is complete.
2. If no, contact the next level of technical support for assistance.

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27. DemXPIC LOS

27.1 Purpose

This procedure describes how to identify and clear the DemXPIC LOS alarm.

27.2 General

The DemXPIC LOS condition indicates that a bad or missing cable or cable connection at the XPIC connector on the MPT-HLC, MPT-HLS, UBT-S, UBT-m, and UBT-T (not in W18).

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: DemXPIC LOS alarms provides information about DemXPIC LOS alarms.

Table 28: DemXPIC LOS alarms

Friendly name/entity	Equipment type
r01/subrack#1[7-8]	MPT-HLC connected to CorEvo
r01/subrack#[3-8][5-8]	MPT-HLC connected to EASv2
r01s[101-102]/board#[1-10]	MPT-HLS
N/A	UBT-S, UBT-m, and UBT-T (not in W18), UBT-I

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27.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: DemXPIC LOS alarms for information.

3 Inspect the XPIC cable is properly terminated, securely connected, and the cable is not damaged.

4 Is the XPIC cable not properly terminated, connected, or the cable is damaged?

   1. If yes, go to step step 5. Repair or replace the XPIC cable.
   2. If no, go to step step 8. Reseat the XPIC cable on the horizontal and vertical MPT-HLC/MPT-HLS.

5 Repair or replace the XPIC cable.

6 Retrieve the alarms using the Craft Terminal's Active Alarms.

7 Did the DemXPIC LOS alarm clear?

   1. If yes, the procedure is complete.
   2. If no, go to step step 8. Reseat the XPIC cable on the horizontal and vertical MPT-HLC/MPT-HLS.

8 Reseat the XPIC cable on the horizontal and vertical MPT-HLC/MPT-HLS.

9 Retrieve the alarms using the Craft Terminal's Active Alarms.

10 Did the DemXPIC LOS alarm clear?

   1. If yes, the procedure is complete.
   2. If no, go to step step 11. Replace the XPIC cable between the horizontal and vertical MPT-HLC/MPT-HLS.

11 Replace the XPIC cable between the horizontal and vertical MPT-HLC/MPT-HLS.

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12 Retrieve the alarms using the Craft Terminal's Active Alarms.

13 Did the DemXPIC LOS alarm clear?

1. If yes, the procedure is complete.
2. If no, go to step step 14. Replace the horizontal MPT-HLC/MPT-HLS. See the Wavence Hardware Installation and Replacement Manual to replace the MPT-HLC/MPT-HLS.

14 Replace the horizontal MPT-HLC/MPT-HLS. See the Wavence Hardware Installation and Replacement Manual to replace the MPT-HLC/MPT-HLS.

15 Retrieve the alarms using the Craft Terminal's Active Alarms.

16 Did the DemXPIC LOS alarm clear?

1. If yes, the procedure is complete.
2. If no, go to step .step 17. Replace the vertical MPT-HLC/MPT-HLS. See Wavence Hardware Installation and Replacement Manual to replace the MPT-HLC/MPT-HLS

17 Replace the vertical MPT-HLC/MPT-HLS. See Wavence Hardware Installation and Replacement Manual to replace the MPT-HLC/MPT-HLS.

18 Retrieve the alarms using the Craft Terminal's Active Alarms.

19 Did the DemXPIC LOS alarm clear?

1. If yes, the procedure is complete.
2. If no, contact the next level of technical support for assistance.

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28. Demodulator fail alarms

28.1 Purpose

This procedure describes how to identify and clear demodulator fail alarms.

28.2 General

A Demodulator Fail alarm indicates that a loss of receive signal, bad receive signal, or internal receive circuit failure.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Demodulator fail alarms provides information about demodulator fail alarms.

Table 29: Demodulator fail alarms

Friendly name/entity	Facility type
Radio/Dir#1.[7-8]/ Slot#1/Port#[7-8]/Ch#[0-1]	MPT-HLC or MPT-HLS connected to CorEvo
Radio/Dir#[3-8].[5-8]/ Slot#[3-8]/Port#[5-8]/Ch#[0-1]	MPT-HLC or MPT-HLS connected to EASV2
N/A	UBT-S UBT-m UBT-T channel A UBT-T channel B

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28.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Demodulator fail alarms for information.

3 Verify that there are no weather-related conditions.

4 Do weather-related conditions exist?

   1. If yes, wait for the weather to clear and recheck alarms. The procedure is complete.
   2. If no, go to step 5.

5 Verify that the RF path is clear.

6 Is the RF path clear?

   1. If yes, go to step 7.
   2. If no, contact the next level of technical support for assistance.

7 Verify that the RSL is above RX threshold.

8 Is the RSL above the RX threshold?

   1. If yes, go to step 12.
   2. If no, go to step 9.

9 Check the upstream transmitter output power and troubleshoot accordingly.

10 Retrieve the alarms using the Craft Terminal's Active Alarms.

11 Did the Demodulator Fail alarm clear?

   1. If yes, the procedure is complete.
   2. If no, go to step 12.

12 Replace the MPT-HL, or MPT-HLC Demodulator Fail declared against. See the Wavence

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Hardware Installation and Replacement Manual.

13 Retrieve the alarms using the Craft Terminal's Active Alarms.

14 Did the Demodulator Fail alarm clear?

1. If yes, the procedure is complete.
2. If no, contact the next level of technical support for assistance.

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29. Dialog failure alarms

29.1 Purpose

This procedure describes how to identify and clear Dialog Failure alarms.

29.2 General

A Dialog Failure alarm manifests itself in two unique time periods.

The first time period a Dialog Failure condition may be declared is during initial turn-up of a MPT-HLS/HLC/UBT radio. It is an indication that communications between the MPT-HLS/HLC/UBT and the MSS shelf have been established, but one or more radio parameters has not been provisioned by the user. During initial turn-up, treat Dialog Failure as an indication one or more radio parameters require provisioning.

The second time period Dialog Failure condition may be declared is after communications with the radio has been established and the radio has been provisioned by the user. It indicates that a temporary loss of communication between the MSS shelf and MPT-HLS/HLC/UBT-T/UBT-S/UBT-m exist. A Dialog Failure condition is declared until communications are reestablished or a time-out period expires. After the time-out period expires, a Dialog Failure condition is cleared and an Internal Communication Problem condition is declared.

> **Note:**
> In case of UBT connected to EASv2 and/or UBT and MPT-HLS/HLC connected to EAC, Dialogue Failure alarm is not reported.

The Dialog Failure time-out durations varies based on the system configuration. The time-out durations are:

- 1+0: 20 minutes
- 1+1 HSB: 5 minutes

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- 1+1 FD: 20 minutes

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Dialog failure alarms provides information about dialog failure alarms.

Table 30: Dialog failure alarms

Friendly name/entity	Equipment type
r01/subrack#1[7-8]	MPT-HLC connected to CorEvo
r01s[101-102]/board#[1-10]	MPT-HLS
N/A	UBT-S UBT-m UBT-T

29.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Dialog failure alarms for information.

3 Is the traffic affected?

   1. If yes, go to step step 11. Verify radio port provisioning against site documentation.
   2. If no, go to step step 4. Has the appropriate time-out period expired?.

4 Has the appropriate time-out period expired?

   1. If yes, go to step step 5. Retrieve the alarms using the Craft Terminal's Active Alarms.
   2. If no, wait for the appropriate time-out period to expire before proceeding to step step 5. Retrieve the alarms using the Craft Terminal's Active Alarms.

5 Retrieve the alarms using the Craft Terminal's Active Alarms.

6 Did the Dialog Failure alarm clear?

1. If yes, go to step step 7. Is an Internal Communication Problem alarm declared?.
2. If no, go to step .step 8. Wait for the status LED to turn steady green or amber

7 Is an Internal Communication Problem alarm declared?

1. If yes, see Internal Communication Problem alarms.
2. If no, contact the next level of technical support for assistance.

8 Wait for the status LED to turn steady green or amber.

9 Retrieve the alarms using the Craft Terminal's Active Alarms.

10 Did the Dialog Failure alarm clear?

1. If yes, the procedure is complete.
2. If no, contact the next level of technical support for assistance.

11 Verify radio port provisioning against site documentation.

12 Is the radio port provisioned per site documentation?

1. If yes, go to step step 13. Verify radio port fiber cabling per site documentation.
2. If no, go to step step 18. Reprovision the radio port per site documentation.

13 Verify radio port fiber cabling per site documentation.

14 Is the radio port fiber cabling correct and connected to the correct MPT type?

1. If yes, go to step step 21. Retrieve the alarms using the Craft Terminal's Active Alarms.
2. If no, go to step step 15. Correct MPT fiber cable connections per site documentation.

15 Correct MPT fiber cable connections per site documentation.

16 Retrieve the alarms using the Craft Terminal's Active Alarms.

17 Did the Dialog Failure alarm clear?

1. If yes, the procedure is complete.
2. If no, go to step step 21. Retrieve the alarms using the Craft Terminal's Active Alarms.

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18 Reprovision the radio port per site documentation.

19 Retrieve the alarms using the Craft Terminal's Active Alarms.

20 Did the Dialog Failure alarm clear?

1. If yes, the procedure is complete.
2. If no, go to step step 13. Verify radio port fiber cabling per site documentation.

21 Retrieve the alarms using the Craft Terminal's Active Alarms.

22 Is a Replaceable Unit Problem alarm declared against the MPT-HLC/MPT-HLS in the protection-pair?

1. If yes, see Alarm and abnormal condition clearing procedures to clear the Replaceable Unit Problem.
2. If no, go to step step 23. Verify that power is properly supplied to the MPT-HLC/MPT-HLS in alarm. Correct if required.

23 Verify that power is properly supplied to the MPT-HLC/MPT-HLS in alarm. Correct if required.

24 Was a power problem corrected?

1. If yes, go to step step 25. Retrieve the alarms using the Craft Terminal's Active Alarms.
2. If no, go to step step 27. Is the MPT-HLC/MPT-HLS in a protected system?.

25 Retrieve the alarms using the Craft Terminal's Active Alarms.

26 Did the Dialog Failure alarm clear?

1. If yes, the procedure is complete.
2. If no, go to step step 27. Is the MPT-HLC/MPT-HLS in a protected system?.

27 Is the MPT-HLC/MPT-HLS in a protected system?

1. If yes, go to step step 28. Is the MPT-HLC/MPT-HLS in alarm the main or spare?.
2. If no, go to step step 36. Reseat the RJ45/SFP at the CorEvo/EASv2 that is associated with the MPT-HLC/MPT-HLS in alarm.

28 Is the MPT-HLC/MPT-HLS in alarm the main or spare?

⚠ CAUTION!

Forced EPS and HSB switching traffic causes a disruption to traffic.

1. If main, go to step step 29. Perform a Forced EPS Switch using the Craft Terminal.
2. If spare, go to step step 33. Perform a Lockout EPS Switch using the Craft Terminal.

29 Perform a Forced EPS Switch using the Craft Terminal.

30 Perform a Forced HSB Switch using the Craft Terminal.

31 Perform a Forced Rx Radio Switch using the Craft Terminal.

32 Go to step step 36. Reseat the RJ45/SFP at the CorEvo/EASv2 that is associated with the MPT-HLC/MPT-HLS in alarm.

33 Perform a Lockout EPS Switch using the Craft Terminal.

34 Perform a Lockout HSB Switch using the Craft Terminal.

35 Perform a Lockout Rx Radio Switch using the Craft Terminal.

36 Reseat the RJ45/SFP at the CorEvo/EASv2 that is associated with the MPT-HLC/MPT-HLS in alarm.

⚠ CAUTION!

Unseating and reseating the communication link between the MPT-HLC and the CorEvo in an unprotected facility causes a loss of traffic on ALL facilities that are associated with the MPT-HLC. Schedule an appropriate maintenance window per local practices and procedures.

37 Reseat the SFP at the MPT-HLC/MPT-HLS in alarm.

38 Retrieve the alarms using the Craft Terminal's Active Alarms.

39 Did the Dialog Failure alarm clear?

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1. If yes, go to step step 51. Is the MPT-HLC/MPT-HLS in a protected system?.
2. If no, go to stepstep 40. Position the MPT-HLC/MPT-HLS PA switch to the OFF ‘0" position
.

40 Position the MPT-HLC/MPT-HLS PA switch to the OFF ‘0" position.

> ⚠ **CAUTION!**
>
> Reseating an MPT-HLC/MPT-HLS in an unprotected facility causes a loss of traffic on ALL facilities that are associated with the MPT-HLS/MPT-HLC reseat. Schedule an appropriate maintenance window per local practices and procedures.

41 Remove the fuse in the power distribution panel that is suppling battery power to the MPT -HLC/MPT-HLS.

42 Install the fuse in the power distribution panel that is suppling battery power to the MPT -HLC/MPT-HLS.

43 Position the MPT-HLC/MPT-HLS PA switch to the ON ‘1" position.

44 Wait for the status LED to turn steady green or amber.

45 Retrieve the alarms using the Craft Terminal’s Active Alarms.

46 Did the Dialog Failure alarm clear?

1. If yes, go to step step 51. Is the MPT-HLC/MPT-HLS in a protected system?.
2. If no, go to step step 47. Replace the card in alarm. See Wavence Hardware Installation and Replacement Manual.

47 Replace the card in alarm. See Wavence Hardware Installation and Replacement Manual.

> ⚠ **CAUTION!**
>
> Replacing an MPT-HLC/MPT-HLS in an unprotected facility causes a loss of traffic on ALL facilities that are associated with the MPT-HLC/MPT-HLS being replaced. Schedule an appropriate maintenance window per local practices and procedures.

48 Wait for the status LED to turn steady green or amber.

49 Retrieve the alarms using the Craft Terminal's Active Alarms.

50 Did the Dialog Failure alarm clear?

1. If yes, go to step step 51. Is the MPT-HLC/MPT-HLS in a protected system?.
2. If no, contact the next level of technical support for assistance.

51 Is the MPT-HLC/MPT-HLS in a protected system?

1. If yes, go to step step 52. Release the Forced/Lockout EPS Switch using the Craft Terminal.
2. If no, the procedure is complete.

52 Release the Forced/Lockout EPS Switch using the Craft Terminal.

53 Release the Forced/Lockout HSB Switch using the Craft Terminal.

54 Release the Forced/Lockout Rx Radio Switch using the Craft Terminal. The procedure is complete.

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30. Duplicate Address Detected alarms

This procedure describes how to identify and clear Duplicate Address Detected alarms.

30.1 General

Duplicate address detected alarm is raised when a duplicate IPv6 address is detected on a TMN LAN interface. The following TMN LAN interfaces are checked for a duplicate address:

- IPv6 Global Unicast address assigned to:
  - TMN Local Ethernet interface
  - TMN User Ethernet Port 4 interface
  - TMN In-Band port #1/2 interface
- TMN-RF Point-to-Point interface

Note:

If access to the NE with the incorrect IPv6 address is through the TMN port in alarm, communication with the NE will be lost when the address is changed. The TMN address must be changed locally or a different TMN interface must be established to communicate with the NE.

Note:

When a different TMN interface is not available o communicate with the NE with the incorrect IPv6 address, it is required to correct the IPv6 address on-site.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Duplicate address detected alarms provides information about duplicate address detected alarms.

Table 31: Duplicate address detected alarms

Friendly name/entity	Equipment type	See
TMNEthernet/MngtPort	CorEvo/TMN MGMT Port	Duplicate address detected - TMN Ethernet MGMT Port
TMNEthernet/Port#4	CorEvo/TMN Port#4	Duplicate address detected - Port #4 TMN Ethernet
TMNInBand#[1-2]	CorEvo/TMN In-Band Port [1-2]	Duplicate address detected - TMN In-Band Port #1/2
PPP-IP/RF#[3-8][5-8]	MPT-HLC or MPT-HLS connected to EASv2	Duplicate address detected - PPP IP/RF Port
PPP-IP/RF#1[7-8]	MPT-HLC or MPT-HLS connected to CorEvo	
PPP-IP/RF/LAG#[1-14]	MPT-HLC or MPT-HLS	
N/A	VPRN Interface	
N/A	TWAMP Light Reflector	

30.2 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Duplicate address detected alarms for information about Duplicate address detected alarms.

> **Note:**
>
> Raising or clearing a DAD alarm is not successful when UBT is muted.
>
> DAD alarm is not raised during a cold restart or a power cycle.

3 Troubleshoot any of the following:

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1. Duplicate address detected - TMN Ethernet MGMT Port
2. Duplicate address detected - Port #4 TMN Ethernet
3. Duplicate address detected - TMN In-Band Port #1/2
4. Duplicate address detected - PPP IP/RF Port

30.2.1 Duplicate address detected - TMN Ethernet MGMT Port

Procedure

1 Using the Craft Terminal, retrieve the provisioned IPv6 TMN Ethernet Configuration value for the NE declaring duplicate address detected alarm.

2 Verify the desired IPv6 TMN Ethernet Configuration value for the NE according to site engineering documentation.

3 Is the provisioned IPv6 TMN Ethernet value correct?

1. If no, go to step 4.
2. If yes, go to step 12.

4 Reprovision IPv6 TMN Ethernet Configuration value for the NE according to site engineering documentation.

⚠ CAUTION!

The following Cautions apply:

- If access to the NE with the incorrect IPv6 address is through the TMN port in alarm, communication with the NE will be lost when the address is changed. The TMN address must be changed locally or a different TMN interface must be established to communicate with the NE.
- When a different TMN interface is not available to communicate with the NE with the incorrect IPv6 address, it is required to correct the IPv6 address on-site. If the IPv6 address is corrected remotely, communication with the NE will be lost.

5 Was the IPv6 address changed on-site?

1. If yes, go to step 6 .
2. If no, go to step 8 .

6 Disconnect the TMN Ethernet interface and reconnect.

7 Go to step 10.

8 Disable TMN Ethernet Configuration port in alarm.

9 Enable TMN Ethernet Configuration port in alarm.

10 Retrieve the alarms using the Craft Terminal's Active Alarms.

11 Did the duplicate address detected alarm clear?

1. If yes, the procedure is complete.
2. If no, contact the next level of technical support for assistance.

12 Verify the desired IPv6 TMN Ethernet Configuration value for the other devices connected to the TMN port according to site engineering documentation.

13 Reprovision IPv6 TMN Ethernet Configuration value on the incorrect device with the correct IPv6 address.

14 Was an IPv6 address changed on the IPv6 TMN Ethernet network?

1. If yes, go to step 15 .
2. If no, go to step 17 .

15 Disconnect the TMN Ethernet interface and reconnect.

16 Go to step 19.

17 Disable TMN Ethernet Configuration port in alarm.

18 Enable TMN Ethernet Configuration port in alarm.

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19 Retrieve the alarms using the Craft Terminal's Active Alarms.

20 Did the duplicate address detected alarm clear?

1. If yes, the procedure is complete.
2. If no, contact the next level of technical support for assistance.

30.2.2 Duplicate address detected - Port #4 TMN Ethernet

Procedure

1 Using the Craft Terminal, retrieve the provisioned IPv6 Port #4 TMN Ethernet Configuration value for the NE declaring duplicate address detected alarm.

2 Verify the desired IPv6 Port #4 TMN Ethernet Configuration value for the NE according to site engineering documentation.

3 Is the provisioned IPv6 Port #4 TMN Ethernet value correct?

1. If no, go to step 4.
2. If yes, go to step 12.

4 Reprovision IPv6 Port #4 TMN Ethernet Configuration value for the NE according to site engineering documentation.

⚠ CAUTION!

The following Cautions apply:

- If access to the NE with the incorrect IPv6 address is through the Port #4 TMN port in alarm, communication with the NE will be lost when the address is changed. The TMN address must be changed locally or a different TMN interface must be established to communicate with the NE.
- When a different TMN interface is not available to communicate with the NE with the incorrect IPv6 address, it is required to correct the IPv6 address on-site. If the IPv6 address is corrected remotely, communication with the NE will be lost.

5 Was the IPv6 address changed on-site?

1. If yes, go to step 6 .
2. If no, go to step 8 .

6 Disconnect the Port #4 TMN Ethernet interface and reconnect.

7 Go to step 10.

8 Disable Port #4 TMN Ethernet Configuration port in alarm.

9 Enable Port #4 TMN Ethernet Configuration port in alarm.

10 Retrieve the alarms using the Craft Terminal's Active Alarms.

11 Did the duplicate address detected alarm clear?

1. If yes, the procedure is complete.
2. If no, contact the next level of technical support for assistance.

12 Verify the desired IPv6 Port #4 TMN Ethernet Configuration value for the far end NE according to site engineering documentation.

13 Reprovision the far end NE with the correct IPv6 address.

14 Was the IPv6 address changed on-site at the near end or far end NE?

1. If yes, go to step 15.
2. If no, go to step 17.

15 Disconnect the TMN Ethernet interface and reconnect.

16 Go to step 19.

17 Disable Port #4 TMN Ethernet Configuration port.

18 Enable Port #4 TMN Ethernet Configuration port.

19 Retrieve the alarms using the Craft Terminal's Active Alarms.

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20 Did the duplicate address detected alarm clear?

1. If yes, the procedure is complete.
2. If no, contact the next level of technical support for assistance.

30.2.3 Duplicate address detected - TMN In-Band Port #1/2

Procedure

1 Using the Craft Terminal, retrieve the provisioned IPv6 TMN In-Band Port #1/2 value for the NE declaring duplicate address detected alarm.

2 Verify the desired IPv6 TMN In-Band Port #1/2 value for the NE according to site engineering documentation.

3 Is the provisioned IPv6 TMN In-Band Port #1/2 value correct?

1. If no, go to step 4 .
2. If yes, go to step 9 .

4 Reprovision IPv6 TMN In-Band Port #1/2 value for the NE according to site engineering documentation.

⚠ CAUTION!

The following Cautions apply:

- If access to the NE with the incorrect IPv6 address is through the TMN In-Band Port #1/2 in alarm, communication with the NE will be lost when the address is changed. The TMN In-Band Port #1/2 address must be changed locally or a different TMN interface must be established to communicate with the NE.
- When a different TMN interface is not available to communicate with the NE with the incorrect IPv6 address, it is required to correct the IPv6 address on-site. If the IPv6 address is corrected remotely, communication with the NE will be lost.

5 Disable TMN In-Band Port #1/2.

6 Enable TMN In-Band Port #1/2.

7 Retrieve the alarms using the Craft Terminal's Active Alarms.

8 Did the duplicate address detected alarm clear?

1. If yes, the procedure is complete.
2. If no, contact the next level of technical support for assistance.

9 Verify the desired IPv6 TMN In-Band Port #1/2 Ethernet Configuration value for the far end NE according to site engineering documentation.

10 Reprovision the far end NE with the correct IPv6 address.

11 Disable TMN In-Band Port #1/2.

12 Enable TMN In-Band Port #1/2.

13 Retrieve the alarms using the Craft Terminal's Active Alarms.

14 Did the duplicate address detected alarm clear?

1. If yes, the procedure is complete.
2. If no, contact the next level of technical support for assistance.

30.2.4 Duplicate address detected - PPP IP/RF Port

Procedure

1 Using the Craft Terminal, retrieve the provisioned Local IPv6 Address value for the NE declaring duplicate address detected alarm.

2 Verify the desired Local IPv6 Address value for the NE according to site engineering documentation.

3 Is the provisioned Local IPv6 Address value correct?

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1. If no, go to step 4.
2. If yes, go to step 7.

4 Reprovision Local IPv6 Address value for the NE according to site engineering documentation.

5 Retrieve the alarms using the Craft Terminal's Active Alarms.

6 Did the duplicate address detected alarm clear?

1. If yes, the procedure is complete.
2. If no, contact the next level of technical support for assistance.

7

> **Note:**
> To perform steps 7 and 8 a redundant network path to the far end must exist or an operator must be on-site at the far end.

Verify the desired Local IPv6 Address value for the far end NE according to site engineering documentation.

7 Reprovision the far end NE Local IPv6 Address value with the correct IPv6 address.

7 Retrieve the alarms using the Craft Terminal's Active Alarms.

> **Note:**
> After the far end Local IPv6 Address is corrected, a PPP Fail condition will be raised and clear the duplicate address detected alarm at the site declaring duplicate address detected.

7 Did the duplicate address detected alarm clear?

1. If yes, the procedure is complete.
2. If no, go to step 11 .

7 Verify the near end PPP-RF port provisioning.

7 Was the near end PPP-RF port provisioned correctly?

1. If yes, go to step 17.
2. If no, go to step 13.

7 Reprovision near end PPP-RF port provisioning.

7 Disable near end PPP RF port.

7 Enable near end PPP RF port.

7 Retrieve the alarms using the Craft Terminal's Active Alarms.

7 Did the duplicate address detected alarm clear?

1. If yes, the procedure is complete.
2. If no, go to step 18 .

7 Verify the far end PPP-RF port provisioning.

7 Was the far end PPP-RF port provisioned correctly?

⚠ CAUTION!

If the far end PPP-RF port provisioning is corrected/disabled remotely, communication with the far end NE will be lost and on-site provisioning will be required to re-establish communications.

1. If yes, go to step 21.
2. If no, go to step 20.

7 Reprovision far end PPP-RF port provisioning.

7 Disable far end PPP RF port.

7 Enable far end PPP RF port.

7 Retrieve the alarms using the Craft Terminal's Active Alarms.

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7 Did the duplicate address detected alarm clear?

1. If yes, the procedure is complete.
2. If no, contact the next level of technical support for assistance.

31. Early warning indication alarms

31.1 Purpose

This procedure describes how to identify and clear Early Warning Indication alarms.

31.2 General

An Early Warning Indication alarm indicates that a 10E-9 Bit Error Rate detected.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Early Warning Indication alarms provides information about Early warning indication alarms.

Table 32: Early Warning Indication alarms

Friendly name/entity	Facility type
Radio/Dir#1.[7-8]/Slot#1/Port#[7-8]/Ch#[0-1]	MPT-HLC or MPT-HLS connected to CorEvo
Radio/Dir#[3-8].[5-8]/Slot#[3-8]/Port#[5-8]/Ch#[0-1]	MPT-HLC or MPT-HLS connected to EASv2
N/A	UBT-S UBT-m UBT-T channel A UBT-T channel B UBT-I

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31.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Early Warning Indication alarms for information about radio alarms.

3 No action is required at this time. Monitor the receive signal for increased degrading. The procedure is complete.

32. ETH OAM loss of continuity alarms

32.1 Purpose

This procedure describes how to identify and clear Eth OAM Loss of Continuity alarms.

32.2 General

An Eth OAM Loss of Continuity alarm indicates that a MEP received no CCM frames from a peer MEP during an interval equal to 3.5 times the CCM transmission interval configured at the MEP. This is caused by a failure in the network located between the local and far end of the MEP.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Eth OAM Loss of Continuity alarms provides information about Eth OAM Loss of Continuity alarms.

Table 33: Eth OAM Loss of Continuity alarms

Friendly name/entity	Equipment type
MEP#*MDID*[1-16]#*MA ID*[1-512]#*MEPID*[1-8191]	CorEvo

32.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

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2 Review the output. See Table: Eth OAM Loss of Continuity alarms for information.

3 Locate the failure in the network.

1. If MIPs were created on the MEP, use the Link Trace command using WT CLI Tool. See the Wavence User Manual, WT CLI Tool Reference Guide for more information.
2. Otherwise, Retrieve the alarms using the Craft Terminal's Active Alarms and locate alarms associated with the path between the local and far end of the MEP.

4 Were alarms identified and cleared between the local and far end of the MEP?

1. If yes, go to step 5.
2. If no, contact the next level of technical support for assistance.

5 Retrieve the alarms using the Craft Terminal's Active Alarms.

6 Did the Eth OAM Loss of Continuity Condition alarm clear?

1. If yes, the procedure is complete.
2. If no, contact the next level of technical support for assistance.

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33. Eth OAM Mismerge Condition alarms

33.1 Purpose

This procedure describes how to identify and clear Eth OAM Mismerge Condition alarms.

33.2 General

An Eth OAM MisMerge Condition alarm indicates that a MEP received a CCM frame with expected MD Level but unexpected MA ID. The MA ID is the *MD Name* appended with the *MA Format*.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Eth OAM MisMerge Condition alarms provides information about Eth OAM MisMerge Condition alarms.

Table 34: Eth OAM MisMerge Condition alarms

Friendly name/entity	Equipment type
MEP#*MDID*[1-16]#*MA ID*[1-512]#*MEPID*[1-8191]	CorEvo

33.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Eth OAM MisMerge Condition alarms for information about

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ECFM alarms.

3 Retrieve MA ID for both ends of the MEP using WT CLI Tool. See the Wavence WebCT User Manual, WT CLI Tool Reference Guide for more information.

4 Determine which MA ID value(s) are incorrect.

5 Reprovision the MA ID value(s) according to the results of step 4.

6 Retrieve the alarms using the Craft Terminal's Active Alarms.

7 Did the Eth OAM MisMerge Condition alarm clear?

1. If yes, the procedure is complete.
2. If no, contact the next level of technical support for assistance.

34. Eth OAM Remote Defect Indication alarms

34.1 Purpose

This procedure describes how to identify and clear Eth OAM Remote Defect Indication alarms.

34.2 General

An Eth OAM Remote Defect Indication alarm indicates that a MEP received a CCM frame with the RDI field set.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Eth OAM Remote Defect Indication alarms describes Eth OAM Remote Defect Indication alarms.

Table 35: Eth OAM Remote Defect Indication alarms

Friendly name/entity	Equipment type
MEP#*MDID*[1-16]#*MA ID*[1-512]#*MEPID*[1-8191]	CorEvo

34.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Eth OAM Remote Defect Indication alarms for information.

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3 Retrieve the alarms on the far end of the MEP using Craft Terminal's Active Alarms.

4 Check the far end for the one or more of the above RDI set conditions and troubleshoot accordingly.

5 Were the RDI set condition cleared at the far end of the MEP?

1. If yes, go to step 6.
2. If no, contact the next level of technical support for assistance.

6 Retrieve the alarms using the Craft Terminal's Active Alarms.

7 Did the Eth OAM Remote Defect Indication Level alarm clear?

1. If yes, the procedure is complete.
2. If no, contact the next level of technical support for assistance.

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35. Eth OAM Unexpected Level alarms

35.1 Purpose

This procedure describes how to identify and clear Eth OAM Unexpected Level alarms.

35.2 General

An Eth OAM Unexpected Level alarm indicates that a MEP received a CCM frame with the incorrect MD Level. The MEP expects to receive a MD level [0-7] either equal to or higher than the provisioned MD Level.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Eth OAM Unexpected Level alarms provides information about Eth OAM Unexpected Level alarms.

Table 36: Eth OAM Unexpected Level alarms

Friendly name/entity	Equipment type
MEP#*MDID*[1-16]#*MA ID*[1-512]#*MEPID*[1-8191]	CorEvo

35.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

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2 Review the output. See Table: Eth OAM Unexpected Level alarms for information about ECFM alarms.

3 Retrieve the MD Level for both ends of the MEP using WT CLI Tool. See the Wavence WebCT User Manual, WT CLI Tool Reference Guide for more information.

4 Determine which MD Level values are incorrect.

5 Reprovision the MD Level values according to the results of step 4.

6 Retrieve the alarms using the Craft Terminal's Active Alarms.

7 Did the Eth OAM Unexpected Level alarm clear?

1. If yes, the procedure is complete.
2. If no, contact the next level of technical support for assistance.

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36. Eth OAM UnexpectedMEPID alarms

36.1 Purpose

This procedure describes how to identify and clear Eth OAM UnexpectedMEPID alarms.

36.2 General

An Eth OAM UnexpectedMEPID alarm indicates that received MEPID value is different than the expected MEPID [1-8191]. The MEP expects to receive the provisioned MEPID.

Eth OAM Unexpected Period alarm indicates that received MEP Transmission Interval value is different than the expected MEP Transmission Interval [*300-hundred-hertz*, *ten-ms*, *hundred-ms*, *one-sec*, *ten-sec*, *one-min*, *ten-min*]. The MEP expects to receive the same value inside the CCM frame as the provisioned Transmission Interval.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Eth OAM UnexpectedMEPID alarms provides information about Eth OAM UnexpectedMEPID alarms.

Table 37: Eth OAM UnexpectedMEPID alarms

Friendly name/entity	Equipment type
MEP#*MDID*[1-16]#*MA ID*[1-512]#*MEPID*[1-8191]	CorEvo

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36.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Eth OAM UnexpectedMEPID alarms for information about ECFM alarms.

3 Retrieve MEPID for both ends of the MEP using WT CLI Tool. See the Wavence WebCT User Manual, WT CLI Tool Reference Guide for more information.

4 Determine which MEPID value(s) are incorrect.

5 Reprovision the MEPID value(s) according to the results of step 4.

6 Retrieve the alarms using the Craft Terminal's Active Alarms.

7 Did the Eth OAM UnexpectedMEPID alarm clear?

   1. If yes, the procedure is complete.
   2. If no, contact the next level of technical support for assistance.

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37. Eth OAM Unexpected Period alarms

37.1 Purpose

This procedure describes how to identify and clear Eth OAM Unexpected Period alarms.

37.2 General

Eth OAM Unexpected Period alarm indicates that received MEP Transmission Interval value is different than the expected MEP Transmission Interval [*300-hundred-hertz*, *ten-ms*, *hundred-ms*, *one-sec*, *ten-sec*, *one-min*, *ten-min*]. The MEP expects to receive the same value inside the CCM frame as the provisioned Transmission Interval.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Eth OAM Unexpected Period alarms describes Eth OAM Unexpected Period alarms.

Table 38: Eth OAM Unexpected Period alarms

Friendly name/entity	Equipment type
MEP#*MDID*[1-16]#*MA ID*[1-512]#*MEPID*[1-8191]	CorEvo

37.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

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2 Review the output. See Table: Eth OAM Unexpected Period alarms for information about ECFM alarms.

3 Retrieve the MEP Transmission Interval for both ends using WT CLI Tool. See the Wavence WebCT User Manual, WT CLI Tool Reference Guide for more information.

4 Determine which MEP Transmission Interval values are incorrect.

5 Reprovision the MEP Transmission Interval according to the results of step 4.

> **Note:**
> Eth OAM Unexpected Period alarm requires 3.5 times the MEP Transmission Interval to clear the alarm. For MEP Transmission Interval provisioned one-sec, the alarm requires 3.5 seconds to clear. For MEP Transmission Interval provisioned 10-min, the alarm requires 35 minutes to clear.

6 Retrieve the alarms using the Craft Terminal's Active Alarms.

7 Did the Eth OAM Unexpected Period alarm clear?

1. If yes, the procedure is complete.
2. If no, contact the next level of technical support for assistance.

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38. Excessive Environmental Temperature alarms

38.1 Purpose

This procedure describes how to identify and clear Excessive Environmental Temperature alarms.

38.2 General

An Excessive Environmental Temperature alarm indicates that excessive temperature on MSS-4/8, or MPT-HLC, MPT-HLS has been detected, or fan unit has failed, or is missing on MPT-HL.

See the Wavence Product Information Manual to determine card location and address and for information about supported temperatures.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Equipment alarms provides information about Excessive Environmental Temperature alarms.

Table 39: Equipment alarms

Friendly name/entity	Equipment type
r01/subrack#[3-8][5-8]	MPT-HLC connected to EASv2
r01/subrack#1[7-8]	MPT-HLC connected to CorEvo
r01s[101-102]/board#[1-10]	MPT-HLS
MSS shelf	MSS-4/8

For an MSS-E the temperature alarm is raised when the MSS-E Temperature is equal or above 67.2 °C and is cleared when temperature drops below 59,1 °C

For an MSS-HE/XE the temperature Alarm is raised when the MSS-HE/XE Temperature is equal or

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above 88 °C and is cleared when temperature drops below 78 °C

For MSS-4/8, the temperature alarm is raised when MSS temperature is equal or above 95°C degrees and is cleared when temperature drops below 85°C.

38.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Equipment alarms for information.

3 Verify that the MPT-HLC/MPT-HLS fan card is equipped and operating properly.

4 Is the MPT-HLC/MPT-HLS fan equipped and operating properly?

   1. If yes, go to step 9.
   2. If no, go to step 5.

5 Is the traffic affected?

   1. If yes, go to step 6.
   2. If no, contact the next level of technical support for assistance.

6 Replace the unit. See the Wavence Hardware Installation and Replacement Manual.

7 Retrieve the alarms using the Craft Terminal's Active Alarms.

8 Did the Excessive Environmental Temperature alarm clear?

   1. If yes, the procedure is complete.
   2. If no, go to step 9.

9 Verify that the building temperature is within acceptable temperature range.

10 Verify the condition of MPT-HLC/MPT-HLS shelf air filter and replace if air filter condition indicates that filter replacement is required. See Wavence Hardware Installation and

Replacement Manual to replace the air filter.

11 Verify that no obstruction occurred that prevents proper air flow through the MPT-HLC/MPT -HLS unit.

12 Retrieve the alarms using the Craft Terminal's Active Alarms.

13 Did the Excessive Environmental Temperature alarm clear?

1. If yes, the procedure is complete.
2. If no, contact the next level of technical support for assistance.

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39. Fan Communication Problem (MPT-HLS)

39.1 Purpose

This procedure describes how to identify and clear Fan Communication Problem alarm on MPT -HLS.

39.2 General

A Fan Communication Problem alarm indicates that a problem communicating with the MPT-HLS fan has been detected.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Fan fail alarm provides information about Fan communication problem alarms.

Table 40: Fan fail alarm

Name	Equipment type
r01s[101-102]/board#[1-10]	MPT-HLS

39.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Fan fail alarm for information.

3 Verify that ALL the MPT-HLS fans are equipped and operating properly.

4 Is the traffic affected?

1. If yes, go to step 6.
2. If no, go to step 5.

5 Is it appropriate to proceed with the Fan Fail procedure?

1. If yes, go to step 6.
2. If no, the procedure is complete.

6 Is the radio direction in a protected configuration?

1. If yes, go to step 7.
2. If no, go to step 16.

7 Verify that the MPT-HLS in alarm status is *Standby*.

8 Is the MPT-HLS in alarm status *Standby*?

1. If yes, go to step 9.
2. If no, the following Caution applies.

⚠ CAUTION!

Forced/Lockout switching traffic causes a disruption to traffic.

9 Perform a Forced EPS Switch using the Craft Terminal.

10 Perform a Forced HSB Switch using the Craft Terminal.

11 Perform a Forced Rx Radio Switch using the Craft Terminal.

12 Go to step 16

13 Perform a Lockout EPS Switch using the Craft Terminal.

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14 Perform a Lockout HSB Switch using the Craft Terminal.

15 Perform a Lockout Rx Radio Switch using the Craft Terminal.

16 Replace the MPT-HLS. See the Wavence Hardware Installation and Replacement Manual to replace MPT-HLS.

> **⚠ CAUTION!**
>
> Physically replacing an unprotected MPT-HLS will causes a loss of traffic on ALL facilities that are associated with the MPT-HLS being replaced. Schedule an appropriate maintenance window per local practices and procedures.

17 Wait for the status LED to turn steady amber or green.

18 Retrieve the alarms using the Craft Terminal's Active Alarms.

19 Did the Fan Fail alarm clear?

1. If yes, go to step 20.
2. If no, contact the next level of technical support for assistance.

20 Is the radio direction in a protected configuration?

1. If yes, go to step 21.
2. If no, the procedure is complete.

21 Release the Forced/Lockout EPS Switch using the Craft Terminal.

22 Release the Forced/Lockout HSB Switch using the Craft Terminal.

23 Release the Forced/Lockout Rx Radio Switch using the Craft Terminal.

The procedure is complete.

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40. Fan Fail (MPT-HLC)

40.1 Purpose

This procedure describes how to identify and clear Fan Fail alarm on MPT-HLC.

40.2 General

A Fan Fail alarm indicates that MPT-HLC card has detected a MPT-HLC fan unit has failed.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Fan fail alarm provides information about Fan fail alarms.

Table 41: Fan fail alarm

Name	Equipment type
r01/subrack#1[7-8]	MPT-HLC connected to CorEvo
r01/subrack#[3-8][5-8]	MPT-HLC connected to EASv2
n/a	UBT-I

40.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

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2 Review the output. See Table: Fan fail alarm for information about equipment alarms.

3 Verify that ALL the MPT-HLC fans are equipped and operating properly.

4 Is the traffic affected?

4.1 If yes, go to step 6.

4.2 If no, go to step 5.

5 Is it appropriate to proceed with the Fan Fail procedure?

5.1 If yes, go to step 6.

5.2 If no, the procedure is complete.

6 Is the radio direction in a protected configuration?

6.1 If yes, go to step 7.

6.2 If no, go to step 16.

7 Verify that the MPT-HLC in alarm status is *Standby*.

8 Is the MPT-HLC in alarm main or spare ?

8.1 If main, go to step 9.

8.2 If spare, go to step 13.

⚠ CAUTION!

Forced/Lockout switching traffic causes a disruption to traffic.

9 Perform a Forced EPS Switch using the Craft Terminal.

10 Perform a Forced HSB Switch using the Craft Terminal.

11 Perform a Forced Rx Radio Switch using the Craft Terminal.

12 go to step 16

13 Perform a Lockout EPS Switch using the Craft Terminal.

14 Perform a Lockout HSB Switch using the Craft Terminal.

15 Perform a Lockout Rx Radio Switch using the Craft Terminal.

16 Replace the MPT-HLC. See the Wavence Hardware Installation and Replacement Manual to replace MPT-HLC.

> **⚠ CAUTION!**
>
> Physically replacing an unprotected MPT-HLC will causes a loss of traffic on ALL facilities that are associated with the MPT-HLC being replaced. Schedule an appropriate maintenance window per local practices and procedures.

17 Wait for the status LED to turn steady amber or green.

18 Retrieve the alarms using the Craft Terminal's Active Alarms.

19 Did the Fan Fail alarm clear?

19.1 If yes, go to step 20.

19.2 If no, contact the next level of technical support for assistance.

20 Is the radio direction in a protected configuration?

20.1 If yes, go to step 21.

20.2 If no, the procedure is complete.

21 Release the Forced/Lockout EPS Switch using the Craft Terminal.

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22 Release the Forced/Lockout HSB Switch using the Craft Terminal.

23 Release the Forced/Lockout Rx Radio Switch using the Craft Terminal.

The procedure is complete.

41. High BER alarms on radios

41.1 Purpose

This procedure describes how to identify and clear High BER alarms on radios. For High BER alarms on SDH facilities, see High BER alarms on SDH cards.

41.2 General

A High BER alarm indicates that Bit Error Rate threshold of 10E-4 has been exceeded on receiver input circuits.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: High BER alarms on radio equipment provides information about High BER alarms.

Table 42: High BER alarms on radio equipment

Friendly name/entity	Facility type
Radio/Dir#1.[7-8]/Slot#1/Port#[7-8]/Ch#[0-1]	MPT-HLC or MPT-HLS connected to CorEvo
Radio/Dir#[3-8].[5-8]/Slot#[3-8]/Port#[5-8]/Ch#[0-1]	MPT-HLC or MPT-HLS connected to EASv2
N/A	UBT-S UBT-m UBT-T channel A UBT-T channel B UBT-I

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41.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: High BER alarms on radio equipment for information about radio alarms.

3 Verify that there are no weather-related conditions.

4 Do weather-related conditions exist?

   4.1 If yes, wait for the weather to clear and recheck the alarms. The procedure is complete.

   4.2 If no, go to step 5.

5 Verify that the RF path is clear.

6 Is the RF path clear?

   6.1 If yes, go to step 7.

   6.2 If no, contact the next level of technical support for assistance.

7 Verify that the RSL is above the RX threshold.

8 Is the RSL above the RX threshold?

   8.1 If yes, contact the next level of technical support for assistance.

   8.2 If no, go to step 9.

9 Check the upstream transmitter output power and troubleshoot accordingly.

10 Retrieve the alarms using the Craft Terminal's Active Alarms.

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11 Did the High BER alarm clear?

11.1 If yes, the procedure is complete.

11.2 If no, go to step 12.

12 Verify that the antenna is aligned.

13 Is the antenna aligned?

13.1 If yes, contact the next level of technical support for assistance.

13.2 If no, align the antenna. See Wavence Turn-up Manual, Antenna Polarization Verification Antenna Feed-horn Verification - XPD Procedure.

14 Retrieve the alarms using the Craft Terminal's Active Alarms.

15 Did the High BER alarm clear?

15.1 If yes, the procedure is complete.

15.2 If no, contact the next level of technical support for assistance.

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42. High BER alarms on SDH cards

42.1 Purpose

This procedure describes how to identify and clear High BER alarms on SDH facilities. For High BER alarms on radios, see High BER alarms on radios.

42.2 General

A High BER (on Ele/OptSdh-in) alarm indicates that Bit Error Rate threshold of 10E-5 has been exceeded on receiver input circuits from the client to the SDHACC card.

A High BER (on Ele/OptSdh-out) alarm indicates that Bit Error Rate threshold of 10E-5 has been exceeded on transmit output circuits to the client from the SDHACC card.

A High BER (on channelized E1-in) alarm indicates that Bit Error Rate threshold of 10E-5 has been exceeded on receiver input circuits from the client to the SDHACC card.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: High BER alarms on SDH facilities provides information about High BER alarms.

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Table 43: High BER alarms on SDH facilities

Friendly name/entity	Facility type
EleSdh-in/Slot#[3-8]/Port[1-2]	STM-1/OC-3
OptSdh-in/Slot#[3-8]/Port[1-2]	
EleSdh-out/Slot#[3-8]/Port[1-2]	
OptSdh-out/Slot#[3-8]/Port[1-2]	
MS/slot#[3-8]/Port#1	STM-1

42.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: High BER alarms on SDH facilities for information.

3 Troubleshoot any of the following:

   1. High BER on EleSdh/OptSdh-in on STM-1/OC-3 facility
   2. High BER on EleSdh/OptSdh-out on STM-1/OC-3 facility
   3. High BER on Channelized E1-in on STM-1 facility

42.3.1 High BER on EleSdh/OptSdh-in on STM-1/OC-3 facility

Procedure

1 Troubleshoot the source of input that is associated with the input signal to the SDHACC card from the demarcation panel.

2 Verify the cabling that is associated with the input signal to the SDHACC card.

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3 Retrieve the alarms using the Craft Terminal's Active Alarms .

4 Did the High BER (on EleSdh/OptSdh-in) alarm clear?

   1. If yes, the procedure is complete.
   2. If no, contact the next level of technical support for assistance.

42.3.2 High BER on EleSdh/OptSdh-out on STM-1/OC-3 facility

Procedure

1 Troubleshoot the upstream source of the incoming STM-1/OC-3.

2 Retrieve the alarms using the Craft Terminal's Active Alarms .

3 Did the High BER (on EleSdh/OptSdh-out) alarm clear?

   1. If yes, the procedure is complete.
   2. If no, contact the next level of technical support for assistance.

42.3.3 High BER on Channelized E1-in on STM-1 facility

Procedure

1 Troubleshoot the source of input that is associated with the input signal to the SDHACC card from the demarcation panel.

2 Verify the cabling that is associated with the input signal to the SDHACC card.

3 Retrieve the alarms using the Craft Terminal's Active Alarms .

4 Did the High BER (on Channelized E1-in) alarm clear?

1. If yes, the procedure is complete.
2. If no, contact the next level of technical support for assistance.

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43. Housekeeping alarms

43.1 Purpose

This procedure describes how to identify and clear Housekeeping alarms (External Point Input) on a FAN card.

43.2 General

Housekeeping Alarm inputs are external alarms that are monitored by the NE. Housekeeping alarms can be provisioned for alarm state active low or high.

Housekeeping Alarm Input 1-6/11-14 (or 11-18 when E-FAN is configured) indicates that the external alarm point is active, improperly configured, or improperly provisioned.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Housekeeping alarms provides information about Housekeeping alarms.

Table 44: Housekeeping alarms

Name	Equipment type
ExtPt/Input#[11-14]	A-FAN
ExPt/Input#[11-18]	E-FAN

43.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Housekeeping alarms for information.

3 Verify the equipment that is associated with housekeeping alarm input which is in the alarm state.

4 Is the equipment that is associated with housekeeping alarm input 1-6/11-14 (or 11-18 when E-FAN is configured) in the alarm state?

   4.1 If yes, go to step 5.

   4.2 If no, go to step 7.

5 Resolve the problem with external equipment generating the housekeeping alarm.

6 Did the Housekeeping Alarm clear?

   6.1 If yes, the procedure is complete.

   6.2 If no, go to step 7.

7 Verify the Housekeeping Alarm Input provisioning for housekeeping alarm input in alarm.

8 Correct the provisioning as required.

9 Retrieve the alarms using the Craft Terminal's Active Alarms.

10 Did the Housekeeping Alarm clear?

   10.1 If yes, the procedure is complete.

   10.2 If no, go to step 11.

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11 Verify the Housekeeping Alarm Input configuration for housekeeping alarm input in alarm.

12 Correct the configuration as required.

13 Retrieve the alarms using the Craft Terminal's Active Alarms.

14 Did the Housekeeping Alarm clear?

14.1 If yes, the procedure is complete.

14.2 If no, contact the next level of technical support for assistance.

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44. Incompatible frequency alarms

44.1 Purpose

This procedure describes how to identify and clear Incompatible Frequency alarms.

44.2 General

An Incompatible Frequency alarm indicates that the configured frequency is outside the supported frequency range.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Incompatible Frequency alarms provides information about Incompatible Frequency alarms.

Table 45: Incompatible Frequency alarms

Friendly name/entity	Facility type
Radio/Dir#1.[7-8]/Slot#1/Port#[7-8]/Ch#1	MPT-HLC or MPT-HLS connected to CorEvo
Radio/Dir#[3-8].[5-8]/Slot#[3-8]/Port#[5-8]/Ch#[0-1]	MPT-HLC or MPT-HLS connected to EASv2
N/A	UBT-S UBT-m UBT-T channel A UBT-T channel B UBT-I

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44.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Incompatible Frequency alarms for information.

3 Using the Craft Terminal, retrieve the provisioned frequency value and supported frequency of the MPT-HLC, or MPT-HLS.

4 Verify the required frequency of the microwave link.

5 Is the required microwave link frequency supported by the MPT-HLC, or MPT-HLS?

   5.1 If yes, go to step 6.

   5.2 If no, go to step 9.

6 Reprovision the frequency of the MPT-HLC, or MPT-HLS to the required frequency of the microwave link.

7 Retrieve the alarms using the Craft Terminal's Active Alarms.

8 Did the Incompatible Frequency alarm clear?

   8.1 If yes, the procedure is complete.

   8.2 If no, go to step 9.

9 Replace the MPT-HLC, or MPT-HLS with the version that supports the required frequency of the microwave link. See the Wavence Hardware Installation and Replacement Manual.

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⚠ CAUTION!

Replacing the MPT-HLC, or MPT-HLS in an unprotected facility causes a loss of traffic on ALL facilities that are associated with the MPT-HLC, or MPT-HLS being replaced. Schedule an appropriate maintenance window per local practices and procedures.

10 Retrieve the alarms using the Craft Terminal's Active Alarms.

11 Did the Incompatible Frequency alarm clear?

11.1 If yes, the procedure is complete.

11.2 If no, contact the next level of technical support for assistance.

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45. Incompatible modulation alarms

45.1 Purpose

This procedure describes how to identify and clear radio alarms.

45.2 General

An Incompatible Modulation alarm indicates that the configured modulation is not supported by MPT-HLC/MPT-HLS/UBT-S/UBT-T (Channel A/B)/ UBT-m in alarm.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Incompatible modulation alarms provides information about Incompatible modulation alarms.

Table 46: Incompatible modulation alarms

Friendly name/entity	Facility type
Radio/Dir#1.[7-8]/Slot#1/Port#[7-8]/Ch#1	MPT-HLC or MPT-HLS connected to CorEvo
Radio/Dir#[3-8].[5-8]/Slot#[3-8]/Port#[5-8]/Ch#[0-1]	MPT-HLC or MPT-HLS connected to EASv2
N/A	UBT-S UBT-m UBT-T channel A UBT-T channel B UBT-I

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45.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Incompatible modulation alarms for information.

3 Using the Craft Terminal, retrieve the provisioned modulation value and supported modulation of the MPT-HLC/MPT-HLS.

4 Verify the required modulation of the microwave link.

5 Is the required microwave link modulation supported by the MPT-HLC/MPT-HLS?

   5.1 If yes, go to step 6.

   5.2 If no, go to step 9.

6 Reprovision the modulation of the MPT-HLC/MPT-HLS to the required modulation of the microwave link.

7 Retrieve the alarms using the Craft Terminal's Active Alarms.

8 Did the Incompatible Modulation alarm clear?

   8.1 If yes, the procedure is complete.

   8.2 If no, go to step 9.

9 Replace the MPT-HLC/MPT-HLS with the version that supports the required modulation of the microwave link. See the Wavence Hardware Installation and Replacement Manual.

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⚠ CAUTION!

Replacing the MPT-HLC, or MPT-HLS in an unprotected facility causes a loss of traffic on ALL facilities that are associated with the MPT-HLC, or MPT-HLS being replaced. Schedule an appropriate maintenance window per local practices and procedures.

10 Retrieve the alarms using the Craft Terminal's Active Alarms.

11 Did the Incompatible Modulation alarm clear?

11.1 If yes, the procedure is complete.

11.2 If no, contact the next level of technical support for assistance.

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46. Incompatible Shifter alarms

46.1 Purpose

This procedure describes how to identify and clear radio alarms.

46.2 General

An Incompatible Shifter alarm indicates that the configured shifter value is not supported by the MPT-HLC/MPT-HLS/UBT-T (Channel A/B)/ UBT-S/ UBT-m.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Incompatible Shifter alarms provides information about Incompatible Shifter alarms.

Table 47: Incompatible Shifter alarms

Friendly name/entity	Facility type
Radio/Dir#1.[7-8]/Slot#1/Port#[7-8]/Ch#[0-1]	MPT-HLC or MPT-HLS connected to CorEvo
Radio/Dir#[3-8].[5-8]/Slot#[3-8]/Port#[5-8]/Ch#[0-1]	MPT-HLC or MPT-HLS connected to EASv2
N/A	UBT-S UBT-m UBT-T channel A UBT-T channel B UBT-I

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46.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Incompatible Shifter alarms for information about radio alarms.

3 Using the Craft Terminal, retrieve the provisioned shifter value and supported shifter levels of the MPT-HLC, or MPT-HLS.

4 Verify the required shifter value of the microwave link.

5 Is the required microwave link shifter level supported by the MPT-HLC, or MPT-HLS?

   5.1 If yes, go to step 6.

   5.2 If no, go to step 9.

6 Reprovision the shifter value of the MPT-HLC/MPT-HLS to the required shifter value of the microwave link.

7 Retrieve the alarms using the Craft Terminal's Active Alarms.

8 Did the Incompatible Shifter alarm clear?

   8.1 If yes, the procedure is complete.

   8.2 If no, go to step 9

9 Replace the MPT-HLC/MPT-HLS with version that supports the required shifter value of the microwave link. See the Wavence Hardware Installation and Replacement Manual.

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⚠ CAUTION!

Replacing the MPT-HLC/MPT-HLS in an unprotected facility causes a loss of traffic on ALL facilities that are associated with the MPT-HLC/MPT-HLS being replaced. Schedule an appropriate maintenance window per local practices and procedures.

10 Retrieve the alarms using the Craft Terminal's Active Alarms.

11 Did the Incompatible Shifter alarm clear?

11.1 If yes, the procedure is complete.

11.2 If no, contact the next level of technical support for assistance.

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47. Incompatible Power alarms

47.1 Purpose

This procedure describes how to identify and clear Incompatible Power alarms.

47.2 General

An Incompatible Power/Incompatible TX Power alarm indicates that the configured RTPC/ATPC Tx power value or ATPC Rx power threshold value is not supported by the MPT-HLC/MPT-HLS/UBT -T(Channel A/B)/UBT-S/UBT-m.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Incompatible Power alarms provides information about Incompatible Power alarms.

Note:

Disconnecting the fiber of a radio (MPT/UBT) and reconnecting it within 1-2 minutes, clears the alarm from the system but the Tx power still has to be configured within the expected range for the system to properly function.

If the fiber is disconnected and connected again after 2 minutes, the alarm is maintained.

Table 48: Incompatible Power alarms

Friendly name/entity	Facility type
Radio/Dir#1.[7-8]/Slot#1/Port#[7-8]/Ch#[0-1]	MPT-HLC, MPT-HLS connected to CorEvo
Radio/Dir#[3-8].[5-8]/Slot#[3-8]/Port#[5-8]/Ch#[0-1]	MPT-HLC or MPT-HLS connected to EASv2
N/A	UBT-S UBT-m UBT-T channel A UBT-T channel B UBT-I

47.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Incompatible Power alarms for information.

3 Using the Craft Terminal, retrieve the provisioned RTPC/ATPC Tx power value and ATPC Rx power threshold value and supported transmit power levels of the MPT-HLC/MPT-HLS.

4 Verify the required transmit power level of the microwave link.

5 Are the required microwave link RTPC/ATPC Tx power levels and ATPC Rx power threshold levels supported by the MPT-HLC/MPT-HLS?

   5.1 If yes, go to step 6.

   5.2 If no, go to step 9

6 Reprovision the RTPC/ATPC Tx power levels and/or ATPC Rx power threshold level of the MPT-HLC/MPT-HLS to the required transmit power level of the microwave link.

7 Retrieve the alarms using the Craft Terminal's Active Alarms.

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8 Did the Incompatible TX Power alarm clear?

8.1 If yes, the procedure is complete.

8.2 If no, go to step 9

9 Replace the MPT-HLC/MPT-HLS with the version that supports the required transmit power level of the microwave link. See the Wavence Hardware Installation and Replacement Manual.

⚠ CAUTION!

Replacing the MPT-HLC/MPT-HLS in an unprotected facility causes a loss of traffic on ALL facilities that are associated with the MPT-HLC/MPT-HLS being replaced. Schedule an appropriate maintenance window per local practices and procedures.

10 Retrieve the alarms using the Craft Terminal's Active Alarms.

11 Did the Incompatible TX Power alarm clear?

11.1 If yes, the procedure is complete.

11.2 If no, contact the next level of technical support for assistance.

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48. Internal Communication Problem alarms

48.1 Purpose

This procedure describes how to identify and clear Internal Communication Problem condition on the radio.

48.2 General

An Internal Communication Problem condition indicates that a loss of communications with the MPT-HLC/MPT-HLS and UBT-S/UBT-T/UBT-m.

For UBT, this alarm indicates the loss of Communication between mated UBTs applicable for XPIC configuration only. For clearing procedure for UBT, see Procedure (UBT)

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Internal communication problem alarms provides information about Internal Communication Problem alarms.

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Table 49: Internal communication problem alarms

Friendly name/entity	Equipment type
r01/subrack#1[7-8]	MPT-HLC connected to CorEvo
r01/subrack#[3-8][5-8]	MPT-HLC connected to EASv2
r01s[101-102]/board#[1-10]	MPT-HLS
r01/subrack#[3-8]	ODU connected to EASv2
N/A	UBT-S UBT-m UBT-T UBT-I

48.3 Procedure (MPT)

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Internal communication problem alarms for information about internal communication problem alarms.

3 Is traffic affected?

3.1 If yes, go to step step 7. Verify radio port provisioning against site documentation.

3.2 If no, go to step step 4. Has the Internal Communication Problem alarm been active for more than five minutes?.

4 Has the Internal Communication Problem alarm been active for more than five minutes?

4.1 If yes, go to step step 7. Verify radio port provisioning against site documentation.

4.2 If no, wait five minutes, then go to step step 5. Retrieve the alarms using the Craft Terminal's Active Alarms.

5 Retrieve the alarms using the Craft Terminal's Active Alarms.

6 Did the Internal Communication Problem alarm clear?

6.1 If yes, the procedure is complete.

6.2 If no, go to step step 7. Verify radio port provisioning against site documentation.

7 Verify radio port provisioning against site documentation.

8 Is the radio port provisioned per site documentation?

8.1 If yes, go to step step 9. Verify radio port fiber cabling per site documentation.

8.2 If no, go to step step 14. Reprovision the radio port per site documentation.

9 Verify radio port fiber cabling per site documentation.

10 Is the radio port fiber cabling correct and connected to the correct MPT type?

10.1 If yes, go to step step 17. Is the radio facility protected?.

10.2 If no, go to step step 11. Correct MPT fiber cable connections per site documentation.

11 Correct MPT fiber cable connections per site documentation.

12 Retrieve the alarms using the Craft Terminal's Active Alarms.

13 Did the Internal Communication Problem alarm clear?

13.1 If yes, the procedure is complete.

13.2 If no, go to step step 17. Is the radio facility protected?.

14 Reprovision the radio port per site documentation.

15 Retrieve the alarms using the Craft Terminal's Active Alarms.

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16 Did the Internal Communication Problem alarm clear?

16.1 If yes, the procedure is complete.

16.2 If no, go to step step 9. Verify radio port fiber cabling per site documentation.

17 Is the radio facility protected?

17.1 If yes, go to step step 18. Is the MPT-HLC/MPT-HLS in alarm main or spare?.

17.2 If no, go to step step 26. Unseat and reseat the communication link between the MPT-HLC/MPT-HLS and CorEvo, EASv2 card.

18 Is the MPT-HLC/MPT-HLS in alarm main or spare?

18.1 If main, go to step step 19. Perform a Forced EPS Switch using the Craft Terminal.

18.2 If spare, go to step step 23. Perform a Lockout EPS Switch using the Craft Terminal.

⚠ CAUTION!

Forced switching traffic causes a disruption to traffic.

19 Perform a Forced EPS Switch using the Craft Terminal.

20 Perform a Forced HSB Switch using the Craft Terminal.

21 Perform a Forced Rx Radio Switch using the Craft Terminal.

22 Go to step step 26. Unseat and reseat the communication link between the MPT-HLC/MPT-HLS and CorEvo, EASv2 card.

23 Perform a Lockout EPS Switch using the Craft Terminal.

24 Perform a Lockout HSB Switch using the Craft Terminal.

25 Perform a Lockout Rx Radio Switch using the Craft Terminal.

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26 Unseat and reseat the communication link between the MPT-HLC/MPT-HLS and CorEvo, EASv2 card.

> **⚠ CAUTION!**
>
> Unseating and reseating the communication link between the MPT-HLC/MPT-HLS and the CorEvo, EASv2 card in an unprotected facility causes a loss of traffic on ALL facilities that are associated with the MPT-HLC/MPT-HLS. Schedule an appropriate maintenance window per local practices and procedures.

27 Retrieve the alarms using the Craft Terminal's Active Alarms.

28 Did the Internal Communication Problem alarm clear?

28.1 If yes, go to step step 40. Release the Forced/Lockout EPS Switch using the Craft Terminal

28.2 If no, go to step step 29. Replace the communication link between the MPT-HLC/MPT-HLS and CorEvo, EASv2 card

29 Replace the communication link between the MPT-HLC/MPT-HLS and CorEvo, EASv2 card.

> **⚠ CAUTION!**
>
> Replacing the communication link between the MPT-HLC/MPT-HLS and the EASv2, CorEvo card in an unprotected facility causes a loss of traffic on ALL facilities that are associated with the MPT-HLC/MPT-HLS. Schedule an appropriate maintenance window per local practices and procedures.

30 Retrieve the alarms using the Craft Terminal's Active Alarms.

31 Did the Internal Communication Problem alarm clear?

31.1 If yes, go to step step 40. Release the Forced/Lockout EPS Switch using the Craft Terminal

31.2 If no, go to step step 32. Position the PT-HLC/MPT-HLS PA switch to the OFF '0"

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position

32 Position the PT-HLC/MPT-HLS PA switch to the OFF ‘0" position.

> **⚠ CAUTION!**
>
> Reseating an MPT-HL/MPT-HLC/MPT-HLS in an unprotected facility causes a loss of traffic on ALL facilities that are associated with the MPT-HL/MPT-HLC/MPT-HLS reseat. Schedule an appropriate maintenance window per local practices and procedures.

33 Remove the fuse in the power distribution panel supplying battery power to the MPT -HLC/MPT-HLS.

34 Install the fuse in the power distribution panel supplying battery power to the MPT-HLC/MPT -HLS.

35 Position the MPT-HLC/MPT-HLS PA switch to the ON ‘1" position.

36 Did the Internal Communication Problem alarm clear?

36.1 If yes, go to step step 40. Release the Forced/Lockout EPS Switch using the Craft Terminal

36.2 If no, go to step step 37. Replace the MPT-HL/MPT-HLC/MPT-HLS in alarm. See the Wavence Hardware Installation and Replacement Manual

37 Replace the MPT-HL/MPT-HLC/MPT-HLS in alarm. See the Wavence Hardware Installation and Replacement Manual.

> **⚠ CAUTION!**
>
> Replacing an MPT-HLC/MPT-HLS in an unprotected facility causes a loss of traffic on ALL facilities that are associated with the MPT-HLC/MPT-HLS being replaced. Schedule an appropriate maintenance window per local practices and procedures.

38 Retrieve the alarms using the Craft Terminal's Active Alarms.

39 Did the Internal Communication Problem alarm clear?

39.1 If yes, go to step step 40. Release the Forced/Lockout EPS Switch using the Craft Terminal

39.2 If no, contact the next level of technical support for assistance.

40 Release the Forced/Lockout EPS Switch using the Craft Terminal.

> **⚠ CAUTION!**
>
> Releasing Forced/Lockout switch may cause a disruption to traffic.

41 Release the Forced/Lockout HSB Switch using the Craft Terminal.

42 Release the Forced/Lockout Rx Radio Switch using the Craft Terminal.

The procedure is complete.

48.4 Procedure (UBT)

Procedure

1 Check other alarms presence that identify an issue on UBT traffic port.

If such alarms exist, check cable/fibers and traffic port SFP.

2 If there are no traffic port alarms, restart both UBT channels H and V.

For more information, see *WebCT User Manual.*

3 If Inter Radio Unit Communication Failure alarm has been raised, check if the radios in XPIC have been port segregated and de-provision port segregation.

4 If the alarm is not cleared, restart the NE.

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49. LAG port down alarms

49.1 Purpose

This procedure describes how to identify and clear LAG Port Down alarms.

49.2 General

A LAG Port Down alarm indicates the indited radio channel was removed from a L1 LAG group.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

provides information about LAG Port Down alarms.

Table 50: LAG Port Down alarms

Friendly name/entity	Equipment type
Radio/Dir#[3-8].[5-8]/Slot#[3-8]/Port#[5-8]/Ch#1	MPT-HLC or MPT-HLS connected to EASv2
N/A	UBT-S, UBT-m, UBT-T, UBT-I

49.3 Procedure

Procedure

1 Check from the LAG port status the Active alarms and Alarm history (save for escalation).

2 From the Monitoring and Maintenance page: Capture RSL data for all members involved in

LAG from both sides.

3 Export I&C Parameters from the Monitoring and Maintenance page.

4 Review for Fades, EW, Clock Fail conditions that might have been transient (save for escalation).

5 If there are not LAG Down Alarms, proceed to isolating loss of traffic:

5.1 Check Ethernet Stats that pertain to LAG traffic (save for escalation).

TX LAG and Ethernet access ports can also be included.

5.2 Capture PM data regarding BBE, ES, SES for both sides (save for escalation).

5.3 Get the LAG debug tool. Run once on both sides. Capture and save for review. After issue presents itself again, run tool again, capture and save for review.

> **Note:**
> Do this before troubleshooting. Keep to provide to escalation if needed.

5.4 To isolate or determine the member, it is best to mute and unmute a single channel at a time. Not required to mute both ends as a single-sided mute will cause the member on both sides to be taken out of the LAG.

6

6 Get configuration files and release identification (save for escalation).

6 Get remote inventory for sites involved (save for escalation).

If unable to isolate where the fragments are getting lost, contact TAC.

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50. LAG Rx Problem

50.1 Purpose

This mpprocedure describes how to identify and clear LAG Rx Problem alarm.

50.2 General

In this case there is ot a LAG traffic loss alarm. It is just reported as traffic loss.

50.3 Procedure

Procedure

1 Check from the LAG port status the Active alarms and Alarm history (save for escalation).

2 From the Monitoring and Maintenance page: Capture RSL data for all members involved in LAG from both sides.

3 Export I&C Parameters from the Monitoring and Maintenance page.

4 Review for Fades, EW, Clock Fail conditions that might have been transient (save for escalation).

5 If there are not LAG Down Alarms, proceed to isolating loss of traffic:

5.1 Check Ethernet Stats that pertain to LAG traffic (save for escalation).

TX LAG and Ethernet access ports can also be included.

5.2 Capture PM data regarding BBE, ES, SES for both sides (save for escalation).

5.3 Get the LAG debug tool. Run once on both sides. Capture and save for review. After

issue presents itself again, run tool again, capture and save for review.

> **Note:**
>
> Do this before troubleshooting. Keep to provide to escalation if needed.

5.4 To isolate or determine the member, it is best to mute and unmute a single channel at a time. Not required to mute both ends as a single-sided mute will cause the member on both sides to be taken out of the LAG.

6

6 Get configuration files and release identification (save for escalation).

6 Get remote inventory for sites involved (save for escalation).

If unable to isolate where the fragments are getting lost, contact TAC.

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51. LAG Traffic Down

51.1 Purpose

This proceure describes how to identify and clear a LAG traffic Down alarm.

51.2 General

In this case there is Not a LAG traffic loss alarm. It is just reported as traffic loss.

51.3 Procedure

Procedure

1 Check from the LAG port status the Active alarms and Alarm history (save for escalation).

2 From the Monitoring and Maintenance page: Capture RSL data for all members involved in LAG from both sides.

3 Export I&C Parameters from the Monitoring and Maintenance page.

4 Review for Fades, EW, Clock Fail conditions that might have been transient (save for escalation).

5 If there are not LAG Down Alarms, proceed to isolating loss of traffic:

5.1 Check Ethernet Stats that pertain to LAG traffic (save for escalation).

TX LAG and Ethernet access ports can also be included.

5.2 Capture PM data regarding BBE, ES, SES for both sides (save for escalation).

5.3 Get the LAG debug tool. Run once on both sides. Capture and save for review. After

issue presents itself again, run tool again, capture and save for review.

> **Note:**
>
> Do this before troubleshooting. Keep to provide to escalation if needed.

5.4 To isolate or determine the member, it is best to mute and unmute a single channel at a time. Not required to mute both ends as a single-sided mute will cause the member on both sides to be taken out of the LAG.

6

6 Get configuration files and release identification (save for escalation).

6 Get remote inventory for sites involved (save for escalation).

If unable to isolate where the fragments are getting lost, contact TAC.

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52. LAG Traffic Loss

52.1 Purpose

This procedure describes how to identify and clear LAG Traffic Loss.

52.2 General

Reports packet loss due to an extreme latency difference over L1 LAG members or due to errors on the data path. It is linked to missorder or fragments for example one fragment has been lost.

52.3 Procedure

Procedure

1 Check from the LAG port status the Active alarms and Alarm history (save for escalation).

2 From the Monitoring and Maintenance page: Capture RSL data for all members involved in LAG from both sides.

3 Export I&C Parameters from the Monitoring and Maintenance page.

4 Review for Fades, EW, Clock Fail conditions that might have been transient (save for escalation).

5 If there are not LAG Down Alarms, proceed to isolating loss of traffic:

   5.1 Check Ethernet Stats that pertain to LAG traffic (save for escalation).

   TX LAG and Ethernet access ports can also be included.

   5.2 Capture PM data regarding BBE, ES, SES for both sides (save for escalation).

5.3 Get the LAG debug tool. Run once on both sides. Capture and save for review. After issue presents itself again, run tool again, capture and save for review.

> **Note:**
>
> Do this before troubleshooting. Keep to provide to escalation if needed.

5.4 To isolate or determine the member, it is best to mute and unmute a single channel at a time. Not required to mute both ends as a single-sided mute will cause the member on both sides to be taken out of the LAG.

6

6 Get configuration files and release identification (save for escalation).

6 Get remote inventory for sites involved (save for escalation).

If unable to isolate where the fragments are getting lost, contact TAC.

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53. LAG TX Problem

53.1 Purpose

This procedure describes how to identify and clear LAG TX Problem alarm.

53.2 General

This alarm reports traffic loss due to issues at TX blocks or to malformed packets.

53.3 Procedure

Procedure

1 Check from the LAG port status the Active alarms and Alarm history (save for escalation).

2 From the Monitoring and Maintenance page: Capture RSL data for all members involved in LAG from both sides.

3 Export I&C Parameters from the Monitoring and Maintenance page.

4 Review for Fades, EW, Clock Fail conditions that might have been transient (save for escalation).

5 If there are not LAG Down Alarms, proceed to isolating loss of traffic:

5.1 Check Ethernet Stats that pertain to LAG traffic (save for escalation).

TX LAG and Ethernet access ports can also be included.

5.2 Capture PM data regarding BBE, ES, SES for both sides (save for escalation).

5.3 Get the LAG debug tool. Run once on both sides. Capture and save for review. After

issue presents itself again, run tool again, capture and save for review.

> **Note:**
>
> Do this before troubleshooting. Keep to provide to escalation if needed.

5.4 To isolate or determine the member, it is best to mute and unmute a single channel at a time. Not required to mute both ends as a single-sided mute will cause the member on both sides to be taken out of the LAG.

6

6 Get configuration files and release identification (save for escalation).

6 Get remote inventory for sites involved (save for escalation).

If unable to isolate where the fragments are getting lost, contact TAC.

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54. Link Identifier Mismatch alarms

54.1 Purpose

This procedure describes how to identify and clear Link Identifier Mismatch alarms.

54.2 General

A Link Identifier Mismatch alarm indicates that the link identifier number provisioned on MPT -HLC/MPT-HLS/UBT-T/UBT-S/UBT-m settings screen is different from the link identifier number provisioned at the far end of the hop.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Link Identifier Mismatch alarms provides information about Link Identifier Mismatch alarms.

Table 51: Link Identifier Mismatch alarms

Friendly name/entity	Facility type
AirFrame/Dir#1.[7-8]/MSS/CORE/Port#[7-8]/Ch#1 AirFrame/Dir#1.[7-8]/MSS/CORE/Port#[7-8]	MPT-HLC or MPT-HLS connected to CorEvo
AirFrame/Dir#[3-8].[5-8]/Slot#[3-8]/Port#[5-8]/Ch#1 AirFrame/Dir#[3-8].[5-8]/Slot#[3-8]/Port#[5-8]	MPT-HLC or MPT-HLS connected to EASv2
N/A	UBT-S UBT-m UBT-T channel A UBT-T channel B UBT-I

54.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Link Identifier Mismatch alarms for information about radio alarms.

3 Using the Craft Terminal, retrieve the provisioned Link Identifier values for the near end and far end of the hop for the alarmed MPT-HLC/MPT-HLS.

4 Verify the required Link Identifier value of the microwave hop.

5 Determine which Link Identifier values are incorrect.

6 Reprovision the Link Identifier values according to the results of step 5.

7 Retrieve the alarms using the Craft Terminal's Active Alarms.

8 Did the Link Identifier Mismatch alarm clear?

   1. If yes, the procedure is complete.
   2. If no, contact the next level of technical support for assistance.

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55. Log Upload failed

55.1 Purpose

This procedure describes how to identify and clear a Log upload failed alarm.

55.2 General

Reported when SFTP failed or due to internal SW error.

55.3 Procedure

Procedure

1 Restart the NE.

2 Check the network connectivity.

3 Check the external server.

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56. Loop Problem alarms

56.1 Purpose

This procedure describes how to identify and clear Loop Problem alarms on radios.

56.2 General

A Loop Problem alarm indicates that loss of ATPC command path between the far end transmitter and local receiver.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Loop Problem alarms provides information about Loop Problem alarms.

Table 52: Loop Problem alarms

Friendly name/entity	Equipment type
Radio/Dir#1.[7-8]/Slot#1/Port#[7-8]/Ch#1	MPT-HLC or MPT-HLS connected to CorEvo
Radio/Dir#[3-8].[5-8]/Slot#[3-8]/Port#[5-8]/Ch#[0-1]	MPT-HLC or MPT-HLS connected to EASv2

56.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarm.

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2 Review the output. See Table: Loop Problem alarms for information.

3 Is the system configured with protection?

3.1 If yes, go to step 4.

3.2 If no, go to step 14.

4 Is the card in alarm the main or spare?

4.1 If main, go to step 5.

4.2 If spare, go to step 9.

5 Perform a Forced EPS Switch at the near end transmitter using the Craft Terminal.

> **⚠ CAUTION!**
>
> Forced/Lockout switching traffic causes a disruption to traffic.

6 Perform a Forced HSB Switch at the near end transmitter using the Craft Terminal.

7 Perform a Forced Rx Radio Switch at the near end transmitter using the Craft Terminal.

8 Go to step 12.

9 Perform a Lockout EPS Switch at the near end transmitter using the Craft Terminal.

> **⚠ CAUTION!**
>
> Forced/Lockout switching traffic causes a disruption to traffic.

10 Perform a Lockout HSB Switch at the near end transmitter using the Craft Terminal.

11 Perform a Lockout Rx Radio Switch at the near end transmitter using the Craft Terminal.

12 Retrieve the alarms using the Craft Terminal's Active Alarms.

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13 Did the Loop Problem alarm clear?

13.1 If yes, go to step 14.

13.2 If no, go to step 26.

14 Replace the MPT-HLC, or MPT-HLS in alarm at the near end. See the Wavence Hardware Installation and Replacement Manual.

> **⚠ CAUTION!**
>
> Replacing a card in an unprotected facility causes a loss of traffic on ALL facilities that are associated with the card being replaced. Schedule an appropriate maintenance window per local practices and procedures.

15 Is the system configured with protection?

15.1 If yes, go to step 16.

15.2 If no, go to step 24.

16 Is the card in alarm the main or spare?

16.1 If main, go to step 17.

16.2 If spare, go to step 21.

17 Release the Forced EPS Switch at the near end transmitter using the Craft Terminal.

18 Release the Forced HSB Switch at the near end transmitter using the Craft Terminal.

19 Release the Forced Rx Radio Switch at the near end transmitter using the Craft Terminal.

20 Go to step 24.

21 Release the Lockout EPS Switch at the near end transmitter using the Craft Terminal.

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22 Release the Lockout HSB Switch at the near end transmitter using the Craft Terminal.

23 Release the Lockout Rx Radio Switch at the near end transmitter using the Craft Terminal.

24 Retrieve the alarms using the Craft Terminal's Active Alarms.

25 Did the Loop Problem alarm return?

25.1 If yes, go to step 26.

25.2 If no, the procedure is complete.

26 Is the system configured with protection?

26.1 If yes, go to step 27.

26.2 If no, go to step 37.

27 Is the card in alarm main or spare?

27.1 If main, go to step 28.

27.2 If spare, go to step 32.

⚠ CAUTION!

Forced/Lockout switching traffic causes a disruption to traffic.

28 Perform a Forced EPS Switch at the far end transmitter using the Craft Terminal.

29 Perform a Forced HSB Switch at the far end transmitter using the Craft Terminal.

30 Perform a Forced Rx Radio Switch at the far end transmitter using the Craft Terminal.

31 Go to step 35.

32 Perform a Lockout EPS Switch at the far end transmitter using the Craft Terminal.

33 Perform a Lockout HSB Switch at the far end transmitter using the Craft Terminal.

34 Perform a Lockout Rx Radio Switch at the far end transmitter using the Craft Terminal.

35 Retrieve the alarms using the Craft Terminal's Active Alarms.

36 Did the Loop Problem alarm clear?

36.1 If yes, go to step 37.

36.2 If no, contact the next level of technical support for assistance.

37 Replace the MPT-HLC, or MPT-HLS at the far end. See the Wavence Hardware Installation and Replacement Manual.

⚠ CAUTION!

Replacing a card in an unprotected facility causes a loss of traffic on ALL facilities that are associated with the card being replaced. Schedule an appropriate maintenance window per local practices and procedures.

38 Is the system configured with protection?

38.1 If yes, go to step 39.

38.2 If no, go to step 47.

39 Is the card in alarm the main or spare?

39.1 If main, go to step 40.

39.2 If spare, go to step 44.

40 Release the Forced EPS Switch at the far end transmitter using the Craft Terminal.

41 Release the Forced HSB Switch at the far end transmitter using the Craft Terminal.

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42 Release the Forced Rx Radio Switch at the far end transmitter using the Craft Terminal.

43 Go to step 47.

44 Release the Lockout EPS Switch at the far end transmitter using the Craft Terminal.

45 Release the Lockout HSB Switch at the far end transmitter using the Craft Terminal.

46 Release the Lockout Rx Radio Switch at the far end transmitter using the Craft Terminal.

47 Retrieve the alarms using the Craft Terminal's Active Alarms.

48 Did the Loop Problem alarm return?

48.1 If yes, contact the next level of technical support for assistance.

48.2 If no, the procedure is completed.

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57. Loss of alignment alarms

57.1 Purpose

This procedure describes how to identify and clear Loss Of Alignment alarm on the radio.

57.2 General

A Loss Of Alignment alarm indicates that a delay/loss of alignment between the main and protect RF paths is detected.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Loss of alignment alarms provides information about Loss of Alignment alarms.

Table 53: Loss of alignment alarms

Friendly name/entity	Equipment type
Hitless/Dir#1.[7-8]	MPT-HLC or MPT-HLS connected to CorEvo
Hitless/Dir#[3-8].[5-8]	MPT-HLC or MPT-HLS connected to EASv2

57.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

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2 Review the output. See Table: Loss of alignment alarms for information.

3 Verify that the MPT-HLC/MPT-HLS in alarm status is *Standby*.

4 Is the MPT-HLC/MPT-HLS in alarm main or spare?

4.1 If main, go to step step 5. Perform a Forced EPS Switch using the Craft Terminal.

4.2 If spare, go to step step 9. Perform a Lockout EPS Switch using the Craft Terminal.

5 Perform a Forced EPS Switch using the Craft Terminal.

⚠ CAUTION!

Forced/Lockout switching traffic causes a disruption to traffic.

6 Perform a Forced HSB Switch using the Craft Terminal.

7 Perform a Forced Rx Radio Switch using the Craft Terminal.

8 Go to step step 12. For MPT-HLC/MPT-HLS.

9 Perform a Lockout EPS Switch using the Craft Terminal.

⚠ CAUTION!

Forced/Lockout switching traffic causes a disruption to traffic.

10 Perform a Lockout HSB Switch using the Craft Terminal.

11 Perform a Lockout Rx Radio Switch using the Craft Terminal.

12 For MPT-HLC/MPT-HLS

Position the MPT-HLC/MPT-HLS Transceiver PA switch to the OFF ‘0" position.

13 Remove the fuse in the power distribution panel that supplies battery power to the MPT

-HLC/MPT-HLS transceiver.

14 Install the fuse in the power distribution panel that supplies battery power to the MPT -HLC/MPT-HLS transceiver.

15 Position the MPT-HLC/MPT-HLS Transceiver PA switch to the ON ‘1" position.

16 Wait for the status LED to turn steady amber.

17 Retrieve the alarms using the Craft Terminal’s Active Alarms.

18 Did the Loss Of Alignment alarm clear?

18.1 If yes, go to step step 23. Release the Forced/Lockout EPS Switch using the Craft Terminal.

18.2 If no, go to step step 19. Replace the MPT-HLC/MPT-HLS in alarm. See the Wavence Hardware Installation and Replacement Manual to replace the MPT-HLC/MPT-HLS.

19 Replace the MPT-HLC/MPT-HLS in alarm. See the Wavence Hardware Installation and Replacement Manual to replace the MPT-HLC/MPT-HLS.

20 Wait for the status LED to turn steady amber.

21 Retrieve the alarms using the Craft Terminal’s Active Alarms.

22 Did the Loss Of Alignment alarm clear?

22.1 If yes, go to step step 23. Release the Forced/Lockout EPS Switch using the Craft Terminal.

22.2 If no, contact the next level of technical support for assistance.

23 Release the Forced/Lockout EPS Switch using the Craft Terminal.

24 Release the Forced/Lockout HSB Switch using the Craft Terminal.

25 Release the Forced/Lockout Rx Radio Switch using the Craft Terminal.

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The procedure is completed.

58. Loss Ethernet Synchronization Message Channel alarms

58.1 Purpose

This procedure describes how to identify and clear Loss Ethernet Synchronization Message Channel (ESMC) alarms.

58.2 General

A Loss ESMC alarm indicates a loss of communication with the far end synchronization source or a provisioning mismatch with the far end Ethernet port or radio channel. Loss ESMC is applicable to the CorEvo Ethernet ports and Radio interfaces (MPT-HLC/MPT-HLS/UBT-T/UBT-S/UBT-m) configured with SSM support enabled and is assigned as a synchronous clock source input.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Loss ESMC alarms provides information about Loss ESMC alarms.

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Table 54: Loss ESMC alarms

Friendly name/entity	Equipment type
MAU/Slot#[1-2]/Port#[1-8]	CorEvo
MAU/Slot#[3-8]/Port#[1-8]	EASv2
RadioLAG#[1-14]	MPT-HLC, MPT-HLS
Radio/Dir#1.[7-8]/Slot#1/Port#[7-8]/Ch#1	MPT-HLC or MPT-HLS connected to CorEvo
Radio/Dir#[3-8].[5-8]/Slot#[3-8]/Port#[5-8]/Ch#[0-1]	MPT-HLC or MPT-HLS connected to EASv2
	UBT-S UBT-m UBT-T UBT-I
Ethernet/Slot#[1]/Port#[1-5]	MSS-E User Ethernet Ports
Ethernet/Slot#[1]/Port#[1-8]	MSS-HE/XE User Ethernet Ports

58.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Loss ESMC alarms for information.

3 Retrieve the alarms using the Craft Terminal's Active Alarms.

4 Verify no radio or Ethernet alarms are declared against the radio direction declaring Loss ESMC.

5 Are radio or Ethernet alarms declared against the radio direction or Ethernet port?

   5.1 If yes, see Alarm and abnormal condition clearing procedures to clear radio or Ethernet alarms declared against the radio direction or Ethernet port.

5.2 If no, go to step 6.

6 Using the Craft Terminal retrieve the provisioned Synchronization SSM values for both near end and far end of the Ethernet port or radio hop for the alarmed synchronization port supported on a CorEvo/EASv2/MPT-HLC/MPT-HLS/MSS-E/MSS-HE/XE.

7 Verify the desired Synchronization SSM value of the microwave hop.

8 Determine which Synchronization SSM value is correct.

9 Reprovision the Synchronization SSM value according to the results of step 8.

10 Retrieve the alarms using the Craft Terminal's Active Alarms.

11 Did the Loss ESMC alarm clear?

11.1 If yes, the procedure is complete.

11.2 If no, contact the next level of technical support for assistance.

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59. Loss Of CESoETH Frame alarms

59.1 Purpose

This procedure describes how to identify and clear Loss Of CESoETH Frame alarms.

59.2 General

A Loss Of CESoETH Frame (CircuitTDM) indicates packets are not being received by the emulation circuits from the switch matrix.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence User Manual for information about how to execute the commands that are described in this procedure.

Table: Loss Of CESoETH Frame alarms provides information about Loss Of CESoETH Frame alarms.

Table 55: Loss Of CESoETH Frame alarms

Friendly name/entity	Facility type
CircuitTDM/Slot#[3-8]/Port#[1-32]	DS1 on P32E1DS1
CircuitTDM/Slot#[3-8]/Port#[1-32]	E1 on P32E1DS1
CircuitTDM/Slot#[3-8]/Port#[1-2]	DS3
CircuitTDM/Slot#[3-8]/Port#[1-2]	STM-1/OC-3
CircuitTDM#[1-3]-[1-7]-[1-3]/Slot#[3-8]/Port#1	Channelized E1
DS1-out/Slot#[1]/Port#[1-16]	DS1 on MSS-E, MSS-HE, MSS-XE shelf
E1-out/Slot#[1]/Port#[1-16]	E1 on MSS-E, MSS-HE, MSS-XE shelf

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59.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Loss Of CESoETH Frame alarms for information about facility alarms.

3 Verify that there are no current path or circuit alarms associated with the facility in alarm for the near end, the far end, and all intermediate NEs.

4 Verify that the radio path PM is not taking errors to traffic.

5 Are there path or circuit alarms, or path PM errors at the near end or far end NE?

   5.1 If yes, clear the path or circuit alarms, or path PM errors, then go to step 6.

   5.2 If no, go to step 8.

6 Retrieve the alarms using the Craft Terminal's Active Alarms.

7 Did the Loss Of CESoETH Frame alarm clear?

   7.1 If yes, the procedure is complete.

   7.2 If no, go to step 8.

8 Verify that the circuit provisioning for near end, the far end, and all intermediate NEs:

   8.1 Verify that the Signal Mode is set the same and is correct.

   8.2 Verify that the Line Coding is set the same and is correct.

   8.3 Verify that the Flow ID is set the same and is correct.

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8.4 Verify that the Service Profile is set the same and is correct.

9 Is the Service Profile TDM2ETH?

9.1 If yes, go to step 10.

9.2 If no, go to step 12.

10 Verify that the TDM2ETH provisioning for the near end, far end, and all intermediate NEs:

10.1 Verify that the Payload Size is set the same and is correct.

10.2 Verify that the TDM Clock Source is set the same and is correct.

10.3 Verify that the near end ECID Tx and far end ECID Rx are set the same and are correct.

10.4 Verify that the far end ECID Tx and near end ECID Rx are set the same and are correct.

11 Verify that the cross-connection provisioning for the near end, far end, and all intermediate NEs:

12 Retrieve the alarms using the Craft Terminal's Active Alarms.

13 Did the Loss Of CESoETH Frame alarm clear?

13.1 If yes, the procedure is complete.

13.2 If no, go to step 14.

14 Is the facility associated with a LAG?

14.1 If yes, go to step 15.

14.2 If no, go to step 18.

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15 Verify that there are no active LAG alarms at all NEs in the LAG associated with the facility in alarm.

16 Retrieve the alarms using the Craft Terminal's Active Alarms.

17 Did the Loss Of CESoETH Frame alarm clear?

17.1 If yes, the procedure is complete.

17.2 If no, go to step 18.

18 Is the facility associated with an ERP (Ring)?

18.1 If yes, go to step 19.

18.2 If no, go to step 22.

19 Verify that there are no active RING alarms at all NEs in the ring associated with the facility in alarm.

20 Retrieve the alarms using the Craft Terminal's Active Alarms.

21 Did the Loss Of CESoETH Frame alarm clear?

21.1 If yes, the procedure is complete.

21.2 If no, go to step 22.

22 Is it appropriate to proceed with the Loss Of CESoETH Frame procedure?

22.1 If yes, go to step 23.

22.2 If no, contact the next level of technical support for assistance.

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⚠ CAUTION!

The following steps cause a loss of traffic on ALL facilities supported by the card, associated with the facility in alarm. Schedule an appropriate maintenance window per local practices and procedures.

23 Is the facility protected?

23.1 If yes, go to step 36.

23.2 If no, go to step 24.

24 For the P32E1DS1 or P2E3DS3 cards, reseat the card at the near end. Physically remove the card and reinsert.

⚠ CAUTION!

Reseating a card in an unprotected facility causes a loss of traffic on ALL facilities that are associated with the card being reseated. Schedule an appropriate maintenance window per local practices and procedures.

25 Retrieve the alarms using the Craft Terminal's Active Alarms.

26 Did the Loss Of CESoETH Frame alarm clear?

26.1 If yes, the procedure is complete.

26.2 If no, go to step 27.

27 Replace the card that reported the Loss Of CESoETH Frame alarm. See the Wavence Hardware Installation and Replacement Manual.

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⚠ CAUTION!

Replacing a card in an unprotected facility causes a loss of traffic on ALL facilities that are associated with the card being replaced. Schedule an appropriate maintenance window per local practices and procedures.

28 Retrieve the alarms using the Craft Terminal's Active Alarms.

29 Did the Loss Of CESoETH Frame alarm clear?

29.1 If yes, the procedure is complete.

29.2 If no, go to step 30.

30 For the P32E1DS1 or P2E3DS3 cards, reseat the card at the far end. Physically remove the card and reinsert.

⚠ CAUTION!

Reseating a card in an unprotected facility causes a loss of traffic on ALL facilities that are associated with the card being reseated. Schedule an appropriate maintenance window per local practices and procedures.

31 Retrieve the alarms using the Craft Terminal's Active Alarms.

32 Did the Loss Of CESoETH Frame alarm clear?

32.1 If yes, the procedure is complete.

32.2 If no, go to step 33.

33 Replace the far end card. See the Wavence Hardware Installation and Replacement Manual.

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⚠ CAUTION!

Replacing a card in an unprotected facility causes a loss of traffic on ALL facilities that are associated with the card being replaced. Schedule an appropriate maintenance window per local practices and procedures.

34 Retrieve the alarms using the Craft Terminal's Active Alarms.

35 Did the Loss Of CESoETH Frame alarm clear?

35.1 If yes, the procedure is complete.

35.2 If no, contact the next level of technical support for assistance.

36 Is the DS1/DS3/OC-3 facility in alarm associated with the main or spare?

36.1 If main, perform a Forced EPS Switch at the near end using a Craft Terminal.

36.2 If spare, perform a Lockout EPS Switch at the near end using a Craft Terminal.

⚠ CAUTION!

Forced/Lockout switching traffic causes a disruption to traffic.

37 Reseat the card in alarm at the near end. Physically remove the card and reinsert.

38 Retrieve the alarms using the Craft Terminal's Active Alarms.

39 Did the Loss Of CESoETH Frame alarm clear?

39.1 If yes, go to step 51.

39.2 If no, go to step 40.

40 Replace the card that reported the Loss Of CESoETH Frame. See the Wavence Hardware Installation and Replacement Manual.

41 Retrieve the alarms using the Craft Terminal's Active Alarms.

42 Did the Loss Of CESoETH Frame condition clear?

42.1 If yes, go to step 51.

42.2 If no, go to step 43.

43 Is the card in alarm the main or spare?

43.1 If main, perform a Forced EPS Switch at the far end using a Craft Terminal.

43.2 If spare, perform a Lockout EPS Switch at the far end using a Craft Terminal.

⚠ CAUTION!

Forced/Lockout switching traffic causes a disruption to traffic.

44 Reseat the card in alarm at the far end. Physically remove the card and reinsert.

45 Retrieve the alarms using the Craft Terminal's Active Alarms.

46 Did the Loss Of CESoETH Frame alarm clear?

46.1 If yes, go to step 50.

46.2 If no, go to step 47.

47 Replace the far end card. See the Wavence Hardware Installation and Replacement Manual.

48 Retrieve the alarms using the Craft Terminal's Active Alarms.

49 Did the Loss Of CESoETH Frame condition clear?

49.1 If yes, go to step 50.

49.2 If no, contact the next level of technical support for assistance.

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50 Release the Forced/Lockout EPS Switch at the far end using a Craft Terminal.

> **⚠ CAUTION!**
>
> Releasing Forced/Lockout switch may cause a disruption to traffic.

51 Release the Forced/Lockout EPS Switch at the near end using a Craft Terminal.

> **⚠ CAUTION!**
>
> Releasing Forced/Lockout switch may cause a disruption to traffic.

52 The procedure is complete.

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60. Loss Of Frame alarms on facilities

This procedure describes how to identify and clear Loss Of Frame alarms on facilities. For Loss of Frame alarms on radios, see Loss of frame alarms on radios.

60.1 General

A Loss Of Frame (E1/DS1/3-in) indicates no E1, DS1 or DS3 frame is detected on the input from the client to the P32E1DS1 or P2E3DS3 cards, MSS-E or MSS-HE/XE shelf.

A Loss Of Frame (E1/DS1/3-out) indicates no E1, DS1, or DS3 frame is detected on an E1/DS1 or DS3 facility from the switch matrix.

A Loss Of Frame (EleSdh-in) indicates no STM-1 or OC-3 frame is detected on the input from the client to the SDHACC card.

A Loss Of Frame (EleSdh-out) indicates no STM-1 or OC-3 frame is detected on an STM-1 or OC-3 facility from the switch matrix.

A Loss Of Frame (MS) indicates no STM-1 frame is detected on the input from the client to the SDHACC card on channelized E1 facility.

A Loss Of Frame (E1-out) indicates no E1 frame is detected on a channelized E1 facility from the switch matrix.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence User Manual for information about how to execute the commands that are described in this procedure.

Table: Loss Of Frame alarms provides information about Loss Of Frame alarms on facilities.

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Table 56: Loss Of Frame alarms

Friendly name/entity	Facility type
DS1-in/Slot#[3-8]/Port#[1-32]	DS1 on P32E1DS1
E1-in/Slot#[3-8]/Port#[1-32]	E1 on P32E1DS1
DS3-in/Slot#[3-8]/Port#[1-2]	DS3
EleSdh-in/Slot#[3-8]/Port[1-2]	STM-1/OC-3
OptSdh-in/Slot#[3-8]/Port[1-2]	
MS/Slot#[3-8]/Port#1	STM-1
DS1-out/Slot#[3-8]/Port#[1-32]	DS1 on P32E1DS1
E1-out/Slot#[3-8]/Port#[1-32]	E1 on P32E1DS1
DS3-out/Slot#[3-8]/Port#[1-2]	DS3
EleSdh-out/Slot#[3-8]/Port[1-2]	STM-1/OC-3
OptSdh-out/Slot#[3-8]/Port[1-2]	
E1-out#[1-3]-[1-7]-[1-3]/Slot#[3-8]/Port#1	Channelized E1
DS1-out/Slot#[1]/Port#[1-16]	DS1 on MSS-E / MSS-HE/XE shelf
E1-out/Slot#[1]/Port#[1-16]	E1 on MSS-E / MSS-HE/XE shelf
DS1-in/Slot#[1]/Port#[1-16]	DS1 on MSS-E / MSS-HE/XE shelf
E1-in/Slot#[1]/Port#[1-16]	E1 on MSS-E / MSS-HE/XE shelf

60.2 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Loss Of Frame alarms for information about facility alarms.

3 Troubleshoot any of the following:

1. Loss of frame on E1/DS1/DS3/EleSdh/OptSdh-in facility
2. Loss of frame on DS1/DS3/E1/Ele/OptSdh-out facility

60.2.1 Loss of frame on E1/DS1/DS3/EleSdh/OptSdh-in facility

Procedure

1 Verify that there are no active circuit alarms associated with the facility in alarm at the near end and client end NE of the facility.

2 Are there active circuit alarms at the near end or client end of the facility?

2.1 If yes, clear circuit alarms, then go to step 3.

2.2 If no, go to step 5.

3 Retrieve the alarms using the Craft Terminal's Active Alarms.

4 Did the Loss Of Frame alarm the clear?

4.1 If yes, the procedure is complete.

4.2 If no, go to step 5.

5 Verify that the circuit provisioning for both the near end and client end of the facility in alarm:

5.1 Verify that the Signal Mode is set the same and is correct.

5.2 Verify that the Line Coding is set the same and is correct.

5.3 Verify that the Flow ID is set the same and is correct.

5.4 Verify that the Flow ID is set the same and is correct.

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5.5 Verify that the LBO setting is correct for the installation.

6 Retrieve the alarms using the Craft Terminal's Active Alarms.

7 Did the Loss Of Frame alarm clear?

7.1 If yes, the procedure is complete.

7.2 If no, go to step 8.

8 Verify that the Source Signal is error-free with a test set configured to match the provisioning of the facility.

9 Retrieve the alarms using the Craft Terminal's Active Alarms.

10 Did the Loss Of Frame alarm clear?

10.1 If yes, the procedure is complete.

10.2 If no, go to step 11.

11 Verify the cabling:

11.1 Verify the cabling from the customer interconnect to the patch panel or card as appropriate. Confirm that the connectors are correctly seated and secured.

11.2 Verify the cabling from the patch panel to the module if appropriate. Confirm that the connectors are correctly seated and secured.

12 Retrieve the alarms using the Craft Terminal's Active Alarms.

13 Did the Loss Of Frame alarm clear?

13.1 If yes, the procedure is complete.

13.2 If no, go to step 14.

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14 Is it appropriate to proceed with the Loss Of Frame procedure?

14.1 If yes, go to step 15.

14.2 If no, contact the next level of technical support for assistance.

> **⚠ CAUTION!**
>
> The following steps cause a loss of traffic on ALL facilities that are supported by the card, associated with the facility in alarm. Schedule an appropriate maintenance window per local practices and procedures.

15 Is the facility protected?

15.1 If yes, go to step 22.

15.2 If no, go to step 16.

16 For the P32E1DS1,P2E3DS3, or SDHACC cards, reseat the card at the near end. Physically remove the card and reinsert.

> **⚠ CAUTION!**
>
> Reseating a card in an unprotected facility causes a loss of traffic on ALL facilities that are associated with the card being reseated. Schedule an appropriate maintenance window per local practices and procedures.

17 Retrieve the alarms using the Craft Terminal's Active Alarms.

18 Did the Loss Of Frame alarm clear?

18.1 If yes, the procedure is complete.

18.2 If no, go to step 19.

19 Replace the near end card or shelf. See the Wavence Hardware Installation and Replacement

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Manual.

> **⚠ CAUTION!**
>
> Replacing a card in an unprotected facility causes a loss of traffic on ALL facilities that are associated with the card being replaced. Schedule an appropriate maintenance window per local practices and procedures.

20 Retrieve the alarms using the Craft Terminal's Active Alarms.

21 Did the Loss Of Frame alarm clear?

21.1 If yes, the procedure is complete.

21.2 If no, contact the next level of technical support for assistance.

22 Is the DS1/DS3/E1/OC-3/STM-1 facility in alarm associated with the main or spare?

22.1 If main, perform a Forced EPS Switch at the near end using a Craft Terminal.

22.2 If spare, perform a Lockout EPS Switch at the near end using a Craft Terminal.

> **⚠ CAUTION!**
>
> Forced/Lockout switching traffic causes a disruption to traffic.

23 Reseat the card in alarm at the near end. Physically remove card and reinsert.

24 Retrieve the alarms using the Craft Terminal's Active Alarms.

25 Did the Loss Of Frame alarm clear?

25.1 If yes, go to step 29.

25.2 If no, go to step 26.

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26 Replace the near end card. See the Wavence Hardware Installation and Replacement Manual.

27 Retrieve the alarms using the Craft Terminal's Active Alarms.

28 Did the Loss Of Frame condition clear?

28.1 If yes, go to step 29.

28.2 If no, contact the next level of technical support for assistance.

29 Release the Forced/Lockout EPS Switch at the near end using the Craft Terminal.

> **⚠ CAUTION!**
>
> Releasing Forced/Lockout switch may cause a disruption to traffic.

30 The procedure is complete.

60.2.2 Loss of frame on DS1/DS3/E1/Ele/OptSdh-out facility

Procedure

1 Check the far end facility for Loss of Frame (DS1/3-in, E1, Ele/OptSdh-in) that is associated with the facility in alarm.

2 Retrieve the alarms using the Craft Terminal's Active Alarms.

3 Did the Loss Of Frame alarm clear?

3.1 If yes, the procedure is complete.

3.2 If no, go to step 4.

4 Verify that there are no active path or circuit alarms associated with the facility in alarm for the near end or far end NE.

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5 Verify that the radio path PM is not taking errors to traffic.

6 Are the path or circuit alarms, or path PM errors existing at the near end or far end NE?

6.1 If yes, clear path or circuit alarms or path PM errors, then go to step 7.

6.2 If no, go to step 9.

7 Retrieve the alarms using the Craft Terminal's Active Alarms.

8 Did the Loss Of Frame alarm clear?

8.1 If yes, the procedure is complete.

8.2 If no, go to step 9.

9 Check the circuit provisioning for the near end and far end facility in alarm and that they are correct, as required:

9.1 Verify that the Signal Mode is set the same and is correct.

9.2 Verify that the Line Coding is set the same and is correct.

9.3 Verify that the Flow ID is set the same and is correct.

9.4 Verify that the Service Profile is set the same and is correct.

10 Is the Service Profile TDM2ETH?

10.1 If yes, go to step 11.

10.2 If no, go to step 12.

11 Check the TDM2ETH provisioning for the near end and far end facility in alarm and that it is correct, as required:

11.1 Verify that the Payload Size is set the same and is correct.

11.2 Verify that the TDM Clock Source is set the same and is correct.

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11.3 Verify that the TDM Clock Source is set the same and is correct.

11.4 Verify that the far end ECID Tx and near end ECID Rx are set the same and are correct.

12 Retrieve the alarms using the Craft Terminal's Active Alarms.

13 Did the Loss Of Frame alarm clear?

13.1 If yes, the procedure is complete.

13.2 If no, go to step 14.

14 Check the cross-connection provisioning at the near end, far end, and all intermediated NEs that are associated with the facility in alarm are correct, as required:

14.1 Verify that the Flow ID is set the same and is correct.

14.2 Verify that the MAC address is set the same and is correct.

14.3 Verify that the TDM profile is set correct

14.4 Verify that the TDM Clock Source is set correct.

15 Retrieve the alarms using the Craft Terminal's Active Alarms.

16 Did the Loss Of Frame alarm clear?

16.1 If yes, the procedure is complete.

16.2 If no, go to step 17.

17 Is it appropriate to proceed with the Loss Of Frame procedure?

17.1 If yes, go to step 18.

17.2 If no, contact the next level of technical support for assistance.

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⚠ CAUTION!

The following steps cause a loss of traffic on ALL facilities supported by the card associated with the facility in alarm. Schedule an appropriate maintenance window per local practices and procedures.

18 Is the facility protected?

18.1 If yes, go to step 31.

18.2 If no, go to step 19.

19 For the SDHACC, P32E1DS1, P2E3DS3, or SDHACC cards, reseat the card at the near end. Physically remove the card and reinsert.

⚠ CAUTION!

Reseating a card in an unprotected facility causes a loss of traffic on ALL facilities that are associated with the card being reseated. Schedule an appropriate maintenance window per local practices and procedures.

20 Retrieve the alarms using the Craft Terminal's Active Alarms.

21 Did the Loss Of Frame alarm clear?

21.1 If yes, the procedure is complete.

21.2 If no, go to step 22.

22 Replace the near end shelf or card. See the Wavence Hardware Installation and Replacement Manual.

⚠ CAUTION!

Replacing a card in an unprotected facility causes a loss of traffic on ALL facilities that are associated with the card being replaced. Schedule an appropriate maintenance window per local practices and procedures.

23 Retrieve the alarms using the Craft Terminal's Active Alarms.

24 Did the Loss Of Frame alarm clear?

24.1 If yes, the procedure is complete.

24.2 If no, go to step 25.

25 For the P32E1DS1, P2E3DS3, or SDHACC cards, reseat the card at the far end. Physically remove the card and reinsert.

⚠ CAUTION!

Reseating a card in an unprotected facility causes a loss of traffic on ALL facilities that are associated with the card being reseated. Schedule an appropriate maintenance window per local practices and procedures.

26 Retrieve the alarms using the Craft Terminal's Active Alarms.

27 Did the Loss Of Frame alarm clear?

27.1 If yes, the procedure is complete.

27.2 If no, go to step 28.

28 Replace the far end card or shelf. See the Wavence Hardware Installation and Replacement Manual.

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⚠ CAUTION!

Replacing a card in an unprotected facility causes a loss of traffic on ALL facilities that are associated with the card being replaced. Schedule an appropriate maintenance window per local practices and procedures.

29 Retrieve the alarms using the Craft Terminal's Active Alarms.

30 Did the Loss Of Frame alarm clear?

30.1 If yes, the procedure is complete.

30.2 If no, contact the next level of technical support for assistance.

31 Is DS1/DS3/E1/OC-3/STM-1 facility in alarm associated with main or spare?

31.1 If main, perform a Forced EPS Switch at the near end using the Craft Terminal.

31.2 If spare, perform a Lockout EPS Switch at the near end using the Craft Terminal.

⚠ CAUTION!

Forced/Lockout switching traffic causes a disruption to traffic.

32 Reseat card in alarm at the near end. Physically remove card and reinsert.

33 Retrieve the alarms using the Craft Terminal's Active Alarms.

34 Did the Loss Of Frame alarm clear?

34.1 If yes, go to step 46.

34.2 If no, go to step 35.

35 Replace the near end card. See the Wavence Hardware Installation and Replacement Manual.

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36 Retrieve the alarms using the Craft Terminal's Active Alarms.

37 Did the Loss Of Frame condition clear?

37.1 If yes, go to step 46.

37.2 If no, go to step 38.

38 Is card in alarm main or spare?

38.1 If main, perform a Forced EPS Switch at the far end using the Craft Terminal.

38.2 If spare, perform a Lockout EPS Switch at the far end using the Craft Terminal.

⚠ CAUTION!

Forced/Lockout switching traffic causes a disruption to traffic.

39 Reseat the card in alarm at the far end. Physically remove the card and reinsert.

40 Retrieve the alarms using the Craft Terminal's Active Alarms.

41 Did the Loss Of Frame alarm clear?

41.1 If yes, go to step 45.

41.2 If no, go to step 42.

42 Replace the far end card. See the Wavence Hardware Installation and Replacement Manual.

43 Retrieve the alarms using the Craft Terminal's Active Alarms.

44 Did the Loss Of Frame condition clear?

44.1 If yes, go to step 45.

44.2 If no, contact the next level of technical support for assistance.

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45 Release the Forced/Lockout EPS Switch at the far end using the Craft Terminal.

> **⚠ CAUTION!**
>
> Releasing a Forced/Lockout switch may cause a disruption to traffic.

46 Release the Forced/Lockout EPS Switch at the near end using the Craft Terminal.

> **⚠ CAUTION!**
>
> Releasing a Forced/Lockout switch may cause a disruption to traffic.

47 The procedure is complete.

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61. Loss of frame alarms on radios

This procedure describes how to identify and clear Loss Of Frame alarms on radios. For Loss of Frame alarms on facilities, see Loss Of Frame alarms on facilities

61.1 General

A Loss Of Frame alarm indicates the BER has increased to the point that frames are being lost. Probable causes include far end transmitter problems, RF path problems, or local card failures.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Loss of frame radio alarms provides information about Loss of Frame radio alarms.

Table 57: Loss of frame radio alarms

Friendly name/entity	Equipment type
Radio/Dir#1.[7-8]/Slot#1/Port#[7-8]/Ch#[0-1]	MPT-HLC or MPT-HLS connected to CorEvo
Radio/Dir#[3-8].[5-8]/Slot#[3-8]/Port#[5-8]/Ch#[0-1]	MPT-HLC or MPT-HLS connected to EASv2

61.2 Procedure

Procedure

1 Use the Craft Terminal's Active Alarms to retrieve all alarms.

2 Review the output. See Table: Loss of frame radio alarms for information.

3 Verify that there are no weather-related problems.

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4 Do weather-related problems exist?

4.1 If yes, wait for the weather to clear and recheck alarms. The procedure is complete.

4.2 If no, go to step step 5. Verify that the RF path is clear.

5 Verify that the RF path is clear.

6 Is the RF path clear?

6.1 If yes, go to step step 7. Verify that the RSL is above the RX threshold.

6.2 If no, contact the next level of technical support for assistance.

7 Verify that the RSL is above the RX threshold.

8 Is the RSL above the RX threshold?

8.1 If yes, contact the next level of technical support for assistance.

8.2 If no, go to step step 9. Check the upstream transmitter output power and troubleshoot accordingly.

9 Check the upstream transmitter output power and troubleshoot accordingly.

10 Retrieve the alarms using the Craft Terminal's Active Alarms.

11 Did the Loss Of Frame alarm clear?

11.1 If yes, the procedure is complete.

11.2 If no, go to step step 12. Is the system configured with protection?.

12 Is the system configured with protection?

12.1 If yes, go to step step 33. Is the traffic affected?.

12.2 If no, go to step step 13. Unprotected radio configuration. Is the traffic affected?.

13 Unprotected radio configuration. Is the traffic affected?

13.1 If yes, go to step step 14. Position the local MPT-HLC/MPT-HLS Transceiver PA switch to the OFF ‘0" position.

13.2 If no, contact the next level of technical support for assistance.

14 Position the local MPT-HLC/MPT-HLS Transceiver PA switch to the OFF ‘0" position.

15 Remove fuse in power distribution panel suppling battery power to local MPT-HLC/MPT-HLS Transceiver.

16 Install fuse in power distribution panel suppling battery power to local MPT-HLC/MPT-HLS Transceiver.

17 Position the local MPT-HLC/MPT-HLS Transceiver PA switch to the ON ‘1" position.

18 Did the Loss Of Frame alarm clear?

18.1 If yes, the procedure is complete.

18.2 If no, go to step step 19. Position the far end MPT-HLC/MPT-HLS Transceiver PA switch to the OFF ‘0" position.

19 Position the far end MPT-HLC/MPT-HLS Transceiver PA switch to the OFF ‘0" position.

20 Remove fuse in power distribution panel suppling battery power to far end MPT-HLC/MPT -HLS Transceiver.

21 Install fuse in power distribution panel suppling battery power to far end MPT-HLC/MPT-HLS Transceiver.

22 Position the far end MPT-HLC/MPT-HLS Transceiver PA switch to the ON ‘1" position.

23 Did the Loss Of Frame alarm clear?

23.1 If yes, the procedure is complete.

23.2 If no, go to step step 24. Replace the local MPT-HLC/MPT-HLS. See the Wavence

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Hardware Installation and Replacement Manual to replace the MPT-HLC/MPT-HLS Transceiver.

24 Replace the local MPT-HLC/MPT-HLS. See the Wavence Hardware Installation and Replacement Manual to replace the MPT-HLC/MPT-HLS Transceiver.

25 Did the Loss Of Frame alarm clear?

25.1 If yes, the procedure is complete.

25.2 If no, go to step step 26. Replace the far end MPT-HL. See the Wavence Hardware Installation and Replacement Manual to replace the MPT-HLC/MPT-HLS Transceiver.

26 Replace the far end MPT-HL. See the Wavence Hardware Installation and Replacement Manual to replace the MPT-HLC/MPT-HLS Transceiver.

27 Did the Loss Of Frame alarm clear?

27.1 If yes, the procedure is complete.

27.2 If no, go to step step 28. Verify antenna is aligned.

28 Verify antenna is aligned.

29 Is the antenna aligned?

29.1 If yes, contact the next level of technical support for assistance.

29.2 If no, align the antenna. See Wavence Turn-up Manual, Antenna Polarization Verification Procedure.

30 Retrieve the alarms using the Craft Terminal's Active Alarms.

31 Did the Loss Of Frame alarm clear?

31.1 If yes, the procedure is complete.

31.2 If no, contact the next level of technical support for assistance.

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32 Protected radio configuration. Determine the status of the MPT-HLC/MPT-HLS declaring the Loss Of Frame alarm.

32.1 If Active, go to step step 33. Is the traffic affected?.

32.2 If Standby, go to step step 34. Is the MPT-HLC/MPT-HLS declaring the Loss Of Frame alarm the main or spare?.

33 Is the traffic affected?

33.1 If yes, go to step step 34. Is the MPT-HLC/MPT-HLS declaring the Loss Of Frame alarm the main or spare?.

33.2 If no, contact the next level of technical support for assistance.

34 Is the MPT-HLC/MPT-HLS declaring the Loss Of Frame alarm the main or spare?

34.1 If main, go to step step 35. Perform a Forced Rx Radio Switch using the Craft Terminal on the local MPT-HLC/MPT-HLS.

34.2 If spare, go to step step 39. Perform a Lockout Rx Radio Switch using the Craft Terminal on the local MPT-HLC/MPT-HLS.

35 Perform a Forced Rx Radio Switch using the Craft Terminal on the local MPT-HLC/MPT-HLS.

36 Perform a Forced Equipment Switch using the Craft Terminal on the local MPT-HLC/MPT-HLS.

37 Perform a Forced HSB Switch using the Craft Terminal on the far end MPT-HLC/MPT-HLS.

38 Go to step step 42. MPT-HLC/MPT-HLS. Position the local MPT-HLC/MPT-HLS Transceiver PA switch to the OFF ‘0" position.

39 Perform a Lockout Rx Radio Switch using the Craft Terminal on the local MPT-HLC/MPT-HLS.

40 Perform a Lockout Equipment Switch using the Craft Terminal on the local MPT-HLC/MPT -HLS.

41 Perform a Lockout HSB Switch using the Craft Terminal on the far end MPT-HLC/MPT-HLS.

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42 MPT-HLC/MPT-HLS. Position the local MPT-HLC/MPT-HLS Transceiver PA switch to the OFF ‘0" position.

43 Remove the fuse in the power distribution panel that is suppling battery power to the local MPT-HLC/MPT-HLS transceiver.

44 Install the fuse in the power distribution panel that is suppling battery power to the local MPT-HLC/MPT-HLS transceiver.

45 Position the local MPT-HLC/MPT-HLS Transceiver PA switch to the ON ‘1" position.

46 Did the Loss Of Frame alarm clear?

46.1 If yes, go to step step 60. Release the Forced/Lockout Rx Radio Switch using the Craft Terminal on the local MPT-HLC/MPT-HLS.

46.2 If no, go to step step 47. Position the far end MPT-HLC/MPT-HLS Transceiver PA switch to the OFF ‘0" position.

47 Position the far end MPT-HLC/MPT-HLS Transceiver PA switch to the OFF ‘0" position.

48 Remove the fuse in the power distribution panel that is supplying battery power to the far end MPT-HLC/MPT-HLS transceiver.

49 Install the fuse in the power distribution panel that is suppling battery power to the far end MPT-HLC/MPT-HLS transceiver.

50 Position the far end MPT-HLC/MPT-HLS Transceiver PA switch to the ON ‘1" position.

51 Did the Loss Of Frame alarm clear?

51.1 If yes, go to step step 60. Release the Forced/Lockout Rx Radio Switch using the Craft Terminal on the local MPT-HLC/MPT-HLS.

51.2 If no, go to step step 52. Replace the local MPT-HLC/MPT-HLS. See Wavence Hardware Installation and Replacement Manual to replace the MPT-HLC/MPT-HLS Transceiver.

52 Replace the local MPT-HLC/MPT-HLS. See Wavence Hardware Installation and Replacement

Manual to replace the MPT-HLC/MPT-HLS Transceiver.

53 Did the Loss Of Frame alarm clear?

53.1 If yes, go to step step 60. Release the Forced/Lockout Rx Radio Switch using the Craft Terminal on the local MPT-HLC/MPT-HLS.

53.2 If no, go to step step 54. Replace the far end MPT-HLC/MPT-HLS. See Wavence Hardware Installation and Replacement Manual to replace the MPT-HLC/MPT-HLS Transceiver.

54 Replace the far end MPT-HLC/MPT-HLS. See Wavence Hardware Installation and Replacement Manual to replace the MPT-HLC/MPT-HLS Transceiver.

55 Did the Loss Of Frame alarm clear?

55.1 If yes, go to step step 60. Release the Forced/Lockout Rx Radio Switch using the Craft Terminal on the local MPT-HLC/MPT-HLS.

55.2 If no, go to step step 56. Verify antenna is aligned.

56 Verify antenna is aligned.

57 Is the antenna aligned?

57.1 If yes, contact the next level of technical support for assistance.

57.2 If no, align the antenna. See Wavence Turn-up Manual, Antenna Polarization Verification Procedure.

58 Retrieve the alarms using the Craft Terminal's Active Alarms.

59 Did the Loss Of Frame alarm clear?

59.1 If yes, go to step step 60. Release the Forced/Lockout Rx Radio Switch using the Craft Terminal on the local MPT-HLC/MPT-HLS.

59.2 If no, contact the next level of technical support for assistance.

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60 Release the Forced/Lockout Rx Radio Switch using the Craft Terminal on the local MPT -HLC/MPT-HLS.

61 Release the Forced/Lockout Equipment Switch using the Craft Terminal on the local MPT -HLC.

62 Release the Forced/Lockout HSB Switch using the Craft Terminal on the far end MPT-HLC.

63 The procedure is complete.

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62. Loss of multi-frame alarms

62.1 Purpose

This procedure describes how to identify and clear Loss of Multi-frame alarms.

62.2 General

A Loss of Multi-frame alarm on channelized E1 has a similar root cause to the AU4-AIS and TU12 -AIS alarms on channelized E1.These alarms are reported if other alarms are not present.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence User Manual for information about how to execute the commands that are described in this procedure.

Table: Loss of multi-frame alarms provides information about Loss of Multi-frame alarms.

Table 58: Loss of multi-frame alarms

Friendly name/entity	Facility type
AU4/Slot#[3-8]/Port#1	Channelized E1

62.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Loss of multi-frame alarms for information.

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3 Troubleshoot the source of input STM-1 signal that is associated with the channelized E1 to the SDHCHAN card.

4 Verify the cabling that is associated with the input STM-1 signal to the SDHCHAN card.

5 Retrieve the alarms using the Craft Terminal's Active Alarms.

6 Did the Loss of Multi-frame alarm clear?

1. If yes, the procedure is complete.
2. If no, contact the next level of technical support for assistance.

63. Loss of pointer alarms

63.1 Purpose

This procedure describes how to identify and clear Loss of Pointer alarms.

63.2 General

A Loss of Pointer alarm on channelized E1 has a similar root cause to the AU4-AIS and TU12-AIS alarms on channelized E1.These alarms are reported if other alarms are not present.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence User Manual for information about how to execute the commands that are described in this procedure.

Table: Loss of Pointer alarms provides information about Loss of Pointer alarms.

Table 59: Loss of Pointer alarms

Friendly name/entity	Facility type
AU4/Slot#[3-8]/Port#1	Channelized E1
TU12#[1-3]-[1-7]-[1-3]/Slot#[3-8]/Port#1	Channelized E1

63.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

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2 Review the output. See Table: Loss of Pointer alarms for information.

3 Troubleshoot the source of input STM-1 signal that is associated with the channelized E1 input signal to the SDHCHAN card.

4 Verify the cabling that is associated with the input STM-1 signal to the SDHCHAN card.

5 Retrieve the alarms using the Craft Terminal's Active Alarms.

6 Did the Loss of Pointer alarm clear?

1. If yes, the procedure is complete.
2. If no, contact the next level of technical support for assistance.

64. Loss of protection alarms

64.1 Purpose

This procedure describes how to identify and clear Loss Of Protection alarms.

64.2 General

A Loss Of Protection alarm indicates that a loss the radio direction is affected by a failure that radio protection (RPS) cannot recover. The alarm is only reported if the RPS is provisioned.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Loss Of Protection alarms provides information about Loss Of Protection alarms.

Table 60: Loss Of Protection alarms

Friendly name/entity	Facility type
Radio/Dir#1.[7-8]/ Slot#1/Port#[7-8]/Ch#[0-1]	MPT-HLS connected to CorEvo
Radio/Dir#[3-8].[5-8]/Slot#[3-8]/Port#[5-8]/Ch#1	MPT-HLS connected to EASv2

64.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

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2 Review the output. See Table: Loss Of Protection alarms for information.

3 Retrieve the alarms for the local end of the radio direction using the Craft Terminal's Active Alarms.

4 Determine all of the radio alarms that are declared against the radio direction declaring the Loss Of Protection and troubleshoot accordingly.

5 Retrieve the alarms using the Craft Terminal's Active Alarms.

6 Did the Loss Of Protection alarm clear?

6.1 If yes, the procedure is complete.

6.2 If no, go to step 7.

7 Retrieve the alarms for the far end of the radio direction using the Craft Terminal's Active Alarms.

8 Determine all of the radio alarms that are declared against the radio direction declaring the Loss Of Protection and troubleshoot accordingly.

9 Retrieve the alarms using the Craft Terminal's Active Alarms.

10 Did the Loss Of Protection alarm clear?

10.1 If yes, the procedure is complete.

10.2 If no, contact the next level of technical support for assistance.

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65. Loss of signal alarms on facilities

65.1 Purpose

This procedure describes how to identify and clear Loss Of Signal alarms on PDH or SDH facilities. For other Loss of Signal alarms, see one of the following:

- Loss of signal alarms on Ethernet
- Loss of signal alarms on radios
- Loss of signal synchronization alarms
- Loss of signal alarms on LAG

65.2 General

A Loss Of Signal (E1/DS1/3-in) indicates no DS1, DS3, E1, STM-1, or OC-3 signal is detected on the input of the P32E1DS1, P2E3DS3, SDHACC, cards or MSS-HE/XE, MSS-E shelf.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence User Manual for information about how to execute the commands that are described in this procedure.

Table: Loss Of Signal alarms provides information about Loss Of Signal alarms.

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Table 61: Loss Of Signal alarms

Friendly name/entity	Facility type
DS1-in/Slot#[3-8]/Port#[1-32]	DS1 P32E1DS1
DS3-in/Slot#[3-8]/Port#[1-2]	DS3
E1-in/Slot#[3-8]/Port#[1-32]	E1 P32E1DS1
EleSdh-in/Slot#[3-8]/Port[1-2]	STM-1/OC-3
OptSdh-in/Slot#[3-8]/Port[1-2]	
MS/slot#[3-8]/Port1	STM-1
DS1-in/Slot#[1]/Port#[1-16]	DS1 on MSS-E, MSS-HE/XE shelf
E1-in/Slot#[1]/Port#[1-16]	E1 on MSS-E, MSS-HE/XE shelf

65.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Loss Of Signal alarms for information about facility alarms.

3 Troubleshoot source of input to MSS-E/HE/XE shelf, P32E1DS1, P2E3DS3, or SDHACC card.

4 Verify cabling associated with the input signal to the MSS-E/HE/XE shelf, P32E1DS1, P2E3DS3, or SDHACC card.

5 Retrieve alarms using the Craft Terminal.

6 Did the Loss Of Signal alarm clear?

   6.1 If yes, the procedure is complete.

   6.2 If no, go to step 7.

7 Retrieve alarms using the Craft Terminal. Determine if Loss Of Signal alarm is declared against ALL facilities associated with the MSS-E/HE/XE shelf, P32E1DS1, P2E3DS3, or SDHACC card.

8 Are ALL facilities associated with the MSS-E/HE/XE shelf, P32E1DS1, P2E3DS3, or SDHACC card declaring Loss Of Signal?

8.1 If yes, go to step 9.

8.2 If no, contact the next level of technical support for assistance.

9 Is the facility protected?

9.1 If yes, go to step 16.

9.2 If no, go to step 10.

10 Reseat the card that declares the Loss Of Signal alarm. Physically remove the card and reinsert.

⚠ CAUTION!

Reseating a card in an unprotected facility causes a loss of traffic on ALL facilities that are associated with the card being reseated. Schedule an appropriate maintenance window per local practices and procedures.

11 Retrieve the alarms using the Craft Terminal's Active Alarms.

12 Did the Loss Of Signal alarm clear?

12.1 If yes, the procedure is complete.

12.2 If no, go to step 13.

13 Replace the near end card. See the Wavence Hardware Installation and Replacement Manual.

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⚠ CAUTION!

Replacing a card in an unprotected facility causes a loss of traffic on ALL facilities that are associated with the card being replaced. Schedule an appropriate maintenance window per local practices and procedures.

14 Retrieve the alarms using the Craft Terminal's Active Alarms.

15 Did the Loss Of Signal alarm clear?

15.1 If yes, the procedure is complete.

15.2 If no, contact the next level of technical support for assistance.

16 Is the card in alarm the main or spare?

16.1 If main, perform a Forced EPS Switch at the near end using the Craft Terminal.

16.2 If spare, perform a Lockout EPS Switch at the near end using the Craft Terminal.

⚠ CAUTION!

Forced EPS switching traffic causes a disruption to traffic.

17 Verify that the card in alarm status is *Standby*.

18 Reseat the card that declared the Loss Of Signal alarm. Physically remove the card and reinsert.

19 Retrieve the alarms using the Craft Terminal's Active Alarms.

20 Did the Loss Of Signal alarm clear?

20.1 If yes, go to step 24.

20.2 If no, go to step 21.

21 Replace the near end card. See the Wavence Hardware Installation and Replacement Manual.

22 Retrieve the alarms using the Craft Terminal's Active Alarms.

23 Did the Loss Of Signal alarm clear?

23.1 If yes, go to step 24.

23.2 If no, contact the next level of technical support for assistance.

24 Release the Forced/Lockout EPS Switch at the near end using the Craft Terminal.

⚠ CAUTION!

Releasing EPS switch may cause a disruption to traffic.

25 The procedure is complete.

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66. Loss of signal alarms on Ethernet

66.1 Purpose

This procedure describes how to identify and clear Loss Of Signal on Ethernet ports. For other Loss of Signal alarms, see one of the following:

- Loss of signal alarms on facilities
- Loss of signal alarms on radios
- Loss of signal synchronization alarms
- Loss of signal alarms on LAG

66.2 General

Loss Of Signal alarm indicates that a loss of Ethernet signal is detected on the port or a communication problem with the remote peer; for example, a link down condition.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Loss of Signal alarms on Ethernet provides information about Loss of Signal alarms on Ethernet.

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Table 62: Loss of Signal alarms on Ethernet

Friendly name/entity	Equipment type
TMNEthernet/MngtPort	TMN MGMT Port on CorEvo, MSS-E, MSS-HE, MSS-XE
TMNEthernet/MngtPort#4	TMN Port#4 CorEvo
MAU/Slot#[1]/Port#[1-8]	CorEvo
MAU/Slot#[3-8]/Port#[1-8]	EASv2
Ethernet/Port#2	Optical user Eth (ifMauType.70002)
N/A	Coupler Ethernet Interface: UBT-T UBT-S UBT-m UBT-I

66.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Loss of Signal alarms on Ethernet for information about the Ethernet alarms.

3 Verify that the link partner is operating normally.

4 Retrieve the alarms using the Craft Terminal's Active Alarms.

5 Did the Loss Of Signal alarm clear?

   5.1 If yes, the procedure is complete.

   5.2 If no, go to step 6.

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6 Verify the link partner cabling between the CorEvo/EASv2 card and the link partner.

7 Retrieve the alarms using the Craft Terminal's Active Alarms.

8 Did the Loss Of Signal alarm clear?

8.1 If yes, the procedure is complete.

8.2 If no, contact the next level of technical support for assistance.

67. Loss of signal alarms on radios

67.1 Purpose

This procedure describes how to identify and clear Loss Of Signal alarms on radios. For other Loss of Signal alarms, see one of the following:

- Loss of signal alarms on facilities
- Loss of signal alarms on Ethernet
- Loss of signal synchronization alarms
- Loss of signal alarms on LAG

67.2 General

A Loss Of Signal alarm indicates that a loss of Ethernet signal is detected on the MPT-HLS coupler port.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Loss of Signal alarms on radios provides information about Loss of Signal alarms on radios.

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Table 63: Loss of Signal alarms on radios

Friendly name/entity	Facility type
MAU/subrack[#3-8].[5-8] Port#n	MPT-HLC connected to EASv2
r01s1[101-102]b[1-10]/daughter#03	RPS SFP on MPT-HLS
Ethernet/Slot#[1]/Port#[1-5]	MSS-E, MSS-HE, MSS-XE User Ethernet Ports
Ethernet/Slot#[1]/Port#[1-8]	MSS-E, MSS-HE, MSS-XE User Ethernet Ports

67.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Loss of Signal alarms on radios for information.

3 Troubleshoot any of the following:

4 Verify that the CorEvo/EASv2 card or MSS-E/HE/XE shelf Ethernet port is operating normally.

5 Retrieve the alarms using the Craft Terminal's Active Alarms.

6 Did the Loss Of Signal alarm clear?

   6.1 If yes, the procedure is complete.

   6.2 If no, go to step 7.

7 Verify the link cabling between the CorEvo/EASv2 card or MSS-E/HE/XE shelf Ethernet port and the MPT-HLC/MPT-HLS.

8 Retrieve the alarms using the Craft Terminal's Active Alarms.

9 Did the Loss Of Signal alarm clear?

9.1 If yes, the procedure is complete.

9.2 If no, contact the next level of technical support for assistance.

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68. Loss of signal synchronization alarms

68.1 Purpose

This procedure describes how to identify and clear Loss Of Signal synchronization alarms. For other Loss of Signal alarms, see one of the following:

- Loss of signal alarms on facilities
- Loss of signal alarms on Ethernet
- Loss of signal alarms on radios
- Loss of signal alarms on LAG

68.2 General

A Loss Of Signal alarm indicates that no signal is present on the incoming Sync-in port.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Loss Of Signal Sync alarms describes Loss of Signal synchronization alarms.

Table 64: Loss Of Signal Sync alarms

Friendly name/entity	Equipment type
Sync-in/Port#[7-8]	CorEvo
Sync-in/Port#[1-2]	MSS-E, MSS-HE, MSS-XE
N/A	UBT-S, UBT-m, UBT-T, UBT-I

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68.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Loss Of Signal Sync alarms for information about synchronization alarms.

3 Verify sync source provisioning at both the sync source and the alarming card.

4 Retrieve the alarms using the Craft Terminal's Active Alarms.

5 Did the Loss Of Signal alarm clear?

   5.1 If yes, the procedure is complete.

   5.2 If no, go to step 6.

6 Verify Sync reference source (GPS or other sync source) is functioning properly.

7 Verify Sync-In cabling and connections to sync-in port on CorEvo card or MSS-E/HE/XE shelf (Main and Spare if equipped).

8 Retrieve the alarms using the Craft Terminal's Active Alarms.

9 Did the Loss Of Signal alarm clear?

   9.1 If yes, the procedure is complete.

   9.2 If no, go to step 10.

10 Is traffic affected?

   10.1 If yes, go to step 11.

   10.2 If no, contact the next level of technical support for assistance.

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11 What is the unit declaring Loss of Signal?

11.1 If CorEvo card or MSS-E/HE/XE shelf, go to step 12.

12 Reseat the sync SFP.

> **⚠ CAUTION!**
>
> Reseating an sync SFP will cause a hit to traffic.

13 Retrieve the alarms using the Craft Terminal's Active Alarms.

14 Did the Loss Of Signal alarm clear?

14.1 If yes, the procedure is complete.

14.2 If no, go to step 15.

15 Replace the Sync SFP declaring Loss Of Signal. See the Wavence Hardware Installation and Replacement Manual.

> **⚠ CAUTION!**
>
> Replacing a sync SFP will cause a hit to traffic.

16 Retrieve the alarms using the Craft Terminal's Active Alarms.

17 Did the Loss Of Signal alarm clear?

17.1 If yes, the procedure is complete.

17.2 If no, contact the next level of technical support for assistance.

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69. Loss of signal alarms on LAG

69.1 Purpose

This procedure describes how to identify and clear Loss of Signal alarms on an Ethernet or radio LAG. For other Loss of Signal alarms, see one of the following:

- Loss of signal alarms on facilities
- Loss of signal alarms on Ethernet
- Loss of signal alarms on radios
- Loss of signal synchronization alarms

69.2 General

A Loss of Signal alarm indicates that ALL members of an Ethernet or Radio LAG have no activity. A Loss of Signal is declared when the number of active ports in the LAG equals zero.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Loss of Signal alarms on LAG provides information about Loss of Signal alarms on LAG.

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Table 65: Loss of Signal alarms on LAG

Friendly name/entity	Equipment type
EthLAG#[1-14]	CorEvo
RadioLAG#[1-14]	MPT-HLC
N/A	UBT-S UBT-m UBT-T UBT-I

69.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve ALL alarms.

2 Review the output. See Table: Loss of Signal alarms on LAG for information.

3 Retrieve the alarms using the Craft Terminal's Active Alarms.

4 Identify the alarms that are declared against members of the LAG group. If alarms exists, see Alarm and abnormal condition clearing procedures and troubleshoot accordingly.

5 Did the alarms exist on the LAG members and were they cleared?

   5.1 If yes, go to step 6.

   5.2 If no, contact the next level of technical support for assistance.

6 Retrieve the alarms using the Craft Terminal's Active Alarms.

7 Did the Loss of Signal alarm clear?

   7.1 If yes, the procedure is complete.

7.2 If no, contact the next level of technical support for assistance.

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70. Low input voltage alarms

70.1 Purpose

This procedure describes how to identify and clear Low Input Voltage alarms.

70.2 General

A Low Input Voltage alarm indicates that the input voltage has dropped below the configured threshold.

See the Wavence Technical Specifications for Split Mount Applications for information about supported input voltages.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Low Input Voltage alarms provides information about Low Input Voltage alarms.

Table 66: Low Input Voltage alarms

Friendly name/entity	Equipment type

70.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Low Input Voltage alarms for information.

3 Using site documentation determine the power source of the radio in alarm.

4 Measure the voltage of the power feeds for the power source of the radio declaring the low input voltage alarm.

5 Is the measured voltage within the supported voltage range?

5.1 If yes, go to step step 9. Using the Craft Terminal, determine the input voltage, current, and voltage threshold for the radio in alarm.

5.2 If no, go to step step 6. Troubleshoot the power feed voltage problem.

6 Troubleshoot the power feed voltage problem.

7 Retrieve the alarms using the Craft Terminal's Active Alarms.

8 Did the Low Input Voltage condition clear?

8.1 If yes, the procedure is complete.

8.2 If no, go to step step 9. Using the Craft Terminal, determine the input voltage, current, and voltage threshold for the radio in alarm.

9 Using the Craft Terminal, determine the input voltage, current, and voltage threshold for the radio in alarm.

10 Are the measured input voltage and current levels reported reasonable for the radio installation (cable type and length) and the input voltage measured in step step 4. Measure the voltage of the power feeds for the power source of the radio declaring the low input voltage alarm?

10.1 If yes, go to step step 11. Re-provision the input voltage threshold based upon site documentation and the results of step.

10.2 If no, go to step step 14. Is the radio in alarm in a protected system?.

11 Re-provision the input voltage threshold based upon site documentation and the results of step step 10. Are the measured input voltage and current levels reported reasonable for the

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radio installation (cable type and length) and the input voltage measured in step ?.

> **Note:**
>
> The hysteresis associated with the input voltage alarm is 3 VDC.

12 Retrieve the alarms using the Craft Terminal's Active Alarms.

13 Did the Low Input Voltage condition clear?

13.1 If yes, the procedure is complete.

13.2 If no, go to step step 14. Is the radio in alarm in a protected system?.

14 Is the radio in alarm in a protected system?

14.1 If yes, go to step step 15. Is the radio in alarm main or spare?.

14.2 If no, go to step step 24. Inspect the power supply cable and cable connections for the radio port in alarm.

15 Is the radio in alarm main or spare?

15.1 If main, go to step 16.

15.2 If spare, go to step 20.

> **CAUTION!**
>
> Removing power to an in-service radio hop in a protected link requires switching the traffic onto the standby radio hop.

16 Perform the Forced EPS Switch using the Craft Terminal.

⚠ CAUTION!

Forced EPS/HSB/RX Radio switching traffic causes a disruption to traffic.

17 Perform the Forced HSB Switch using the Craft Terminal.

18 Perform the Forced Rx Radio Switch using the Craft Terminal.

19 Go to step step 23. Verify radio in alarm status is Standby.

20 Perform the Lockout EPS Switch using the Craft Terminal.

21 Perform the Lockout HSB Switch using the Craft Terminal.

22 Perform the Lockout Rx Radio Switch using the Craft Terminal.

23 Verify radio in alarm status is *Standby*.

24 Inspect the power supply cable and cable connections for the radio port in alarm.

⚠ CAUTION!

Removing power to an in-service radio in an unprotected link causes loss of traffic. Removing power to an in-service radio hop in a protected link requires switching the traffic onto the standby radio hop.

25 Is the power supply cable damaged or improperly connected?

25.1 If yes, go to step step 26. Re-terminate, repair, or replace power supply cable as required..

25.2 If no, go to step step 29. Which unit is providing power to the radio declaring Low Input Voltage alarm?.

26 Re-terminate, repair, or replace power supply cable as required.

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27 Retrieve the alarms using the Craft Terminal's Active Alarms.

28 Did the Low Input Voltage alarm clear?

28.1 If yes, go to step step 82. Were Forced/Lockout EPS/HSB/Rx Radio switches established related to CorEvo/EASv2 cards providing power to the radio in alarm?.

28.2 If no, go to step step 29. Which unit is providing power to the radio declaring Low Input Voltage alarm?.

29 Which unit is providing power to the radio declaring Low Input Voltage alarm?

29.1 If EASv2, go to step step 30. Determine the EASv2 configuration.

29.2 If CorEvo, go to step step 50. Determine the CorEvo configuration.

29.3 If Power Injector Box, go to step step 62. Does the Power Injector Box/MPT Power Unit/MPT Extended Power Unit support one or more radio channels?.

29.4 If MPT Power Unit, go to step step 62. Does the Power Injector Box/MPT Power Unit/MPT Extended Power Unit support one or more radio channels?.

29.5 If MPT Extended Power Unit, go to step step 62. Does the Power Injector Box/MPT Power Unit/MPT Extended Power Unit support one or more radio channels?.

29.6 If Direct connection to Battery, go to step step 72. Inspect the power supply cable connection at the UBTODU.

30 Determine the EASv2 configuration.

30.1 Does the EASv2 support more than one radio direction?

30.2 Are the radio directions protected or unprotected?

30.3 Are Ethernet ports configured and carrying traffic?

31 Based on the answers to step step 30. Determine the EASv2 configuration , verify that the appropriate maintenance window is scheduled per local practices and procedures.

32 Is it appropriate to proceed with the Low Input Voltage procedure?

32.1 If yes, go to step step 33. Is the radio in alarm in a protected system?.

32.2 If no, contact the next level of technical support for assistance.

33 Is the radio in alarm in a protected system?

33.1 If yes, go to step step 34. Is the radio in alarm main or spare?.

33.2 If no, go to step step 43. Repeat steps through for all radios supported by the EASv2 card.

34 Is the radio in alarm main or spare?

34.1 If main, go to step step 35. Perform the Forced EPS Switch using the Craft Terminal.

34.2 If spare, go to step step 39. Perform the Lockout EPS Switch using the Craft Terminal.

⚠ CAUTION!

The following Cautions apply:

- Removing an in-service card in a protected link requires switching the traffic onto the protect hop.
- In protected EASv2 radio configurations, the Main EASv2 based radio resides in slots 3, 5, and/or 7. The spare EASv2 based radio resides in slots 4, 6, and/or 8.

35 Perform the Forced EPS Switch using the Craft Terminal.

⚠ CAUTION!

Forced EPS/HSB/RX Radio switching traffic causes a disruption to traffic.

36 Perform the Forced HSB Switch using the Craft Terminal.

37 Perform the Forced Rx Radio Switch using the Craft Terminal.

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38 Go to step step 42. Verify radio in alarm status is Standby.

39 Perform the Lockout EPS Switch using the Craft Terminal.

> **⚠ CAUTION!**
>
> Lockout EPS/HSB/RX Radio switching traffic causes a disruption to traffic.

40 Perform the Lockout HSB Switch using the Craft Terminal.

41 Perform the Lockout Rx Radio Switch using the Craft Terminal.

42 Verify radio in alarm status is *Standby*.

43 Repeat steps step 33. Is the radio in alarm in a protected system? through step 42. Verify radio in alarm status is Standby for all radios supported by the EASv2 card.

44 Physically reseat card in alarm and wait for the status LED to turn steady green or amber.

> **⚠ CAUTION!**
>
> The following Cautions apply:
>
> - Removing an in-service EASv2 card WILL cause a loss of ALL unprotected radio traffic and ALL Ethernet traffic associated with the EASv2 card.
> - Removing an in-service radio in an unprotected link causes loss of traffic. Removing an in-service card in a protected link requires switching the traffic onto the protect hop.

45 Retrieve the alarms using the Craft Terminal's Active Alarms.

46 Did the Low Input Voltage alarm clear?

46.1 If yes, go to step step 82. Were Forced/Lockout EPS/HSB/Rx Radio switches established related to CorEvo/EASv2 cards providing power to the radio in alarm?.

46.2 If no, go to step step 47. See Wavence Hardware Installation and Replacement Manual to replace EASv2.

47 See Wavence Hardware Installation and Replacement Manual to replace EASv2.

48 Retrieve the alarms using the Craft Terminal's Active Alarms.

49 Did the Low Input Voltage alarm clear?

49.1 If yes, go to step step 82. Were Forced/Lockout EPS/HSB/Rx Radio switches established related to CorEvo/EASv2 cards providing power to the radio in alarm?.

49.2 If no, go to step step 72. Inspect the power supply cable connection at the UBTODU.

50 Determine the CorEvo configuration.

50.1 Is the CorEvo configured in a protected core configuration?

50.2 Does the CorEvo support more than one radio direction?

50.3 Are Ethernet ports configured and carrying traffic?

51 Based on the answers to step step 50. Determine the CorEvo configuration, verify that the appropriate maintenance window is scheduled per local practices and procedures.

52 Is it appropriate to proceed with the Low Input Voltage procedure?

52.1 If yes, go to step step 53. Is the CorEvo in a protected system?.

52.2 If no, contact the next level of technical support for assistance.

53 Is the CorEvo in a protected system?

53.1 If yes, perform the Forced EPS Switch using the Craft Terminal. go to step .step 54. Perform the Forced EPS Switch using the Craft Terminal

53.2 If no, go to step .step 56. Physically reseat the CorEvo card associated with the radio in alarm and wait for the status LED to turn steady green or amber

54 Perform the Forced EPS Switch using the Craft Terminal.

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⚠ CAUTION!

The following Cautions apply: In protected CorEvo configuration, the Main CorEvo resides in slot 1. The spare CorEvo resides in slot 2.

55 Verify the main CorEvo status is *Standby*.

56 Physically reseat the CorEvo card associated with the radio in alarm and wait for the status LED to turn steady green or amber.

⚠ CAUTION!

Removing an In-service CorEvo card WILL cause a loss of ALL radio traffic and ALL unprotected Ethernet traffic associated with the CorEvo card.

57 Retrieve the alarms using the Craft Terminal's Active Alarms.

58 Did the Low Input Voltage alarm clear?

58.1 If yes, go to step step 82. Were Forced/Lockout EPS/HSB/Rx Radio switches established related to CorEvo/EASv2 cards providing power to the radio in alarm?.

58.2 If no, go to step step 59. See Wavence Hardware Installation and Replacement Manual to replace CorEvo.

59 See Wavence Hardware Installation and Replacement Manual to replace CorEvo.

60 Retrieve the alarms using the Craft Terminal's Active Alarms.

61 Did the Low Input Voltage alarm clear?

61.1 If yes, go to step step 82. Were Forced/Lockout EPS/HSB/Rx Radio switches established related to CorEvo/EASv2 cards providing power to the radio in alarm?.

61.2 If no, go to step step 72. Inspect the power supply cable connection at the UBTODU.

62 Does the Power Injector Box/MPT Power Unit/MPT Extended Power Unit support one or more radio channels?

63 Based on the answer to step step 62. Does the Power Injector Box/MPT Power Unit/MPT Extended Power Unit support one or more radio channels? , verify that the appropriate maintenance window is scheduled per local practices and procedures.

64 Is it appropriate to proceed with the Power Injector Box/MPT Power Unit/MPT Extended Power Unit Low Input Voltage procedure?

64.1 If yes, go to step step 65. Remove power to the Power Injector Box/MPT Power Unit/MPT Extended Power Unit in alarm.

64.2 If no, contact the next level of technical support for assistance.

65 Remove power to the Power Injector Box/MPT Power Unit/MPT Extended Power Unit in alarm.

⚠ CAUTION!

Power cycling an in-service Power Injector Box/MPT Power Unit/MPT Extended Power Unit shelf WILL cause a loss of ALL traffic associated with the Power Injector Box/MPT Power Unit/MPT Extended Power Unit.

66 Wait ten seconds and apply power removed in step step 65. Remove power to the Power Injector Box/MPT Power Unit/MPT Extended Power Unit in alarm.

67 Retrieve the alarms using the Craft Terminal's Active Alarms.

68 Did the Low Input Voltage alarm clear?

68.1 If yes, the procedure is complete.

68.2 If no, go to step step 69. See Wavence Hardware Installation and Replacement Manual to replace Power Injector Box/MPT Power Unit/MPT Extended Power Unit.

69 See Wavence Hardware Installation and Replacement Manual to replace Power Injector Box/MPT Power Unit/MPT Extended Power Unit.

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70 Retrieve the alarms using the Craft Terminal's Active Alarms.

71 Did the Low Input Voltage alarm clear?

71.1 If yes, the procedure is complete.

71.2 If no, go to step step 72. Inspect the power supply cable connection at the UBTODU.

72 Inspect the power supply cable connection at the UBT ODU.

73 Is power supply cable damaged or improperly connected?

73.1 If yes, go to step step 74. Re-terminate, repair, or replace power supply cable as required..

73.2 If no, go to step step 77. Inspect the power supply cable run between the CorEvo/EASv2/Power Injector Box/MPT Power Unit/MPT Extended Power Unit/direct Connection to DC Power and UBTODU.

74 Re-terminate, repair, or replace power supply cable as required.

75 Retrieve the alarms using the Craft Terminal's Active Alarms.

76 Did the Low Input Voltage alarm clear?

76.1 If yes, go to step step 82. Were Forced/Lockout EPS/HSB/Rx Radio switches established related to CorEvo/EASv2 cards providing power to the radio in alarm?.

76.2 If no, go to step step 77. Inspect the power supply cable run between the CorEvo/EASv2/Power Injector Box/MPT Power Unit/MPT Extended Power Unit/direct Connection to DC Power and UBTODU.

77 Inspect the power supply cable run between the CorEvo/EASv2/Power Injector Box/MPT Power Unit/MPT Extended Power Unit/direct Connection to DC Power and UBT ODU.

78 Is the power supply cable damaged?

78.1 If yes, go to step step 79. Repair or replace the power supply cable as required.

78.2 If no, contact the next level of technical support for assistance.

79 Repair or replace the power supply cable as required.

80 Retrieve the alarms using the Craft Terminal's Active Alarms.

81 Did the Low Input Voltage alarm clear?

81.1 If yes, go to step step 82. Were Forced/Lockout EPS/HSB/Rx Radio switches established related to CorEvo/EASv2 cards providing power to the radio in alarm?.

81.2 If no, contact the next level of technical support for assistance.

82 Were Forced/Lockout EPS/HSB/Rx Radio switches established related to CorEvo/EASv2 cards providing power to the radio in alarm?

82.1 If yes, go to step step 83. Release the Forced/Lockout EPS Switch using the Craft Terminal.

82.2 If no, the procedure is complete.

83 Release the Forced/Lockout EPS Switch using the Craft Terminal.

⚠ CAUTION!

Releasing Forced/Lockout EPS/HSB/RX Radio switch causes a disruption to traffic.

84 Release the Forced/Lockout HSB Switch using the Craft Terminal.

85 Release the Forced/Lockout Rx Radio Switch using the Craft Terminal.

86 Repeat steps step 83. Release the Forced/Lockout EPS Switch using the Craft Terminal through step 85. Release the Forced/Lockout Rx Radio Switch using the Craft Terminal until all protection switches are released.

87 The procedure is complete.

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71. MAC Status defect

71.1 Purpose

This procedure describes how to identify and clear MAC Status defect alarms.

71.2 General

This alarm occurs when a MEP received a CCM frame with correct MD/MEG Level, correct MAID/MEG, correct MEP ID and correct CCM Interval, but reports a failure in its Port Status TLV or Interface Status TLV.

Port Status TLV and Interface Status TLV are used in Connectivity Fault Management (CFM) to monitor link status. A defect occurs when the received TLV indicates a status other than Up.

71.3 Procedure

Purpose

Procedure

1 Use tools or dashboards to monitor the Connectivity Fault Management (CFM) status.

2 Check for any reported defects, for example RDI (Remote Defect Indication) or CCM (Continuity Check Message) errors

3 Verify that the physical connections between devices are intact and functioning properly. Loose cables or faulty hardware can cause TLV failures.

4 Perform connectivity tests between devices to ensure proper communication. Use diagnostic tools to identify any issues.

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72. Missing Diversity Alignment Configuration

72.1 Purpose

This procedure describes how to identify and clear Missing Diversity Alignment Configuration alarm.

72.2 General

Reported when not having a calibrated state and Diversity is enabled. This alarm indicates that the calibration of space diversity needs to be completed.

72.3 Procedure

Procedure

1 Configure the path for 4QAM modulation through WebCT.

2 Check the RSL on both Local Main and Local Div (>-50dBm having an MSE better than -30dB).

3 In case of XPIC, mute the remote transmitter of alternate polar.

4 Apply the Diversity Alignment Button associated with the Radio having the alarm.

Refer to *WebCT User Manual* chapter *Aligning Space Diversity* for more information.

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73. Memory Card Forbidden to use

73.1 Purpose

This chapter describes details about the Memory Card Forbidden alarm.

73.2 General

This alarm is a warning about the fact that the NE is in an inconsistent state and it prevents any further configuration changes.

The alarm is permanent, meaning that the alarm is never cleared and it's raised again by the NE after an NE Restart or power off/power on cycle. Flash cannot be used.

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74. OAM discovery not completed alarms

74.1 Purpose

This procedure describes how to identify and clear OAM discovery not completed alarms.

74.2 General

A OAM discovery not completed alarm indicates the Ethernet port (EFM) OAM discovery process is not in operational status.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: OAM discovery not completed alarms provides information about OAM discovery not completed alarms.

Table 67: OAM discovery not completed alarms

Friendly name/entity	Equipment type
ETHLinkOAM/Slot#1/Ports[1-8]	Ethernet port on CorEvo
ETHLinkOAM/Slot#[3-8]/Ports[1-8]	Ethernet port on EASv2

74.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

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2 Review the output. See Table: OAM discovery not completed alarms for information.

3 Verify Ethernet first mile feature is desired on the Ethernet link in alarm.

4 Is Ethernet first mile feature desired on the Ethernet link in alarm?

4.1 If yes, go to step 5.

4.2 If no, go to step 13.

5 Using the Craft Terminal, verify/provision Ethernet first mile feature on the Ethernet port in alarm.

6 Verify/provision Ethernet first mile feature on the Ethernet port link partner associated with the Ethernet port in alarm.

7 Retrieve the alarms using the Craft Terminal's Active Alarms.

8 Did the OAM discovery not completed alarm clear?

8.1 If yes, the procedure is complete.

8.2 If no, go to step 9.

9 Retrieve the alarms using the Craft Terminal's Active Alarms. Verify the operational status of the Ethernet port and troubleshoot accordingly.

10 Verify the operational status of the link partner and troubleshoot accordingly.

11 Retrieve the alarms using the Craft Terminal's Active Alarms.

12 Did the OAM discovery not completed alarm clear?

12.1 If yes, the procedure is complete.

12.2 If no, contact the next level of technical support for assistance.

13 Using the Craft Terminal, disable Ethernet first mile on the Ethernet port in alarm.

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14 Retrieve the alarms using the Craft Terminal's Active Alarms.

15 Did the OAM discovery not completed alarm clear?

15.1 If yes, the procedure is complete.

15.2 If no, contact the next level of technical support for assistance.

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75. Power supply failure alarms

75.1 Purpose

This procedure describes how to identify and clear Power Supply Failure alarms.

75.2 General

A Power Supply Failure alarm indicates one or more of the following:

- hardware failure:
  - on the EASv2 card supplying power to an UBT ODU
  - on the EAC/EAC1G2G5 card supplying power to a UBT ODU
  - on the MSS-E Ethernet port 5, supplying power to an UBT
  - on the MSS-HE/XE Ethernet ports 7-8, supplying power to an UBT
- bad cable, or an improperly terminated cable between:
  - the EASv2 and UBT ODU
  - the EAC/EAC1G2G5 and UBT ODU

The EASv2 card provides power to the UBT ODU using the PFoE supported RJ45 connectors.

The EAC/EAC1G2G5 card provides power to the MPT/UBT using the PFoE supported RJ45 or QMA coaxial connector(s).

See the Wavence Product Information Manual to determine card location and address, and for information about supported voltage.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Power Supply Failure alarms provides information about Power Supply Failure alarms.

Table 68: Power Supply Failure alarms

Friendly name/entity	Equipment type
r01/subrack#[3-8][1-4]	EASv2
N/A	EAC, EAC1G2G5
UBT-[S,T,m,C]/Slot#1/Port#5	MSS-E
UBT-[S,T,m]/Slot#1/Port#[7-8]	MSS-HE/XE

75.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Power Supply Failure alarms for information.

3 Inspect the power supply cable connection for radio port in alarm:

   3.1 For EASv2, verify RJ45 connector for PFoE port 1, 2, 3, or 4.

   3.2 For EAC/EAC1G2G5, verify RJ45 connector for PFoE port or Coax connector for QMA port.

4 Is the power supply cable damaged or improperly connected?

   4.1 If yes, go to step step 5. Terminate, repair, or replace power supply cable as required..

   4.2 If no, go to step step 8. Which card is Power Supply Failure alarm declared?.

5 Terminate, repair, or replace power supply cable as required.

6 Retrieve the alarms using the Craft Terminal's Active Alarms.

7 Did the Power Supply Failure alarm clear?

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7.1 If yes, the procedure is complete.

7.2 If no, go to step step 8. Which card is Power Supply Failure alarm declared?.

8 Which card is Power Supply Failure alarm declared?

8.1 If EAC/EAC1G2G5, go to step step 9. Determine the EAC/EAC1G2G5 configuration.

8.2 If EASv2, go to step step 28. Determine the EASv2 configuration.

9 Determine the EAC/EAC1G2G5 configuration.

9.1 Does the EAC/EAC1G2G5 support one or two radio channels?

9.2 Are the radio channels protected?

9.3 Is protection available for both radio channels?

10 Verify that radio protection is available for all traffic that is supported by the EAC/EAC1G2G5 to reseat/replace or appropriate maintenance window is scheduled per local practices and procedures.

11 Is it appropriate to proceed with the EAC/EAC1G2G5 Power Supply Failure procedure?

11.1 If yes, go to step step 12. Is the EAC/EAC1G2G5 in alarm in a protected system?.

11.2 If no, contact the next level of technical support for assistance.

12 Is the EAC/EAC1G2G5 in alarm in a protected system?

12.1 If yes, go to step step 13. Is the card in alarm main or spare?.

12.2 If no, go to step step 22. Physically reseat card in alarm and wait for the status LED to turn steady green or amber.

13 Is the card in alarm main or spare?

13.1 If main, go to step step 14. Perform the Forced EPS Switch using the Craft Terminal.

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13.2 If spare, go to step step 18. Perform the Lockout EPS Switch using the Craft Terminal.

⚠ CAUTION!

The following Cautions apply:

- Removing an in-service card in an unprotected link causes loss of traffic. Removing an in-service card in a protected link requires switching the traffic onto the protected channel.
- In redundant EAC/EAC1G2G5 protected radio configurations, the Main EAC/EAC1G2G5 card(s) resides in slots 3, 5, and/or 7. The spare EAC/EAC1G2G5 cards reside in slots 4, 6, and/or 8.
- In single EAC/EAC1G2G5 protected radio configurations, the Main UBT ODU and the spare UBT ODU are supported by a single EAC/EAC1G2G5 card. Reseating/Replacing an EAC/EAC1G2G5 card in this configuration causes a loss of traffic. Schedule an appropriate maintenance window prior to EAC/EAC1G2G5 replacement per local practices and procedures.

14 Perform the Forced EPS Switch using the Craft Terminal.

⚠ CAUTION!

Forced EPS/HSB/RX Radio switching traffic causes a disruption to traffic.

15 Perform the Forced HSB Switch using the Craft Terminal.

16 Perform the Forced Rx Radio Switch using the Craft Terminal.

17 Go to step step 21. Verify card in alarm status is Standby.

18 Perform the Lockout EPS Switch using the Craft Terminal.

19 Perform the Lockout HSB Switch using the Craft Terminal.

20 Perform the Lockout Rx Radio Switch using the Craft Terminal.

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21 Verify card in alarm status is *Standby*.

22 Physically reseat card in alarm and wait for the status LED to turn steady green or amber.

> **⚠ CAUTION!**
>
> Removing an in-service card in an unprotected link causes loss of traffic.

23 Retrieve the alarms using the Craft Terminal's Active Alarms.

24 Did the Power Supply Failure alarm clear?

24.1 If yes, go to step step 59. Is EASv2/EAC/EAC1G2G5 in alarm in a protected system?.

24.2 If no, go to step step 25. See Wavence Hardware Installation and Replacement Manual to replace EAC/EAC1G2G5.

25 See Wavence Hardware Installation and Replacement Manual to replace EAC/EAC1G2G5.

26 Retrieve the alarms using the Craft Terminal's Active Alarms.

27 Did the Power Supply Failure alarm clear?

27.1 If yes, go to step step 59. Is EASv2/EAC/EAC1G2G5 in alarm in a protected system?.

27.2 If no, go to step step 48. Inspect the power supply cable connection at UBTODU.

28 Determine the EASv2 configuration.

28.1 Does the EASv2 support more than one radio direction?

28.2 Are the radio directions protected or unprotected?

28.3 Are Ethernet ports configured and carrying traffic?

29 Based on the answers to step step 28. Determine the EASv2 configuration, verify that the appropriate maintenance window is scheduled per local practices and procedures.

30 Is it appropriate to proceed with the EASv2 Power Supply Failure procedure?

30.1 If yes, go to step step 31. Is the radio in alarm in a protected system?.

30.2 If no, contact the next level of technical support for assistance.

31 Is the radio in alarm in a protected system?

31.1 If yes, go to step step 32. Is the radio in alarm main or spare?.

31.2 If no, go to step step 42. Physically reseat card in alarm and wait for the status LED to turn steady green or amber.

32 Is the radio in alarm main or spare?

32.1 If main, go to step step 33. Perform the Forced EPS Switch using the Craft Terminal.

32.2 If spare, go to step step 37. Perform the Lockout EPS Switch using the Craft Terminal.

⚠ CAUTION!

The following Cautions apply:

- Removing an in-service card in a protected link requires switching the traffic onto the protect hop.
- In protected EASv2 radio configurations, the Main EASv2 based radio resides in slots 3, 5, and/or 7. The spare EASv2 based radio resides in slots 4, 6, and/or 8.

33 Perform the Forced EPS Switch using the Craft Terminal.

⚠ CAUTION!

Forced EPS/HSB/RX Radio switching traffic causes a disruption to traffic.

34 Perform the Forced HSB Switch using the Craft Terminal.

35 Perform the Forced Rx Radio Switch using the Craft Terminal.

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36 Go to step step 40. Verify radio in alarm status is Standby.

37 Perform the Lockout EPS Switch using the Craft Terminal.

> **⚠ CAUTION!**
>
> Lockout EPS/HSB/RX Radio switching traffic causes a disruption to traffic.

38 Perform the Lockout HSB Switch using the Craft Terminal.

39 Perform the Lockout Rx Radio Switch using the Craft Terminal.

40 Verify radio in alarm status is *Standby*.

41 Repeat steps step 32. Is the radio in alarm main or spare? through step 40. Verify radio in alarm status is Standby for all radios supported by the EASv2 card.

42 Physically reseat card in alarm and wait for the status LED to turn steady green or amber.

> **⚠ CAUTION!**
>
> The following Cautions apply:
>
> - Removing an in-service EASv2 card WILL cause a loss of ALL unprotected radio traffic and ALL Ethernet traffic associated with the EASv2 card.
> - Removing an in-service radio in an unprotected link causes loss of traffic. Removing an in-service card in a protected link requires switching the traffic onto the protect hop.

43 Retrieve the alarms using the Craft Terminal's Active Alarms.

44 Did the Power Supply Failure alarm clear?

44.1 If yes, go to step step 60. Release the Forced/Lockout EPS Switch using the Craft Terminal.

44.2 If no, go to step step 45. See Wavence Hardware Installation and Replacement Manual to replace EASv2.

45 See Wavence Hardware Installation and Replacement Manual to replace EASv2.

46 Retrieve the alarms using the Craft Terminal's Active Alarms.

47 Did the Power Supply Failure alarm clear?

47.1 If yes, go to step step 60. Release the Forced/Lockout EPS Switch using the Craft Terminal.

47.2 If no, go to step step 48. Inspect the power supply cable connection at UBTODU.

48 Inspect the power supply cable connection at UBT ODU.

49 Is power supply cable damaged or improperly connected?

49.1 If yes, go to step step 50. Terminate, repair, or replace power supply cable as required..

49.2 If no, go to step step 53. Inspect the power supply cable run between the EAC/EAC1G2G5/EASv2 and UBTODU.

50 Terminate, repair, or replace power supply cable as required.

51 Retrieve the alarms using the Craft Terminal's Active Alarms.

52 Did the Power Supply Failure alarm clear?

52.1 If yes, go to step step 59. Is EASv2/EAC/EAC1G2G5 in alarm in a protected system?.

52.2 If no, go to step step 53. Inspect the power supply cable run between the EAC/EAC1G2G5/EASv2 and UBTODU.

53 Inspect the power supply cable run between the EAC/EAC1G2G5/EASv2 and UBT ODU.

54 Is the power supply cable damaged?

54.1 If yes, go to step step 55. Repair or replace the power supply cable as required.

54.2 If no, contact the next level of technical support for assistance.

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55 Repair or replace the power supply cable as required.

56 Retrieve the alarms using the Craft Terminal's Active Alarms.

57 Did the Power Supply Failure alarm clear?

57.1 If yes, go to step step 58. Which card is Power Supply Failure alarm declared?.

57.2 If no, contact the next level of technical support for assistance.

58 Which card is Power Supply Failure alarm declared?

58.1 If EASv2/EAC/EAC1G2G5, go to step step 59. Is EASv2/EAC/EAC1G2G5 in alarm in a protected system?.

58.2 The procedure is complete.

59 Is EASv2/EAC/EAC1G2G5 in alarm in a protected system?

59.1 If yes, go to step step 60. Release the Forced/Lockout EPS Switch using the Craft Terminal.

59.2 If no, the procedure is complete.

60 Release the Forced/Lockout EPS Switch using the Craft Terminal.

⚠ CAUTION!

Releasing Forced EPS/HSB/RX Radio switch causes a disruption to traffic.

61 Release the Forced/Lockout HSB Switch using the Craft Terminal.

62 Release the Forced/Lockout Rx Radio Switch using the Craft Terminal.

63 The procedure is complete.

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76. PPP fail alarms

76.1 Purpose

This procedure describes how to identify and clear PPP Fail alarms.

76.2 General

A PPP Fail alarm indicates that TMN over PPP has failed.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: PPP Fail alarms provides information about PPP Fail alarms.

Table 69: PPP Fail alarms

Friendly name/entity	Facility type
PPP-IP/RF#1[7-8]	MPT-HLC or MPT-HLS connected to CorEvo
PPP-IP/RF#[3-8][5-8]	MPT-HLC or MPT-HLS connected to EASv2
N/A	UBT-S, UBT-m, UBT-T, UBT-I

76.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

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2 Review the output. See Table: PPP Fail alarms for information.

3 Retrieve the alarms using the Craft Terminal's Active Alarms, verify that no other radio path alarms (such as Demodulator Fail) exist, and troubleshoot accordingly.

4 Did other radio path alarms exist and were cleared?

4.1 If yes, go to step 5.

4.2 If no, go to step 7.

5 Retrieve the alarms using the Craft Terminal's Active Alarms.

6 Did the PPP Fail alarm clear?

6.1 If yes, the procedure is complete.

6.2 If no, go to step 7.

7 Check the provisioning of the far end radio card that is associated with card in alarm, and verify that the PPP RF is enabled.

8 Is the far end radio card PPP RF disabled?

8.1 If yes, go to step 9.

8.2 If no, go to step 14.

9 Enable the PPP RF at the far end radio card.

10 Retrieve the alarms using the Craft Terminal's Active Alarms.

11 Did the PPP Fail alarm clear?

11.1 If yes, the procedure is complete.

11.2 If no, go to step 14.

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12 Verify that the Link Identifiers match at the local and far end radio cards.

13 Do the local and far end Link Identifiers match?

13.1 If yes, the procedure is complete.

13.2 If no, go to step 14.

14 Determine which Link Identifier is correct.

15 Reprovision the incorrect Link Identifier.

16 Retrieve the alarms using the Craft Terminal's Active Alarms.

17 Did the PPP Fail alarm clear?

17.1 If yes, the procedure is complete.

17.2 If no, contact the next level of technical support for assistance.

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77. Provisioning mismatch alarms

77.1 Purpose

This procedure describes how to identify and clear Provisioning Mismatch alarms.

77.2 General

A Provisioning Mismatch alarm is declared against a CorEvo card indicates that has an incorrect flash card installed on the CorEvo card.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Provisioning Mismatch alarms provides information about Provisioning Mismatch alarms.

Table 70: Provisioning Mismatch alarms

Friendly name/entity	Equipment type
r01s1/board#[1-2]	CorEvo
MSS-E/Slot#[1]	MSS-E
MSS-HE/XE Slot#[1]	MSS-HE/XE

77.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Provisioning Mismatch alarms for information.

3 Replace the CorEvo/MSS-E/MSS-HE/MSS-XE flash card with the correct flash card for the license or upgrade the license. See Wavence Hardware Installation and Replacement Manual.

4 Retrieve the alarms using the Craft Terminal's Active Alarms.

5 Did the Provisioning Mismatch alarm clear?

1. If yes, the procedure is complete.
2. If no, contact the next level of technical support for assistance.

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78. Radio encryption mismatch alarms

78.1 Purpose

This procedure describes how to identify and clear radio encryption mismatch alarms.

78.2 General

A Radio encryption mismatch alarm indicates either radio encryption is enabled only on one end of the radio hop, or the radio encryption key provisioned on one end of the hop is does not match the key provisioned on the other end of the hop.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Radio encryption mismatch alarms provides information about Radio encryption mismatch alarms.

Table 71: Radio encryption mismatch alarms

Friendly name/entity	Equipment type
Radio/Dir#1.[7-8]/Slot#1/Port#[7-8]/Ch#1	MPT-HLC or MPT-HLS connected to CorEvo
Radio/Dir#[3-8].[5-8]/Slot#[3-8]/Port#[5-8]/Ch#1	MPT-HLC or MPT-HLS connected to EASv2

78.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Radio encryption mismatch alarms for information about radio alarms.

3 Verify radio encryption is desired on the microwave link in alarm.

4 Is radio encryption desired on the microwave link in alarm?

   4.1 If yes, go to step 5.

   4.2 If no, go to step 12.

5 Using site documentation determine the desired encryption key for the radio link.

6 Using the Craft Terminal, provision radio encryption on the far end using the encryption key identified in step 5.

7 Retrieve the alarms using the Craft Terminal's Active Alarms.

8 Did the radio encryption mismatch alarm clear?

   8.1 If yes, the procedure is complete.

   8.2 If no, go to step 9.

9 Using the Craft Terminal, provision radio encryption on the near end using the encryption key identified in step 5.

10 Retrieve the alarms using the Craft Terminal's Active Alarms.

11 Did the radio encryption mismatch alarm clear?

   11.1 If yes, the procedure is complete.

   11.2 If no, contact the next level of technical support for assistance.

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12 Using the Craft Terminal, disable radio encryption on the far end.

13 Using the Craft Terminal, disable radio encryption on the near end.

14 Retrieve the alarms using the Craft Terminal's Active Alarms.

15 Did the radio encryption mismatch alarm clear?

15.1 If yes, the procedure is complete.

15.2 If no, contact the next level of technical support for assistance.

79. Rate mismatch alarms

79.1 Purpose

This procedure describes how to identify and clear rate mismatch alarms.

79.2 General

A rate mismatch alarm indicates the configured speed of an 10 GbEth/1 GbEth User Ethernet interface is not compatible with the equipped SFP transceiver.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Rate mismatch alarms provides information about rate mismatch alarms.

Table 72: Rate mismatch alarms

Friendly name/entity	Equipment type
MAU/Slot#[1-2]/Port#[5-8]	CorEvo
Ethernet/Slot#[1]/Port#[1-2]	MSS-E SFP User Ethernet Ports
Ethernet/Slot#[1]/Port#[1-6]	MSS-HE/XE SFP Radio Ports
Ethernet/Slot#[1]/Port#[1-2]	MSS-E SFP User Ethernet Ports
Ethernet/Slot#[1]/Port#[1-6]	MSS-HE/XE SFP Radio Ports

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79.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Rate mismatch alarms for information.

3 Using the Craft Terminal, retrieve the provisioned Ethernet interface value for the Ethernet interface in alarm.

4 Verify the required Ethernet interface rate for the Ethernet interface.

5 Determine if the equipped SFP supports the required Ethernet interface rate.

6 Does the equipped SFP support the required Ethernet interface rate?

   6.1 If yes, go to step 7.

   6.2 If no, go to step 10.

7 Using the Craft Terminal, verify/provision Ethernet interface port in alarm.

8 Retrieve the alarms using the Craft Terminal's Active Alarms.

9 Did the rate mismatch alarm clear?

   9.1 If yes, the procedure is complete.

   9.2 If no, go to step 10.

10 Replace the SFP.

11 Retrieve the alarms using the Craft Terminal's Active Alarms.

12 Did the rate mismatch alarm clear?

   12.1 If yes, the procedure is complete.

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12.2 If no, contact the next level of technical support for assistance.

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80. Remote Defect Indication alarms on radios

80.1 Purpose

This procedure describes how to identify and clear Remote Defect Indication alarms on radios. For Remote Defect Indication alarms on SDH facilities, see Remote defect indication alarms on facilities.

80.2 General

A Remote Defect Indication in a received Radio Direction. The alarm is reported when an RDI is received on a given radio direction from the remote end of radio link.

> **Note:**
> When a Tx mute is executed in a local station,a RDI is displayed in the remote station's current alarms whereas only Demodulator Fail should be raised.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Remote Defect Indication alarms provides information about Remote Defect Indication alarms.

Table 73: Remote Defect Indication alarms

Friendly name/entity	Facility type
Radio/Dir#1.[7-8]/Slot#1/Port#[7-8]/Ch#[0-1]	MPT-HLC or MPT-HLS connected to CorEvo
Radio/Dir#[3-8].[5-8]/Slot#[3-8]/Port#[5-8]/Ch#[0-1]	MPT-HLC or MPT-HLS connected to EASv2
N/A	UBT-S UBT-m UBT-T

80.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Remote Defect Indication alarms for information.

3 Retrieve the alarms for the far end of the radio direction using the Craft Terminal's Active Alarms.

4 Determine all of the radio alarms that are declared against the radio direction associated with the radio direction declaring the Remote Defect Indication (RDI) and troubleshoot accordingly.

5 Retrieve the alarms using the Craft Terminal's Active Alarms.

6 Did the RDI alarm clear?

   1. If yes, the procedure is complete.
   2. If no, contact the next level of technical support for assistance.

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81. Remote defect indication alarms on facilities

81.1 Purpose

This procedure describes how to identify and clear Remote Defect Indication alarms on SDH facilities. For Remote Defect Indication alarms on radios, see Remote Defect Indication alarms on radios.

81.2 General

A Remote Defect Indication in a received STM-1/OC-3 facility indicates that an RDI is received on an STM-1/OC-3 from the remote end of the STM-1/OC-3 facility.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence User Manual for information about how to execute the commands that are described in this procedure.

Table: Remote Defect Indication alarms provides information about Remote Defect Indication alarms.

Table 74: Remote Defect Indication alarms

Friendly name/entity	Facility type
EleSdh-in/Slot#[3-8]/Port[1-2]	STM-1/OC-3
OptSdh-in/Slot#[3-8]/Port[1-2]	
MS/slot#[3-8]/Port1]	STM-1

81.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Remote Defect Indication alarms for information about facility alarms.

3 Retrieve the alarms for the far end of the STM-1/OC-3 facility using the Craft Terminal's Active Alarms.

4 Determine all of the facility alarms that are declared against the far end STM-1/OC-3 facility associated with the STM-1/OC-3 facility declaring the Remote Defect Indication (RDI) and troubleshoot accordingly.

5 Retrieve the alarms using the Craft Terminal's Active Alarms.

6 Did the RDI alarm clear?

   1. If yes, the procedure is complete.
   2. If no, contact the next level of technical support for assistance.

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82. Replaceable unit missing alarms

82.1 Purpose

This procedure describes how to identify and clear Replaceable Unit Missing alarms.

82.2 General

A Replaceable Unit Missing alarm indicates that a card/SFP is missing or unreachable from the configured slot/port location.

See the Wavence Product Information Manual to determine card location and address and for information about supported temperatures.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Replaceable Unit Missing alarms provides information about Replaceable Unit Missing alarms.

Table 75: Replaceable Unit Missing alarms

Friendly name/entity	Equipment type
r01s1/board#[1-2]	CorEvo
r01s1/board#[5,9]	FAN, A-FAN, or E-FAN
r01s1/board#[3-8]	EASv2, P2E3DS3, P32E1DS1, SDHACC
r01s1b[1-2]/daughter#[5-8]	SFP on CorEvo
r01s1b[3-8]/daughter#[05-08]	SFP on EASv2
r01s1b[3-8]/daughter#[01-02]	SFP on SDHACC
r01s[101-102]b[1-10]/daughter#3	RPS SFP on MPT-HLS
N/A	SFP Coupler on: UBT-S UBT-T UBT-m UBT-I
MSS-E/SFP/Slot#[1]/Port#[1-2]	SFP on MSS-E
MSS-HE/XE/SFP/Slot#[1]/Port#[1-6]	SFP on MSS-HE/XE

82.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Replaceable Unit Missing alarms for information about equipment alarms.

3 Visually inspect the indicated slot location.

4 Is the slot location empty?

4.1 If yes, go to step 5.

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4.2 If no, go to step 8.

5 Install the card/SFP type corresponding to slot provisioning.

6 Retrieve the alarms using the Craft Terminal's Active Alarms.

7 Did the Replaceable Unit Missing alarm clear?

7.1 If yes, the procedure is complete.

7.2 If no, contact the next level of technical support for assistance.

8 Perform the NE restart on the NE: Supervision/Restart NE. Communication with the NE is lost during the NE restart and you are logged out of the system.

> **⚠ CAUTION!**
>
> Performing the Restart NE command may cause a disruption to traffic.

9 Log into the NE.

10 Retrieve the alarms using the Craft Terminal's Active Alarms.

11 Did the Replaceable Unit Missing alarm clear?

11.1 If yes, the procedure is complete.

11.2 If no, contact the next level of technical support for assistance.

83. Replaceable unit problem alarms

This procedure describes how to identify and clear Replaceable Unit Problem alarms.

83.1 General

A Replaceable Unit Problem alarm indicates that a card/SFP failure has been detected against the indicated card/SFP.

See the Wavence Product Information Manual to determine card location and address and for information about supported temperatures.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Replaceable Unit Problem alarms provides information about Replaceable Unit Problem alarms.

> **Note:**
> When MPT has an internal Tx Power alarm raised - reported as RUP alarm - MPT Tx is not automatically switched off. Manually mute the ODU before any site operation.

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Table 76: Replaceable Unit Problem alarms

Friendly name/entity	Equipment type
r01s1/board#[1-2]	CorEvo
r01s1/board#[5,9]	FAN, A-FAN, or E-FAN
r01s1/board#[3-8]	EASv2, P2E3DS3, P32E1DS1 or SDHACC
r01/subrack#[3-8][5-8]	MPT-HLC connected to EASv2
r01/subrack#1[7-8]	MPT-HLC connected to CorEvo
r01s[101-102]/board#[1-10]	MPT-HLS
r01s[101-102]/board#[11-12]	MPT-HLS Fan card
r01s1b[1-2]/daughter#[05-08]	SFP on CorEvo
r01s1b[3-8]/daughter#[05-08]	SFP on EASv2
r01s1b[3-8]/daughter#[01-02]	SFP for SDHACC
r01s[101-102]b[1-10]/daughter#04	RPS SFP on MPT-HLS
r01s1[7-8]b1/daughter#04	Rx Diversity Module on MPT-HLC connected to CorEvo
r01/subrack#[3-8][5-8]/ daughter#04	Rx Diversity Module on MPT-HLC connected to EASv2
r01s[101-102]b[1-10]/daughter#04	Rx Diversity Module on MPT-HLS
UBT-S UBT-m UBT-T	Card failed in UBT-T, UBT-S, UBT-m
Traffic SFP on: UBT-S UBT-m UBT-T	TX_Fault indication is raised by the SFP (if supported). Hence, the transmission capability of this SFP is problematic.
UBT-T channel A UBT-T channel B	One of the carriers of the TWIN is in failure in UBT-T
MSS-E/Slot#[1]	MSS-E
MSS-HE/XE Slot#[1]	MSS-HE/XE
MSS-E/SFP/Slot#[1]/Port#[1-2]	SFP on MSS-E

Friendly name/entity	Equipment type
MSS-HE/XE/SFP/Slot#[1]/Port#[1-6]	SFP on MSS-HE/XE

83.2 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Replaceable Unit Problem alarms for information.

3 Troubleshoot any of the following:

> **⚠ CAUTION!**
>
> CorEvo, EASV2, P2E3DS3, P32E1DS1 and SDHACCC cards support multiple services in both the protected and not protected configuration, Reseating and replacing any of these cards may cause a loss of traffic on the other services supported by the card. Schedule an appropriate maintenance window prior to reseating or replacing cards per local practices and procedures.

1. Replaceable unit problem on CorEvo
2. Replaceable unit problem on MSS-E/HE/XE
3. Replaceable unit problem on P2E3DS3, P32E1DS1, or SDHACC
4. Replaceable unit problem on EASv2
5. Replaceable unit problem on FAN, A-FAN, E-FAN, or MPT-HLS fan
6. Replaceable unit problem on SFP
7. Replaceable unit problem on MPT-HLC, or MPT-HLS
8. Replaceable unit problem on UBT ODU
9. Replaceable unit problem on RPS SFP on UBT ODU or MPT-HLS
10. Replaceable unit problem on RPS or XPIC module on UBT ODU
11. Replaceable unit problem on Diversity receiver module on MPT-HLC or MPT-HLS

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83.2.1 Replaceable unit problem on CorEvo

Procedure

1 Determine the CorEvo configuration.

1.1 Is the CorEvo configured in a protected core configuration?

1.2 Does the CorEvo support one or more radio channels?

1.3 Are Ethernet ports configured and carrying traffic?

1.4 Is protection available for radio channels?

1.5 Is the power to the radio channels provided by the CorEvo card?

2 Determine if radio protection is available for all traffic supported by the CorEvo to be replaced or an appropriate maintenance window is scheduled per local practices and procedures.

3 Is it appropriate to proceed with the Replaceable Unit Problem on CorEvo procedure?

3.1 If yes, go to step 4.

3.2 If no, contact the next level of technical support for assistance.

4 Is the CorEvo in alarm in a protected configuration?

4.1 If yes, go to step 5.

4.2 If no, go to step 10.

5 Is the CorEvo in alarm main or spare?

5.1 If main, go to step 6.

5.2 If spare, go to step 8.

⚠ CAUTION!

Forced EPS switching traffic causes a disruption to traffic.

6 Perform the Forced EPS Switch using the Craft Terminal.

7 go to step 9.

8 Perform the Lockout EPS Switch using the Craft Terminal.

9 Verify that the CorEvo in alarm status is *Standby*.

10 Physically reseat the CorEvo in alarm and wait for the status LED to turn steady green or amber.

⚠ CAUTION!

Removing an in-service card in an unprotected link causes loss of traffic.

11 Retrieve the alarms using the Craft Terminal's Active Alarms.

12 Did the Replaceable Unit Problem alarm clear?

12.1 If yes, go to step 16.

12.2 If no, go to step 13.

13 Replace the equipment the Replaceable Unit Problem is declared against. See the Wavence Hardware Installation and Replacement Manual.

14 Retrieve the alarms using the Craft Terminal's Active Alarms.

15 Did the Replaceable Unit Problem alarm clear?

15.1 If yes, go to step 16.

15.2 If no, contact the next level of technical support for assistance.

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16 Is the CorEvo in a protected system?

16.1 If yes, go to step 17.

16.2 If no, the procedure is complete.

17 Release the Forced/Lockout EPS Switch using the Craft Terminal.

> **⚠ CAUTION!**
>
> Releasing EPS switch may cause a disruption to traffic.

18 The procedure is complete.

83.2.2 Replaceable unit problem on MSS-E/HE/XE

Procedure

1 Determine the MSS-E/HE/XE configuration.

1.1 Does the MSS-E/HE/XE support one or more radio channels?

1.2 Are Ethernet ports configured and carrying traffic?

1.3 Is protection available for radio channels?

1.4 Is power to the radio channel(s) provided by the MSS-E/HE/XE?

2 Determine if radio protection is available for all traffic supported by the MSS-E/HE/XE to be replaced or an appropriate maintenance window is scheduled per local practices and procedures.

3 Is it appropriate to proceed with the Replaceable Unit Problem on MSS-E/HE/XE procedure?

3.1 If yes, go to step 4.

3.2 If no, contact the next level of technical support for assistance.

4 Remove the fuse(s) in the power distribution panel that is suppling battery power to the MSS-E/HE/XE.

5 Install the fuse in the power distribution panel that is suppling battery power to MSS -E/HE/XE.

6 Retrieve the alarms using the Craft Terminal's Active Alarms.

7 Did the Replaceable Unit Problem alarm clear?

7.1 If yes, the procedure is complete.

7.2 If no, go to step 8.

8 Replace the equipment the Replaceable Unit Problem is declared against. See the Wavence Hardware Installation and Replacement Manual.

9 Retrieve the alarms using the Craft Terminal's Active Alarms.

10 Did the Replaceable Unit Problem alarm clear?

10.1 If yes, the procedure is complete.

10.2 If no, contact the next level of technical support for assistance.

10.3 The procedure is complete.

83.2.3 Replaceable unit problem on P2E3DS3, P32E1DS1, or SDHACC

Procedure

1 Is the card in alarm in a protected system?

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1.1 If yes, go to step 2.

1.2 If no, go to step 7.

2 Is the card in alarm main or spare?

2.1 If main, go to step 3.

2.2 If spare, go to step 5.

> **⚠ CAUTION!**
>
> Forced EPS switching traffic causes a disruption to traffic.

3 Perform the Forced EPS Switch using the Craft Terminal.

4 Go to step 6.

5 Perform the Lockout EPS Switch using the Craft Terminal.

6 Verify that the card in the alarm status is *Standby*.

7 Physically reseat the card in alarm and wait for the status LED to turn steady green or amber.

> **⚠ CAUTION!**
>
> Removing an in-service card in an unprotected link causes loss of traffic.

8 Retrieve the alarms using the Craft Terminal's Active Alarms.

9 Did the Replaceable Unit Problem alarm clear?

9.1 If yes, go to step 13.

9.2 If no, go to step 10.

10 Replace the equipment the Replaceable Unit Problem is declared against. See the Wavence Hardware Installation and Replacement Manual.

11 Retrieve the alarms using the Craft Terminal's Active Alarms.

12 Did the Replaceable Unit Problem alarm clear?

12.1 If yes, go to step 13.

12.2 If no, contact the next level of technical support for assistance.

13 Is the card in alarm in a protected system?

13.1 If yes, go to step 14.

13.2 If no, the procedure is complete.

14 Release the Forced/Lockout EPS Switch using the Craft Terminal.

⚠ CAUTION!

Releasing EPS switch may cause a disruption to traffic.

15 The procedure is complete.

83.2.4 Replaceable unit problem on EASv2

Procedure

1 Determine the EASv2 configuration.

1.1 Is the EASv2 configured in a stacked configuration?

1.2 Does the EASv2 support one or more radio channels?

1.3 Are Ethernet ports configured and carrying traffic?

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1.4 Is protection available for supported radio channels?

1.5 Is power to any radio channel provided by the EASv2 card?

2 Determine if radio protection is available for all traffic supported by the EASv2 to be replaced or an appropriate maintenance window is scheduled per local practices and procedures.

3 Is it appropriate to proceed with the Replaceable Unit Problem on EASv2 procedure?

3.1 If yes, go to step 4.

3.2 If no, contact the next level of technical support for assistance.

4 Are radio channels supported by the EASv2 in alarm in a protected configuration?

4.1 If yes, go to step 5.

4.2 If no, go to step 15.

5 Is the radio channel supported main or spare?

5.1 If main, go to step 6.

5.2 If spare, go to step 10.

⚠ CAUTION!

Forced EPS/HSB/RX Radio switching traffic causes a disruption to traffic.

6 Perform the Forced EPS Switch using the Craft Terminal.

⚠ CAUTION!

Forced EPS/HSB/RX Radio switching traffic causes a disruption to traffic.

7 Perform the Forced HSB Switch using the Craft Terminal.

8 Perform the Forced Rx Radio Switch using the Craft Terminal.

9 go to step 13.

10 Perform the Lockout EPS Switch using the Craft Terminal.

11 Perform the Lockout HSB Switch using the Craft Terminal.

12 Perform the Lockout Rx Radio Switch using the Craft Terminal.

13 Verify that the supported radio channel status is *Standby*.

14 If two or more radio channels are supported by the EASv2/card, then repeat steps 5 through 13 for each supported protected radio channel.

15 Physically reseat the EASv2 in alarm and wait for the status LED to turn steady green or amber.

> **⚠ CAUTION!**
>
> Removing an in-service card in an unprotected link causes loss of traffic.

16 Retrieve the alarms using the Craft Terminal's Active Alarms.

17 Did the Replaceable Unit Problem alarm clear?

17.1 If yes, go to step 21.

17.2 If no, go to step 18.

18 Replace the equipment the Replaceable Unit Problem is declared against. See the Wavence Hardware Installation and Replacement Manual.

19 Retrieve the alarms using the Craft Terminal's Active Alarms.

20 Did the Replaceable Unit Problem alarm clear?

20.1 If yes, go to step 21.

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20.2 If no, contact the next level of technical support for assistance.

21 Are radio channels supported by the EASv2 in a protected configuration?

21.1 If yes, go to step 22.

21.2 If no, the procedure is complete.

22 Release the Forced/Lockout EPS Switch using the Craft Terminal.

> **⚠ CAUTION!**
>
> Releasing Forced EPS/HSB/RX Radio switch causes a disruption to traffic.

23 Release the Forced/Lockout HSB Switch using the Craft Terminal.

24 Release the Forced/Lockout Rx Radio Switch using the Craft Terminal.

25 If two or more radio channels are supported by the EASv2 card, then repeat steps 22 through 24 for the each supported protected radio channel.

26 The procedure is complete.

83.2.5 Replaceable unit problem on FAN, A-FAN, E-FAN, or MPT-HLS fan

Procedure

1 Physically reseat the FAN in alarm and wait for the status LED to turn steady green.

2 Retrieve the alarms using the Craft Terminal's Active Alarms.

3 Did the Replaceable Unit Problem alarm clear?

3.1 If yes, the procedure is complete.

3.2 If no, go to step 4.

4 Replace the FAN Replaceable Unit Problem is declared against. See the Wavence Hardware Installation and Replacement Manual.

5 Retrieve the alarms using the Craft Terminal's Active Alarms.

6 Did the Replaceable Unit Problem alarm clear?

6.1 If yes, the procedure is complete.

6.2 If no, contact the next level of technical support for assistance.

83.2.6 Replaceable unit problem on SFP

Procedure

1 Is the SFP (Ethernet/OC-3/Radio equipment) in alarm protected?

1.1 If yes, go to step 2.

1.2 If no, go to step 11.

2 Is the SFP in alarm the main or spare?

2.1 If main, go to step 3.

2.2 If spare, go to step 7.

⚠ CAUTION!

Forced/Lockout switching traffic causes a disruption to traffic.

3 Perform the Forced EPS Switch using the Craft Terminal.

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4 Perform the Forced HSB Switch using the Craft Terminal (radio SFP only).

5 Perform the Forced Rx Radio Switch using the Craft Terminal (radio SFP only).

6 Go to step 10.

7 Perform the Lockout EPS Switch using the Craft Terminal.

8 Perform the Lockout HSB Switch using the Craft Terminal (radio SFP only).

9 Perform the Lockout Rx Radio Switch using the Craft Terminal (radio SFP only).

10 Verify that the SFP in alarm status is *Standby*.

11 Physically reseat the SFP in alarm.

> **⚠ CAUTION!**
>
> Removing SFP in an unprotected system causes a loss of traffic on ALL facilities that are associated with the SFP removed. Schedule an appropriate maintenance window per local practices and procedures.

12 Retrieve the alarms using the Craft Terminal's Active Alarms.

13 Did the Replaceable Unit Problem alarm clear?

13.1 If yes, go to step 17.

13.2 If no, go to step 14.

14 See Wavence Hardware Installation and Replacement Manual to replace SFP in alarm.

15 Retrieve the alarms using the Craft Terminal's Active Alarms.

16 Did the Replaceable Unit Problem alarm clear?

16.1 If yes, go to step 17.

16.2 If no, contact the next level of technical support for assistance.

17 Is the SFP protected?

17.1 If yes, go to step 18.

17.2 If no, the procedure is complete.

18 Release the Forced/Lockout EPS Switch using the Craft Terminal.

> **⚠ CAUTION!**
>
> Releasing Forced/Lockout switch may cause a disruption to traffic.

19 Release the Forced/Lockout HSB Switch using the Craft Terminal (radio SFP only).

20 Release the Forced/Lockout Rx Radio Switch using the Craft Terminal (radio SFP only).

21 The procedure is complete.

83.2.7 Replaceable unit problem on MPT-HLC, or MPT-HLS

Procedure

1 Verify that the MPT-HLC/MPT-HLS PA switch is in the ON ‘1" position.

2 Is the MPT-HLC/MPT-HLS PA switch in the ON ‘1" position?

2.1 If yes, go to step 8.

2.2 If no, go to step 3.

3 Ready to move the MPT-HLC/MPT-HLS PA switch to the ON ‘1" position?

3.1 If yes, go to step 5.

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3.2 If no, go to step 4.

4 When the MPT-HLC/MPT-HLS PA is in the OFF ‘0" position, the Replaceable Unit Problem condition is raised and Tx Mute > Local > Auto > Enable is displayed on the EAS Main View Settings tab.

The procedure is complete.

5 Position the MPT-HLC/MPT-HLS PA switch to the ON ‘1" position.

6 Retrieve the alarms using the Craft Terminal’s Active Alarms.

7 Did the Replaceable Unit Problem alarm clear?

7.1 If yes, the procedure is complete.

7.2 If no, go to step 8.

8 Is the MPT-HLC/MPT-HLS in alarm in a protected system?

8.1 If yes, go to step 9.

8.2 If no, go to step 18.

9 Is the MPT-HLC/MPT-HLS in alarm main or spare?

9.1 If main, go to step 10.

9.2 If spare, go to step 14.

⚠ CAUTION!

Performing a Manual EPS/HSB/Rx Radio switch causes a disruption to traffic.

10 Perform the Forced EPS Switch using the Craft Terminal.

11 Perform the Forced HSB Switch using the Craft Terminal.

12 Perform the Forced Rx Radio Switch using the Craft Terminal.

13 Go to step 17.

14 Perform the Lockout EPS Switch using the Craft Terminal.

15 Perform the Lockout HSB Switch using the Craft Terminal.

16 Perform the Lockout Rx Radio Switch using the Craft Terminal.

17 Verify card in alarm status is *Standby*.

18 On the MPT-HLC with status of Standby, position the PA switch to the OFF ‘0" position.

> **⚠ CAUTION!**
>
> Power cycling an in-service transceiver in an unprotected link causes loss of traffic.

19 Remove the fuse in the power distribution panel that is suppling battery power to the MPT -HLC/MPT-HLS with a status of Standby.

20 Install the fuse in the power distribution panel that is suppling battery power to MPT -HLC/MPT-HLS with a status of Standby.

21 On the MPT-HLC/MPT-HLS with status of Standby, position the PA switch to the ON ‘1" position.

22 Wait for the status LED to turn steady green or amber.

23 Retrieve the alarms using the Craft Terminal’s Active Alarms.

24 Did the Replaceable Unit Problem alarm clear?

24.1 If yes, go to step 28.

24.2 If no, go to step 25.

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25 See Wavence Hardware Installation and Replacement Manual to replace MPT-HLC/MPT-HLS.

> **⚠ CAUTION!**
>
> Removing an in-service transceiver in an unprotected link causes loss of traffic.

26 Retrieve the alarms using the Craft Terminal's Active Alarms.

27 Did the Replaceable Unit Problem alarm clear?

27.1 If yes, go to step 28.

27.2 If no, contact the next level of technical support for assistance.

28 Is the MPT-HLC/MPT-HLS in alarm in a protected system?

28.1 If yes, go to step 29.

28.2 If no, go to step 37.

29 Is the protection scheme configured as Not Revertive?

29.1 If yes, go to step 34.

29.2 If no, go to step 30.

30 Release the Forced/Lockout EPS Switch using the Craft Terminal.

> **⚠ CAUTION!**
>
> Releasing Forced EPS/HSB/Rx Radio switch causes a disruption to traffic.

31 Release the Forced/Lockout HSB Switch using the Craft Terminal.

32 Release the Forced/Lockout Rx Radio Switch using the Craft Terminal.

33 go to step 37.

34 Perform the Manual EPS Switch using the Craft Terminal.

> **⚠ CAUTION!**
>
> Performing a Manual EPS/HSB/Rx Radio switch causes a disruption to traffic.

35 Perform the Manual HSB Switch using the Craft Terminal.

36 Perform the Manual Rx Radio Switch using the Craft Terminal.

37 Verify that the replaced MPT-HLC/MPT-HLS status is *Active*.

38 Verify that the local and far end radio direction are alarm free and error-free.

39 Are the local and far end alarm free and error-free?

39.1 If yes, go to step 40.

39.2 If no, contact the next level of technical support for assistance.

40 Is the protection scheme configured as Not Revertive?

40.1 If yes, go to step 41.

40.2 If no, the procedure is complete.

41 Release the Manual EPS Switch using the Craft Terminal.

42 Release the Manual HSB Switch using the Craft Terminal.

43 Release the Manual Rx Radio Switch using the Craft Terminal.

The procedure is complete.

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83.2.8 Replaceable unit problem on UBT ODU

Procedure

1 Is the UBT ODU in alarm in a protected system?

1.1 If yes, go to step 2.

1.2 If no, go to step 11.

2 Is the UBT ODU in alarm the main or spare?

2.1 If main, go to step 3.

2.2 If spare, go to step 7.

3 Perform the Forced EPS Switch using the Craft Terminal.

> **⚠ CAUTION!**
>
> Forced EPS/HSB/RX Radio switching traffic causes a disruption to traffic.

4 Perform the Forced HSB Switch using the Craft Terminal.

5 Perform the Forced Rx Radio Switch using the Craft Terminal.

6 Go to step 10.

7 Perform the Lockout EPS Switch using the Craft Terminal.

8 Perform the Lockout HSB Switch using the Craft Terminal.

9 Perform the Lockout Rx Radio Switch using the Craft Terminal.

10 Verify that the UBT ODU in the alarm status is *Standby*.

11 Power cycle the UBT ODU in alarm.

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⚠ CAUTION!

Removing an in-service UBT ODU in an unprotected link causes loss of traffic.

12 Retrieve the alarms using the Craft Terminal's Active Alarms.

13 Did the Replaceable Unit Problem alarm clear?

13.1 If yes, go to step 17.

13.2 If no, go to step 14.

14 Replace the UBT ODU the Replaceable Unit Problem is declared against. See the Wavence Hardware Installation and Replacement Manual.

15 Retrieve the alarms using the Craft Terminal's Active Alarms.

16 Did the Replaceable Unit Problem alarm clear?

16.1 If yes, go to step 17.

16.2 If no, contact the next level of technical support for assistance.

17 Is the UBT ODU in alarm in a protected system?

17.1 If yes, go to step 18.

17.2 If no, the procedure is complete.

18 Release the Forced/Lockout EPS Switch using the Craft Terminal.

⚠ CAUTION!

Releasing Forced/Lockout EPS/HSB/RX Radio switch causes a disruption to traffic.

19 Release the Forced/Lockout HSB Switch using the Craft Terminal.

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20 Release the Forced/Lockout Rx Radio Switch using the Craft Terminal.

21 The procedure is complete.

83.2.9 Replaceable unit problem on RPS SFP on UBT ODU or MPT-HLS

Procedure

1 Physically reseat the RPS cable at both ends that are associated with RPS SFP in alarm.

2 Retrieve the alarms using the Craft Terminal's Active Alarms.

3 Did the Replaceable Unit Problem alarm clear?

3.1 If yes, the procedure is complete.

3.2 If no, go to step 4.

4 Physically reseat the RPS SFP in alarm.

5 Retrieve the alarms using the Craft Terminal's Active Alarms.

6 Did the Replaceable Unit Problem alarm clear?

6.1 If yes, the procedure is complete.

6.2 If no, go to step 7.

7 See Wavence Hardware Installation and Replacement Manual to replace the RPS SFP in alarm.

8 Retrieve the alarms using the Craft Terminal's Active Alarms.

9 Did the Replaceable Unit Problem alarm clear?

9.1 If yes, the procedure is complete.

9.2 If no, go to step 10.

10 Replace the RPS cable that is associated with the RPS SFP in alarm.

11 Retrieve the alarms using the Craft Terminal's Active Alarms.

12 Did the Replaceable Unit Problem alarm clear?

12.1 If yes, the procedure is complete.

12.2 If no, go to step 13.

13 See Wavence Hardware Installation and Replacement Manual to replace UBT ODU associated with RPS SFP in alarm.

14 Retrieve the alarms using the Craft Terminal's Active Alarms.

15 Did the Replaceable Unit Problem alarm clear?

15.1 If yes, the procedure is complete.

15.2 If no, contact the next level of technical support for assistance.

83.2.10 Replaceable unit problem on RPS or XPIC module on UBT ODU

Procedure

1 See Wavence Hardware Installation and Replacement Manual to replace RPS/XPIC module in alarm.

2 Retrieve the alarms using the Craft Terminal's Active Alarms.

3 Did the Replaceable Unit Problem alarm clear?

3.1 If yes, the procedure is complete.

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3.2 If no, go to step 4.

4 See Wavence Hardware Installation and Replacement Manual to replace UBT ODU associated with RPS/XPIC module in alarm.

5 Retrieve the alarms using the Craft Terminal's Active Alarms.

6 Did the Replaceable Unit Problem alarm clear?

6.1 If yes, the procedure is complete.

6.2 If no, contact the next level of technical support for assistance.

83.2.11 Replaceable unit problem on Diversity receiver module on MPT-HLC or MPT-HLS

Procedure

1 Is the MPT-HLC/MPT-HLS in alarm in a protected system?

1.1 If yes, go to step 2.

1.2 If no, go to step 11.

2 Is the MPT-HLC/MPT-HLS in alarm main or spare?

2.1 If main, go to step 3.

2.2 If spare, go to step 7.

⚠ CAUTION!

Performing a Manual EPS/HSB/Rx Radio switch causes a disruption to traffic.

3 Perform the Forced EPS Switch using the Craft Terminal.

4 Perform the Forced HSB Switch using the Craft Terminal.

5 Perform the Forced Rx Radio Switch using the Craft Terminal.

6 go to step 10.

7 Perform the Lockout EPS Switch using the Craft Terminal.

8 Perform the Lockout HSB Switch using the Craft Terminal.

9 Perform the Lockout Rx Radio Switch using the Craft Terminal.

10 Verify card in alarm status is *Standby*.

11 On the MPT-HLC/MPT-HLS with status of Standby, position the PA switch to the OFF ‘0" position.

> **⚠ CAUTION!**
>
> Power cycling an in-service transceiver in an unprotected link causes loss of traffic.

12 Remove the fuse in the power distribution panel that is suppling battery power to the MPT -HLC/MPT-HLS with a status of Standby.

13 Install the fuse in the power distribution panel that is suppling battery power to MPT -HLC/MPT-HLS with a status of Standby.

14 On the MPT-HLC/MPT-HLS with status of Standby, position the PA switch to the ON ‘1" position.

15 Wait for the status LED to turn steady green or amber.

16 Retrieve the alarms using the Craft Terminal’s Active Alarms.

17 Did the Replaceable Unit Problem alarm clear?

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17.1 If yes, go to step 21.

17.2 If no, go to step 18.

18 See Wavence Hardware Installation and Replacement Manual to replace MPT-HLC/MPT-HLS.

> **⚠ CAUTION!**
>
> Removing an in-service transceiver in an unprotected link causes loss of traffic.

19 Retrieve the alarms using the Craft Terminal's Active Alarms.

20 Did the Replaceable Unit Problem alarm clear?

20.1 If yes, go to step 21.

20.2 If no, contact the next level of technical support for assistance.

21 Is the MPT-HLC/MPT-HLS in alarm in a protected system?

21.1 If yes, go to step 22.

21.2 If no, go to step 30.

> **Note:**
>
> To verify that the replaced MPT-HLC/MPT-HLS is capable of carrying traffic, the replaced MPT-HLC/MPT-HLS must be placed in the Active state.

22 Is the protection scheme configured as Not Revertive?

22.1 If yes, go to step 27.

22.2 If no, go to step 23.

23 Release the Forced/Lockout EPS Switch using the Craft Terminal.

⚠ CAUTION!

Releasing Forced EPS/HSB/Rx Radio switch causes a disruption to traffic.

24 Release the Forced/Lockout HSB Switch using the Craft Terminal.

25 Release the Forced/Lockout Rx Radio Switch using the Craft Terminal.

26 Go to step 30.

27 Perform the Manual EPS Switch using the Craft Terminal.

⚠ CAUTION!

Performing a Manual EPS/HSB/Rx Radio switch causes a disruption to traffic.

28 Perform the Manual HSB Switch using the Craft Terminal.

29 Perform the Manual Rx Radio Switch using the Craft Terminal.

30 Verify that the replaced MPT-HLC/MPT-HLS status is *Active*.

31 Verify that the local and far end radio direction are alarm free and error-free.

32 Are the local and far end alarm free and error-free?

32.1 If yes, go to step 33.

32.2 If no, contact the next level of technical support for assistance.

33 Is the protection scheme configured as Not Revertive?

33.1 If yes, go to step 34.

33.2 If no, the procedure is complete.

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34 Release the Manual EPS Switch using the Craft Terminal.

35 Release the Manual HSB Switch using the Craft Terminal.

36 Release the Manual Rx Radio Switch using the Craft Terminal.

The procedure is complete.

84. Replaceable unit type mismatch alarms

84.1 Purpose

This procedure describes how to identify and clear Replaceable Unit Type Mismatch alarms.

84.2 General

A Replaceable Unit Type Mismatch alarm indicates that a card is present in a slot/port provisioned for a different card/SFP type.

Note:

If an Enhanced Fan W/Alarms (A-FANS) is equipped and configured as an E-FANS, after an upgrade to R07.01.00 a Replaceable Unit Type Mismatch alarm condition will be raised. While this alarm is standing, the housekeeping alarm inputs and outputs are not managed.

To avoid this condition, perform one of the following prior to the software upgrade:

1. The existing A-FANS must be reconfigured from E-FANS to A-FANS, OR
2. Install an E-FANS unit which will enable the management of the 8 housekeeping input points.

Note that housekeeping alarm configuration parameters will be lost when the E-FANS are reconfigured. It is recommended to record the housekeeping input provisioning before disabling the E-FANS to aid in reconfiguring the housekeeping alarm inputs.

Note:

Disabling a card/SFP will cause a loss of configuration parameters. It is recommended to record card/SFP provisioning prior to disabling the card/SFP to aid in reconfiguring the card/SFP.

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See the Wavence Product Information Manual to determine card location and address and for information about supported temperatures.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Replaceable Unit Type Mismatch alarms provides information about Replaceable Unit Type Mismatch alarms.

Table 77: Replaceable Unit Type Mismatch alarms

Friendly name/entity	Equipment type
r01s1/board#[1-2]	CorEvo
r01s1/board#[5,9]	FAN, A-FAN, or E-FAN
r01s1/board#[3-8]	EASv2, P2E3DS3, P32E1DS1, SDHACC
r01/subrack#[3-8][5-8]	MPT-HLC connected to EASV2
r01/subrack#1[7-8]	MPT-HLC connected to CorEvo
r01s[101-102]/board#[1-10]	MPT-HLS
N/A	UBT-S UBT-m UBT-T UBT-I
r01s1/board#[7-8]	SFP
N/A	Coupling port SFP on: UBT-S UBT-T UBT-m UBT-I
MSS-E/SFP/Slot#[1]/Port#[1-2]	SFP on MSS-E
MSS-HE/XE/SFP/Slot#[1]/Port#[1-2]	SFP on MSS-HE/XE

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84.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Replaceable Unit Type Mismatch alarms for information about equipment alarms.

3 Visually inspect the indicated slot location.

4 Is the card or radio in the indicated slot location the correct card or radio type?

   4.1 If yes, go to step 8.

   4.2 If no, go to step 5.

5 Replace the installed card or radio with the card or radio type provisioned in Core memory. See the Wavence Hardware Installation and Replacement Manual.

6 Retrieve the alarms using the Craft Terminal's Active Alarms.

7 Did the Replaceable Unit Type Mismatch alarm clear?

   7.1 If yes, the procedure is complete.

   7.2 If no, contact the next level of technical support for assistance.

8 Reprovision the slot to the correct card or radio type.

9 Retrieve the alarms using the Craft Terminal's Active Alarms.

10 Did the Replaceable Unit Type Mismatch alarm clear?

   10.1 If yes, the procedure is complete.

   10.2 If no, contact the next level of technical support for assistance.

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85. Retiming buffer overflow alarms

85.1 Purpose

This procedure describes how to identify and clear Retiming Buffer Overflow alarms.

85.2 General

A Retiming Buffer Overflow alarm on channelized E1 indicates overflow or underflow conditions of the buffer used to perform the retiming of the circuit emulated E1 when Node Timing TDM clock source is selected. The alarm is reported if there are no defects related to the Loss of CESoETH Frame alarm or Rx AIS alarm.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence User Manual for information about how to execute the commands that are described in this procedure.

Table: Retiming Buffer Overflow alarms provides information about Retiming Buffer Overflow alarms.

Table 78: Retiming Buffer Overflow alarms

Friendly name/entity	Facility type
E1-in/Slot#[3-8]/Port#[1-32]	E1 on P32E1DS1
E1-out#[1-3]-[1-7]-[1-3]/Slot#[3-8]/Port#1	Channelized E1

85.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Retiming Buffer Overflow alarms for information about facility alarms.

3 Troubleshoot any possible synchronization alarms; see Alarm and abnormal condition clearing procedures.

4 Retrieve the alarms using the Craft Terminal's Active Alarms.

5 Did the synchronization alarms clear?

   5.1 If no, continue troubleshooting the synchronization alarms or contact technical support for assistance.

   5.2 If yes, proceed to step 6.

6 Retrieve the alarms using the Craft Terminal's Active Alarms.

7 Did the Retiming Buffer Overflow alarm clear?

   7.1 If yes, the procedure is complete.

   7.2 If no, contact the next level of technical support for assistance.

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86. RF Switch Fail alarms

86.1 Purpose

This procedure describes how to identify and clear RF Switch Fail alarms.

86.2 General

A RF Switch Fail alarm indicates that a failure is detected on the transmit RF switch.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: RF Switch Fail alarms provides information about RF Switch Fail alarms.

Table 79: RF Switch Fail alarms

Friendly name/entity	Equipment type
r01/subrack#[3-8][5-8]	MPT-HLC connected to EASv2
r01s[101-102]b[1-10]	MPT-HLS
n/a	UBT-I

86.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: RF Switch Fail alarms for information.

3 Inspect the cabling to/from the transmit RF switch:

3.1 MPT-HLC/MPT-HLS shelf switch power connection

3.2 MPT-HLC/MPT-HLS shelf TX main and Tx spare connections

3.3 Transmit RF switch antenna port connection

4 Were issues identified with the cabling to/from the transmit RF switch?

4.1 If yes, go to step 5.

4.2 If no, go to step 9.

5 Mute both the main and spare transmitters.

⚠ CAUTION!

Muting both the main and spare MPT-HLC/MPT-HLS will causes a loss of traffic on ALL facilities that are associated with the MPT-HLC/MPT-HLS with RF Switch Fail alarm. Schedule an appropriate maintenance window per local practices and procedures.

6 Correct all cable and cable connections as required.

7 Retrieve the alarms using the Craft Terminal's Active Alarms.

8 Did the RF Switch Fail alarm clear?

8.1 If yes, go to step 13.

8.2 If no, go to step 10.

9 Mute both the main and spare transmitters.

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⚠ CAUTION!

Muting both the main and spare MPT-HLC/MPT-HLS will causes a loss of traffic on ALL facilities that are associated with the MPT-HLC/MPT-HLS with RF Switch Fail alarm. Schedule an appropriate maintenance window per local practices and procedures.

10 Replace the transmit RF switch.

11 Retrieve the alarms using the Craft Terminal's Active Alarms.

12 Did the RF Switch Fail alarm clear?

12.1 If yes, go to step 13.

12.2 If no, contact the next level of technical support for assistance.

13 Unmute the main transmitter.

14 Verify that the main MPT-HLC/MPT-HLS switch status is Active

15 Verify traffic is running error free at the far end associated with the main transmitter.

16 Unmute the spare transmitter.

17 Perform a Forced HSB Switch using the Craft Terminal.

18 Verify that the spare MPT-HLC/MPT-HLS switch status is Active

19 Verify traffic is running error free at the far end associated with the spare transmitter.

20 Did traffic run error-free on both the main and spare transmitters and is the direction operating normally?

20.1 If yes, go to step 21.

20.2 If no, contact the next level of technical support for assistance.

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21 Release the Forced HSB Switch using the Craft Terminal.

The procedure is complete.

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87. RPS path fail

87.1 Purpose

This procedure describes how to identify and clear an RPS Path Fail alarm.

87.2 General

An RPS Path Fail alarm declared against the MPT-HLC indicates that a failure is detected on the RPS coupler port between the MPT-HLC protection pair.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: RPS path fail alarms provides information about RPS Path Fail alarms.

Table 80: RPS path fail alarms

Friendly name/entity	Equipment type
RadioProt/Dir#1[7-8]	MPT-HLC connected to CorEvo
RadioProt/Dir#[3-8][5-8]	MPT-HLC connected to EASv2

87.3 Procedure

Procedure

1 Use the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: RPS path fail alarms for information.

3 Use the Craft Terminal, determine which MPT-HLC status is Standby.

4 Determine the active radio for the MPT-HLC protection pair declaring the RPS Path Fail alarm.

4.1 If main, go to step step 9. Perform a Lockout EPS Switch using the Craft Terminal.

4.2 If spare, go to step step 5. Perform a Forced EPS Switch using the Craft Terminal.

5 Perform a Forced EPS Switch using the Craft Terminal.

⚠ CAUTION!

Forced/Lockout switching traffic causes a disruption to traffic.

6 Perform a Forced HSB Switch using the Craft Terminal.

7 Perform a Forced Rx Radio Switch using the Craft Terminal.

8 Go to step step 12. On the MPT-HLC with status of Standby, position the PA switch to the OFF ‘0" position.

9 Perform a Lockout EPS Switch using the Craft Terminal.

⚠ CAUTION!

Forced/Lockout switching traffic causes a disruption to traffic.

10 Perform a Lockout HSB Switch using the Craft Terminal.

11 Perform a Lockout Rx Radio Switch using the Craft Terminal.

12 On the MPT-HLC with status of Standby, position the PA switch to the OFF ‘0" position.

13 Remove the fuse in the power distribution panel that is suppling battery power to the MPT -HLC with a status of Standby.

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14 Install the fuse in the power distribution panel that is suppling battery power to MPT-HLC with a status of Standby.

15 On the MPT-HLC with status of Standby, position the PA switch to the ON ‘1" position.

16 Wait for the status LED to turn steady amber.

17 Retrieve the alarms using the Craft Terminal's Active Alarms.

18 Did the RPS Path Fail alarm clear?

18.1 If yes, go to step step 42. Release the Forced/Lockout EPS Switch using the Craft Terminal.

18.2 If no, go to step step 19. Replace the MPT-HLC with status of Standby. See the Wavence Hardware Installation and Replacement Manual to replace MPT-HLC.

19 Replace the MPT-HLC with status of Standby. See the Wavence Hardware Installation and Replacement Manual to replace MPT-HLC.

20 Wait for the status LED to turn steady amber.

21 Retrieve the alarms using the Craft Terminal's Active Alarms.

22 Did the RPS Path Fail alarm clear?

22.1 If yes, go to step step 42. Release the Forced/Lockout EPS Switch using the Craft Terminal.

22.2 If no, go to step step 23. Is the active radio for the MPT-HLC protection pair declaring the RPS Path Fail alarm main or spare?.

23 Is the active radio for the MPT-HLC protection pair declaring the RPS Path Fail alarm main or spare?

23.1 If main, go to step step 24. Perform a Forced EPS Switch using the Craft Terminal.

23.2 If spare, go to step step 28. Perform a Lockout EPS Switch using the Craft Terminal.

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24 Perform a Forced EPS Switch using the Craft Terminal.

> **⚠ CAUTION!**
>
> Forced/Lockout switching traffic causes a disruption to traffic.

25 Perform a Forced HSB Switch using the Craft Terminal.

26 Perform a Forced Rx Radio Switch using the Craft Terminal.

27 Go to step step 31. On the MPT-HLC with status of Standby, position the PA switch to the OFF ‘0" position.

28 Perform a Lockout EPS Switch using the Craft Terminal.

> **⚠ CAUTION!**
>
> Forced/Lockout switching traffic causes a disruption to traffic.

29 Perform a Lockout HSB Switch using the Craft Terminal.

30 Perform a Lockout Rx Radio Switch using the Craft Terminal.

31 On the MPT-HLC with status of Standby, position the PA switch to the OFF ‘0" position.

32 Remove the fuse in the power distribution panel that is suppling the battery power to the MPT-HLC with a status of Standby.

33 Install the use in the power distribution panel that is suppling battery power to the MPT-HLC with a status of Standby.

34 On the MPT-HLC with status of Standby, position the PA switch to the ON ‘1" position.

35 Wait for the status LED to turn steady amber.

36 Retrieve the alarms using the Craft Terminal's Active Alarms.

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37 Did the RPS Path Fail alarm clear?

37.1 If yes, go to step step 42. Release the Forced/Lockout EPS Switch using the Craft Terminal.

37.2 If no, go to step step 38. Replace MPT-HLC with status of Standby. See the Wavence Hardware Installation and Replacement Manual to replace the MPT-HLC.

38 Replace MPT-HLC with status of Standby. See the Wavence Hardware Installation and Replacement Manual to replace the MPT-HLC.

39 Wait for the status LED to turn steady amber.

40 Retrieve the alarms using the Craft Terminal's Active Alarms.

41 Did the RPS Path Fail alarm clear?

41.1 If yes, go to step step 42. Release the Forced/Lockout EPS Switch using the Craft Terminal.

41.2 If no, contact the next level of technical support for assistance.

42 Release the Forced/Lockout EPS Switch using the Craft Terminal.

⚠ CAUTION!

Releasing Forced/Lockout switch may cause a disruption to traffic.

43 Release the Forced/Lockout HSB Switch using the Craft Terminal.

44 Release the Forced/Lockout Rx Radio Switch using the Craft Terminal.

45 The procedure is complete.

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88. Rx diversity missing alarms

88.1 Purpose

This procedure describes how to identify and clear Rx Diversity Missing alarms.

88.2 General

A Rx Diversity Missing alarm indicates that the MPT-HLC or MPT-HLS has been configured for a diversity receiver configuration but the system does not detect the diversity/combiner module on the MPT-HLC or MPT-HLS.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Rx diversity missing alarm provides information about Rx Diversity Missing alarms.

Table 81: Rx diversity missing alarm

Friendly name/entity	Equipment type
r01s1[7-8]b1/daughter#04	MPT-HLC connected to CorEvo
r01/subrack#[3-8][5-8]/daughter#04	MPT-HLC connected to EASv2
r01s[101-102]b[1-10]/daughter#04	MPT-HLS
n/a	UBT-I

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88.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms , retrieve all alarms.

2 Review the output. See Table: Rx diversity missing alarm for information.

3 Using site engineering documentation determine if the radio direction is configured for a diversity receiver?

   3.1 If yes, go to step 8.

   3.2 If no, go to step 4.

4 Using the Craft Terminal, from the radio main view, Settings tab, select the SD check-box (unchecked).

5 Click on the Apply button.

6 Retrieve the alarms using the Craft Terminal's Active Alarms .

7 Did the Rx Diversity Missing alarm clear?

   7.1 If yes, the procedure is complete.

   7.2 If no, contact the next level of technical support for assistance.

8 Using the Craft Terminal, from the remote inventory drop down, determine the MPT-HLC or MPT-HLS type.

9 Is the traffic affected?

   9.1 If yes, go to step 11.

   9.2 If no, go to step 10.

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10 Is it appropriate to proceed with the Rx Diversity Missing alarm procedure?

10.1 If yes, go to step 11.

10.2 If no, contact the next level of technical support for assistance.

11 Is the radio direction in a protected configuration?

11.1 If yes, go to step 12.

11.2 If no, go to step 21.

12 Verify that the MPT-HLC or MPT-HLS in alarm status is *Standby*.

13 Is the MPT-HLC or MPT-HLS in alarm main or spare?

13.1 If main, go to step 14.

13.2 If spare, go to step 18.

14 Perform a Forced EPS Switch using the Craft Terminal.

⚠ CAUTION!

Forced/Lockout switching traffic causes a disruption to traffic.

15 Perform a Forced HSB Switch using the Craft Terminal.

16 Perform a Forced Rx Radio Switch using the Craft Terminal.

17 Go to step 21.

18 Perform a Lockout EPS Switch using the Craft Terminal.

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⚠ CAUTION!

Forced/Lockout switching traffic causes a disruption to traffic.

19 Perform a Lockout HSB Switch using the Craft Terminal.

20 Perform a Lockout Rx Radio Switch using the Craft Terminal.

21 Replace the MPT-HLC or MPT-HLS. Verify that the replacement MPT-HLC or MPT-HLS type supports the diversity receiver as described in site engineering documentation. See the Wavence Hardware Installation and Replacement Manual to replace the MPT-HLC or MPT -HLS.

⚠ CAUTION!

Physically replacing an unprotected MPT-HLC or MPT-HLS will causes a loss of traffic on ALL facilities that are associated with the MPT-HLC or MPT-HLS being replaced. Schedule an appropriate maintenance window per local practices and procedures.

22 Wait for the status LED to turn steady amber or green.

23 Retrieve the alarms using the Craft Terminal's Active Alarms .

24 Did the Rx Diversity Missing alarm clear?

24.1 If yes, go to step 25.

24.2 If no, contact the next level of technical support for assistance.

25 Is the radio direction in a protected configuration?

25.1 If yes, go to step 26.

25.2 If no, the procedure is complete.

26 Release the Forced/Lockout EPS Switch using the Craft Terminal.

27 Release the Forced/Lockout HSB Switch using the Craft Terminal.

28 Release the Forced/Lockout Rx Radio Switch using the Craft Terminal.

The procedure is complete.

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89. Rx Overload alarms

89.1 Procedure

Rx Overload alarm clearing procedure for UBT-S, UBT-m, UBT-T channel A, and UBT-T channel B.

Purpose

To clear the Rx Overload alarm. This alarm indicates that input overload condition at the RF front end is detected.

Procedure

1 Reduce `TX Power`

Reduce `TX Power` of remote transmitter for both RTPC or ATPC cases.

90. Rx problem alarms

90.1 Purpose

This procedure describes how to identify and clear Rx Problem alarms.

90.2 General

A Rx Problem alarm indicates a loss of traffic due to issues at the RX blocks or to malformed packets on the L1 LAG data path.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Rx Problem alarms provides information about Rx Problem alarms.

Table 82: Rx Problem alarms

Friendly name/entity	Equipment type
RadioLAG#[1-14]	MPT-HLC, or MPT-HLS
N/A	RLAG: UBT-T, UBT-S, UBT-m, UBT-I

90.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms , retrieve ALL alarms.

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2 Review the output. See Table: Rx Problem alarms for information.

3 Retrieve the alarms using the Craft Terminal's Active Alarms .

4 Identify the alarms that are declared against members of the LAG group. If alarms exists, see Alarm and abnormal condition clearing procedures and troubleshoot accordingly.

5 Did alarms exist on the LAG members and were they cleared?

5.1 If yes, go to step 6.

5.2 If no, contact the next level of technical support for assistance.

6 Retrieve the alarms using the Craft Terminal's Active Alarms .

7 Did the Rx problem alarm clear?

7.1 If yes, the procedure is complete.

7.2 If no, contact the next level of technical support for assistance.

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91. Secondary battery failure alarms

91.1 Purpose

This procedure describes how to identify and clear power alarms that are associated with the MPT-HLS, MSS-E, MSS-HE, MSS-XE and MSS-8 shelf.

91.2 General

A Secondary Battery Failure alarm indicates that either the B battery feed has failed or is missing, when battery protection is available.

See the Wavence Technical Specifications for Full Indoor Applications for information about supported input voltages.

See the Wavence Product Information Manual to determine card location and address, and for information about supported voltage.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Secondary Battery Failure alarms provides information about Secondary Battery Failure alarms.

Table 83: Secondary Battery Failure alarms

Friendly name/entity	Equipment type
r01s[101-102]board#[1-10]	MPT-HLS
MSS-E (Battery B)	MSS-E
MSS-HE/XE (Battery B)	MSS-HE/XE

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91.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms , retrieve all alarms.

2 Review the output. See Table: Secondary Battery Failure alarms for information.

3 Verify that there is no blown fuse providing power to the B-power feed.

4 Verify the shelf power cable connections between the MSS-8/E/HE/XE or MPT-HLS and the power distribution panel.

5 Verify the B-power connections between the BDFB and power distribution panel.

6 Verify that the B-power BDFB breakers are in the ON position.

7 Retrieve the alarms using the Craft Terminal's Active Alarms .

8 Did the Secondary Battery Failure clear?

   8.1 If yes, the procedure is complete.

   8.2 If no, go to step 9.

9 Measure the voltage of the B-power feed.

10 Is the measured voltage within the supported voltage range?

   10.1 If yes, contact the next level of technical support for assistance.

   10.2 If no, go to step 11.

11 Troubleshoot the B-power feed voltage problem.

12 Retrieve the alarms using the Craft Terminal's Active Alarms .

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13 Did the Secondary Battery Failure condition clear?

13.1 If yes, the procedure is complete.

13.2 If no, contact the next level of technical support for assistance.

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92. Service selective drop alarms

92.1 Purpose

This procedure describes how to identify and clear Service Selective Drop alarms.

92.2 General

A Service Selective Drop alarm indicates an SDH Flow is selectively dropped due to a reduction of Radio L1 LAG throughput.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Service Selective Drop alarms provides information about Service Selective Drop alarms.

Table 84: Service Selective Drop alarms

Friendly name/entity	Equipment type
RadioLAG#[1-14]/SDHFlowID#[2-4080]	STM-1/OC-3

92.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms , retrieve ALL alarms.

2 Review the output. See Table: Service Selective Drop alarms for information.

3 Retrieve the alarms using the Craft Terminal's Active Alarms .

4 Identify the alarms that are declared against members of the LAG group associated with the affected STM-1/OC-3. If alarms exists, see Alarm and abnormal condition clearing procedures and troubleshoot accordingly.

5 Did alarms exist on the LAG members and were they cleared?

5.1 If yes, go to step 6.

5.2 If no, contact the next level of technical support for assistance.

6 Retrieve the alarms using the Craft Terminal's Active Alarms .

7 Did the Service Selective Drop alarm clear?

7.1 If yes, the procedure is complete.

7.2 If no, contact the next level of technical support for assistance.

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93. Signal Label Mismatch alarms

93.1 Purpose

This procedure describes how to identify and clear Signal Label Mismatch alarms.

93.2 General

A high order Signal Label Mismatch alarm on channelized E1 is declared when the value of C2 byte in VC-4 POH is different between received VC-4 and Tx VC-4. The vale of C2 byte should be 0x02 for E1 mapping for channelized SDH. The alarm becomes significant and must be reported if STM-1 LOS, STM-1 LOF, STM-1 TIM, MS-AIS, AU4-AIS or AU4-LOP alarms are not present. As a consequent action, any E1 extracted from STM-1 will be replaced with AIS (and then circuit emulated according the provisioned settings).

A low order Signal Label Mismatch alarm on channelized E1 is declared when the value of V5 [bit5-7] in VC-12 POH is different from the expected value (010: asynchronous mapping). The alarm becomes significant and must be reported if STM-1 LOS, STM-1 LOF, STM-1 TIM, MS-AIS, AU4-AIS, AU4-LOP, VC4-SLM, VC4-LOM, TU12-AIS or TU12- LOP alarms are not present. As a consequent action the relevant E1 will be replaced with AIS (and then circuit emulated according the provisioned settings).

See the Wavence Product Information Manual to determine card location and address.

See the Wavence User Manual for information about how to execute the commands that are described in this procedure.

Table: Signal Label Mismatch alarms provides information about Loss of Pointer alarms.

Table 85: Signal Label Mismatch alarms

Friendly name/entity	Facility type
AU4/Slot#[3-8]/Port#1	Channelized E1
TU12#[1-3]-[1-7]-[1-3]/Slot#[3-8]/Port#1	Channelized E1

93.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Signal Label Mismatch alarms for information.

3 Troubleshoot the source of input STM-1 signal that is associated with the channelized E1 input signal to the SDHCHAN card.

4 Retrieve the alarms using the Craft Terminal's Active Alarms.

5 Did the Signal Label Mismatch alarm clear?

   1. If yes, the procedure is complete.
   2. If no, contact the next level of technical support for assistance.

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94. Software download failed

94.1 Purpose

This procedure describes how to identify and clear Software Download Failed alarm.

94.2 General

This alarm is reported when SFTP failed or due o inernal SW error.

94.3 Procedure

Procedure

1 Redownload the software package off the support portal and retry. Check the network connectivity before trying to redownload from the support portal.

2 Check if the Software package is he correct one for the equipment.

3 If flash usage has been exceeded, contact TAC.

4 If flash corruption has been occurred, contact TAC.

> **Note:**
> Before contacting TAC, download the Zipped Debug File Logs via the Monitoring and Maintenance Debug Screen.

95. Standby version mismatch alarms

95.1 Purpose

This procedure describes how to identify and clear Standby Version mismatch alarms.

95.2 General

A Standby Version Mismatch alarm indicates that the software version on MPT-HLC/MPT-HLS does not match the software version on the CorEvo shelf.

A Standby Version Mismatch alarm indicates that the software version on UBT-T/UBT-S/UBT-m does not match software version on Core.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Standby Version Mismatch alarms provides information about Standby Version Mismatch alarms.

Table 86: Standby Version Mismatch alarms

Friendly name/entity	Equipment type
r01/subrack#1[7-8]	MPT-HLC connected to CorEvo
r01/subrack#[3-8][5-8]	MPT-HLC connected to EASv2
r01s[101-102]/board#[1-10]	MPT-HLS
N/A	UBT-S, UBT-T, UBT-m, UBT-I

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95.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Standby Version Mismatch alarms for information about License/Software Mismatch alarms.

3 Retrieve the alarms using the Craft Terminal's Active Alarms.

4 Verify that the Firmware Download On Going is declared against the MPT-HLC/MPT-HLS.

5 Is the Firmware Download On Going declared against the MPT-HLC/MPT-HLS?

   5.1 If yes, go to step 6.

   5.2 If no, contact the next level of technical support for assistance.

6 A Firmware Download On Going alarm indicates that the CorEvo is downloading the correct software to the MPT-HLC/MPT-HLS. Wait at least 30 minutes for the downloading process to complete.

7 Retrieve the alarms using the Craft Terminal's Active Alarms.

8 Did the Standby Version Mismatch alarm clear?

   8.1 If yes, the procedure is complete.

   8.2 If no, contact the next level of technical support for assistance.

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96. Sync Packet Loss

96.1 Purpose

This procedure describes how to identify and clear Sync Packet Loss alarm.

96.2 General

A Sync Packet Loss alarm indicates that either a Boundary Clock port working as a slave or an OC Slave has stopped receiving PTP Sync messages from either the Boundary Clock port working as a Master or OC Master.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Sync Packet Loss alarms describes the equipment that can generate the Sync Packet Loss alarm.

Table 87: Sync Packet Loss alarms

Friendly name/entity	Facility/equipment
PTPClock/PTPPort#[1-16]/Peer#[1-16]	CorEvo

96.3 Procedure

Procedure

1 Determine the location of either the Boundary Clock or OC Master, and the path from either the Boundary Clock or OC Slave in the alarm, to either the Boundary Clock or OC Master.

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2 Using the Craft Terminal's Active Alarms, retrieve all alarms associated with the path to either the Boundary Clock or OC Master.

3 Verify that no radio path alarms exist along the path to either the Boundary Clock or OC Master. If radio path alarms exist, troubleshoot accordingly.

4 Did radio path alarms exist and were they cleared?

4.1 If yes, go to step 5.

4.2 If no, go to step 7.

5 Retrieve the alarms using the Craft Terminal's Active Alarms.

6 Did the Sync Packet Loss alarm clear?

6.1 If yes, the procedure is complete.

6.2 If no, go to step 7.

7 Using the craft terminal, determine the VLAN associated with either the Boundary Clock or OC Slave. Retrieve all alarms associated with the VLAN along the path from either the Boundary Clock or OC Master, to either the Boundary Clock or OC Slave.

8 Verify that no alarms exist along the path to either the Boundary Clock or OC Master. If alarms exist, troubleshoot accordingly.

9 Did alarms exist and were they cleared?

9.1 If yes, go to step 10.

9.2 If no, go to step 12.

10 Retrieve the alarms using the Craft Terminal's Active Alarms.

11 Did the Sync Packet Loss alarm clear?

11.1 If yes, the procedure is complete.

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11.2 If no, go to step 12.

12 Using the craft terminal, retrieve Boundary Clock, OC Master, and Slave provisioning. Using site documentation determine the correct provisioning. If provisioning errors exist, correct accordingly.

13 Did provisioning errors exist and were they corrected?

13.1 If yes, go to step 14.

13.2 If no, contact the next level of technical support for assistance.

14 Did the Sync Packet Loss alarm clear?

14.1 If yes, the procedure is complete.

14.2 If no, contact the next level of technical support for assistance.

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97. Synchronization signal fail alarms

97.1 Purpose

This procedure describes how to identify and clear Synchronization Signal Fail alarms.

97.2 General

A Synchronization Signal Fail alarm indicates that the received source signal is not suitable as a Sync Source.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Synchronization Signal Fail alarms provides information about Synchronization Signal Fail alarms.

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Table 88: Synchronization Signal Fail alarms

Friendly name/entity	Equipment type
Sync-in/Port#[5-8]	CorEvo
SyncSource/ETH/Slot#1/Port#[1-4,7-8]	CorEvo user Ethernet Port
SyncSource/ETH/Slot#[3-8]/Port#[1-8]	EASv2 user Ethernet Port
Radio/Dir#1.[7-8]/Slot#1/Port#[7-8]/Ch#1	MPT-HLC or MPT-HLS connected to CorEvo
Radio/Dir#[3-8].[5-8]/Slot#[3-8]/Port#[5-8]/Ch#1	MPT-HLC or MPT-HLS connected to EASv2
RadioLAG#[1-14]	MPT-HLC, or MPT-HLS
DS1-in/Slot#[3-8]/Port#[1-32]	DS1 on P32E1DS1
E1-in/Slot#[3-8]/Port#[1-32]	E1 on P32E1DS1
EleSdh-in/slot#[3-8]/Port[1-2]	STM-1/OC-3
OptSdh-in/slot#[3-8]/Port[1-2]	
MS/Slot[3-8]/Port#1	STM-1
N/A	UBT-S UBT-m UBT-T
Sync-in/Port#[1-2]	MSS-E, MSS-HE/XE
Ethernet/Slot#[1]/Port#[1-5]	MSS-E User Ethernet Ports
Ethernet/Slot#[1]/Port#[1-8]	MSS-HE/XE User Ethernet Ports
DS1-in/Slot#[1]/Port#[1-16]	DS1 Tributary on MSS-E, MSS-HE/XE
E1-in/Slot#[1]/Port#[1-16]	E1 Tributary on MSS-E, MSS-HE/XE

97.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

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2 Review the output. See Table: Synchronization Signal Fail alarms for information.

3 Retrieve the alarms using the Craft Terminal's Active Alarms.

4 Verify Degrade, Loss ESMC, or Loss of Signal are not also declared against Sync source declaring Synchronization Signal Fail.

5 Are Degrade, Loss ESMC, or Loss of Signal also declared against synch source?

1. If yes, see Table: Synchronization Signal Fail alarms and clear other Sync alarms before proceeding to step 6.
2. If no, go to step 8.

6 Retrieve the alarms using the Craft Terminal's Active Alarms.

7 Did the Synchronization Signal Fail alarm clear?

1. If yes, the procedure is complete.
2. If no, go to step 8.

8 What is the sync source?

1. If `Sync-in`, go to step 9.
2. If `Radio port`, go to step 14.
3. If `DS1, E1, OC-3 or STM-1 port`, go to step 21.

9 Sync-in

Verify Sync reference source (GPS or other sync source) is functioning properly.

10 Verify Sync-In cabling and connections to sync SFP on CorEvo card or MSS-E/HE/XE shelf (Main and Spare if equipped).

11 Verify Sync cabling and connections to sync-out port on sync source.

12 Retrieve the alarms using the Craft Terminal's Active Alarms.

13 Did the Synchronization Signal Fail alarm clear?

1. If yes, the procedure is complete.
2. If no, go to step 30.

14 Radio port

Retrieve the alarms using the Craft Terminal's Active Alarms.

15 Verify no radio alarms are declared against the radio direction declaring Synchronization Signal Fail.

16 Are radio alarms declared against the radio direction?

1. If yes, see Alarm and abnormal condition clearing procedures to clear radio alarms declared against the radio direction.
2. If no, go to step 17.

17 Verify no other radio path alarms exist. If radio path alarms exist, troubleshoot accordingly.

18 Did the other radio path alarms exist and were they cleared?

1. If yes, go to step 19.
2. If no, go to step 30.

19 Retrieve the alarms using the Craft Terminal's Active Alarms.

20 Did the Synchronization Signal Fail alarm clear?

1. If yes, the procedure is complete.
2. If no, go to step 30.

21 DS1, E1, OC-3 or STM-1 port

Retrieve the alarms using the Craft Terminal's Active Alarms.

22 Verify no LOS alarm exists on sync reference DS1, E1, OC-3, or STM-1 port. If LOS alarm exists, troubleshoot accordingly.

23 Did the an LOS alarm exist on sync reference DS1, E1, OC-3, or STM-1 port and was it cleared?

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1. If yes, go to step 24.
2. If no, go to step 26.

24 Retrieve the alarms using the Craft Terminal's Active Alarms.

25 Did the Synchronization Signal Fail alarm clear?

1. If yes, the procedure is complete.
2. If no, go to step 26.

26 Verify sync reference DS1, E1, STM-1, or OC-3 port frequency is correct. If sync reference frequency is not correct, troubleshoot accordingly.

27 Was a sync reference DS1, E1, STM-1, or OC-3 port frequency problem identified and was it corrected?

1. If yes, go to step 28.
2. If no, go to step 30.

28 Retrieve the alarms using the Craft Terminal's Active Alarms.

29 Did the Synchronization Signal Fail alarm clear?

1. If yes, the procedure is complete.
2. If no, go to step 30.

30 Verify sync source provisioning at both the sync source and the alarming card.

31 Retrieve the alarms using the Craft Terminal's Active Alarms.

32 Did the Synchronization Signal Fail alarm clear?

1. If yes, the procedure is complete.
2. If no, go to step 33.

33 Replace the card the Synchronization Signal Fail alarm is declared against. See the Wavence Hardware Installation and Replacement Manual.

34 Retrieve the alarms using the Craft Terminal's Active Alarms.

35 Did the Synchronization Signal Fail alarm clear?

1. If yes, the procedure is completed.
2. If no, contact the next level of technical support for assistance.

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98. Threshold crossing alarms

98.1 Purpose

This procedure describes how to identify Threshold Crossing Alarms (TCA).

98.2 General

TCA alarms are associated with Performance Monitoring (PM). TCA thresholds are default values or values configured by the user. Depending on the configured TCA threshold values provisioned, the severity of the TCA alarms are determined. Due to the nature and flexibility of TCA alarms, only general recommendations can be provided as a course of action to clear TCA alarms. Follow local practices and procedures in assessing the severity/importance of TCA alarms.

A TCA declared against a MPT-HLC/MPT-HLS indicates that the alarm threshold has been exceeded; applies to the PM parameters with an accumulation period of 15 minutes and/or 24 hours on the MPT-HLC/MPT-HLS. The problem is associated with the radio facility.

A TCA declared against a UBT indicates that the alarm threshold has been exceeded; applies to the PM parameters with an accumulation period of 15 minutes and/or 24 hours on UBT-T (Channel A/B), UBT-S, UBT-m. The problem is associated with the radio facility.

A TCA (on DS1/E1/OptSdh/EleSdh-in) declared against a P32E1DS1/SDHACC/DS1 or E1 on MSS -E/HE/XE shelf indicates that the alarm threshold has been exceeded; applies to the PM parameters with an accumulation period of 15 minutes and/or 24 hours on the P32E1DS1/SDHACC/DS1 or E1 on MSS-E/HE/XE shelf. The problem is associated with the E1, DS1, OC-3, or STM-1 input signal to the P32E1DS1 or SDHACC cards or MSS-E/HE/XE shelf.

A TCA (on DS1/E1/OptSdh/EleSdh-out) declared against a P32E1DS1/SDHACC/DS1 indicates that the alarm threshold has been exceeded; applies to the PM parameters with an accumulation period of 15 minutes and/or 24 hours on the P32E1DS1/SDHACC/DS1. The problem is associated with the radio/Ethernet facility associated to the P32E1DS1 or SDHACC cards or MSS-E, MSS -HE/XE Synchronization Signal Fail.

Single level threshold crossing mechanism is utilized with 24 hour PM TCA alarms. TCA alarm is

raised when 24 hour threshold is exceeded. Implicit clearing of 24 hour TCA alarms occurs at the beginning of each 24 hour period. No TCA alarm clear message is sent by the NE.

Dual level threshold crossing mechanism is utilized with 15 minute PM TCA alarms. A TCA alarm is raised when the high threshold is exceeded. An explicit clearing of the TCA alarm occurs at the end of a 15 minute period where the current data value does not exceed the low threshold, and the monitoring period does not contain a suspect interval or an unavailable period. When this occurs, the NE sends TCA alarm clear message.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Threshold crossing alarms provides information about Threshold Crossing alarms.

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Table 89: Threshold crossing alarms

Friendly name/entity	Equipment type
Hop/Dir#1[7-8]/Slot#1/Port#[7-8]/Ch#1-PM15min	MPT-HLC or MPT-HLS connected to CorEvo
Hop/Dir#1[7-8]/Slot#1/Port#[7-8]/Ch#1-PM1day	
Link/Dir#1[7-8]-PM15min	
Link/Dir#1[7-8]-PM1day	
Hop/Dir#[3-8][5-8]/Slot#[3-8]/Port#[5-8]/Ch#[0-1]-PM15min	MPT-HLC or MPT-HLS connected to EASv2
Hop/Dir#[3-8][5-8]/Slot#[3-8]/Port#[5-8]/Ch#[0-1]-PM1day	
Link/Dir#[3-8][5-8]-PM15min	
Link/Dir#[3-8][5-8]-PM1day	
DS1-in/Slot#[3-8]/Port#[1-32]-PM15min	DS1 on P32E1DS1
DS1-in/Slot#[3-8]/Port#[1-32]-PM1day	
DS1-out/Slot#[3-8]/Port#[1-32]-PM15min	
DS1-out/Slot#[3-8]/Port#[1-32]-PM1day	
E1-in/Slot#[3-8]/Port#[1-32]-PM15min	E1 on P32E1DS1
E1-in/Slot#[3-8]/Port#[1-32]-PM1day	
E1-out/Slot#[3-8]/Port#[1-32]-PM15min	
E1-out/Slot#[3-8]/Port#[1-32]-PM1day	
EleSdh-in/Slot#[3-8]/Port#[1-2]-PM15min	STM-1/OC-3
EleSdh-in/Slot#[3-8]/Port#[1-2]-PM1day	
EleSdh-out/Slot#[3-8]/Port#[1-2]-PM15min	
EleSdh-out/Slot#[3-8]/Port#[1-2]-PM1day	
OptSdh-in/Slot#[3-8]/Port#[1-2]-PM15min	
OptSdh-in/Slot#[3-8]/Port#[1-2]-PM1day	
OptSdh-out/Slot#[3-8]/Port#[1-2]-PM15min	
OptSdh-out/Slot#[3-8]/Port#[1-2]-PM1day	

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Friendly name/entity	Equipment type
N/A	UBT-S, UBT-m, UBT-T channel A, UBT-T channel B, UBT-I
E1-out/Slot#[1]/Port#[1-16]-PM15min	E1 on MSS-E, MSS-HE/XE
E1-out/Slot#[1]/Port#[1-16]-PM1day	
E1-in/Slot#[1]/Port#[1-16]-PM15min	
E1-in/Slot#[1]/Port#[1-16]-PM1day	
DS1-out/Slot#[1]/Port#[1-16]-PM15min	DS1 on MSS-E, MSS-HE/XE
DS1-out/Slot#[1]/Port#[1-16]-PM1day	
DS1-in/Slot#[1]/Port#[1-16]-PM15min	
DS1-in/Slot#[1]/Port#[1-16]-PM1day	

98.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Threshold crossing alarms for information about TCA alarms.

3 Verify the PM data using the Craft Terminal.

4 Troubleshoot any of the following:

   1. TCA on radio facility
   2. TCA on DS1/E1/OptSdh/EleSdh-in
   3. TCA on DS1/E1/OptSdh/EleSdh-out

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98.3.1 TCA on radio facility

Procedure

1 Verify the radio channel power levels.

2 Retrieve the alarms using the Craft Terminal's Active Alarms.

3 Determine the local weather conditions that could affect radio performance.

4 Following local practices and procedures, determine whether corrective actions are required.

5 Are corrective actions required?

   5.1 If yes, go to step 6.

   5.2 If no, the procedure is complete.

6 Perform corrective actions per local practices and procedures.

7 Retrieve the alarms using the Craft Terminal's Active Alarms.

8 Did the Threshold Crossed alarm clear?

   8.1 If yes, the procedure is complete.

   8.2 If no, contact the next level of technical support for assistance.

98.3.2 TCA on DS1/E1/OptSdh/EleSdh-in

Procedure

1 Troubleshoot the source of input that is associated with the input signal to the P32E1DS1 or SDHACC cards.

2 Verify the cabling that is associated with the input signal to the P32E1DS1 or SDHACC cards.

3 Retrieve the alarms using the Craft Terminal's Active Alarms.

4 Did the TCA (on DS1/E1/OptSdh/EleSdh-in) alarm clear?

1. If yes, the procedure is complete.
2. If no, contact the next level of technical support for assistance.

98.3.3 TCA on DS1/E1/OptSdh/EleSdh-out

Procedure

1 Verify the radio channel power levels associated with the facility declaring TCA.

2 Retrieve the alarms using the Craft Terminal's Active Alarms.

3 Determine the local weather conditions that could affect radio performance.

4 Following local practices and procedures, determine whether corrective actions are required.

5 Are corrective actions required?

5.1 If yes, go to step 6.

5.2 If no, the procedure is complete.

6 Perform corrective actions per local practices and procedures.

7 Retrieve the alarms using the Craft Terminal's Active Alarms.

8 Did the Threshold Crossed alarm clear?

8.1 If yes, the procedure is complete.

8.2 If no, contact the next level of technical support for assistance.

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99. Trace identifier mismatch alarms

99.1 Purpose

This procedure describes how to identify and clear Trace Identifier Mismatch alarms.

99.2 General

A Trace Identifier Mismatch indicates a J0 byte value received on the STM-1/OC-3 facility is different from the provisioned expected value for the STM-1/OC-3 facility.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence User Manual for information about how to execute the commands that are described in this procedure.

Table: Trace Identifier Mismatch alarms provides information about Trace Identifier Mismatch alarms.

Table 90: Trace Identifier Mismatch alarms

Friendly name/entity	Facility type
EleSdh-in/Slot#[3-8]/Port[1-2]	STM-1, OC-3
OptSdh-in/Slot#[3-8]/Port[1-2]	
MS/Slot#[3-8]/Port#1	STM-1

99.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve all alarms.

2 Review the output. See Table: Trace Identifier Mismatch alarms for information about facility alarms.

3 Using the Craft Terminal, retrieve the provisioned Trace Identifier values for the alarmed STM-1/OC-3 facility.

> **Note:**
> The section trace identifier (J0) value is not generated within the Wavence network. The received J0 value is verified against the provisioned expected value. To determine the provisioned J0 value the operator must query the originating SDH/OC-3 equipment.

4 Determine the origination of the STM-1/OC-3 facility and retrieve the provisioned Trace Identifier parameters.

5 Verify that the desired Trace Identifier parameters of the STM-1/OC-3 facility.

6 Determine which Trace Identifier parameters are incorrect.

7 Provision the Trace Identifier parameters according to the results of step 6.

8 Retrieve the alarms using the Craft Terminal's Active Alarms.

9 Did the Trace Identifier Mismatch alarm clear?

   1. If yes, the procedure is complete.
   2. If no, contact the next level of technical support for assistance.

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100. Traffic down alarms

100.1 Purpose

This procedure describes how to identify and clear Traffic Down alarms.

100.2 General

A Traffic Down alarm indicates a loss of traffic and the L1 LAG data path is unavailable.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Traffic Down alarms provides information about Traffic Down alarms.

Table 91: Traffic Down alarms

Friendly name/entity	Equipment type
RadioLAG#[1-14]	MPT-HLC, or MPT-HLS RLAG: UBT-T, UBT-S, UBT-m, UBT-I

100.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve ALL alarms.

2 Review the output. See Table: Traffic Down alarms for information about LAG alarms.

3 Retrieve the alarms using the Craft Terminal's Active Alarms.

4 Identify the alarms that are declared against members of the LAG group. If alarms exists, see Alarm and abnormal condition clearing procedures and troubleshoot accordingly.

5 Did alarms exist on the LAG members and were they cleared?

5.1 If yes, go to step 6.

5.2 If no, contact the next level of technical support for assistance.

6 Retrieve the alarms using the Craft Terminal's Active Alarms.

7 Did the traffic down alarm clear?

7.1 If yes, the procedure is complete.

7.2 If no, contact the next level of technical support for assistance.

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101. Traffic loss alarms

101.1 Purpose

This procedure describes how to identify and clear Traffic Loss alarms.

101.2 General

A Traffic Loss alarm indicates packet loss due to extreme latency difference between L1 LAG members or when traffic errors occur on the L1 LAG data path.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Traffic Loss alarms provides information about Traffic Loss alarms.

Table 92: Traffic Loss alarms

Friendly name/entity	Equipment type
RadioLAG#[1-14]	MPT-HLC, or MPT-HLS RLAG: UBT-T, UBT-S, UBT-m, UBT-I

101.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms, retrieve ALL alarms.

2 Review the output. See Table: Traffic Loss alarms for information.

3 Using site documentation determine the expected LAG membership for both ends of the LAG.

4 Using the Craft Terminal, verify LAG membership for both ends of the LAG.

5 Does the LAG membership agree with site documentation and does the LAG membership align at both ends of the LAG?

5.1 If yes, go to step 9.

5.2 If no, go to step 6.

6 Determine the desired LAG membership and provision LAG membership accordingly.

7 Retrieve the alarms using the Craft Terminal's Active Alarms.

8 Did the traffic loss alarm clear?

8.1 If yes, the procedure is complete.

8.2 If no, go to step 9.

9 Retrieve the alarms using the Craft Terminal's Active Alarms.

10 Identify the alarms that are declared against members of the LAG group. If alarms exists, see Alarm and abnormal condition clearing procedures and troubleshoot accordingly.

11 Did alarms exist on LAG members and were they cleared?

11.1 If yes, go to step 12.

11.2 If no, contact the next level of technical support for assistance.

12 Retrieve the alarms using the Craft Terminal's Active Alarms.

13 Did the traffic loss alarm clear?

13.1 If yes, the procedure is complete.

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13.2 If no, contact the next level of technical support for assistance.

102. Unavailable time alarms

102.1 Purpose

This procedure describes how to identify and clear Unavailable Time alarms.

102.2 General

An Unavailable Time alarm is declared against a MPT-HLC/MPT-HLS/P32E1DS1/SDHACC/DS1 which indicates that ten consecutive SES events have occurred; applies to the PM parameters on the MPT-HLC/MPT-HLS/P32E1DS1/SDHACC/DS1 on MSS-E or MSS-HE/XE shelf.

An Unavailable Time alarm is declared against a UBT-T (Channel A/B), UBT-S, UBT-m indicates that ten consecutive SES events have occurred.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Unavailable Time alarms provides information about Unavailable Time alarms.

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Table 93: Unavailable Time alarms

Friendly name/entity	Equipment type
Hop/Dir#1[7-8]/Slot#1/Port#[7-8]/Ch#1-PM15min	MPT-HLC or MPT-HLS connected to CorEvo
Hop/Dir#1[7-8]/Slot#1/Port#[7-8]/Ch#1-PM1day	
Link/Dir#1[7-8]/Slot#1/Port#[7-8]/Ch#1-PM15min	
Link/Dir#1[5-6]/Slot#1/Port#[5-6]/Ch#1-PM1day	
Hop/Dir#[3-8][5-8]/Slot#[3-8]/Port#[5-8]/Ch#[0-1]-PM15min	MPT-HLC or MPT-HLS connected to EASv2
Hop/Dir#[3-8][5-8]/Slot#[3-8]/Port#[5-8]/Ch#[0-1]-PM1day	
Link/Dir#[3-8][5-8]-PM15min	
Link/Dir#[3-8][5-8]-PM1day	
DS1-in/Slot#[3-8]/Port#[1-32]-PM15min	DS1 on P32E1DS1
DS1-in/Slot#[3-8]/Port#[1-32]-PM1day	
DS1-out/Slot#[3-8]/Port#[1-32]-PM15min	
DS1-out/Slot#[3-8]/Port#[1-32]-PM1day	
E1-in/Slot#[3-8]/Port#[1-32]-PM15min	E1 on P32E1DS1
E1-in/Slot#[3-8]/Port#[1-32]-PM1day	
E1-out/Slot#[3-8]/Port#[1-32]-PM15min	
E1-out/Slot#[3-8]/Port#[1-32]-PM1day	
EleSdh-in/Slot#[3-8]/Port#[1-2]-PM15min	STM-1/OC-3
EleSdh-in/Slot#[3-8]/Port#[1-2]-PM1day	
EleSdh-out/Slot#[3-8]/Port#[1-2]-PM15min	
EleSdh-out/Slot#[3-8]/Port#[1-2]-PM1day	
OptSdh-in/Slot#[3-8]/Port#[1-2]-PM15min	
OptSdh-in/Slot#[3-8]/Port#[1-2]-PM1day	
OptSdh-out/Slot#[3-8]/Port#[1-2]-PM15min	
OptSdh-out/Slot#[3-8]/Port#[1-2]-PM1day	

Friendly name/entity	Equipment type
N/A	UBT-S, UBT-m, UBT-T channel A, UBT-T channel B, UBT-I
E1-out/Slot#[1]/Port#[1-16]-PM15min	E1 on MSS-E, MSS-HE/XE
E1-out/Slot#[1]/Port#[1-16]-PM1day	
E1-in/Slot#[1]/Port#[1-16]-PM15min	
E1-in/Slot#[1]/Port#[1-16]-PM1day	
DS1-out/Slot#[1]/Port#[1-16]-PM15min	DS1 on MSS-E, MSS-HE/XE
DS1-out/Slot#[1]/Port#[1-16]-PM1day	
DS1-in/Slot#[1]/Port#[1-16]-PM15min	
DS1-in/Slot#[1]/Port#[1-16]-PM1day	

102.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms , retrieve all alarms.

2 Review the output. See Table: Unavailable Time alarms for information.

3 Verify the PM data using the Craft Terminal.

4 Verify the radio channel power levels.

5 Retrieve the alarms using the Craft Terminal's Active Alarms .

6 Determine local weather conditions that could affect radio performance.

7 Following local practices and procedures, determine whether corrective actions are required.

8 Are corrective actions required?

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8.1 If yes, go to step 9.

8.2 If no, the procedure is complete.

9 Perform corrective actions per local practices and procedures.

10 Retrieve the alarms using the Craft Terminal's Active Alarms .

11 Did the Unavailable Time alarm clear?

11.1 If yes, the procedure is complete.

11.2 If no, contact the next level of technical support for assistance.

103. Tx problem alarms

103.1 Purpose

This procedure describes how to identify and clear Tx Problem alarms.

103.2 General

A Tx Problem alarm indicates a loss of traffic due to issues at the TX blocks or to malformed packets on the L1 LAG data path.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Tx Problem alarms provides information about Tx Problem alarms.

Table 94: Tx Problem alarms

Friendly name/entity	Equipment type
RadioLAG#[1-14]	MPT-HLC, or MPT-HLS
N/A	RLAG: UBT-T, UBT-S, UBT-m, UBT-I

103.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms , retrieve all alarms.

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2 Review the output. See Table: Tx Problem alarms for information.

3 Retrieve the alarms using the Craft Terminal's Active Alarms .

4 Identify the alarms that are declared against members of the LAG group. If alarms exists, see Alarm and abnormal condition clearing procedures and troubleshoot accordingly.

5 Did the alarms exist on the LAG members and were they cleared?

5.1 If yes, go to step 6.

5.2 If no, contact the next level of technical support for assistance.

6 Retrieve the alarms using the Craft Terminal's Active Alarms .

7 Did the Tx problem alarm clear?

7.1 If yes, the procedure is complete.

7.2 If no, contact the next level of technical support for assistance.

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104. Unconfigured equipment present alarms

104.1 Purpose

This procedure describes how to identify and clear Unconfigured Equipment present alarms on cards and radios.

104.2 General

An Unconfigured Equipment Present alarm indicates that a card/SFP is present in an unprovisioned slot/port.

See the Wavence Product Information Manual to determine card location and address and for information about supported temperatures.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Unconfigured equipment present alarms provides information about Unconfigured Equipment Present alarms.

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Table 95: Unconfigured equipment present alarms

Friendly name/entity	Equipment type
r01s1/board#[2]	CorEvo
r01s1/board#[5,9]	FAN, E-FAN, or A-FAN
r01s1/board#[3-8]	EASv2, P2E3DS3, P32E1DS1, SDHACC
r01s1b[1-2]/daughter#[05-08]	SFP on CorEvo
r01s1b[3-8]/daughter#[05-08]	SFP on EASv2
r01s1b[3-8]/daughter#[01-02]	SFP on SDHACC
MSS-E/Slot#[1]/Port#[1-2]	SFP on MSS-E
MSS-HE/XE/Slot#[1]/Port#[1-6]	SFP on MSS-HE/XE

104.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms , retrieve all alarms.

2 Review the output. See Table: Unconfigured equipment present alarms for information.

3 Visually inspect the indicated slot/SFP location.

4 Is the indicated slot location equipped with a card/SFP?

4.1 If yes, go to step 5.

4.2 If no, contact the next level of technical support for assistance.

5 Is the card/SFP that is equipped in the slot location expected to be there?

5.1 If yes, go to step 6.

5.2 If no, go to step 9.

6 Provision the slot to the correct card/SFP type.

7 Retrieve the alarms using the Craft Terminal's Active Alarms .

8 Did the Unconfigured Equipment Present alarm clear?

8.1 If yes, the procedure is complete.

8.2 If no, contact the next level of technical support for assistance.

9 Remove the card/SFP from the slot.

10 Install the blank filler panel or face plate into the slot that the card was removed.

11 Retrieve the alarms using the Craft Terminal's Active Alarms .

12 Did the Unconfigured Equipment Present alarm clear?

12.1 If yes, the procedure is complete.

12.2 If no, contact the next level of technical support for assistance.

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105. Underlying resource unavailable alarms

105.1 Purpose

This procedure describes how to identify and clear Underlying Resource Unavailable alarms.

105.2 General

An Underlying Resource Unavailable alarm indicates that transmission resources are affected by an equipment failure.

See the Wavence Product Information Manual to determine card location and address and for information about supported temperatures.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Underlying Resource Unavailable alarms provides information about Underlying Resource Unavailable alarms.

Table 96: Underlying Resource Unavailable alarms

Friendly name/entity	Equipment type
ETH/Slot#1/Port#[1-8]	User Ethernet port on CorEvo
ETH/Slot#[3-8]/Port#[1-8]	User Ethernet port on EASv2
r01s1b[1-2]/daughter#[05-08]	SFP User Ethernet port on CorEvo
r01s1b[3-8]/daughter#[05-08]	SFP User Ethernet port on EASv2
E1-in#[1-3]-[1-7]-[1-3]/Slot#[3-8]/Port#1	Channelized E1
E1-in/Slot#[3-8]/Port#[1-32]	E1 on P32E1DS1
EleSdh-in/Slot#[3-8]/Port#[1-2]	STM-1
OptSdh-in/Slot#[3-8]/Port#[1-2]	
Radio/Dir#1.[7-8]/Slot#1/Port#[7-8]/Ch#[0-1]	MPT-HLC or MPT-HLS connected to CorEvo
Radio/Dir#[3-8].[5-8]/Slot#[3-8]/Port#[5-8]/Ch#[0-1]	MPT-HLC or MPT-HLS connected to EASv2

105.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms , retrieve all alarms.

2 Review the output. See Table: Underlying Resource Unavailable alarms for information.

3 Retrieve the alarms using the Craft Terminal's Active Alarms .

4 Evaluate and troubleshoot other alarms on the equipment reporting the Underlying Resource Unavailable alarm.

5 Did the Underlying Resource Unavailable alarm clear?

   1. If yes, the procedure is complete.

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2. If no, contact the next level of technical support for assistance.

106. Underlying resource unavailable service alarms

106.1 Purpose

This procedure describes how to identify and clear Underlying Resource Unavailable Service alarms.

106.2 General

An Underlying Resource Unavailable Service alarm indicates that transmission resources are affected by an equipment failure.

See the Wavence Product Information Manual to determine card location and address and for information about supported temperatures.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Underlying Resource Unavailable Service alarms provides information about Underlying Resource Unavailable Service alarms.

Table 97: Underlying Resource Unavailable Service alarms

Friendly name/entity	Equipment type
E1-in#[1-3]-[1-7]-[1-3]/Slot#[3-8]/Port#1	Channelized E1
E1-in/Slot#[3-8]/Port#[1-32]	E1 on P32E1DS1
Radio/Dir#1.[7-8]/Slot#1/Port#[7-8]/Ch#[0-1]	MPT-HLC or MPT-HLS connected to CorEVO
Radio/Dir#[3-8].[5-8]/Slot#[3-8]/Port#[5-8]/Ch#[0-1]	MPT-HLC or MPT-HLS connected to EASv2

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106.3 Procedure

Procedure

1 Using the Craft Terminal's Active Alarms , retrieve all alarms.

2 Review the output. See Table: Underlying Resource Unavailable Service alarms for information about equipment alarms.

3 Retrieve the alarms using the Craft Terminal's Active Alarms .

4 Evaluate and troubleshoot other alarms on the equipment reporting the Underlying Resource Unavailable service alarm.

5 Did the Underlying Resource Unavailable service alarm clear?

   1. If yes, the procedure is complete.
   2. If no, contact the next level of technical support for assistance.

107. Version mismatch alarms

This procedure describes how to identify and clear Version Mismatch alarms.

107.1 General

A Version Mismatch alarm declared against a CorEvo indicates that the software license string is not loaded on the CorEvo card.

A Version Mismatch alarm declared against the MPT-HLC/MPT-HLS/UBT indicates that software version on the MPT-HLC/MPT-HLS/UBT-T/UBT-S/UBT-m does not match the software version on the CorEvo.

See the Wavence Product Information Manual to determine card location and address.

See the Wavence WebCT User Manual for information about how to execute the commands that are described in this procedure.

Table: Version Mismatch alarms provides information about Version Mismatch alarms.

Table 98: Version Mismatch alarms

Friendly name/entity	Equipment type	See
r01/subrack#1[7-8]	MPT-HLC connected to CorEvo	Version mismatch (Software)
r01/subrack#[3-8][5-8]	MPT-HLC connected to EASv2	
r01s[101-102]/board#[1-10]	MPT-HLS	
N/A	UBT-S, UBT-m, UBT-T, UBT-I	

107.2 Procedure

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Procedure

1 Using the Craft Terminal's Active Alarms , retrieve all alarms.

2 Review the output. See Table: Version Mismatch alarms for information about License/Software Mismatch alarms.

3 Troubleshoot any of the following:

   1. Version mismatch (License)
   2. Version mismatch (Software)

107.2.1 Version mismatch (License)

Procedure

1 Retrieve the alarms using the Craft Terminal's Active Alarms .

2 Verify the license string that is installed on the NE. Contact the next level of technical support for assistance.

107.2.2 Version mismatch (Software)

Procedure

1 Retrieve the alarms using the Craft Terminal's Active Alarms .

2 Verify that a Firmware Download On Going alarm is declared against the MPT-HLC/MPT-HLS.

3 Is the Firmware Download On Going declared against the MPT-HLC/MPT-HLS?

   3.1 If yes, go to step 4.

   3.2 If no, contact the next level of technical support for assistance.

4 A Firmware Download On Going alarm indicates the CorEvo/MSS/CORE is downloading the correct software to the MPT-HLC/MPT-HLS. Wait at least 30 minutes for the downloading process to complete.

5 Retrieve the alarms using the Craft Terminal's Active Alarms .

6 Did the Version Mismatch or Standby Version Mismatch alarm clear?

6.1 If yes, the procedure is complete.

6.2 If no, contact the next level of technical support for assistance.

107.3 XPIC Cable Loss

107.4 Purpose

This procedure describes how to identify and clear XPIC Cable Loss alarm.

107.5 General

This alarm occurs, when a XPIC cable connection defect is detected. In cases where both radios in a mated XPIC pair are reporting the alarm, the XPIC auto-mute procedure will auto-mute the far -end Horizontal Channel (so the Vertical channel will continue to operate without Horizontal channel interference). In cases where only one radio in a mated XPIC pair is reporting the alarm, the XPIC auto-mute procedure will not occur (the radio without the alarm should be operating and canceling the Cross-polarization correctly).

To troubleshoot, check which radios have the cable loss alarm.

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107.6 Procedure

Procedure

1 Check which radios have the cable loss alarm.

2 Replace the XPIC cable between the radios and the cable.

3 If only one of the radios in a mated XPIC pair is reporting the alarm it can be a unidirectional cable problem or an XPIC hardware failure on the radio reporting the alarm. In this case, contact TAC.